PROJECT TECHNICAL DOCUMENTATION

BioSpecInfo

Web platform for computational chemistry, physics and mathematics

Full dossier

Complete collection of the project documents, in numbering order. Each document is also available as a separate PDF.

DOCUMENT CONTROL

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BioSpecInfo — Technical Dossier

Open-access web platform for computational chemistry and the interactive consultation of chemical data on any device.

Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Profile	Cheminformatics Engineer / Scientific Software Developer
Software type	Progressive Web App (PWA), client-side, local-first, offline-first
Live demo	https://samupropio1-ship-it.github.io/BioSpecInfo-v11/
Repository	github.com/samupropio1-ship-it/BioSpecInfo-v11

> This dossier summarises the application and is the cover of the full technical documentation (SAD, Validation & Benchmark, Security & Compliance, Licensing) attached below.

1. Executive summary

BioSpecInfo is an **interactive cheminformatics platform** that runs all of its scientific logic **in the client browser**, with no server: molecular analysis, spectral prediction, 2D/3D modelling and educational tools — all **free, installable as an app and working offline**.

Its distinctive value is the **rare combination of advanced chemistry and modern software engineering** (WebAssembly, WebGL, PWA), with spectral predictors **quantitatively validated** against the official SDBS and NIST databases.

2. Product factsheet

Value proposition — A **≈4.5 MB** web platform (main file) that runs molecular analysis, spectral prediction and 3D visualisation at **zero server cost** and **100% offline-ready**.

Technical architecture	Content & capabilities
Client-side / Local-First (no backend)	**RDKit** cheminformatics engine (WebAssembly)
Single-file HTML5 / ES6, installable as **PWA**	**Spectra Centre**: IR, $^{1}H/^{13}C$-NMR, MS + isotope pattern, UV-Vis, Raman
2D (SVG/Canvas) and **3D** (three.js, 3Dmol.js) rendering	**Organic Chemistry**: 25 interactive modules
Service Worker offline-first, versioned	**Retrosynthesis**: 46 strategies + animated 2D/3D mechanisms

Technical architecture	Content & capabilities
Zero copyleft dependencies (MIT/BSD-3 only)	**3D Drug Atlas**: 63 diseases (23 tumours), 3D organs, PubMed
Local storage (localStorage, SQLite via sql.js)	3D astrochemistry, Quantum Lab, Academy (SM-2 quizzes)

Quantitative validation (thesis benchmark)

- **^{1}H-NMR** vs SDBS → **MAE 0.31 ppm** (89% of signals within ±0.5 ppm)
- **IR (C=O band)** vs NIST WebBook → **MAE 11 cm^{-1}** (90% within ±30 cm^{-1})
- Interactive performance **30–60 fps** from iPhone/iPad to Raspberry Pi 4

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3. Verified capabilities (current code state)

Spectra Centre (molecular editor & analysis) — draw from scratch (full periodic table, templates, drag-to-bond) or load via SMILES/library →

- All chemical notations: formula, canonical SMILES, CXSMILES, SMARTS, InChI,

InChIKey, mol block, common/IUPAC name, CAS (one-click copy).

- Predicted spectra with bibliographic sources; **isotope pattern** (IUPAC

abundances) and MS **adduct masses**.

- Functional groups and properties (H donors/acceptors, aromatic, rotatable,

stereocentres) **highlightable on the 2D and 3D structure**.

- Drug-likeness: **Lipinski, Veber, Egan, Ghose, Muegge**, ESOL solubility,

QED, **Brenk/PAINS structural alerts**, physicochemical radar.

- **Tanimoto** similarity (Morgan ECFP4 fingerprint), elemental analysis, SMARTS

substructure search, SVG/PNG/MOL export.

Synthesis & Retrosynthesis — 46 strategies with an **animated mechanism player** in **2D and 3D**: *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

Organic Chemistry — 25 modules (fundamentals → heterocycles, pericyclic & Woodward–Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups).

Data Science — 3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), ML labs, SQL playground.

Other — 3D astrochemistry, Quantum Lab, Academy with spaced repetition (SM-2), interactive simulations, databases of elements/drugs/biomolecules.

> Overall metrics declared in the thesis (historical data assets): 324 organic reactions in vector SVG graphics, 26 biosynthetic pathways, 140+ drugs, 80+ biomolecules, 118 elements, 10 quantum simulators.

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4. Technology stack

`HTML5` · `ES6/ESNext` · `WebAssembly` (RDKit MinimalLib, sql.js/SQLite) · `WebGL` (three.js, 3Dmol.js) · `SVG` / `Canvas 2D` · `Service Worker` / `PWA` · public API integration (PubChem PUG-REST, NIST, SDBS, MassBank, nmrshiftdb2, NASA/ESA).

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5. Author's declaration (self-attestation)

The undersigned **Samuele Pio Provenzano** declares that:

1. The software **BioSpecInfo** is the author's **original work**, except for the

third-party open-source libraries (RDKit, 3Dmol.js, three.js, SmilesDrawer, sql.js) used under their respective permissive licenses and documented in the *Open-Source Licensing & IP Matrix*.

1. The reported validation data (MAE 0.31 ppm for ^{1}H-NMR vs SDBS; MAE 11 cm^{-1} for

the IR C=O band vs NIST) were measured by the author as part of the thesis work; the predictors are **heuristic group-additivity** models, suitable for educational use and qualitative functional-group identification, with the limitations stated in the *Validation Report*.

1. The architecture is **local-first**: no user personal data is transmitted to

author-controlled servers (alignment with GDPR principles).

> This is an **author's declaration**, not a certification issued by a third party.

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6. Attached documentation

1. **Software Architecture Document (SAD)** — architecture, components, technologies.
2. **Verification & Validation & Benchmark Report** — test method, benchmarks, limitations.
3. **Security, Privacy & Compliance** — local-first, GDPR, OWASP, GAMP 5 (alignment).
4. **Open-Source Licensing & IP Matrix** — dependencies, licenses, intellectual property.

Software license: proprietary — "All rights reserved" (see `LICENSE`).

Software Architecture Document (SAD) — BioSpecInfo

Field	Value
Project	BioSpecInfo
Author	Samuele Pio Provenzano
Thesis advisor	Prof. Savino Longo — University of Bari Aldo Moro
Type	Progressive Web App (PWA), client-side, local-first, offline-first
Platform	Modern browsers (Chromium, Firefox, Safari/WebKit) · Desktop and Mobile
Repository	`samupropio1-ship-it/BioSpecInfo-v11`

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1. System overview

BioSpecInfo is a **monolithic yet modular** web platform that runs all of its application logic inside the client browser, with no application server. It is distributed as an installable **Progressive Web App** (`display: standalone`) and works **offline** thanks to a Service Worker that precaches the heavy assets.

Architectural pattern: Single-Page/Single-File Application + satellite sub-apps in iframes, with a WebAssembly-based cheminformatics *core* decoupled from the presentation logic.

1.1 Guiding principles

- **Local-First:** data and computations live on the device; no mandatory transmission to third parties.
- **Zero-backend:** no server database, no dynamic hosting cost (deployable on static hosting, e.g. GitHub Pages).
- **Progressive enhancement:** online features (PubChem search, real 3D models, astronomical imagery) degrade gracefully when offline.

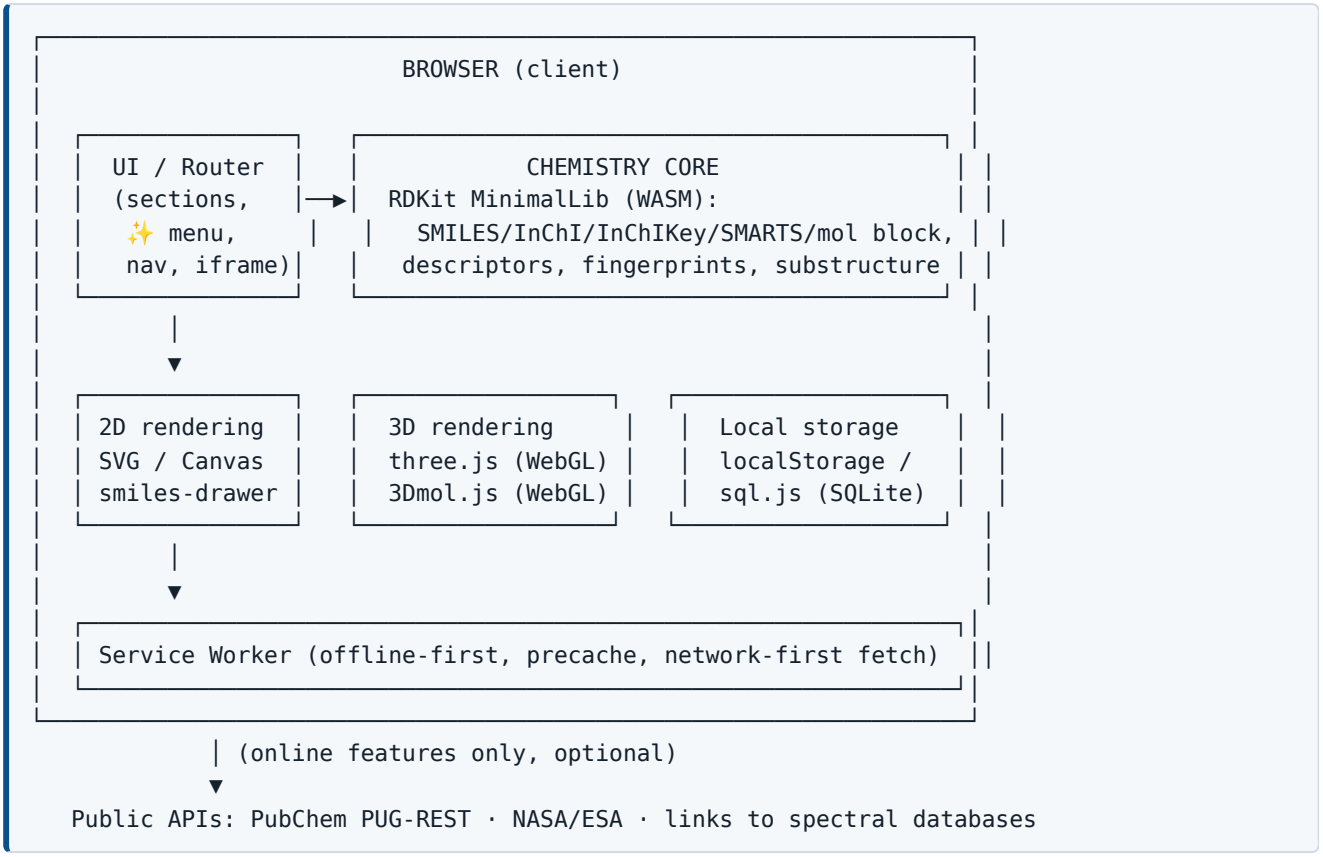
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2. File structure

File	Role
`index.html`	Main application (chemistry core, Spectra Centre, Data Science, ✨ menu). ~4.5 MB, over 43,000 lines.
`chimorga.html`	Organic Chemistry course (25 modules).
`astro.html`	3D Astrochemistry (three.js).

File	Role
`pro.html`, `rdkit_lab.html`, `accademia.html`, `simulazioni.html`, `guidaret.html`, `file_manager.html`, `Biochimica_Guida_Definitiva.html`, `sr_completo.html`, `sr_essenziale.html`, `download.html`, `changelog_tesi.html`	Thematic sub-apps loaded in iframes with `?v=timestamp` cache-busting.
`manifest.json`	PWA configuration (icons, theme, standalone).
`sw.js`	Service Worker: *network-first* strategy with cache fallback; precache of heavy libraries/assets; cache version bumped on every release.
`RDKit_minimal.js` / `RDKit_minimal.wasm`	Cheminformatics engine (RDKit MinimalLib).
`3Dmol-min.js`	3D viewer for molecules/macromolecules.
`three.min.js`, `three_bloom.js`, `gltf_loader.js`	WebGL 3D rendering (anatomy, astrochemistry, mechanisms).
`smiles-drawer.min.js`	2D depiction of SMILES.
`lib/sql-wasm.js` / `lib/sql-wasm.wasm`	SQLite in WebAssembly (SQL playground).
`models/`, `textures/`	3D assets (organs, écorché, planets) and textures.

3. Component architecture



3.1 Module Router / UI

Navigation between sections via CSS classes (`.section.on`) and a ✨ menu (FAB) that opens the special tools (Spectra Centre, Retrosynthesis, Quantum Lab, Data Science, etc.). Sub-apps are isolated in versioned `iframe`s.

3.2 WebAssembly Engine (RDKit MinimalLib)

Loaded on demand (`window.bsiLoadRDKit`) with `locateFile` pointing to the local `.wasm` files and a CDN as a last-resort fallback. It exposes:

- **Identifiers:** canonical/isomeric SMILES, CXSMILES, InChI, InChIKey, SMARTS,

MDL V2000 mol block.

- **Descriptors:** molecular weight, monoisotopic exact mass, LogP

(Wildman–Crippen), TPSA, HBD/HBA, rotatable bonds, aromatic/aliphatic rings, Fsp^3, molar refractivity, stereocentres.

- **Fingerprints & similarity:** Morgan ECFP4 (1024 bit) and the Tanimoto index.
- **Substructure matching** via SMARTS (`get_qmol` + `get_substruct_matches`).
- **2D depiction** (`get_svg`, `get_svg_with_highlights`) and 2D coordinate

generation (mol block) re-importable into the editor.

3.3 Rendering

- **2D:** proprietary SVG (reaction mechanisms with Bézier arrows), Canvas 2D (molecule editor, CIP stereochemistry viewer), smiles-drawer.
- **3D:** `three.js` (WebGL) for anatomy (Drug Atlas), astrochemistry and 3D mechanisms; `3Dmol.js` for molecular models from PubChem SDF.

3.4 Storage Layer

- `localStorage` (with a **safety guard** and in-memory fallback for incognito / blocked-storage scenarios).
- `sql.js` (SQLite compiled to WASM) for the Data Science SQL playground.

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4. Implemented features (current state)

4.1 Spectra Centre (molecular editor & analysis)

Draw from scratch (full periodic table, ring/group templates, drag-to-bond with snapping) or load via SMILES / library (63 molecules) → full analysis:

- **All chemical notations** (formula, canonical SMILES, CXSMILES, SMARTS, InChI, InChIKey, mol block, common/IUPAC name, CAS) with one-click copy.
- **Predicted spectra** as sub-panels: IR, ^{1}H-NMR, ^{13}C-NMR, MS with **isotope pattern **(IUPAC abundances) and** adduct masses** ($[M+H]^+$, $[M+Na]^+$…), UV-Vis, Raman — each with a bibliographic source.
- **Functional groups** (~30, via SMARTS) and **properties** (H donors/acceptors, aromatic, rotatable, stereocentres) **highlightable on the 2D and 3D structure**.
- **Drug-likeness / ADMET:** Lipinski, Veber, Egan, Ghose, Muegge rules; estimated ESOL solubility; QED (when exposed by the engine); **Brenk/PAINS structural alerts**; physicochemical radar profile.
- **Elemental analysis** (weight %) and **Tanimoto similarity** against the library.
- **3D models** (ball & stick, spacefill, wireframe) from PubChem; SVG/PNG/MOL export.

4.2 Synthesis & Retrosynthesis

46 disconnection strategies with an **animated mechanism player** in **2D and 3D**: curved *electron-pushing* arrows, reaction intermediates, and a retrosynthesis mode showing **synthon formation**.

4.3 Organic Chemistry (`chimorga.html`)

25 modules: fundamentals, stereochemistry, alkenes/alkynes, aromatics & EAS, SN/E, carbonyls, amines, carbohydrates, amino acids, lipids, nucleotides, animated syntheses, named reactions, heterocycles, pericyclic & Woodward–Hoffmann, organometallics & cross-coupling, radicals, spectroscopy, retrosynthesis & protecting groups.

4.4 Data Science

3D Drug Atlas (13 body regions, 63 diseases of which 23 tumours, 3D organ models, PubMed links), interactive ML labs, SQL playground (sql.js), course roadmap.

4.5 Other sections

3D astrochemistry, Quantum Lab, Academy (gamified quizzes with SM-2 spaced repetition), interactive simulations, guides, databases of elements/drugs/ biomolecules.

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5. Architectural roadmap (NOT yet implemented)

The following items belong to the *enterprise/HPC* vision and require external binaries, models or build pipelines that cannot be hosted inside a single-file PWA:

- HPC parallelism via Web Workers + `SharedArrayBuffer` / `Atomics` .
- WebGPU acceleration for dense/sparse algebra and GNN inference.
- ML predictors in ONNX Runtime WebGPU (GNN for NMR such as ShiftML2/DeepNMR;

hERG, BBB, CYP450, LogS).

- In-browser molecular docking (AutoDock Vina/Webina, PDBQT format).
- Semi-empirical quantum chemistry (xTB/GFN2-xTB in WASM).
- Rule-based computational retrosynthesis engine (SMIRKS).
- SQLite-WASM database at the 1,000,000-structure scale and SDF batch processing

via Web Workers.

- Immersive WebXR (VR/AR) visualisation.
- Native packaging (Android TWA, iOS, Electron/Tauri) and CI/CD pipelines.

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6. Requirements and constraints

- **Browser:** WebAssembly and WebGL support; JavaScript enabled.
- **Network:** not required for the core; recommended for PubChem search and real

3D models.

- **Performance:** target 30–60 fps for interactive rendering; heavy computations

are delegated to RDKit WASM and kept responsive.

Verification & Validation (V&V) & Benchmark Report — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Purpose	Document the verification method, validation data for the spectral predictors, cross-device performance and declared model limitations.

1. Scope and V&V strategy

- **Verification:** the software is built correctly (syntax, no runtime errors, expected component behaviour).
- **Validation:** the scientific results (spectral predictors, descriptors) are compared against reference experimental databases.

2. Software verification (QA)

Technique	Description	Status
Syntax checking	`node --check` on the critical JavaScript blocks on every change.	✅ In use
End-to-end (E2E) testing	Automation with **Playwright** on headless Chromium: open sections, draw/load molecules, verify panels (notations, spectra, functional groups, ADMET, similarity), 2D/3D animations. Counting of console/page JavaScript errors (acceptance criterion: **0 non-network errors**).	✅ In use
HTTP-server verification	RDKit WASM requires an HTTP context (CORS): tests run on a local server to reproduce the real environment.	✅ In use
Manual cross-device testing	Testing on real devices (see §4).	✅ In use
Automated unit-test suite (Vitest/Jest)	—	🔜 Roadmap
Coverage ≥ 90%	—	🔜 Roadmap

> Every release is verified E2E with Playwright under a "zero JS errors" criterion before merging to the main branch.

3. Scientific validation of the spectral predictors

BioSpecInfo's spectral predictors use **heuristic group-additivity / structural rule** models (literature reference tables), chosen to guarantee **instant in-browser response** on any device.

3.1 ^{1}H-NMR benchmark vs SDBS

Metric	Reported value
Reference database	**SDBS** (Spectral Database for Organic Compounds, AIST)
Mean Absolute Error (MAE)	**0.31 ppm**
Signals within ±0.5 ppm	**89%**

3.2 IR benchmark (C=O band) vs NIST

Metric	Reported value
Reference database	**NIST Chemistry WebBook**
Mean Absolute Error (MAE)	**11 cm^{-1}**
Bands within ±30 cm^{-1}	**90%**

3.3 Physicochemical properties

Descriptors (molecular weight, formula, LogP, TPSA, HBD/HBA, rings, stereocentres) are computed with **RDKit MinimalLib** and comparable with **PubChem** and **DrugBank**. Reproducible verification example: *cholesterol → $C_{27}H_{46}O$, 4 rings, 8 stereocentres*.

> **Benchmark data source:** measurements carried out by the author as part of the undergraduate thesis. The values are to be understood as validation for **educational and qualitative functional-group identification** purposes, not as a substitute for reference quantum-mechanical calculations.

4. Cross-device performance

Smoothness tests of the interactive renderers (3D viewers, animations, editor) across a range of devices:

Device	Result
iPhone / iPad	30–60 fps
Windows / macOS laptop	60 fps
Raspberry Pi 4	30–60 fps

Size of the main `index.html` file: **≈ 4.5 MB** (the full package, including WASM modules and 3D assets, is larger but remains entirely static and cacheable offline).

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5. Declared model limitations

In line with the principle of scientific transparency:

1. **Additivity model.** The NMR/IR predictors sum group contributions and do

not fully model extended chemical-environment effects (long conjugation, anisotropic effects, intramolecular hydrogen bonds). Larger deviations are expected e.g. for **strongly conjugated carbonyls** or edge cases such as **formic acid**.

1. **Qualitative/educational spectra.** The simulated spectra serve to recognise

functional groups and understand structure–spectrum relationships, not to replace certified experimental data.

1. **Network dependency for some functions.** IUPAC/common name, CAS number, GHS

pictograms and real 3D models come from PubChem: without network the app shows a fallback and remains fully functional on local data.

1. **Two-way accuracy.** For rigorous use, direct links to the official databases

(NIST, SDBS, MassBank, nmrshiftdb2) are integrated into the interface.

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6. Release acceptance criteria

- ✅ `node --check` passes without errors on the modified modules.
- ✅ Playwright E2E: 0 non-network JS errors.
- ✅ No visible regression in the touched sections.
- ✅ Service Worker cache version bumped.
- ✅ `tools/verifica-farmaci.js`: no defects; every deviation recorded with its reason.
- ✅ `verifica_guida`: whatever the documentation promises exists in the code.

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7. Verification battery — state at version `bsi-v250`

Verification is no longer only functional end-to-end testing: it includes benches dedicated to the properties a functional test does not observe (stability over time, network degradation, data accuracy).

7.1 Composition

Family	Benches	Subject
Scientific data	`verifica-farmaci`, `test_spettri`, `test_assi`, `test_assi_canvas`, `test_costanti`, `audit_dati`, `test_v237`, `test_3d238`	structure vs molecular weight, functional-group recognition, axis conventions, physical constants; the twenty astrophysics, physics and mathematics sections against known values or an independent second method; the 3D tabs of 29 sections (311 computation checks, every tab opened at 1280 and 390 px)
AI agent	`test_ko`, `test_404`, `test_503`, `test_firma`, `test_attesa`, `test_attesalunga`, `test_tetto`, `test_nucleo`, `browser_ko`, `browser_prova`	retired model, CORS, overload, reasoning signatures, token ceiling, long waits, memory of unreachable providers
Stability	`audit_stabilita`, `audit_promesse`, `audit_quota`, `test_filemanager`, `test_doppioinvio`, `test_sw`, `test_visore3d`	long sessions, rejected promises, exhausted storage, unavailable archive, overlapping submissions, degraded network, WebGL contexts
Interface	`browser_reset`, `browser_proxy`, `browser_proxyui`, `browser_rdkit`, `browser_lab`, `browser_frontiera`, `test_aggiorna`, `test_controllaagg`, `test_spectra238`, `test_lezione`, `test_spectra_studio`, `test_spectra_rig`, `test_spectra_figura`, `test_guidaproxy`	panels, proxy, RDKit laboratory, updates (including the «Check» button pressed in the three cases: up to date, new version, site unreachable); the Spectra Live console on phone and large screen, navigator, level and concentration, creator recognition, the face's eyes and sleep, a single Spectra button; the guided lesson in all 194 sections (chapters from the tabs, controls shown at work and put back as they were, check question, summary); Spectra Studio: the computations of the three lessons, the body driver on its bench's mannequin, the always-present embedded body (a packed block, not a `.glb`) and the real rigged body, the voice chain, the lessons on two screens and without WebGL; the automatic rig on a synthetic humanoid (skeleton, weights, hood with the face only, night blue); Spectra with the body in the four AI sections, the face-only icon, no substitute, the figure's reactions
Consistency	`verifica_guida`	documentation ⟷ code correspondence, conflict markers across 58 files

7.2 Properties verified that a functional test does not observe

Property	Method	Measured result
Absence of memory leaks	194 sections opened for 5 rounds, DOM node count per round	construction +26,876 nodes on the first round, **+0 on the four that follow**
Survival of exhausted storage	`localStorage.setItem` forced to throw	10 pages out of 10 stay operational

Property	Method	Measured result
Network degradation (≠ no network)	requests held for 20 s, interception at context level	answer served from cache in **3,507 ms** (threshold 3,500 ms)
Spectrum reproducibility	drawn twice, compared byte for byte	identical
WebGL contexts	viewer constructions counted across 10 molecules	from 7 down to **1**

7.3 Notes on method

A bench that measures nothing passes, and that is the most insidious way to obtain a false guarantee. While the battery was being built, several cases of vacuous measurement were found and corrected, among them:

- Playwright's `page.route` does **not** intercept requests originated by the

Service Worker: `context.route` is required. With the former, the "degraded network" test was measuring a perfectly healthy network and concluding in 16 ms.

- Probing `localStorage` exhaustion with a one-byte write proves nothing: after

saturation, one byte always finds room.

- The ticks of an axis drawn on `<canvas>` are not readable as text: they must

be collected by intercepting `fillText`, and the bottom-most line is the axis *title*, not a tick.

Hence the rule adopted: **every bench that simulates a condition must count how many times the simulation actually fired**, and fail if that count is zero.

Security, Privacy & Compliance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Model	Client-side / Local-First PWA

1. Privacy & data protection (GDPR)

1.1 Local-First by design

- **No account, no login, no advertising tracking.**
- All user data (drawn molecules, preferences, quiz history, study progress) is

stored **exclusively locally** on the device (`localStorage` / IndexedDB / `sql.js`), with no transmission to author-controlled servers.

- **There is no application backend** collecting or processing personal data.

1.2 Data handling and GDPR alignment

- **Data minimisation:** the app requires no personal data to function.
- **No transfer to the author's third parties:** data never leaves the device for

infrastructure controlled by the author.

- **Optional, transparent external calls:** only upon a user action (searching a

molecule, opening a 3D model) does the app query public third-party APIs (PubChem, NASA/ESA, spectral databases). Such requests contain **only the chemical identifier** (e.g. SMILES/name) and **no personal data**. Use of these services is subject to their respective privacy policies.

- **Spectra's quality voice (since v239, `bsi-spectra-voce.js`):** if the user

has saved an OpenAI key in the app, the **text Spectra speaks** (up to 3,800 characters at a time: console answers, guided-lesson steps, Spectra Studio texts) is sent directly from the browser to `api.openai.com/v1/audio/speech` (`gpt-4o-mini-tts`, voice «onyx»; `tts-1` as fall-back), with the user's key. This happens **even when the chat uses another provider**, and the request does not go through the proxy. The text may contain what the user asked and what Spectra remembers about the user. The audio received stays only in memory for the session. Without an OpenAI key this flow does not exist: the browser's speech synthesis is used. To avoid it, turn the voice off (🔇) in the console, in the guided lesson and in Spectra Studio, where the choice is remembered (`bsi-spectra-studio-voce`); or do not save an OpenAI key.

- **«🧠 Spiega» and the screen (`bsi-spectra-spiega.js`, v240):** with the

«📺 Includi lo schermo» checkbox checked (it is checked by default when a section is open behind the window), the configured AI provider receives, with its key and together with the question, **the screen text** (section, tab, control values, visible tables and text) and **an image of the visible graphs and 3D scenes**, like a chat attachment; the image is sent in the first turn only. With the checkbox unchecked only the question is sent. **An unchecked box is respected**: even when Spectra opens a section named in the question («spiegami le titolazioni») or chosen from the «🗂️ Tutte le sezioni nell'ologramma» catalogue, the section opens behind the window and is shown in the hologram, but the box stays unchecked and the model receives only the question, with no screen text or image. Only the quick prompts about the screen («📈 Spiegami questo grafico»…) check it again, since they are an explicit choice. Without a key nothing leaves the device. The conversation is kept in memory for the session and is not saved.

- **Spectra's body (since v241):** the body is embedded in the app

(`models/spectra-corpo.bsi`, downloaded from the app's own site and kept in the service worker's cache); the user no longer loads any file and nothing is written to IndexedDB any more. On start an old body stored by a previous version (database `bsi-spectra-modelli`), if any, is deleted; «Clear data → Progress in the rest of the app» still deletes that database.

- **Local right to erasure:** the user can delete data by clearing the site

storage in the browser.

> **Note.** This alignment describes the local-first design properties that favour GDPR compliance. It does not constitute a legal certification; for regulated contexts a dedicated DPIA assessment is recommended.

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2. Application security (OWASP)

2.1 Attack surface

Being a **static application with no backend**, entire classes of server-side vulnerabilities are absent by construction (SQL injection on a server DB, server-side RCE, server authentication/session, IDOR on proprietary APIs).

2.2 Input sanitisation

- Parsing of **SMILES / SMARTS** strings and molecular files is delegated to

RDKit MinimalLib (WASM), which validates the structures and rejects malformed input by returning invalid molecules rather than executing them.

- Strings coming from user input or external APIs, when inserted into the DOM,

are treated as untrusted. Their textual value (names, notations) is shown as text content/`textContent` or with escaping of the `& < >` characters.

2.3 Runtime robustness

- **`localStorage` guard:** every storage access is protected by `try/catch` with

an in-memory fallback, so that incognito mode or blocked storage cannot halt execution (a historical regression that has been fixed).

- **Controlled degradation:** panels never silently stay empty; errors show a

message and, where applicable, a "Retry" button.

- **Sub-app isolation** in `iframe`s.

2.4 Content Security Policy (deployment recommendation)

Since the app can be published on static hosting, it is recommended to serve it with security headers at the hosting/CDN level:

- Restrictive `Content-Security-Policy` (self script/style + WASM),

`frame-ancestors`, `X-Content-Type-Options: nosniff`, `Referrer-Policy: no-referrer`.

- `Cross-Origin-Opener-Policy` / `Cross-Origin-Embedder-Policy` if

`SharedArrayBuffer` and parallel Web Workers are enabled in the future (HPC roadmap).

2.5 Page-side checks: deterrents, not locks

On static hosting every check made in the page can be bypassed by whoever holds the device or reads the source. Two features declare it:

- **File Manager** (`file_manager.html`): the password is compared as a SHA-256

fingerprint; it is a deterrent against passers-by, not protection for confidential documents (`08-Traceability-Matrix.md`, SEC-03).

- **Spectra's creator recognition** (`bsi-spectra-creatore.js`, since version

`bsi-v238`): on «I am your creator» Spectra asks for the same secret phrase as the File Manager, typed into a password field and compared as a SHA-256 fingerprint. The phrase does not appear in the code, is not stored and enters neither the dialogue, nor memory, nor voice, nor the requests to the model; after three wrong attempts there is a one-minute wait. Only the mark `bsi-spectra-creatore-v1` stays in `localStorage`, erasable from «Clear data → Persistent memory». **It is not an access control**: the fingerprint is in the code, the mark can be written by hand by whoever holds the device, and an unsalted fingerprint of a weak phrase can be searched by trial outside the app. It protects no data: it tells Spectra whom it is talking to (`08-Traceability-Matrix.md`, SEC-08; `09-Release-Conformance-Statement.md`, D-13).

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3. Integrity and reproducibility (GAMP 5 alignment)

For use in pharmaceutical/regulated contexts, the architecture is **compatible** with GAMP 5 principles:

- **Determinism:** the cheminformatics computations (RDKit) are deterministic and reproducible for the same input.
- **Version traceability:** code versioning on Git, versioned Service Worker cache, changelog.
- **Method transparency:** the heuristic predictors are documented with their sources and limitations (see the *Validation Report*).

> Again: this is **alignment with the principles**, not a formal qualification (IQ/OQ/PQ), which would require a dedicated validation process at the using organisation.

---

4. Summary

Aspect	Status
Personal data transmitted to author's servers	**None**
Text spoken by Spectra sent to OpenAI for the voice	**Only with a saved OpenAI key** (§1.2)
Backend / server database	**Absent**
Molecular input validation	RDKit WASM
DOM output escaping	Yes (untrusted content)
Storage guard / error degradation	Yes
CSP / security headers	Recommended at hosting level
Offline operation	Yes (Service Worker)

Open-Source Licensing & IP Compliance Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author / owner of proprietary code	Samuele Pio Provenzano

1. Third-party dependency matrix

All bundled third-party libraries are **open-source under permissive licenses** (MIT / BSD-3-Clause). **No viral copyleft dependency (GPL / AGPL / LGPL)** is used, so there are no obligations to redistribute the proprietary code.

Component	Repo file	Use	License
RDKit (MinimalLib / rdkit-js)	`RDKit_minimal.js`, `RDKit_minimal.wasm`	Cheminformatics engine (SMILES/InChI/InChIKey/SMARTS, descriptors, fingerprints, 2D depiction)	**BSD-3-Clause**
3Dmol.js	`3Dmol-min.js`	3D viewer for molecules/macromolecules (WebGL)	**BSD-3-Clause**
three.js	`three.min.js`, `three_bloom.js`, `gltf_loader.js`	WebGL 3D rendering (anatomy, astrochemistry, mechanisms)	**MIT**
SmilesDrawer	`smiles-drawer.min.js`	2D depiction of SMILES strings	**MIT**
sql.js (SQLite compiled to WASM)	`lib/sql-wasm.js`, `lib/sql-wasm.wasm`	Client-side SQL playground	**MIT**
PDF.js	`vendor/pdfjs/`	Opening PDF documents in the spectrum reader: every page drawn and its text layer	**Apache-2.0**

> The exact versions are those of the files included in the repository. Each permissive license requires preserving the original copyright notice, which must be kept in the library files or in a `THIRD_PARTY_NOTICES` file.

2. External services/APIs (not redistributed)

These services are queried over the network at runtime (their data is not part of the package); their use is subject to the respective terms.

Service	Use
PubChem PUG-REST (NIH/NCBI)	IUPAC/common name, CAS, formula, 3D SDF models, GHS/CLP pictograms
NIST WebBook · SDBS · MassBank · nmrshiftdb2 · SpectraBase	Links to reference experimental spectra
NASA / ESA	Astronomical images and data (Astrochemistry section)

3. Intellectual property of the proprietary code

- BioSpecInfo's application code (UI logic, the proprietary SVG mechanism engine,

the CIP stereochemistry viewer in Canvas, the heuristic spectral predictors, the educational content, the internal databases) is the **author's original work**, except for the third-party libraries listed above.

- **No copyleft-licensed code** is embedded in the proprietary source: the author

retains full licensing freedom over their own code.

3.1 Project license

The repository includes a **proprietary — "All rights reserved"** `LICENSE` file: the code is viewable and assessable (personal study and professional/academic evaluation), but **copying, modification, redistribution and commercial use require the Author's written permission**. This choice protects the project from unauthorised reuse while keeping it presentable to recruiters and companies. Third-party components remain governed by their permissive licenses.

4. Compliance summary

Item	Result
Viral copyleft dependencies (GPL/AGPL)	**None**
Dependency licenses	Permissive (MIT / BSD-3-Clause)
Obligation to keep copyright notices	Yes (in library files / NOTICES)
Licensing freedom over proprietary code	Full
Project `LICENSE` file	✅ Present — proprietary "All rights reserved"

AI Agent Architecture "Spectra" — BioSpecInfo

Field	Value
Project	BioSpecInfo — Spectra component (agentic AI copilot)
Author	Samuele Pio Provenzano
Thesis supervisor	Prof. Savino Longo — University of Bari Aldo Moro
Component	`bsi-ai-hub.js` — 6,254 lines, zero runtime dependencies
Type	Multi-provider conversational agent with client-side tool execution
Repository	`samupropio1-ship-it/BioSpecInfo-v11`
Documented version	Service Worker `bsi-v250`

1. Executive summary

Spectra is an **agent**, not a thin wrapper around a language model. The difference is measurable: the model does not answer quantitative questions from memory — it **invokes deterministic tools** that compute the result, and the control loop lets it **chain up to 10 calls**, verifying each step before answering.

The architecture addresses three problems that separate a usable agent from a demo:

1. **Grounding** — a fabricated chemical value can be dangerous. 47 tools cover

computation, public databases and the application's own datasets; the system prompt explicitly forbids quoting numeric values from memory.

1. **Transparency** — the model's reasoning is streamed live and stored

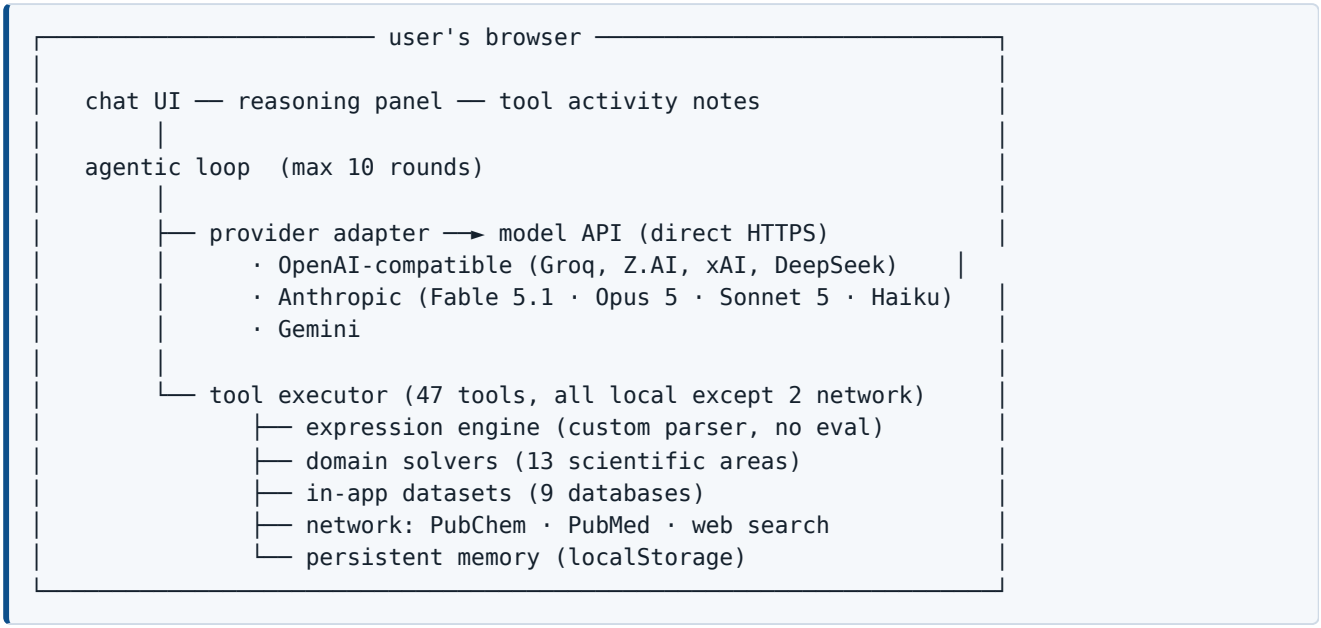
alongside the answer, together with the list of tools actually invoked. The agent's work is auditable after the fact.

1. **Provider portability** — a single tool registry is translated into the

three function-calling formats in use today (OpenAI-compatible, Anthropic, Gemini), so the agent behaves identically across ten model configurations, four of them free.

2. Architecture

2.1 Overview



There is no backend. API keys stay in the browser's `localStorage` and travel only to the selected provider: no data passes through BioSpecInfo servers, which do not exist.

2.2 Agentic loop

Each round: send the conversation history (last 40 turns) plus the system prompt and the schema of all 47 tools → receive the streamed response → if it contains tool calls, execute **all** of them → rebuild the assistant turn in the provider's native format → repeat.

Three exit conditions: no tool calls (final answer), `MAX_ROUNDS = 10` exhausted, or user interruption.

2.3 Provider-neutral tool registry

Tools are declared once in JSON Schema and translated at runtime:

Family	Required shape
Anthropic	`{name, description, input_schema}`
Gemini	`[{functionDeclarations: [...]}]`
OpenAI-compatible	`{type:"function", function:{...}}`

The same applies to messages: turns containing tool calls are serialised into each provider's native form and kept in a `_native` field, so a conversation stays coherent even when switching models.

2.4 No hard-coded model names

A model name written into the code is a time bomb: it works until the provider retires that model, and then the application stops with a 404 the user cannot fix. It has happened twice in this project:

Provider	Error	When
Google	`models/gemini-1.5-flash is not found for API version v1beta`	removed from `v1beta`
Groq	`The model llama-3.3-70b-versatile does not exist or you do not have access to it`	deprecated 17/06/2026

On some services the problem is structural: free model ids change constantly, by design. (That is why OpenRouter was later dropped from the list, see 2.4.)

The fix is not to write the new name — that would be the same bomb with a longer fuse — but to **remove the name** and ask the API, which knows what exists *for that key*. This also resolves the ambiguity in Groq's message: "does not exist" **or** "you do not have access" are different cases, and a per-key listing tells them apart.

Every configuration with `modelliCandidati` starts at `model: null` and is resolved at runtime. The two families use different strategies because their names are different:

Gemini — version scoring. Names follow a regular scheme (`gemini-<major>.<minor>-<family>`), so they can be ordered: newest version first, `flash` family preferred, with penalties for experimental, `preview` and `-lite` variants; models without `streamGenerateContent` are discarded, as are non-conversational ones (embedding, image, native audio, Live API). When Gemini 3.0 ships it will be picked automatically.

OpenAI family — candidates in preference order. Here names are not comparable to each other (`openai/gpt-oss-120b` versus `qwen/qwen3.6-27b`) and automatic scoring would choose badly. `GET /models` is queried and the **first candidate that still exists** wins: the order of the list *is* the preference, and the listing serves to skip the ones that vanished. Only if no candidate survives does a generic score run over the available models — discarding transcription, speech, embedding and guard models, and preferring the larger model all else being equal.

Three fallback levels in both cases: model listing → candidate probing → first candidate, letting the real error from the generation call speak instead of inventing one.

The choice is cached for seven days per provider, keyed to a 32-bit FNV-1a fingerprint of the API key (different keys see different catalogues; the key itself is never duplicated). If the model is retired *while* cached, the `404` — or a `400` whose text mentions the model, as some providers return — invalidates the cache, re-resolves and retries **once**: a flag prevents an infinite loop when the new model fails too.

Anthropic is the deliberate exception: its models are paid, explicitly chosen by the user, and deprecated with long notice.

The three free services, chosen one by one

Free quality is not all alike, and the difference matters: an 8-billion-parameter model does not hold up on a multi-step physical-chemistry problem. The list has been **shortened**, not extended — two services were removed because they made answers worse, not better.

Service	Models	Its role
Groq	GPT-OSS 120B, Qwen3	The workhorse: 131K context, ~30 requests/minute
Google Gemini	Gemini Flash	Up to 1M context: whole documents and PDFs
Z.AI GLM	GLM-4.7-Flash	Free **with no expiry**: the reserve still there when credits are gone

Removed. *GitHub Models*: it was here, and was removed because GitHub **retired it entirely on 30 July 2026** — playground, catalogue and inference API switched off for everyone. It had been added two days earlier on the strength of pages describing the still-live service: the lesson is that verifying *a thing exists* is not verifying *it is still alive*, and the search to run is "<name> retired / shutdown / deprecated <year>". The failure did not stop at the dropdown entry: the rank assigned to it also made it Core Mode's automatic choice.

Mistral: mid-tier quality and a free plan requiring consent to data training — not worth paying for unpublished thesis material when five alternatives carry no such clause. *OpenRouter*: small free models with ids that change constantly, unsuited to an agent chaining ten tool calls. *Cerebras*: since August 2026 there is no card-free plan, and free context is capped at 8K — Spectra sends 32 tool definitions on top of history, which does not fit.

None of the five matches a paid frontier model on the hardest problems; GitHub Models come closest, at the cost of low request ceilings. That is why the choice stays the user's rather than imposing one service.

The paid frontier

Free tiers are enough for studying; for a hard problem they are not. The selection criterion was **GPQA Diamond** — graduate-level physics, chemistry and biology questions — because it is the only benchmark measuring what this app actually asks.

Service	Why it is here
GPT-5.6	94.6% on GPQA Diamond: the highest score available today. ~$4/M input
Gemini 3 Pro	Over 1M context, and **the same key** as free Gemini. ~$2/M
DeepSeek V4	High-tier reasoning at ~$0.66/M: the best quality/price ratio
Grok 4.6	First on agentic tool calling and lowest hallucination rate — exactly the profile an agent chaining tools needs. ~$2/M
Claude (Fable 5.1, Opus 5, Sonnet 5)	One key for all three; the only ones with the Python sandbox in Core Mode

Two configurations can use **the same key**: the four Claude models are one Anthropic account, and Gemini Flash and Gemini 3 Pro one AI Studio key. `chiaveCondivisaCon` points them at the same entry, so the user pastes it once. Clearing it clears it for all twins — leaving one behind would make the key reappear on the other.

The two Gemini entries forced the scorer to be generalised: they share family, endpoint and key but want **opposite** models. The preference is now declared in configuration (`/flash/` versus `/pro/`) and outweighs the rest of the score. Without it the free configuration would have picked `gemini-3-pro` — a newer generation, therefore a higher score, but **paid**: a model the free key cannot use.

The same pass surfaced a second issue: the minor version in the name is **optional**. From generation 3 Google dropped it (`gemini-3-pro`, not `gemini-3.0-pro`), and the regex requiring the dot scored the *newest* models as generic aliases — so they lost to older ones, and the automatic upgrade described in 2.4 would never have triggered.

The provider audit, and why it must be repeated

The system repairs itself on **model names** — it queries the catalogue and picks among what exists. It does not repair itself against a **dead service**: that must be checked by hand, and GitHub Models' retirement showed what not doing so costs.

September 2026 check, provider by provider:

Provider	Result
Groq	Active. `llama-3.3-70b-versatile` **is no longer served** as of August 2026 and `llama-3.1-8b-instant` is deprecated: removed from candidates, leaving Groq's own replacements
Google Gemini	Active
NVIDIA NIM	Removed 09/2026: alive, but without CORS headers it cannot be called from a web page
Z.AI	Active, GLM-4.7-Flash and 4.5-Flash still free
OpenAI, DeepSeek, Anthropic	Active
xAI	Active, but candidates were **two generations** behind: the top is `grok-4.6` since 12/08/2026

Two traps recorded so they are not repeated. First: never put a paid model among a free configuration's candidates — true for `glm-5.x` on Z.AI as it was for `gemini-3-pro`, or the configuration picks something the free key cannot use. Second: a dead provider with a high rank does not merely break a dropdown entry, it **becomes Core Mode's automatic choice**.

When a provider does not answer

A `fetch` that fails **before** receiving a response means one of three things: no network, the service does not accept direct browser calls (CORS), or the service no longer exists. The browser, for security reasons, does not let you tell which — the error is deliberately generic.

This used to stop the turn. But if the three causes cannot be told apart, the right response is the same for all three: **move to another provider** the user holds a key for, exactly as for exhausted quota. It is the behaviour that would have masked GitHub Models' retirement entirely, instead of leaving every request stuck on a dead service.

Errors that would repeat identically everywhere stay out of the fallback — a wrong key, a malformed request: trying them on every provider and then reporting a random last error would hide the real cause.

Removing a service from the list is a breaking operation, and deserves a note: whoever had it selected has that name saved in `localStorage`, and without validation would hit `PROVIDERS[undefined]` — Spectra would no longer open. `getSavedProvider()` always validates against the registry and falls back to an existing service.

2.5 The optional proxy — Spectra without a key

The API key lives in the user's browser: with no backend there is no alternative, and it is the limitation stated openly at the end of this document. Every visitor has to obtain their own free key and paste it in.

`proxy/spectra-proxy.js` is the way out, and it sits **outside** the page: a Cloudflare Worker (free plan) holding the keys as server-side secrets. When it is configured the browser carries no key at all, and anyone opening BioSpecInfo uses Spectra without entering anything.

It is not a plain forwarder:

- **The path passes through untouched.** The route is `/<provider>/<path>` and

the rest is relayed as received: the proxy knows nothing about API shapes and keeps working when Spectra changes model or parameters.

- **Several keys per provider, with automatic takeover.** Each secret is a

comma-separated list; on `429` (quota) or `401/403` (revoked key) it moves to the next one within the same request. Not on `400` — that is a bad request and would fail identically.

- **Streaming is never buffered**: the response body is passed straight

through, otherwise answers would appear all at once at the end.

- **It is not an open relay.** Credentials arriving from the client

(`Authorization` , `x-api-key` , `?key=`) are discarded and replaced: nobody can use the proxy to push a key of their own.

- **Three brakes** against abuse: an allowed-origins list, a per-IP limit and a

daily cap.

On the Spectra side the graft is in one place. `GET /stato` reports which providers the proxy actually covers, so a model whose secret is missing is never offered as "no key needed"; `buildRequest` routes to the proxy **and omits authentication entirely**; `chiaveDaUsare()` returns a placeholder so the "missing key" guards do not block sending — a placeholder that never leaves the browser.

Both paths coexist: without `PROXY_URL` everything works as before, and for providers the proxy does not cover Spectra keeps using the local key.

2.6 Starting over

Everything Spectra remembers lives in `localStorage`: chats, history, API keys, persistent memory, review cards. A "clear history" button that removed only the conversations would not start over — reopening the app would restore the same state.

The delicate part is that keys exist in **two formats**: the current map (`bsi_api_keys`, one key per provider) and the single-key format of earlier versions (`bsi_api_key`). `getKeysMap()` migrates the latter into the former, so clearing only the map **brings the old key back** on the next access. Both must go — along with the resolved-model caches, which mean nothing without a key.

That is why the list of everything the application writes lives in one place, `DATI_CANCELLABILI`, split into five groups with a label and a description. The confirmation panel is built from it: it shows how many entries actually exist per group, disables the empty ones and requires a second explicit click. Only chats and keys are pre-selected; memory, review scheduling and the rest of the app's progress must be chosen deliberately, because they erase work unrelated to starting the conversation over.

Deliberately excluded: the PRO licence, the trial period and the device identity — data a user does not expect to lose by emptying a chat, and a licence must never be deleted by accident. The panel says so.

After deletion the in-process state is reset too — the list of providers the proxy covers, the resolved model per provider — otherwise it would stay valid until the page reloads, and Spectra would keep using a model chosen with a key that no longer exists.

2.7 Making free keys carry real load

Free tiers impose two different limits, and each needs a different answer.

The per-request ceiling. GitHub Models accepts 8,000 input tokens. Spectra's fixed cost is ~8,150 — 2,009 for the system prompt plus **6,128 for tool definitions alone** — so that service would not even start: it would fail before the user typed a word. The measurement came before the fix, and changed what the fix had to be.

The answer is not to trim history — it is to trim **the fixed cost**. Tools are paid on every turn; the conversation is the content. Of the two, the fixed cost must shrink first. `adattaAlBudget()` selects the tools **relevant** to the question instead of all 32, then shortens history with what remains.

Relevance is scored against a keyword map kept **outside** the tool registry: the words people use to *ask* for something are not the words a tool is *documented* with. The description of `astrofisica` talks about Wien and Stefan-Boltzmann; the user writes "what temperature is this nebula". Matching the question against descriptions alone picked `farmacocinetica` for an astrochemistry question — measured, not assumed. Matching is by stem (first five characters) because Italian inflects: "nebulosa" and "nebulose" must count as the same word.

Rate limits. A `429` on a free tier is almost always the *per-minute* limit, not an exhausted quota: waiting a few seconds clears it. The provider says how long in `Retry-After`, or in the error text ("*Please try again in 7.5s*"); when it says nothing, exponential backoff with a little jitter, so every open tab does not retry at the same instant. If the requested wait is absurd, it does not wait at all: better to say "quota exhausted" than to leave the user watching an hourglass for an hour.

Provider fallback. When a quota really is exhausted, the turn is redone on another service the user holds a key for. This is why several free keys together carry load that none of them carries alone.

Two non-negotiable constraints:

- It restarts from the **original** messages, not from half-finished history.

Turns containing tool calls are stored in the previous provider's native format and cannot be handed to another. The turn's work is lost; a correct answer is gained.

- Fallback uses **free services only**. Silently moving to a paid one would

spend the user's money without their say-so; if the chosen service was paid, it still gets the first attempt.

And it moves on **only** for exhausted quota. A `401` or a malformed request would repeat identically everywhere: trying all providers and then reporting a random last error would hide the real cause.

2.8 Core Mode, and saying what is happening

The mode. One switch that changes four things at once, because none of them alone would move the needle: it switches to the most capable configuration among those with a key, raises agentic rounds from 10 to 16 (a long tool chain ran out halfway), requests maximum reasoning depth from models that expose it, and adds a system-prompt instruction to work in verified steps rather than answer fast.

The model switch is **announced in the chat**, and if the new one is paid it says so: switching silently to a metered service would spend the user's money without telling them. For the same reason it is a switch and not the default — it costs more quota, and on a paid service it costs money.

Saying what is happening. Several seconds can pass between sending and the first word of the answer: the model reasons, calls tools, waits on the network. Before, the user saw a mute pause and concluded it had frozen.

The header now carries an atomic core whose orbitals change speed and colour with state — slow and cyan at rest, fast while reasoning, amber while a tool runs — alongside a line saying the same thing in words, for anyone not watching the animation. The states hook into callbacks the agentic loop already emitted: no mechanism was added, only made visible.

Three layout defects fixed in the same pass, all found by measuring rather than looking:

- The input field shared its row with four buttons and stayed 156 px wide: the

placeholder wrapped and was **clipped**. The text now has its own row with the controls below it. The placeholder length was chosen by testing at 360 px width, not by eye.

- The title "Spectra — il copilota AI di BioSpecInfo" did not fit and truncated

mid-word. Only the name remains; the rest became the status line, which now has a purpose.

- The core's orbitals, when rotating, escaped their box and overlapped the name.

2.9 Using what the provider actually offers

Three things the Anthropic API provides that were not being used. Checked against the official reference rather than recalled — two of the three carry a constraint that would not have been guessed.

Prompt caching. The fixed cost of every request is about 8,150 tokens: 2,000 of system prompt plus 6,128 of tool definitions, identical every turn. Marked for caching they are paid once and then read back at a fraction of the price.

The constraint is that caching works by *prefix*: one differing byte at the start invalidates all of it. But Spectra's prompt also carries user memory and question context, which change every message. Concatenated into one string the cache **would never have hit**. The prompt now travels in two blocks — stable and marked, volatile and not — and the marker also goes on the last tool, because composition order is `tools` → `system` → `messages`.

The Python sandbox, and why it excludes web search. `code_execution` gives the model an environment with sympy, numpy, scipy and matplotlib: numerical integration, symbolic algebra, matrix diagonalisation — things the 32 solvers do not cover because they cannot all be anticipated.

But the new-generation web search **already carries an execution environment** (it filters results by writing code), and declaring `code_execution` as well would create a second one, confusing the model. They are alternatives, not complements. The choice follows Core Mode: normally web search and page fetching, in Core Mode the sandbox — because when maximum power is requested the problem is computational, not documentary. A test verifies the two never appear together.

Page fetching. `web_fetch` joins search: if the user pastes an article link, Spectra reads it instead of merely searching for its title.

Sandbox results arrive as `bash_code_execution_tool_result` and `text_editor_code_execution_tool_result` — not a generic `code_execution_tool_result` — and must be kept in the assistant turn like the web-search blocks, or a server-paused turn cannot be resumed.

---

3. The 47 tools

The list is the one **measured** in `window.BSI_AI_TOOLS` on the running application, not transcribed from memory: this document declared 32 when there were already 35, and three — `analizza_molecola`, `disegna_molecola`, `mostra_spettri` — did not appear at all.

Area	Tools
General computation	`calcola`, `risolvi_equazione`, `analisi_dati`
General chemistry	`bilancia_equazione`, `stechiometria`, `massa_molecolare`, `converti_unita`, `costante_fisica`
Physical chemistry	`termodinamica`, `equilibrio_acido_base`, `cinetica`, `gas_e_soluzioni`, `elettrochimica`
Structure & spectra	`spettroscopia`, `quantistica_e_spettroscopia`, `cristallografia`, `analizza_molecola`, `disegna_molecola`, `mostra_spettri`
Life sciences	`biochimica`, `farmacocinetica`, `valuta_druglikeness`
Physics	`astrofisica`, `nucleare`, `statistica_inferenziale`
External databases	`cerca_pubchem` (NIH), `cerca_letteratura` (PubMed)
Internal data	`cerca_nel_database` (9 datasets), `cerca_molecola`
Macro-areas	`risolvi_matematica`, `risolvi_fisica`, `risolvi_chimica` — the same engine as the three areas' assistant: a natural-language question, or an explicit model and data; working, data with their origin, inverse problem
App control	`naviga_sezione` (194 sections), `apri_strumento` (12 labs), `stato_app`
Memory	`ricorda`, `ricordi`
Animations	`apri_animazione` (6 reaction mechanisms)
Screen and page (Spectra Live)	`leggi_schermo` (subject, section, tab, controls with their values, buttons, note text), `imposta_controllo`, `scegli_scheda`, `premi_pulsante`, `cambia_materia`
Study (Spectra Live)	`consulta_kit` (summaries, formula sheet, glossary, quiz of the subject), `stato_studio` (flashcards due, quiz mistakes), `disegna_grafico`, `apri_strumento_rapido`

The nine tools in the last two rows are defined in `bsi-spectra-live.js` and added to the same `window.BSI_AI_TOOLS` registry when the page starts: the agentic loop treats them like the others, both in the chat and in the console.

> **Web search is not among these thirty-five**, and the distinction matters. The thirty-five are executed **by the application**, in the user's browser. On providers that offer them, the agent adds two tools to the schema that are executed **by the provider** — `web_search` and `web_fetch`, with a per-turn usage cap. Counting them with the others would make them look always available, whereas they depend on the provider chosen.

3.1 Internal datasets exposed

297 synthesis reactions · 118 elements · 67 amino acids · 263 drugs · 63 pathologies · 46 retrosynthetic strategies · 36 drug interactions · 29 metabolic pathways · 29 redox potentials.

Drugs and strategies are the two figures that `tools/verifica-affermazioni.js` measures against the running application on every battery run: here they had been left at 143 and 39 while the app counted 178 and 46.

These are the datasets curated by the author for the application: the agent treats them as a primary source and is instructed to **report discrepancies** between the database and its own memory rather than silently smoothing them over.

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4. Notable engineering decisions

This section documents the non-obvious choices and the reasoning behind them.

4.1 Expression engine without `eval()`

Problem. Serving arbitrary physical-chemistry computations requires an expression evaluator. `eval()` would solve it in a few lines.

Why it was rejected. The agent reads untrusted external content (PubChem results, PubMed abstracts, web pages). An injection in that content could induce the model to emit a hostile expression; with `eval()` that string would execute in the page context, **where the user's API key lives**.

Solution. Recursive-descent parser with a *closed* set of 27 functions and constants. It has no access to global scope: it can only do arithmetic. Validated against 10 escape attempts (`window.localStorage`, `constructor`, `this`, `fetch(...)`, calls to non-whitelisted functions) — all rejected, including when executed in a real browser.

4.2 Preserving reasoning blocks

On models with adaptive reasoning, `thinking` blocks are part of the assistant turn and must be **echoed back unchanged, cryptographic signature included**, on the next tool-use round. Dropping them causes the request to be rejected.

The implementation accumulates `thinking_delta` and `signature_delta` from the stream and reinserts them **first** in the reconstructed turn. Verified in integration tests: signature preserved, ordering correct.

4.3 Resuming paused turns (`pause_turn`)

Web search is a server-side tool. When its internal loop exhausts its iterations, the response ends with `stop_reason: "pause_turn"`.

The turn must be resumed by **sending the assistant turn back as-is, without adding any user message**: the server recognises the trailing `server_tool_use` block and continues from there. Appending a "continue" message would break the mechanism.

4.4 Per-model configuration, not global

The four Claude tiers do not accept the same parameters:

Model	`effort`	Web search	Notes
Fable 5.1	`xhigh`	yes (8)	refusal fallbacks; reasoning always on
Opus 5	`high`	yes (6)	default
Sonnet 5	`high`	yes (5)	
Haiku 4.5	**unsupported**	**unsupported**	sending them returns HTTP 400

On all recent models `temperature`, `top_p` and `budget_tokens` were removed from the API and their presence causes an error: none of the four sends them. Every parameter is therefore conditioned on the provider rather than set globally.

4.5 Multimodal input and material reading

The agent accepts images, PDFs and text files, several at once, treated as consecutive pages of a single document. Three constraints shaped the implementation:

- **Images are downscaled to 2000 px** and re-encoded as JPEG before sending.

A phone photo drops from roughly 1 MB to 440 KB: without this step a single image would approach the request ceiling and multiply token cost. 2000 px is the minimum for reading small handwriting — at 1600 px fine script became illegible.

- **The PDF page ceiling depends on the model**, it is not a constant: 600

pages for models with a 1M-token window, 100 for 200K ones. Page counting is done by reading `/Type /Page` objects from the file bytes, without libraries.

- **History stores a 160 px thumbnail**, not the image that was sent. Storing

the large one (440 KB each) filled `localStorage` within a dozen photos, and once the quota is exceeded every subsequent write fails silently.

Six quick actions turn the material into a study artefact (summary, concept map, outline, flashcards, exam questions, transcription) and two translate it. For transcriptions the agent is instructed to write `[illegible]` rather than guess: an invented chemical term is worse

than a declared gap.

4.6 In-house graph renderer, no CDN

Concept maps are produced by the model as Mermaid diagrams and drawn by a renderer written into the application (~190 lines): longest-path layering from the roots with cycle breaking, barycentre reordering over three passes to reduce edge crossings, SVG output with Bézier curves.

Not loading Mermaid from a CDN follows from the app's offline-first nature: a map must be drawable without a network. Unsupported syntaxes are recognised and left as a readable code block instead of being drawn badly.

4.7 Voice command with wake word

User-activated continuous listening: the phrase "Hey Spectra" followed by a command sends it automatically. Two non-obvious details:

- the browser's speech recognition **stops on its own** after a few seconds of

silence and must be restarted, otherwise listening dies at the first pause;

- **all transcription alternatives** are examined, not just the first, and the

most common phonetic variants are accepted: Italian recognition renders "Spectra" in many different ways.

Denying microphone permission stops the loop instead of retrying forever. The word "spettroscopia", extremely frequent in this domain, was verified as non-triggering.

4.8 Guard against non-finite results

A systematic audit revealed that, given legitimate degenerate inputs (zero wavelength, zero volume, zero half-life, a transition between identical levels), eight solvers returned `ok: true` with `Infinity` or `NaN` among their fields.

This is the most insidious failure in this context: an exception is noticed, a formally valid but meaningless value is reported to the user as correct. The fix is a single check at the point where all results pass through: if a numeric field is not finite, the result becomes an explicit error naming the fields and stating the probable cause. It also covers tools added in the future.

4.9 Graceful degradation

- **Models without reliable function calling** (some free tiers): on the first

tool-related error the request is retried once in text-only mode instead of failing.

- **Idle timeout (45 s)**, reset on every byte received: a long answer is never

truncated, but a dead connection is reported immediately with an explicit message.

- `localStorage` **quota**: history is capped (30 conversations, 100 messages)

with progressive pruning. Without this cap, exceeding the quota made every subsequent write fail *silently*, **including saving the API key**.

- **Classifier refusals** (`stop_reason: "refusal"`, HTTP 200): detected and

surfaced, instead of appearing as an empty answer.

4.10 Spectra Live: voice, screen and local commands

`bsi-spectra-live.js` adds a full-screen console. It opens from any screen with Spectra's only button: the face in a circle (`#bsiJV-fab`), which since v239 takes the place of the «Spectra» pill (`#bsi-spectra-fab`). The pill stays in the document, hidden by the `bsiJV-solo-volto` class on `body`, because other modules and the benches use it to open the chat; from the console the chat opens with 💬. The circle appears at once with the vector face and switches to the portrait (`models/spectra-ritratto.jpg`) after the `load` event. The «◈ Live» button in Spectra's header and `Ctrl+Shift+J` remain:

- **hands-free conversation**: the browser's speech recognition listens, the

answer is spoken sentence by sentence with the system's speech synthesis (a male voice is preferred, rate 0.95, pitch 0.7) or, since v239 and only with a saved OpenAI key, with the voice of `bsi-spectra-voce.js` (§4.15); the console then listens again; «stop» or a tap interrupts it. If either capability is missing in the browser, the console says so;

- **local commands**, recognised with regular expressions before the model is

called and therefore available without network and without a key: open a section or switch subject, compute, differentiate and integrate (with the app's symbolic engine), convert units, look up a constant, define a term, plot a graph, quiz, review, move a slider, change tab, pause and resume an animation, tell the time and date, and the mind's commands («what do you remember», «how are you», «forget everything»);

- **screen context**: every question that goes to the model carries a short

summary of what the user is looking at (subject, section, tab, control values, up to 700 characters of visible text). If the question contains words such as «this», «here», «chart», or the user pressed 👁, an image of the tab's visible canvases is attached as well.

Everything that goes to the model takes the same path as the chat, to the provider the user chose. Speech recognition is the browser's: where the browser runs it on a remote service, the audio follows that browser's rules, not BioSpecInfo's.

Layout and navigation (v238). On a phone with the app installed, Android's status bar and keys sit over the page. The console uses `env(safe-area-inset-top/bottom)` when the system reports them and, with the class `.bsiJV-bordi` (full-screen window on a touch device, or installed app), at least 34 px at the top and 46 px at the bottom. Positions are no longer fixed: on every frame the header, telemetry and control bar are measured, and the face, state, thought and dialogue take the remaining space. The 🧭 navigator builds from the bar itself, for each of the three subjects, the list of groups and sections, then the quick tools

(`BSIAttrezzi`) and the locally recognised phrases; the 💬 button switches to the chat; the context shortcuts change with the open section. The telemetry shows USER (guest or creator), LEVEL and FOCUS.

Level and concentration. `difficolta(testo)` starts from the level of the open section, i.e. the first level of the `bsi-percorsi.js` path in which it appears (high school 0, bachelor 1, master 2, PhD 3: value 0.15 + 0.22·level), adds 0.3 for advanced-course words (theorem, Hamiltonian, eigenvalues, …), up to 0.2 for symbols, 0.1 for a long question, subtracts 0.25 for «in simple words», and clamps the result between 0.05 and 1. The value goes to the face through `concentra()`.

Initiative. With the console open, waiting and silent for more than a minute, if the mind has a thought of weight ≥ 2 (a break, mistakes to review, the late hour) Spectra says it, at most once every three minutes and never the same twice, with the matching expression.

4.11 The face

`bsi-spectra-androide.js` shows the «Noir» 3D model by Skyfox13 in the console (Sketchfab, CC BY 4.0, modified; attribution in the console and in `07-SBOM.md`). The model has neither a skeleton nor mouth shapes, so the animation is built in code:

- the **head** is detached from the neck and turns with the gaze and the

breathing;

- the **mouth** opens on the spoken text: the jaw displacement is computed

once per vertex and the shader multiplies it by the current opening;

- the **expressions** are 17 emotions, each a blend of **11 expressive

fields **(smile, rounded lips, wide mouth, raised and furrowed brows, …). WebGL allows no more than 16 attributes per vertex, so the eleven fields are kept in memory and** blended on the CPU** into a single attribute when the expression changes, recomputing only the vertices some field moves;

- the emotion of a sentence comes from Italian and English keywords

(`emozioneDi`); with no signal it is «serena» (calm);

- the **eyes** (version 3.0, v238) have **real eyelids**: the vertices of

each eye are deformed on the CPU around its centre, with the opening and asymmetry of each emotion (joy narrows them, surprise widens them, doubt makes them asymmetric); the gaze moves them; the **tint** changes with the emotion within the theme's blue family (`#1a3050` at rest, brighter in joy and surprise);

- **concentration** (0–1, from `difficolta()`) half closes the eyes, brightens

them, draws the brows together, presses the lips and slows the blinking;

- every facial muscle is a **damped spring**, so expressions have inertia and

a small settling; every few seconds a quarter-second **micro-expression** appears; the mouth uses **coarticulation**: the next letter's viseme starts before the current one ends;

- neutral lights (the saturated blue back-lights tinted the metal), the

screen's real density up to 3× and anisotropic filtering on the metal maps;

- after **two minutes without contact** (touch, voice, answer) the face falls

asleep: heavy eyelids, lowered head, slow breathing; the first touch wakes it with surprise and then a smile. The gaze follows the pointer and, on a phone, the touched point.

Like the mind, it is a **simulation of behaviour, not consciousness**.

Without WebGL, or if the model does not arrive, the console uses the head built by `bsi-spectra-robot.js` and, as a last resort, the vector portrait of `bsi-spectra-volto.js`. With «reduce motion» the head stays still.

4.12 The mind

`bsi-spectra-mente.js` gives Spectra continuity across sessions. **It is a simulation of behaviour, not consciousness**: five numbers between 0 and 1 and a few rules, not subjective states.

- **Memory**: conversation episodes (at most 300, each with time, subject,

section and the first 160 characters of the question), the time spent in each section, the study hours, the emotions detected in the user's sentences (at most 60).

- **Internal state**: curiosity (rises with new questions), attention (rises

when working together, falls during pauses), care (grows with session length and at night), mood (moves with the tone of the user's sentences and drifts back to an equilibrium), bond (grows with every conversation, ever more slowly). The state picks the face's background emotion and the tone of the greeting.

- **Thoughts**: suggestions derived from these data (a break after 90

minutes, reviewing quiz mistakes, a subject left behind).

- **Summary for the model**: every question that goes to the model carries a

summary of study time, frequent sections, recent topics and internal state.

Everything is saved only in the browser's `localStorage`, key `bsi-spectra-mente-v1` (`11-Data-Model.md`); «what do you remember» shows it, «forget everything» erases it.

4.13 The creator

`bsi-spectra-creatore.js` (v238) tells Spectra who created it: Samuele Pio Provenzano. «Who created you?» has a local answer. «I am your creator» opens a proof: the console field becomes a password field for a minute, voice listening stops, and the text sent **does not go through** `elabora()`: no dialogue, memory, voice or model. Its SHA-256 fingerprint (`crypto.subtle`) is computed and compared with the File Manager's; the phrase does not appear in the code. Three wrong attempts lead to a one-minute wait.

Once recognised, Spectra writes only `bsi-spectra-creatore-v1` (`ok`, `dal`, `ultima`) to `localStorage`, greets the author by name when the console opens, shows «CREATOR» in the telemetry and adds to the model's context (`contesto()`, also inside the mind's `coscienza()`) that it is talking with its author. For a guest the context says the user has not been recognised and that the phrase must never be asked for in chat. «Forget that I am your creator» removes the mark; so does «Clear data → Persistent memory».

Limitation. Like the File Manager password it is a deterrent, not an access control: the fingerprint is in the code and the mark can be written by hand by whoever holds the device (`09-Release-Conformance-Statement.md`, D-13).

4.14 The guided lesson

`bsi-lezione.js` (v239, object `BSILezione`) turns every section into a lesson in chapters led by Spectra. It does not use the model: the lesson is built by reading the open section. Only the voice goes online, and only with a saved OpenAI key (§4.15).

- **Construction** (`costruisci(section)`): an opening step with title and

subtitle; one chapter per tab (`.bsiFX-schede button`), whose steps are read when the tab is open: the explanation split into pieces of one or two sentences (at most three pieces, about 260 characters each), the first visible canvas or 3D scene, up to four sliders and selects; for sections without tabs a single chapter with the visible headings and paragraphs and the same controls. Then a **check question** from the subject's study kit (`BSIStudio.dati(subject).quiz`): the one with the most words (longer than four letters) in common with title, subtitle and tabs wins, with double weight for the words of its topic; a score of at least 2 is required, otherwise there is no check. Finally the **summary** of the chapters.

- **Demonstrations**: a slider goes from its current value to the minimum, the

maximum and back to the starting value in about 4 s, firing `input` and `change` events as the user would; a select goes through its first five options and returns to the starting one. Skipping ahead half-way, or closing the lesson, puts the control back as it was.

- **Player**: panel `#bsiLZ` (on the right on a large screen, at the bottom

below 760 px; from 1100 px `#main` shifts left), progress by chapter, text appearing word by word, ◀ ▶, pause, voice, `Esc` to close, arrow keys to move. The spotlight `#bsiLZ-luce` dims the rest and follows the piece being discussed. The voice is Spectra Live's (`BSISpectraLive.parla`) and, if the console is not there, `BSISpectraVoce.parla`; without voice, reading time depends on the length of the text. With «reduce motion» sliders jump to the value, text appears all at once and the spotlight does not pulse.

- **Floating buttons**: they are hidden during the lesson (including those

with `visibility` written inline, whose value is noted and restored on closing).

- **Start**: the «🎓 Guided lesson with Spectra» button the module adds under

the title of every visible section; Spectra Live's local command «give me a lesson» («fammi lezione»), which closes the console and starts the lesson of the open section; the 🎓 shortcut in the console.

The lesson has no keys of its own in `localStorage`; at the end it signals an emotion to Spectra's mind (`sente('fiera', 'lezione')`), which the mind records in `bsi-spectra-mente-v1` like the others.

4.15 Embodied Spectra: the studio, the body, the voice

Since v239 Spectra also has a body, on a page of its own: `spectra-studio.html`, reached from the console's 🧭 navigator. The full design (model requirements, choice of stack, modules, state, lesson cycle, migration, domain coverage) is in the Italian design documents in `docs/spectra/`: analysis and plan, architecture, domain map. What is in the code is summarised here.

- **The studio** (`spectra-studio.html`): a three.js r128 scene (grid floor,

neon lights, holographic bench) with Spectra standing; an accessible panel with all the text. The lessons in `bsi-spectra-studio-lezioni.js` (`BSIStudioLezioni`: `LEZIONI`, `SCENE`, `calcolo`) follow four phases: **predict** (choices with the most common misconceptions and one right answer, each explained), **test** (the simulation with the numbers measured in the scene), **prove** (the derivation step by step), **reveal** (why, connections, exercises on four levels). The computations live in `calcolo` and also run in node: fall with semi-implicit Euler (and optional quadratic drag), VSEPR as the minimum of the energy w_i·w_j/r of domains on the sphere (w = 1 for bonds, 1.25 for lone pairs), partial sums, tail, Fourier series of x^2, Monte Carlo coprimes.

- **The studio's memory**: `bsi-spectra-studio-v1` keeps, per lesson, the

predictions (`quando`, `scelta`, `giusta`) and the exercises opened; the recommended level is the number of right predictions plus exercises opened, up to 3. Each prediction also goes to the mind (`BSISpectraMente.ricorda`). The 🔊/🔇 switch in the action bar turns the studio's voice off and the choice is kept in `bsi-spectra-studio-voce` (`'0'` = off). Both keys are covered by «Clear data → Persistent memory».

- **The body** (`bsi-spectra-corpo.js`, `BSISpectraCorpo`): the driver

recognises bones, ARKit blendshapes, hair and clips of a humanoid GLB; every pose is a direction in the world, converted into the bone's local frame (`orienta`, a minimal inverse kinematics valid for any rig); emotions, lip sync, eyelids, gaze, gestures, walking and sitting are blended with springs. The mannequin from `BSISpectraCorpo.manichino()` (bones with Mixamo names in T-pose, 26 ARKit blendshapes, red hair that the driver turns black) is used **only by the driver's bench** (`opz.manichino === true`): since v240 it is no longer Spectra anywhere. **Spectra's body** is chosen by the shared controller `BSISpectraFigura.Spectra()` (`bsi-spectra-figura.js`), in the studio and in the AI sections. Since v241 the body is **always present**: `carica()` downloads `models/spectra-corpo.bsi`, the «Bounty Hunter» (CGTrader, Royalty Free licence) embedded as a block packed by `tools/impacchetta-corpo.js`, and reopens it with `apri`; the published file is not a reusable

`.glb` (it is not absolute protection). While it arrives, «Spectra sta arrivando…»; if it does not, «↻ Riprova». Only in the studio, in development, if the block is missing `models/spectra-corpo.glb` (out of git) is tried; in the app (`soloDispositivo`) no development file is requested, no 404s. The «🧍 Corpo» button, the user-loaded body in IndexedDB and the `models/spectra-cyberpunk.glb` fallback were removed; an old body in `bsi-spectra-modelli` is deleted on start. No substitute. A body without a skeleton goes through the automatic rig (`bsi-spectra-rig.js`); the «Noir» face (`BSISpectraAndroide.crea(cont, { scena, renderer, senzaBusto: true })`) goes on the collar, aligned by measuring the chin, with only the face visible, «solidale» with the body's head.

- **The figure in the AI sections** (`BSISpectraFigura.monta`, `sezioni`,

v240): one figure, moved into whichever section is open among Spectra AI and the three areas with the assistant, shows Spectra standing in a lab-style frame and **reacts to the app's events**: `bsi-spectra-stato` (thinking, tools, writing) → concentrated, hand on chin, «SPECTRA · PENSA»; `bsi-spectra-risposta` (the model's answer) → explains with the hands, lip sync of body and face from the text, emotion from the content, «SPECTRA · PARLA»; an error → doubt; `bsi-spectra-dice`, dispatched by Spectra Live's `parla()` and by the local assistant (`bsi-assistente.js`) when it solves an exercise («Ecco lo svolgimento: …», or doubt if it did not understand) → the figure says that text. It renders only when visible and the page is in the foreground. The events are one-way: the figure does not change what the model answers.

- **The figure in the Spectra Live console** (v242): with the console open,

`sezioni()` moves the shared figure into `#bsiJV-figura` (between the telemetry and the dialogue; on a large screen on the right), adds the `con-figura` class and hides the bust `#bsiJV-volto`: instead of the face alone, Spectra with the body and the hologram. The console emits `bsi-spectra-stato` at every state change (waiting, listening, thinking, speaking): thinking → concentrated, hand on chin; listening → curious, «SPECTRA · LISTENING TO YOU». The console's event carries the `daLive: true` signature and the console ignores its own events: without the signature the console, which listens to the same event for the other modules, called itself forever (a recursion found by the bench). With the console closed, the figure returns to the section.

- «🧠 **Spiega**» (`bsi-spectra-spiega.js`, `BSISpiega`, v240): the second

tab of the Spectra window, after «💬 Chat», for an **exhaustive explanation** of any chemistry, physics or mathematics question. The system prompt (`sistema(level, subject)`) sets the level — five, from High school to Advanced PhD · frontier, each with its instructions in `PROFONDITA` and «do not simplify below this level and do not leave out anything relevant» — and a **mandatory structure** with `##` titles: in short; what you are looking at (only with the screen: quantitative reading of axes, units, curves and values; for NMR spectra δ, multiplicity, J, integral and assignment of every signal, IR bands and groups, MS molecular ion, fragments and isotopes, UV-Vis λmax, transitions and ε, then the structure and its consistency; for 3D scenes geometry, symmetry, nodes, critical points, isosurfaces and fields; for mathematical graphs domain, asymptotes, zeros, extrema, inflections, convergence); foundations; numbered derivation; computed examples; limiting cases and common errors; links be-

tween the three subjects; applications and frontier (literature only if certain); three self-check questions with answers. **The screen** (`schermo()`, checkbox «📺 Includi lo schermo»): section, tab, every control with value and range, visible tables (up to 4 × 40 rows), visible text, number of graphs; `immagine()` composes the visible canvases (`BSILab3D` scenes re-rendered first, via `BSISpectraLive.cattura`) and the visible SVGs rasterised, at most 1400 px wide, as JPEG. The image goes to models that accept images, with the chat's per-provider conversion, in the first turn only; follow-up questions continue the conversation, which is kept in memory for the session. Provider and key are the chat's (`window.bsiLLM`). The shared figure moves into the tab (priority over the sections), thinks while the model writes, then speaks and gestures. **Without a model it does not pretend**: it says so and shows what the app knows on its own (the local assistant's recognition, the chapters of the section's guided lesson, the screen data as text). With the screen included, text and image go to the configured provider together with the question. **The catalogue** (`catalogo()`): every section button of the bar (`.nav-group-panel [data-s]`) with its group label and subject (`BSIMaterie.materiaDi`), 201 sections since v242 (chemistry 78, physics 63, mathematics 31, shared 29), read from the bar rather than written by hand. The «🗂️ Tutte le sezioni nell'ologramma» card lists them by group, with search and a subject filter; `mostraSezione(id)` opens the section behind the window (`BSISpectraLive.calcolo.apriSezione`, which switches subject if needed), the hologram shows it live and the screen checkbox is checked, unless the user unchecked it; `giro(materia)` («🎞️ Mostrami tutto») shows them in turn every 7 s, with the counter «n/N · name», and `giroFerma()` stops it. `sezioneDi(testo)` finds the section a question names (meaningful words of the names, also by stem; acronyms MS, IR, UV, NMR, MO… weighted more; synonyms «spettro di massa» → MS, «infrarosso» → IR, «risonanza magnetica» → NMR, «orbitali molecolari» → MO; ties go to the shorter name; none if the question names no topic): asking without the screen included, the named section opens, is shown in the hologram and the explanation includes its screen. **A box unchecked by the user is respected** (`ST.schermoNo`, set by the checkbox's `change` event): the section still opens and is shown, but the model receives only the question, with no screen text or image; the quick prompts about the screen check it again (explicit choice). A section without graphs is projected on the hologram as a card (title, tab, up to 6 controls with values, visible text), so every section is showable.

- **The interactive hologram, «⛶ Interagisci»** (`bsi-spectra-olointerattivo.js`, `BSIOloInterattivo`, v242): the button on the shared figure's hologram opens `#bsi-olo-ov`, full screen; `sezioni()` gives precedence to `#bsiOI-figura` while the window is open, so Spectra with the body moves in, and with the window closed returns to the section. The mode comes from the hologram's scene (planets → universe; spectrum and molecule → spectra; surface and orbital → mathematics; wave → physics; the «screen» according to the open section). **No model call**: spectra come from the six reference molecules (typical literature values) or from `leggiTraccia` (candidate assignments from group-frequency, chemical-shift and neutral-loss tables, taking band shape into account); planets from the NASA fact sheet with Kepler and vis-viva computed; mathematics and physics from the local assistant (`bsi-assistente.js`, `bsi-mat-cas.js`), whose working becomes the list of steps and whose result drives the graph (function and derivative with zeros and stationary points, pro-

jectile trajectory, pendulum). Each step has a title and a text; «🔊 Spectra spiega i passaggi» hands them one by one to `BSISpectraLive.parla` and the figure says them with lip-sync and gestures; tapping a peak or a planet calls `scegli(i)`. Only **«🧠 Approfondisci»** goes to a model, through «🧠 Spiega» with the hologram's data, and under its rules.

- **«⚡ Explore with Spectra»** (`bsi-spectra-esplora.js`, `BSIEsplora`,

v242): three sections, one per subject (`sesplchim`, `sesplfis`, `sesplmat`), registered with `BSIFisica.sezione`, in the style of explorable lessons. Ten chapters per subject (`CAPITOLI`, levels 0→4, from high school to advanced PhD · frontier), each with the phenomenon, a four-answer prediction with the reason for each, in 19 chapters a computed slider (`calcola(id, v)`), the linked app section and the hologram scene. The shared figure enters the section with the `bsiEX-palco` class (`position: sticky` under the measured app header; shorter on phones) and stays on stage while scrolling; the chapter crossing the middle of the free space under the figure becomes active: `bsi-spectra-dice` (Spectra tells it, curious) and `olo.mostra(scena)`. The prediction emits `bsi-spectra-dice` with joy or surprise. «📺 Guardalo dal vivo» opens the window on « 🧠 Spiega» and the section behind it (`BSISpiega.mostraSezione`); «🧠 Spiegamelo tutto» calls `BSISpiega.chiedi(title + question, { schermo: true, livello })`, so it goes through the same provider and the same screen rules as «🧠 Spiega»; «🔊 Ascolta Spectra» uses `BSISpectraLive.parla`. The Explore sections count as AI sections: closing the window returns there. No model call without a user gesture; predictions stay in `bsi-esplora-v1`.

- **The voice** (`bsi-spectra-voce.js`, `BSISpectraVoce`): `qualita()` is

`'audio'` with an OpenAI key in `bsi_api_keys` and Web Audio available, `'sistema'` with the browser's speech synthesis only, `'testo'` without. In audio mode the text (up to 3,800 characters) goes to `api.openai.com/v1/audio/speech` (`gpt-4o-mini-tts`, voice «onyx», with instructions on the timbre; `tts-1` if that fails); the decoded audio stays in an in-memory cache per sentence and goes through the `TIMBRO` chain (bass +5 dB below 140 Hz, −3 dB at 350 Hz, +3 dB at 2.8 kHz, ring modulator at 55 Hz at 12 %, 4.3 ms comb, saturation, synthetic 1.4 s reverb at 16 %, compressor). An analyser gives energy and spectral centroid every frame for the lip sync. The text to be spoken leaves the device: see `03-Security-Privacy-Compliance.md` §1.2.

Like the mind and the face, the body is a **simulation of behaviour, not consciousness**.

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5. Verification

Verification was carried out at three levels, with recorded outcomes.

5.1 Numeric correctness — against reference values

Area	Case	Expected	Obtained
Balancing	$KMnO_4$ + HCl → …	2:16:2:2:8:5	✓ exact
Molecular mass	$K_4[Fe(CN)_6]$ (nested)	368.35	368.345

Area	Case	Expected	Obtained
Molecular mass	$CuSO_4 \cdot 5H_2O$ (hydrate)	249.68	249.677
Equilibrium	acetic acid 0.1 M, K_a $1.8 \cdot 10^{-5}$	pH 2.874	2.875
Thermodynamics	ΔG (N_2O_4/NO_2) at 298 K	+4.79 kJ/mol	✓
Kinetics	E_a from k(300 K), k(320 K)	55.3 kJ/mol	55.33
Real gases	van der Waals CO_2	−10 % deviation	−10.13 %
Spectroscopy	bromine M+2	97.3 %	✓
Biochemistry	glycylglycine	132.12 Da	✓
Pharmacokinetics	t½ from V_d 50 L, Cl 5 L/h	6.93 h	✓
Astrophysics	solar emission peak (Wien)	501.5 nm	501.52
Astrophysics	Earth escape velocity	11.19 km/s	11.186
Nuclear	^{56}Fe binding energy	8.79 MeV/nucleon	8.790
Statistics	critical t, 95 %, df = 4	2.7764	✓
Crystallography	copper density (fcc)	8.96 g/cm³	8.935

P-values are not tabulated: they are computed with the regularised incomplete beta function (Lentz continued fraction) and the incomplete gamma function.

5.2 Robustness — cases that must fail

Verified as **rejected**: unbalanceable equations, formulas with unbalanced parentheses or non-existent elements, species absent from the equation, invalid database types, and the 10 injection attempts against the expression engine.

5.3 Integration — real browser

Automated tests with headless Chromium across all 14 application pages: no JavaScript errors, no missing resources. All 194 sections of `index.html` and the 18 tabs of the Astrochemistry module were traversed one by one. Request bodies were inspected for the four Claude tiers, and a full cycle with web search, turn suspension and resumption was simulated.

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6. Quantitative summary

Metric	Value
Component size	6,254 lines
Tools	47 (38 in `bsi-ai-hub.js`, 9 in `bsi-spectra-live.js`)
Model configurations	10 (3 free)
Scientific areas covered	13

Metric	Value
Records in exposed internal datasets	over 800
Max agentic loop rounds	10
History sent to the model	40 turns
Runtime dependencies	none

7. Stated limitations

Documented for technical honesty; these are deliberate choices, not omissions.

- **The API key lives in the browser** — unless the proxy is deployed. In

the default configuration the key never leaves the device, but it is accessible to anyone with access to that browser, and every user must obtain their own. The Worker in section 2.5 removes this limitation by moving keys to a server, at the cost of one component to maintain outside the page: a distribution choice, not a requirement.

- **Network tools depend on service availability.** PubChem and PubMed are

queried directly from the browser; when unreachable, the tool returns an explicit error and the agent is instructed to say so rather than substituting an estimate.

- **Solvers apply simplified models** where the literature offers more refined

ones (e.g. Woodward–Fieser for UV, or ATP yields using average P/O ratios). Each tool states the method used in its own result.

- **The screen context and the mind's summary go to the provider.** When a

question goes to the model, the chosen provider also receives the summary of the open section and, if there is memory, the mind's summary (§4.10, §4.12). Only the answers to local commands stay in the browser.

- **Verification is functional and numeric**, not formal: there is no proof of

solver correctness, but a set of reference cases drawn from the teaching literature.

Companion document to the BioSpecInfo technical dossier. See also: `01-Software-Architecture-Document.md`, `02-Verification-Validation-Report.md`, `03-Security-Privacy-Compliance.md`.

Scientific Accuracy and Data Provenance — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	`bsi-v250`
Purpose	Document how the scientific data shown by the application are generated, by what method they are verified, and what the declared limits are.

> **Why this document exists.** A chemistry teaching application can be impeccable from a software point of view and wrong from a scientific one, and the user has no way of noticing: a plausible molecular weight, a convincing-looking spectrum and a wrong structure all present themselves in the same way. This document describes the checks that make that error **visible and automatic**, and declares what stays beyond their reach.

1. Principle: every scientific datum must be open to contradiction

The criterion adopted throughout the application is that a scientific datum is not considered verified because it *looks* right, but because **a second, independent datum exists that would confirm or refute it**. Where that second datum does not exist, the limit is declared instead of being filled with an estimate presented as a measurement.

Three concrete applications of this principle:

Domain	First statement	Second, independent statement	Check
Drug database	structure (SMILES)	molecular weight from the literature	the weight recomputed from the structure must agree
IR / NMR spectra	functional groups recognised	molecular structure via RDKit	groups are sought on the graph, not on the text of the SMILES
Physical constants	tabulated value	symbol and unit of measure	the lookup refuses ambiguous matches instead of picking one

2. Pharmacological database — 263 entries

2.1 The verification method

Every entry declares **two** properties of the same molecule: the structure, in SMILES notation, and the molecular weight. They are independent statements — the first describes atomic connectivity, the second is a number taken from the literature — and therefore comparable:

```
weight computed by RDKit from the structure   ↔   weight declared in the table
```

A discrepancy above **0.6 u** (a tolerance that covers rounding and small differences between atomic-mass tables, but not a real error, which is worth tens of units) indicates that at least one of the two statements is wrong.

The check is automated (`tools/verifica-farmaci.js` , runnable by anyone holding the repository) and requires no human judgement: it does not ask anyone to trust a source, it compares two numbers.

2.2 What it found

Applied to the 143 pre-existing entries, the check identified **89 errors**. The distribution is significant: the **molecular weights were correct** — taken from the literature — while the **structures were wrong**.

Drug	Structure declared
Ketamine	an unrelated pyrrolidinone
Morphine	non-matching structure (in two separate entries)
Lorazepam	missing the 3-position hydroxyl that defines its identity
Atorvastatin	missing the anilide group
Testosterone	missing the Δ^4 double bond of ring A

Plus **7 duplicated drugs** in different categories, with data that sometimes diverged between the two copies.

2.3 The discrepancy as a diagnostic instrument

The numerical value of the difference is not merely an alarm: it indicates **which** fragment is missing or superfluous.

Δ (u)	Interpretation	Real case
28.05	C_2H_4	ketamine, fentanyl, morphine
16.00	one oxygen atom	tetracycline (one hydroxyl too many)
14.02	a CH_2 group	doxycycline
2.02	a missing double bond	testosterone (Δ^4)

Δ (u)	Interpretation	Real case
0.98	cytosine ⟷ uracil	**sofosbuvir**

The last case deserves a note: a discrepancy of 0.98 u corresponds exactly to replacing an −NH group with an oxygen atom, that is, to the difference between cytosine and uracil. Sofosbuvir is a **uridine** analogue, and the proposed structure contained the wrong base. The number showed where to look.

2.4 Current state and declared limits

Drugs in the database	**178** (it was 143)
With a verified structure	**157** (it was 153)
Defects	**0**
Declared and accepted deviations	**21** (it was 25)

Verification is **conformant**: no entry presents a structure that contradicts its own molecular weight. The 21 deviations are entries **without a structure**, each recorded with its own reason in `../evidence/deviazioni-note.json` and reported in the verification report.

Defect versus deviation. The check distinguishes two situations that must not be confused:

- a structure that **contradicts** its own weight is a defect, and fails

verification;

- an **absent** structure may be an acceptable deviation — but only if

explicitly recorded, with its reason. An entry without a structure and **not** recorded fails verification like any other defect.

Recording a deviation is therefore a deliberate decision, traced in git and visible in the report, not a way of silencing a check.

And it must also be **revoked** when it is no longer needed. From version `bsi-v188` the bench also fails in the opposite case: an entry listed in the registry that **does** have a verified structure is a permission left switched on for nothing, and tomorrow it would silently cover a wrong structure put in its place.

2.4-bis Four entries that left the registry

Of the six entries without a structure, four were closed by taking the structure from **ChEMBL** — a source independent of this project — and verifying it with the same structure ⟷ molecular weight comparison that applies to all the others.

Entry	ChEMBL	Formula	Weight declared	Weight computed
Digoxin	`CHEMBL1751`	$C_{41}H_{64}O_{14}$	780.94	780.95
Vincristine	`CHEMBL90555`	$C_{46}H_{56}N_4O_{10}$	824.96	824.97
Tacrolimus topical (Protopic)	`CHEMBL269732`	$C_{44}H_{69}NO_{12}$	804.02	804.03

Entry	ChEMBL	Formula	Weight declared	Weight computed
Tacrolimus systemic (Prograf)	`CHEMBL269732`	$C_{44}H_{69}NO_{12}$	804.02	804.03

The reason recorded for digoxin said "every structure tried departs by 14-30 u from the literature weight": the declared weight was right, it was the structures tried that were wrong. That the comparison is real and not complacent can be seen by adding a single carbon to the correct structure — the bench reports **Δ 14.04** and fails.

The two that remain are not single molecules, and no SMILES notation represents them:

- **Ivermectin** is a mixture of homologues: at least 80 % B1a ($C_{48}H_{74}O_{14}$) and no

more than 20 % B1b ($C_{47}H_{72}O_{14}$). The declared weight, 875.10, is that of the B1a component alone. Showing its structure would pass off the majority homologue as the whole drug.

- **Artemether/Lumefantrine (Coartem)** is a combination of two distinct active

ingredients.

In neither case is a structure missing: there is no single one to show.

Group	Entries	Nature
D-01 — not single molecules	2	ivermectin (mixture of B1a/B1b homologues) and artemether/lumefantrine (combination of two active ingredients). It was six: the other four were closed with the structure from ChEMBL, see §2.4-bis.
D-02 — not representable in SMILES	19	monoclonal antibodies, proteins and peptides (trastuzumab, pembrolizumab, insulin, semaglutide, ciclosporin A…). The absence is correct, not a gap.

> The choice is deliberate: a wrong datum replaced by another wrong datum is no progress, and a declared absence is preferable to an unverified presence.

---

3. Spectral prediction

3.1 Structural, not textual, recognition

Functional groups are identified with **SMARTS patterns applied to the molecular graph** through RDKit (`bsi-spettri.js`), not with regular expressions applied to the SMILES string.

The difference is not stylistic. The earlier textual method had two structural defects:

1. **Exclusivity.** The checks were organised in `else if` chains, so **only one

functional group per molecule** was recognised.

1. **Blindness to structure.** A regex reads characters, not bonds.

The emblematic case is **acetylsalicylic acid** (`CC(=O)Oc1ccccc1C(=O)O`), which contains both an ester and a carboxylic acid. Textual recognition identified only the ester; the resulting IR spectrum lacked the broad O–H band at ~3000 cm^{-1} and the acid C=O at ~1690 cm^{-1} — the two bands that identify the molecule — and the ^1H NMR spectrum lacked the carboxylic proton signal at 11.6 ppm.

Graph-based recognition is **additive by construction**: a molecule may present every group it possesses.

3.2 Representation conventions

Axis conventions are not uniform across techniques, and are verified automatically (`test_assi.js`, `test_assi_canvas.js`) by extracting the tick labels from the drawing produced:

Technique	Abscissa	Direction	Ordinate
IR	wavenumber (cm^{-1})	decreasing, 4000 → 400	**% transmittance**, bands pointing down
^1H NMR	δ (ppm)	decreasing, 14 → 0	intensity, TMS reference at 0
UV-Vis	λ (nm)	increasing, 200 → 400	absorbance
MS	m/z	increasing	relative abundance

3.3 Determinism

No spectrum contains random components. Earlier versions added pseudo-random noise to the trace on every redraw, with the consequence that the same molecule produced a different spectrum each time and two spectra were not comparable. A computed profile must be **reproducible**: where a realistic baseline is needed, a deterministic function of the wavenumber is used.

Likewise, in the NMR predictor the peak heights were generated at random. In a ^1H spectrum the height **is** the quantitative information — the integration — and inventing it is worse than omitting it. Currently:

- where the number of protons is known, the height is the real integration and

the label reports it (e.g. `7.26 (4H)`);

- where the prediction is a **range** (e.g. "6.5–8.5 ppm"), the band of the range

is drawn with a mark on the typical value and a uniform height: the chart shows *where* the signal falls without asserting how intense it is.

3.4 Chemical effects modelled

Effect	Implementation
Carbonyl conjugation	a ~25 cm^{-1} lowering applied **to the individual carbonyl**, not globally
Phenolic ester	raising to ~1760 cm^{-1}
C–H intensity	proportional to the number of CH_2/CH_3 groups present

Effect	Implementation
Aromatic ring substitution	"out of plane" bands 900–690 cm^{-1} distinguished for ortho/meta/para/mono
Aldehyde Fermi doublet	bands at 2820 and 2720 cm^{-1}
Amide bands I and II	1655 and 1550 cm^{-1}

Verification: acetylsalicylic acid produces an aryl ester at 1760 cm^{-1} and a conjugated acid at 1690 cm^{-1} — the two values that identify it — while acetophenone (conjugated ketone, 1690) and ethyl acetate (unconjugated ester,

1. confirm that the effect is applied selectively and not indiscriminately.

3.5 Declared limits

> The spectra produced are **predictions based on functional-group tables**, not measured spectra nor quantum-mechanical simulations. They indicate where the bands of the groups present fall, with tabulated positions and intensities. **They do not replace an experimental database** (SDBS, NIST) nor an instrumental measurement, and the chart says so.

In particular the following are not modelled: second-order spin-spin couplings, solvent and concentration effects, combination bands and overtones, solid-state polymorphism.

3-bis. Cheminformatics and QSAR models

3-bis.1 Where the methods come from

None of the methods in the **Cheminformatics** section was invented for the occasion. They are the ones in use, with the references that define them.

Method	Reference	How it is implemented here
Tanimoto coefficient	Rogers & Tanimoto, 1960	Bits in common over bits in union, with the 0/0 = 1 convention declared in the code
Morgan / ECFP fingerprints	Rogers & Hahn, *J. Chem. Inf. Model.* 50 (2010) 742	From RDKit MinimalLib, radius 2, 2048 bits
MACCS keys	Durant *et al.*, *J. Chem. Inf. Comput. Sci.* 42 (2002) 1273	From RDKit MinimalLib, 166 keys
Rule of five	Lipinski *et al.*, *Adv. Drug Deliv. Rev.* 23 (1997) 3	Over the RDKit descriptors, with the violation count
Veber filter	Veber *et al.*, *J. Med. Chem.* 45 (2002) 2615	Rotatable bonds ≤ 10, TPSA ≤ 140 $Å^2$
QED	Bickerton *et al.*, *Nat. Chem.* 4 (2012) 90	**Recomputed**: MinimalLib exposes 43 descriptors but not QED. The desirability-function parameters are those of the paper

Method	Reference	How it is implemented here
Butina clustering	Butina, *J. Chem. Inf. Comput. Sci.* 39 (1999) 747	Threshold on the Tanimoto distance, leaders chosen by neighbour count
Bemis–Murcko scaffolds	Bemis & Murcko, *J. Med. Chem.* 39 (1996) 2887	Ring systems plus the bonds that connect them
PAINS	Baell & Holloway, *J. Med. Chem.* 53 (2010) 2719	A subset of the SMARTS patterns, with the reason for the flag
Brenk alerts	Brenk *et al.*, *ChemMedChem* 3 (2008) 435	Likewise
SALI	Guha & Van Drie, *J. Chem. Inf. Model.* 48 (2008) 646	Δactivity / (1 − Tanimoto), over the pairs above the similarity threshold

3-bis.2 The four things that make a QSAR honest

A QSAR model is extremely easy to make look good. The four precautions below are what separate a number from a measurement, and all four are verified by the `test_cheminfo` bench.

Scaffold split. With a random split, close analogues from the same series end up on both sides: the model recovers what it has already seen. The scaffold split groups the molecules by Bemis–Murcko scaffold and assigns **whole scaffolds** to one side only. The R^2 that comes out is lower — and it is the one that survives the new molecule.

Null model by scrambling. The same model is retrained eight times on shuffled labels. If the true score does not exceed the best of the twins, the model has learnt nothing transferable, and the panel says so in those words. It is the check most often missing: without it, an R^2 of 0.4 over thirty molecules is indistinguishable from chance.

Applicability domain. A prediction on a molecule far from everything the model has seen is an extrapolation, not a prediction. The distance to the nearest training neighbour is shown next to every predicted value.

Scaffold-grouped cross-validation. A single split, over a small set, is a noisy number: on 28 molecules at 25 % the test set holds seven, and moving one shifts the R^2 by tenths. The folds group by scaffold — no scaffold sits in two folds — and the result is given as **mean ± deviation**, not as a single figure.

> **How much this matters, measured on the 28-inhibitor example.** The single split gives R^2 **0.861**. Five-fold validation gives **0.605 ± 0.452**, with the individual folds at 0.861 · 0.538 · 0.795 · **−0.150** · 0.980. The single split had drawn the lucky fold. Together the two figures say what neither says alone: the model works, but it depends appreciably on which molecules it gets for training — and on a set of twenty-eight molecules that is exactly what one should expect.

3-bis.3 Declared limits

Limit	Consequence
The example sets are **small** (28 and 40 molecules)	They exist to show the method, not to produce a usable model. On sets that size, the null model is the only serious defence — which is why it is there
The model is **linear in the kernel** (kernel ridge, Tanimoto kernel) or logistic	No neural network, no gradient boosting: on a few hundred molecules these give no demonstrable advantage, and the cost would be a model one cannot inspect
PAINS are not a verdict	A PAINS pattern signals that the compound has often been a false positive in fluorescence assays, not that it is inactive. The panel writes this next to every flag
The **PCA is on the descriptors**, not on the fingerprints	On binary fingerprints PCA is not very informative; loadings on descriptors can be read, and are shown
There is no **large-scale substructure search**	The workbench is sized for the sets one pastes in by hand, not for a library of millions of compounds

---

3-bis.4 Two surfaces, two roles — and a number that was false

The application has **two** places where cheminformatics happens, and they serve different purposes. The distinction is written here because an assessor who opens the wrong page first would form the wrong impression.

	`rdkit_lab.html` — "RDKit Lab"	Cheminformatics section
Who it is for	Someone studying: one molecule at a time, to see what RDKit can say about it	Someone working: a set of molecules, to derive a defensible model
Input	One SMILES	A pasted set, with the measured activity
QSAR	**Empirical** estimates from the descriptors, declared as such in the panel	A model **trained** on the supplied data, with scaffold split, cross-validation and null model
Output	What is on screen	CSV and method report

> **A number that was false, and how the bench caught it.** The lab's "Pharma Pro" panel compared the molecule against eight reference drugs and displayed a similarity percentage. That percentage did not come from a fingerprint: it came from **eight bits of thresholded descriptors** — "has aromatic rings", "HBA > 4", "weight between 200 and 500" — over which a Tanimoto was computed. Measured: **caffeine against metformin gave 0.75**, while on Morgan fingerprints it is **0.024**. Two molecules with no fragments in common, shown at 75 %, on a page called "RDKit Lab" where RDKit was already loaded.
>
> Alongside that, three of the eight reference fingerprints were **written by hand** and had the "aromatic" bit wrong (caffeine, which has two aromatic rings, morphine and amoxicillin which have one), and the **SMILES for omeprazole was not omeprazole**: `COc1ccc2[nH]c(=S)cc2c1OC` is a structure RDKit rejects.

> All corrected: the comparison uses the Morgan fingerprint with the shared engine's Tanimoto, the fingerprints are computed rather than typed in, and omeprazole is omeprazole (monoisotopic mass 345.11, three aromatic rings — both verified). The two functions of the fake similarity were **removed**, not bypassed: leaving them around would have put them back in use at the first edit, and the bench checks that they no longer exist.

One implementation of Tanimoto. The lab had its own, which agreed with the engine's on ordinary cases but answered 0 where the engine answers 1 (two empty fingerprints). The page now loads `bsi-cheminfo.js` and calls that: `test_cheminfo` compares the two routes over all 28 reference pairs and demands a deviation of **exactly zero**.

3-ter. Alternative text for figures is not invented

The biochemistry guide carries **98 figures**. On replacing it, the accessibility bench rejected it: none had an `alt` attribute.

The easy way out was `alt=""`, and the bench would have accepted it — the comment in its code says, correctly, that for a **decorative** image this is the right thing: it tells the screen reader to skip it. But these are biochemistry figures, that is, information. Marking them decorative would have passed the check by telling screen-reader users to ignore 98 diagrams: a hollow green, of the same family as the "contrast 0" that did not cover the new panels (§3-bis) and the 100 % coverage that summed the wrong ranges.

The honest way was available: every image sits inside a `<figure>` with its own `<figcaption>`, and **the captions were written by the document's author**.

Figures	62, holding 98 images
Single-image figures	26 — the alt is the caption, minus the "Figura N." label
Two-panel figures	36
Pair captions split between the two panels	**33 out of 36**
Pairs sharing one caption	3 — they do not split unambiguously

All 36 pair captions distinguish the two panels with "a sinistra" (left) and "a destra" (right): where the split is unambiguous, each image receives its own half of the sentence. Where it is not, the two images share the whole caption — verbose to listen to, but **it cannot be wrong**, and a wrong description of a figure is worse than a verbose one.

> **Why not attempt a more aggressive split.** The remaining 3 could have been squeezed out with looser regular expressions. A split that goes wrong produces "this image shows X" next to an image that shows Y, and the screen reader reads it with the same confidence as the correct ones. The fallback is noisy; the error is invisible.

Measured after the change: **98 out of 98 with alternative text, none empty**, contrast **0** over 1,897 text elements.

3-quater. From prototype to working tool

The functions described in §3-bis were enough to say something defensible about a set of molecules. They were not enough to **work** with it: on day one, a cheminformatics group asks to search a substructure across the whole set, to see the SAR table, and to know what happens to activity when a chlorine is replaced by a methyl.

3-quater.1 What the library had, and what it does not

Before designing anything, MinimalLib 2025.03.4 was **probed at runtime** rather than assumed. Probing first changed the design: three of the capabilities below had gone unused because nobody knew they were there.

Present	What it is for
`SubstructLibrary`	Substructure search with a pattern fingerprint: it discards in bulk the molecules that cannot match, instead of attempting isomorphism on each
`get_rxn` · `Reaction.run_reactants`	Chemical transformations. This is what makes it possible to **cut** a bond, and from there come matched pairs and the SAR table
`generate_aligned_coords`	Aligning molecules to a common core
`get_molblock` · `get_v3Kmolblock`	**SDF** export, the format in which sets are exchanged between groups and programs

Absent	Declared consequence
`cleanup`, `neutralize`	Standardisation stops at the largest fragment and the canonical SMILES: **charges are not neutralised**
`canonical_tautomer`	Two tautomers written differently remain two distinct entries
`FragmentOnBonds`, `RWMol`	No direct molecule editing: cutting goes through a reaction
Maximum common substructure (MCS)	A SAR table's core must be **supplied**, not deduced. The panel offers eight ready cores and a free field

3-quater.2 The cut, and why two rules are needed

A reaction SMARTS breaks a bond and marks both ends with a dummy atom. Verified at runtime: `CCOc1ccccc1` gives three cuts under the fine rule (CH_3–CH_2, CH_2–O, O–aryl) and two under the coarse one; benzene and methane give none.

Rule	Cuts	When it is needed
fine (default)	Any acyclic single bond, **including terminal substituents** Cl, CH_3, OH, F	SAR table and fine transformations (Cl → CH_3)

Rule	Cuts	When it is needed
coarse (BRICS-style)	Excludes terminal atoms	Large sets, where fine cuts are too many; produces larger variable parts ("chlorophenyl → phenyl")

> **The fine rule was born of a defect.** The first draft used only the coarse one, and the SAR table of the test series **lost the chlorine on the ring**: it found the substituent on the nitrogen and missed exactly the column that matters, because chlorine is terminal. The fine rule is a superset of the coarse one: every cut the coarse rule finds, the fine one finds too.

3-quater.3 Matched molecular pairs

Two molecules form a pair when, cutting one bond in each, they are left with the **same context** and two different variable parts. The activity difference is attributed to that substitution and to nothing else — which is why this reading beats a correlation over the whole set, which averages over molecules differing in ten ways.

Verified on a series **built with known effects**: if the analysis does not recover them, it is not usable on real data.

Transformation	Expected	Found	Concordant pairs
chlorophenyl → phenyl	−1.0	**−1.0**	2 of 2
methylphenyl → phenyl	−0.5	**−0.5**	2 of 2
methyl → chlorine	+0.5	**+0.5**	2 of 2
ethylamide → propylamide	+0.2	**+0.2**	3 of 3

On the 28 example inhibitors, taken from ChEMBL: **449 pairs, 98 transformations** seen at least twice, 199 seen once, in 133 ms. At the top, **CF_3 → SO_2NH_2 with a median Δ of −2.65** over two concordant pairs.

Two declared choices:

- the **direction is normalised** on the alphabetical order of the two parts.

Without it the same effect would appear twice, with opposite medians, and neither would carry the right pair count;

- transformations seen **fewer than twice** are excluded from the table and

counted separately. A transformation seen once with $\Delta = +3$ is not a discovery: it is an anecdote, and putting it at the top would have decisions made on it.

> **The "concordant" column is the one that decides whether to trust it.** A median of +1.0 over three pairs one of which is −2.0 is not the same as +1.0 over three pairs all positive, and the median alone does not say so.

3-quater.4 The SAR table, and how the position is found

The hard part is not finding the substituents: it is knowing which **position** each occupies. With no MCS and no molecule editing, the solution exploits a property of the cut: **the fragment containing the core carries the dummy atom exactly at the attachment point**. Matching the core onto that fragment reveals which of its atoms the dummy hangs off, and therefore which position the substituent occupies.

This also yields a necessary and free filter: if the dummy is **not** attached to a core atom, the cut happened inside a substituent and that piece is not the whole substituent. It is discarded.

On the 28 inhibitors, with a benzamide core: **20 molecules with the core, 2 positions, 8 excluded**. The excluded ones do not appear in the table: showing them with empty cells would make them look like part of the series with substituents that were not found.

> **A naming defect, and how it showed itself.** `scaffoldMurcko()` returned its result in a field called `smiles`, but that value **is not a SMILES**: it is a canonical fingerprint of the scaffold, built so that two molecules with the same scaffold produce the same string. It serves to **group** — folds, scaffold split — not to query. The SAR panel took it for a SMILES and proposed `6,6,6,6,7,…|0-13:1,…` as the core; decomposition answered "core not interpretable".
>
> The field is now called `chiave` (key) and declares `eUnoSmiles: false`; `smiles` remains as an alias. The bench pins that equal scaffolds give the same key, that different scaffolds give different keys, and that the key **does not read back as a SMILES**. A field's wrong name is a defect like any other.

3-quater.5 The rigour an assessor demands

Five things a modelling group puts in every report, and whose absence is the first objection in a meeting.

	Method	Why
Model comparison	Trivial reference (predict the mean), kNN over Tanimoto, kernel ridge — all on the **same folds**, grouped by scaffold	Comparing a model evaluated on one split with another evaluated on a different split is not a comparison. And the trivial reference sits at the top of the list, because it is the number every other must beat
Nested validation	Hyperparameters (k, lambda) are chosen on folds **internal to the training set**	Trying ten of them on the test set and reporting the best inflates the score: that number is not a prediction, it is the maximum of ten attempts
Conformal intervals	Split-conformal, quantile $\lceil (n+1)(1-\alpha) \rceil$ of the absolute residuals on a calibration portion split **by scaffold**	"7.2" and "7.2 ± 0.3" are two different pieces of information, and only the second says whether synthesis is worth it. Calibrating on analogues of the training set would give intervals too narrow precisely on new molecules

	Method	Why
Enrichment	EF at 1 %, 5 %, 10 % with the **maximum achievable** alongside; BEDROC (Truchon & Bayly, *J. Chem. Inf. Model.* 47 (2007) 488)	In a screen one does not look at the whole list. A ROC-AUC of 0.80 can hide a top-of-list with not a single active, because AUC also rewards the ordering in the tail, which nobody will buy
Learning curve	Score against training-set size, with the test set **whole** at every point	It answers the only question that matters when the model is mediocre: more chemistry or more data?

The comparison's verdict is proven in both directions, like the null model. On a real signal — activity proportional to halogen count, over ten distinct cores — kernel ridge reaches R^2 0.943 with a margin of 0.943 against a dispersion of 0.035, and the verdict is *beats it*. On the **same molecules with random labels** the best stops at −0.265, margin −0.034, and the verdict is *does not beat it*. A guard that never fires and one that always fires are the same defect.

Enrichment is verified on the three cases where the right value is computable by hand:

Ordering (2 actives of 10)	AUC	EF@20 %	BEDROC
Perfect: both actives on top	1	**5** — and it is the maximum achievable	**1**
Worst: both actives last	0	0	**0**, not a negative
Actives at ranks 2 and 5	**0.75** = 12/16	**2.5** = 1/0.4	0.119

The worst case is the one that counts: without it one would not know whether BEDROC reaches zero or goes below, and a negative BEDROC is the sign that the formula's additive term is missing.

3-quater.6 Limits of this block

Limit	Consequence
Pairs grow with the **square** of the molecule count	Measured: 6 molecules 45 ms, 40 molecules 227 ms with 1,255 pairs. Beyond 300 molecules the panel refuses to start and says so, instead of freezing the page
The cut is **single**	A pair differing by two substitutions at once is not found. This is the limit of the single-cut scheme, shared with the standard tools
The same transformation appears in **several forms**	Cuts at different positions give different contexts for the same chemical change (`*CC>>*CCC`, `*NCC>>*NCCC`, `*C(=O)NCC>>*C(=O)NCCC`). This is expected redundancy in the single-cut scheme, not an error
Charges are **not** neutralised and tautomers not canonicalised	MinimalLib does not allow it (§3-quater.1). A set mixing neutral and ionic forms of the same molecule counts them as distinct entries
The SAR table's core must be **supplied**	No MCS available. The panel proposes nothing automatically, because a wrong proposal is worse than no proposal

3.6 Per-atom NMR prediction: the source and the measured deviation

The $^{1}H/^{13}C$ predictor that assigns every signal to an ATOM (`bsi-nmr.js`) contains no numbers. The estimation tables live in **`bsi-pretsch.js`**, transcribed line by line from:

> E. Pretsch, P. Bühlmann, M. Badertscher, *Structure Determination of Organic Compounds — Tables of Spectral Data*, 4th ed., Springer. §4.1 pp. 82-84 · §4.5 pp. 100-102 · §5.1 p. 170 · §5.2 pp. 178-179 · §5.3 p. 182 · §5.5 pp. 188-189

They sit in a separate file so they can be **checked against the printed page** without reading the code that uses them, and so a wrong line is fixed in one place only.

scheme	formula	rows
monosubstituted benzenes, ^{13}C	$\delta = 128.5 + \Sigma Zi$	91
monosubstituted benzenes, ^{1}H	$\delta = 7.34 + \Sigma Zi$	66
substituted ethylenes, ^{1}H	$\delta = 5.25 + Zgem + Zcis + Ztrans$	42
substituted alkanes, ^{1}H	$\delta = base(CH_3/CH_2/CH) + \Sigma Z\alpha + \Sigma Z\beta$	31
terminal alkynes, ^{1}H	direct value per substituent	30
aliphatics, ^{13}C	$\delta = -2.3 + \Sigma Zi + \Sigma Sj$	24
steric corrections Sj	by degree of the observed C and of the α atom	4×4

The deviation is measured, on molecules that did not choose the parameters. The test molecules are split into two sets: *tuning*, used to choose values and rules, and *validation*, never used for that. The figure declared in the application's panel is the validation one.

	measured mean deviation
^{13}C, tuning (9 molecules)	0.71 ppm
^{13}C, validation (22 molecules)	**0.92 ppm**
^{13}C, worst case	4.9 ppm (cyclohexanone)
^{1}H (14 molecules)	0.06 ppm
^{1}H, aromatics only	0.03 ppm

The bench (`test_nmr`, 56 checks) also requires validation to stay **worse** than tuning: were they to become equal it would mean a molecule had been moved from one set to the other, and the figure would no longer say anything.

What is still wrong has not been corrected after the fact. Biphenyl misses its ipso carbon by 4.6 ppm, because the table's phenyl increment ($Z_1 = 8.1$) does not describe it. That number **has not been adjusted**: changing a transcribed value because a validation molecule does not fit is exactly how the 0.92 ppm declared above would become a lie.

Declared limits. This is not a quantum calculation: it does not predict solvent effects, does not distinguish conformers (the source's conformational corrections K are transcribed but evaluate to 0, because conformation cannot be derived from a SMILES) and does no two-dimensional NMR. Coupling constants come from a table of **typical** values by geometric rela-

tionship, not calculated. **Substituted heteroaromatics** have, in this source, no positional increment table as benzenes do: they remain the module's least reliable prediction, and the elucidation tool declares it by lowering the confidence when it meets one.

3.7 Two-dimensional NMR: COSY, HSQC, HMBC

`bsi-nmr2d.js` holds no shifts: it asks the per-atom predictor for them and builds the correlations by **walking the graph**. This is why it could only be done once the Pretsch tables arrived: it needs the shift of *every* proton and *every* carbon, plus the index of the atom carrying them.

map	topological rule
HSQC	proton and carbon **one** bond apart (^{1}J); inverted sign on CH_2
COSY	two protons **three** bonds apart (H–C–C–H), excluding equivalents
HMBC	proton and carbons **two or three** bonds away (^{2}J, ^{3}J)

Exchangeable hydrogens (O–H, N–H) give no off-diagonal spots: they exchange with the solvent too fast for the coupling to be seen. They do appear on the COSY diagonal, because they are protons like any other.

Verified by counting paths (`test_nmr2d`, 40 checks), in both directions: hexafluorobenzene gives no HSQC spots, benzene gives no off-diagonal COSY spots, methane gives none in HMBC.

Declared limits. This is not a simulation of the experiment: no computed intensities, no artefacts, no residual coupling, no mixing-time dependence. In a real HMBC some two-bond correlations are not seen; here they all are. The shifts inherit the predictor's measured uncertainty (0.9 ppm on ^{13}C, 0.06 on ^{1}H).

3.8 3D geometry: built from the graph, not downloaded

`bsi-geom3d.js` builds coordinates by **distance geometry**: a bounds matrix from bonds, valence angles, ring diagonals and van der Waals contacts; initial positions random but **reproducible**, because the generator is seeded from the SMILES; iterative correction of the violations. Aromatic systems are projected onto their plane and linear triples are straightened by rotating the branches as rigid bodies, because those two things a distance alone cannot impose.

The indices are the NMR predictor's, and hydrogens are appended at the end: this is the condition without which the peak ↔ atom ↔ 3D link would light up the wrong atom. The PubChem-fed 3D viewer could not be used for this — an SDF's numbering bears no relation to RDKit's.

Measured against known values (`test_geom3d`, 29 checks): C–C 1.54 Å, C=C 1.34, C≡C 1.20, aromatic 1.39, C–H 1.09; tetrahedral angle 109.5°, aromatic 120.0°, alkyne 180.0°. **In both directions**: benzene planar (0.00 Å), cyclohexane **not** planar (0.34). On real molecules the residual stays under 0.03 Å up to about thirty atoms and reaches 0.31 Å on a 76-atom molecule.

Declared limits. It is not a force field and does not minimise an energy: it minimises geometric violations. It does not choose the most stable conformer and does not handle stereochemistry — an R centre and its enantiomer come out the same.

3.9 From document to working: what is deduced and what is read

`bsi-documento.js` opens the file; `bsi-quesito.js` recognises the data in the text; `bsi-elucida.js` works through them. They are three separate steps **on purpose**, because they fail in different ways and conflating them hides the error: if recognition reads "1715" as an IR band when it was a mass, the working that follows is impeccable and the conclusion is wrong. This is why the panel shows **the table of data read above the working**, each entry with where it came from.

Recognition is by section, not by number. An IR band at 1738 and a mass at 150 are both numbers: telling them apart from the value alone is impossible. The technique's label is found (IR, MS, ^{1}H NMR, ^{13}C NMR) and what follows is read, up to the end of the paragraph or the next label.

What it does not do. There is no language model: there are regular expressions and a spectroscopic rule engine. If the text writes the data in an unforeseen form, the data are **not** read — and the module says so rather than working through half of them. The working goes as far as the rules go: compatible functional groups, fragments, counts. **The final structure is not proposed**, because proposing it would mean guessing; a structure can be written and the program **compares** it with the data, saying which signals fit and which do not.

There is no optical character recognition. A scanned page contains pixels, not letters: the text cannot be read, and the module declares it rather than returning an empty string as if the document were empty. The pages stay visible, and that is where pixel trace extraction takes over.

3.10 The two places where the predictor gives way, measured

Declaring "mean deviation 1.0 ppm" and stopping there hides what one actually needs to know: *where* it is wrong. There are two places, and they are known by measurement.

a) Substituted heteroaromatics — the table is missing, not the method

For benzenes the source gives positional substituent increments; for furan, thiophene, pyrrole and pyridine it **does not**. A substituted heteroaromatic therefore gets the values of the *un*substituted parent, and its β carbons all come out equal when in reality they are not.

Measured on the 23/04/2024 problem (structure `O=CC(=Cc1ccco1)CSC`):

observed	predicted	deviation	which carbon
192.04	192.0	0.04	CHO
150.77	150.0	0.77	furan α-C, substituted
146.05	142.7	**3.35**	furan α-C

observed	predicted	deviation	which carbon
135.55	135.0	0.55	=C
135.05	135.0	0.05	=C
118.55	109.6	**8.95**	furan β-C
112.97	109.6	**3.37**	furan β-C
26.67	34.0	**7.33**	CH_2 between C=C and S
15.72	15.6	0.12	S–CH_3

The two β carbons come out **identical** (109.6) because without positional increments nothing distinguishes them. Transferring the benzene increments to the heteroaromatic ring **was tried**: on the *ipso* carbon it works (151.6 against 150.77 measured), but on the β carbons it makes things **worse** — C3 would go from 3.4 to 5.7 ppm of deviation. It was not done: a generalisation that gets worse exactly where the problem is, is not a generalisation.

b) An sp^3 carbon with TWO α substituents

The increments are derived from **mono**-substituted compounds. Adding two of them on the same carbon counts twice an effect that in fact saturates, and the prediction comes out high. Measured twice, same magnitude and same direction:

molecule	carbon	observed	predicted	deviation
benzyl acetate	OCH_2 (ester O + ring)	66.3	73.8	**7.5**
the problem above	CH_2 (C=C + S)	26.7	34.0	**7.3**

It is the module's declared worst case, and benzyl acetate entered the validation set precisely to keep it measured.

The comparison tool says so. When a proposed structure contains such a carbon, confidence drops from "high" to "medium" and the reason appears: *a signal that does not fit on that carbon may be a limit of the predictor, not evidence against the structure*. Without that warning, a low score on a **correct** structure would read as a refutation — which is exactly what happened to benzyl acetate, which scores 42 out of 100 while being the answer.

3.11 What could be consulted, and what could not

A table of increments for substituted heteroaromatics would close the weak point described above. **From the machine this project is built on, no primary source is reachable**, and that was verified, not assumed:

route	outcome
`curl` to any domain	the network policy allows **only** package registries (npm, PyPI, crates, Go) — the proxy says so itself
NIST WebBook, SDBS (AIST), PubChem, SpectraBase	**403**, blocked

route	outcome
CSIRO, RSC, ACS, Springer, Wikipedia, LibreTexts	**403**, blocked
the Hans Reich tables (`organicchemistrydata.org`, `chem.wisc.edu`)	**403**, blocked
NMRShiftDB2 (`nmrshiftdb.nmr.uni-koeln.de`)	**403**, blocked
PubMed (through its service)	no results: it indexes biomedicine, not pure chemistry
npm and PyPI	no package redistributes a reference NMR database: `nmr-predictor` **downloads** it from GitHub, which is blocked; `nmr-processing` does not contain it

What text search could establish is *where* the data are, and that is a useful result even without having them in hand:

> M. T. W. Hearn, "Carbon-13 chemical shifts in some substituted furans and thiophens", *Australian Journal of Chemistry* **29**(1), 107–113 (1976). DOI 10.1071/CH9760107

The abstract states that "the substituent effects in these heterocycles **resemble** those reported for substituted benzenes". *Resemble* is not *equal*: transferring the benzene increments was tried (§3.10 a) and on the β carbons it makes things worse. The real table is still needed.

Why it was not written from memory. It would have been easy: those numbers are well known. But a value nobody can trace to a source is exactly what this document exists to prevent, and putting one in a table next to ninety-one rows transcribed from a printed page would make those less credible too. **A declared hole is worth more than a badly patched one.**

One validation row with weaker provenance

For honesty this too must be said: benzyl acetate, which entered the validation set because it shows the two-α-substituent weak point, carries values **not verified against a primary source** from this environment. The phenomenon it shows is independently confirmed by the 23/04/2024 exam problem — a datum with real provenance — but the **declared worst case (7.5 ppm) rests on that row**, and the bench says so in plain words.

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4. Physical constants and tabulated data

The lookup of physical constants proceeds by decreasing specificity (exact match → symbol → significant words → substring of at least 4 characters) and **collects every candidate**: if more than one remains, the request is refused with the list of possibilities rather than arbi-

trarily returning the first. A wrong physical value returned with confidence is worse than a request for clarification.

The exam molecules (`MOLECOLE_ESAME`) are tabulated with verified values; the recognition of groups from SMILES uses a `consuma` field, so that each group removes the portion of structure it has explained and no overlapping counts are produced.

4-bis. Astrophysics, physics and mathematics laboratories, study kits

4-bis.1 Where the data come from

Section	Data used	Source declared in the code
🪐 Solar system 3D, 🔭 Planetarium 3D	J2000 Keplerian elements of the planets and their secular rates	E. M. Standish's table (JPL), «Keplerian elements for the approximate positions of the major planets», valid 1800–2050
🪐 Solar system 3D, 🔭 Planetarium 3D	Julian date, sidereal time, Sun, Moon, Galilean moons	J. Meeus, *Astronomical Algorithms* (chapters 7, 12, 25, 44, 47; lunar theory truncated to the main terms, ~0.1°)
🔭 Planetarium 3D	About 170 bright stars: α, δ (J2000), V, B − V; 28 constellations	Rounded values; proper motions are neglected
🌠 Cosmology 3D	Cosmological parameters	Planck 2018: Ω_m = 0.315, Ω_Λ = 0.685, H_0 = 67.4 km/s/Mpc
🌌 Galaxies 3D	Local Group positions and distances	Compilations by Mateo (1998) and McConnachie (2012), rounded values
⭐ Life of stars 3D, ☢️ Nuclei and reactions	Atomic masses	AME2016/AME2020, rounded
🕳️ Black holes, 🔢 Constants and units	Physical constants	CODATA 2018; those defining the SI since 2019 are marked exact

Declared illustrative, and the sections say so: the exoplanet mass-period population (not a catalogue), the rotation-curve «observations» (the reference model with reproducible noise), the «Milky-Way-like» disc parameters, the cosmic microwave background map (synthesised with spherical harmonics from an illustrative power spectrum, tuned so that the peaks resemble Planck's).

4-bis.2 How it is verified

Each of the twenty new sections has a script in `tools/banchi/verifiche/verifica-<id>.js`, run by `test_v237`, that sets the computations against a known value or an independent second method. Some examples, with the value the script prints:

Section	Comparison	Obtained
🪐 Solar system	Julian date of 1957 Oct 4.81 (Meeus, ex. 7.a)	2436116.31
🪐 Solar system	Mars 2003: closest approach 0.3727 AU on 27 August	0.3730 AU on 27/8
🛰️ Orbital mechanics	Curtis, ex. 4.3: h, i, Ω, e, ω, ν, a from the state	58312, 153.25°, 255.28°, 0.1712, 20.07°, 28.45°, 8788 km
🛰️ Orbital mechanics	elements → state → elements over 2000 random orbits	relative error $2.8 \cdot 10^{-14}$
📐 Celestial mathematics	Laplace limit of the Lagrange series	0.6627434
⭐ Life of stars	Lane–Emden: first zero for n = 3	6.89685 ($\rho_c/\bar{\rho}$ = 54.18)
🕳️ Black holes	perihelion precession of Mercury, 42.98″ per century	42.982″
🕳️ Black holes	light deflection at the Sun's limb, 1.75″	1.7512″
〰️ Gravitational waves	chirp mass of GW150914 (36 + 29 M☉)	28.096 M☉
🌌 Galaxies	Freeman disc: velocity peak at R/R_d = 2.15	2.150
🌠 Cosmology	age of the universe with Planck parameters (13.8 ± 0.1 Gyr)	13.791 Gyr
🔭 Planetarium	Greenwich sidereal time, 1987 Apr 10 0h UT (Meeus, ex. 12.a)	13.1795463 h
🌍 Exoplanets	Earth's equilibrium temperature (A = 0.3)	254.8 K
🌀 Spin and qubits	HZH = X, HXH = Z	deviation $2.2 \cdot 10^{-16}$
🎲 Statistical mechanics	Schottky peak: kT = 0.4168 Δ, C = 0.4392 k	0.41678; 0.43923
☢️ Nuclei	B/A of ^{56}Fe from the masses	8.7903 MeV
☢️ Nuclei	Q of D + T → ^{4}He + n	17.589 MeV
🧱 Materials	I-section: formula against Huygens–Steiner	7.9990 against 7.9990 ($\cdot 10^{7}$ mm⁴)
🔌 Electronics	diode: Newton against the closed form with Lambert's W	0.650463089 V in both
🌊 PDEs	zero j_{01} of the Bessel function	2.404825557695773
🌀 Complex analysis	e^(iπ) + 1 = 0	$1.2 \cdot 10^{-16}$
🔗 Markov	stationary distribution by powers and by Cramer's rule	π_0 = 0.4615384615384615 in both
🧩 SVD	[[1,1],[0,1]]: $\sigma_1 = \varphi$, $\sigma_2 = 1/\varphi$	1.6180339887498947; 0.6180339887498948
📡 Codes	CRC-32("123456789") = 0xCBF43926; Hamming(7,4) corrects all 112 single errors	0xCBF43926; 112 of 112

Altogether the twenty scripts run **705 checks**. The study-kit scripts check the data schema and minimum quantities and, for mathematics, recompute **264 quiz answers** out of 346 questions; the resources script checks the citation formats on hand-built Crossref records and the tables' numeric proofs.

4-bis.3 The 3D model of Spectra's face

The face of the Spectra Live console is the «Noir» model by Skyfox13 (<https://sketchfab.com/podderkuntal25>), published on Sketchfab under the CC BY 4.0 licence, file `models/spectra-noir.glb`. It has been **modified**: textures recompressed, tangents removed, the face rendered as dark grey metal, head and mouth animated by the code. It carries no scientific data; the attribution appears in the console and in `07-SBOM.md`.

4-ter. The v238 3D tabs

Twenty-nine sections receive a 3D tab built **on the same model** as the section: the scene draws computed quantities (the LEPS surface, the cells with their lattice constants, the integrated field lines, the Lorenz solution, the mean-curvature flow of the soap film…), not hand-made figures. Every tab has a script `tools/banchi/verifiche/verifica-<id>-3d.js` (for Topology and Non-Euclidean geometries the section's own script), run by `test_3d238`; **311 checks** in all, described in full in `15-Test-Documentation.md` §3.1-ter.

Kind of comparison	Examples, with the value the script prints
Measured or published value	the LEPS model's ω(H_2) 4396.6 cm^{-1} against 4401 measured; H_2 at 1.4 bohr −1.116714 Eh (Szabo-Ostlund); acentric factor of water and CO_2 0.3443 and 0.2249
Closed form	Larmor radius 0.61538; h*/R = 1.32549 for the catenoid (u* tanh u* = 1); volume of the sphere 4π/3 with Simpson
Second method	cell lists against the sum over all pairs ($5 \cdot 10^{-14}$); sphere rotations against SU(2) Möbius maps ($5.5 \cdot 10^{-16}$); hyperboloid against Poincaré disc ($1.2 \cdot 10^{-15}$)
Statistical quantity within its error	fraction of collisions above threshold = e^(−E_0/kT) within 3σ; K measured in the equilibrium box within 1.5 %

Declared in the code as models or literature values, not measurements by the app: the critical temperature of the 3D Ising model, T_c ≈ 4.5115 J/k, comes from published simulations (there is no exact solution) and is used as a reference; the detector is «like CMS» (3.8 T solenoid, yoke at −1.8 T, 13 tracker layers up to 1.08 m) with Z → μμ, Z → ee and H → γγ events generated by the code (m_H = 125.2 GeV); the molecular and crystal parameters are those already used and verified by the 2D tabs of the same sections.

Fixed before publication. In molecular collisions the collision normal was taken from spheres the discrete step had already overlapped, and grazing collisions came out more head-on: the fraction above threshold at $E_0/kT = 1$ was 0.345 against $e^{-1} = 0.368$. Now the motion is traced back along the relative velocity to the contact: 0.370. On the Riemann

sphere the pole was not recognised for an angle of exactly π, because cos(π/2) in floating point is $6 \cdot 10^{-17}$: now the pole is recognised from the ratio of the Möbius denominator to its numerator.

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5. Automation of the verification

All the checks described are runnable and repeatable. At the time of writing the battery includes, among others:

Bench	Subject	Checks
`audit_farmaci`	structure ⟷ weight, duplicates, formula rebuilt from the graph against ChEMBL, deviations recorded with their reason	263 entries
`test_spettri`	group recognition on reference molecules	36
`test_assi` / `test_assi_canvas`	axis conventions (SVG and canvas)	19
`test_costanti`	physical-constant lookup and refusal of ambiguities	45
`audit_dati`	consistency of the tabulated data	29
`verifica_guida`	correspondence between what the documentation promises and what the code does	71
`test_v237`	the twenty astrophysics, physics and mathematics sections against known values or a second method (§4-bis.2), the study kits and the resources	705 in the twenty scripts, plus the kits, the resources and the in-browser checks
`test_3d238`	the 3D tabs of 29 sections against known values, closed forms or a second method (§4-ter), and every tab opened in the browser at 1280 and 390 px	311 in the 29 scripts, 488 in all

The last bench deserves a mention: it verifies that the statements contained in the user documentation correspond to the real behaviour of the code. A promise the documentation does not keep is a defect on a par with a computational error, and is treated as such.

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6. Transparency statement

This document describes checks that are **actually implemented and runnable**, with the results actually obtained and the limits of the predictors. At version `bsi-v188` the residual errors on the drug database are **zero**: the 21 deviations that remain are entries without a structure, each with its own recorded reason, not errors passed over in silence.

Where a datum is not verifiable with the instruments available, that is declared rather than presented as verified.

Document updated to version `bsi-v250`.

Software Bill of Materials (SBOM)

BioSpecInfo · document generated by `tools/genera-sbom.js`.

Application version	`bsi-v250`
Commit	`2c08fbcb6b0e7b4d5659a8a010c3ef62b4d69a94`
Generated (UTC)	`2026-11-12T06:42:27.511Z`
Machine-readable format	`../evidence/sbom.cdx.json` — CycloneDX 1.5

> The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

Component	Version/File	Licence	Size	Role
RDKit MinimalLib (JS glue) RDKit / Greg Landrum	`RDKit_minimal.js`	BSD-3-Clause	125 kB	Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing.
RDKit MinimalLib (WebAssembly) RDKit / Greg Landrum	`RDKit_minimal.wasm`	BSD-3-Clause	6753 kB	WebAssembly binary of the cheminformatics engine.
three.js mrdoob e collaboratori	`three.min.js`	MIT	589 kB	WebGL rendering engine for the 3D scenes.
three.js — UnrealBloomPass mrdoob e collaboratori	`three_bloom.js`	MIT	25 kB	Post-processing effect for the astronomical visualisations.
three.js — GLTFLoader mrdoob e collaboratori	`gltf_loader.js`	MIT	94 kB	Loading of anatomical and molecular models in glTF format.
3Dmol.js University of Pittsburgh / David Koes	`3Dmol-min.js`	BSD-3-Clause	502 kB	3D molecular visualisation (PDB, SDF, surfaces).

Component	Version/File	Licence	Size	Role
SmilesDrawer Daniel Probst	`smiles-drawer.min.js`	MIT	246 kB	2D structure drawing from SMILES.
PDF.js (build legacy) Mozilla Foundation	`vendor/pdfjs/pdf.min.mjs`	Apache-2.0	511 kB	Opening PDF documents in the spectrum reader: every page drawn and its text layer. It is the legacy build, because the default one uses very recent language features and breaks on browsers that do not have them yet.
PDF.js — worker Mozilla Foundation	`vendor/pdfjs/pdf.worker.min.mjs`	Apache-2.0	1290 kB	The PDF decoding work, off the main thread: a complex page does not freeze the interface.
sql.js (JS glue) sql.js / SQLite	`lib/sql-wasm.js`	MIT	45 kB	In-browser SQLite database for the data exercises.
SQLite (WebAssembly) SQLite Consortium	`lib/sql-wasm.wasm`	Public Domain	644 kB	SQLite engine compiled to WebAssembly.
Dataset dimuoni (CERN Open Data) CERN Open Data Portal	`lib/dimuon.js`	CC0-1.0	48 kB	Experimental data for the particle-physics exercise.
Modello 3D «Noir» (Skyfox13, Sketchfab) Skyfox13 — https://sketchfab.com/podderkuntal25	`models/spectra-noir.glb`	CC-BY-4.0	8154 kB	Spectra's face in the Spectra Live console. Modified: textures recompressed, tangents removed, face in dark grey metal, mouth and head animated. Credited in the console.

Digests

File	SHA-256
`RDKit_minimal.js`	`58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9`
`RDKit_minimal.wasm`	`e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8`

File	SHA-256
three.min.js	9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2
three_bloom.js	42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090
gltf_loader.js	5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6
3Dmol-min.js	06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff
smiles-drawer.min.js	955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2
vendor/pdfjs/pdf.min.mjs	bccc24ea711db8e44503629519904a5292d73b9daaa214bbe7cdcc282b0f4259
vendor/pdfjs/pdf.worker.min.mjs	145d2dd3ab0c86151011dba95acfa2d5336e2accd59388ea43dbee0efddaaec6
lib/sql-wasm.js	77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f
lib/sql-wasm.wasm	438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed
lib/dimuon.js	61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d
models/spectra-noir.glb	b781b54819d1d2aabf751788a813d8da830dabaddb913085352b572140903479

2. Code written for this project

File	Size	Role
index.html	4704 kB	Main application: interface, teaching sections, chemical data.
bsi-ai-hub.js	446 kB	«Spectra» AI agent: agentic loop, tools, provider management.
bsi-spettri.js	36 kB	IR/NMR spectral prediction engine over the molecular graph.
bsi-cheminfo.js	125 kB	Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model.
bsi-pretsch.js	37 kB	The NMR estimation tables transcribed in full from Pretsch–Bühlmann–Badertscher, 4th ed.: ^{13}C and ^{1}H benzenes, ethylenes, alkanes, alkynes, ^{13}C aliphatics, steric corrections, ^{1}J(C,H). Numbers only, no code.
bsi-nmr.js	57 kB	Per-atom ^{1}H and ^{13}C NMR prediction: applies the bsi-pretsch.js tables by walking the ring and measuring distances, with a cyclic reference compound and chemical equivalence by environment code.
bsi-geom3d.js	46 kB	3D coordinates from a SMILES by distance geometry, carrying the NMR predictor's own atom indices: the precondition for the peak↔atom↔3D link.
bsi-fisica-comune.js	33 kB	The shared home of the physics sections: header, tabs, sliders, canvas charts, language.
bsi-fis-schrodinger.js	38 kB	1D Schrödinger equation: finite differences, Sturm bisection, inverse iteration, Crank-Nicolson, Morse for real molecules.
bsi-fis-hartreefock.js	48 kB	Restricted Hartree-Fock in the STO-3G basis: McMurchie-Davidson integrals, Boys function, SCF with DIIS, molecular orbitals.

File	Size	Role
`bsi-fis-rotovib.js`	34 kB	IR and Raman rovibrational spectra of diatomics from molecular constants; combination differences.
`bsi-fis-termostat.js`	11 kB	Statistical thermodynamics: partition functions, entropy, heat capacity, Boltzmann populations.
`bsi-fis-dinamica.js`	26 kB	2D Lennard-Jones molecular dynamics: velocity Verlet, thermostat, virial, radial distribution.
`bsi-fis-matematica.js`	24 kB	Mathematics for chemistry: FFT of the FID, Hückel, Runge-Kutta for kinetics, least squares with errors.
`bsi-fis-cristalli.js`	44 kB	Crystallography: Bravais lattices, Miller indices, powder pattern with structure factor and absences.
`bsi-fis-nucleare.js`	14 kB	Radiochemistry: Bateman equations, radiometric dating, Weizsäcker semi-empirical formula.
`bsi-mat-cas.js`	67 kB	Computer-algebra engine: parser, exact fractions, canonical form, derivatives, verified antiderivatives, truncated series, limits.
`bsi-mat-basi.js`	26 kB	Mathematics basics: significant figures, logarithms and pH, equations with working, practice.
`bsi-mat-funzioni.js`	17 kB	Function analysis: domain, zeros, extrema, inflections, asymptotes; families and transformations.
`bsi-mat-calcolo.js`	34 kB	Derivatives with rules, integrals and Riemann sums, Taylor polynomials, limits.
`bsi-mat-lineare.js`	54 kB	Exact-fraction linear algebra: step-by-step Gauss-Jordan, determinant, inverse, eigenvalues, balancing as a null space.
`bsi-mat-ode.js`	38 kB	Differential equations: direction field, Euler/Heun/RK4, phase plane and equilibria from the Jacobian.
`bsi-mat-multi.js`	38 kB	Functions of several variables: surfaces, critical points from the Hessian, Maxwell relations, exact differentials.
`bsi-mat-complessi.js`	37 kB	Complex numbers (Argand plane, roots, Euler) and Fourier series.
`bsi-mat-risolutore.js`	110 kB	Mathematics exam solver: step-by-step working for 16 exercise types.
`bsi-mat-esame.js`	56 kB	Istituzioni di Matematiche I and II exam: syllabus, solver, solved papers, simulation.
`bsi-fis-esame.js`	135 kB	General Physics I and II exam: syllabus, 48 problem models, solved papers.
`bsi-percorsi.js`	9 kB	Study paths from high school to PhD for chemistry, physics and mathematics.
`bsi-mat-probabilita.js`	34 kB	Probability in motion: Galton, central limit, Monte Carlo, Bayes.
`bsi-mat-visiva.js`	24 kB	Visual mathematics: linear maps, Fourier epicycles, fractals, logistic map.

File	Size	Role
`bsi-fis-em.js`	44 kB	Electromagnetism in motion: charges, induction, generator, EM waves.
`bsi-fis-suono.js`	18 kB	Sound and acoustics with Web Audio: harmonics, timbre, Doppler, decibels.
`bsi-fis-solido.js`	32 kB	Semiconductors: Fermi-Dirac, doping, p-n junction, LEDs.
`bsi-chi-distillazione.js`	17 kB	Distillation: Antoine, van Laar, McCabe-Thiele, Rayleigh.
`bsi-chi-voltammetria.js`	12 kB	Simulated cyclic voltammetry: diffusion and Butler-Volmer.
`bsi-mat-numeri.js`	16 kB	Numbers and cryptography: sieve, Euclid, modular arithmetic, RSA, prime distribution.
`bsi-mat-grafi.js`	22 kB	Graphs and optimisation: Dijkstra, Kruskal, gradient descent, linear programming.
`bsi-mat-serie.js`	32 kB	Sequences and series: ε–N limit, tests, rearrangement, radius of convergence.
`bsi-mat-variazioni.js`	51 kB	Calculus of variations: brachistochrone, catenary, catenoid, Fermat.
`bsi-mat-topologia.js`	57 kB	Topology: Euler characteristic, surfaces, fundamental polygons, knots.
`bsi-mat-algebra.js`	45 kB	Algebraic structures: dihedral groups, permutations, finite fields, Lagrange.
`bsi-mat-giochi.js`	36 kB	Game theory: Nash, iterated dilemma, minimax, Nim.
`bsi-mat-logica.js`	46 kB	Logic and computability: truth tables, circuits, Turing machine, sets.
`bsi-mat-inferenza.js`	39 kB	Statistical inference: intervals, tests and power, regression, bootstrap.
`bsi-mat-noneuclidea.js`	66 kB	Non-Euclidean geometries: sphere, Poincaré disk, curvature and parallel transport.
`bsi-fis-gravita.js`	35 kB	Gravitation: N bodies with Verlet, Lagrange points, gravity assist.
`bsi-fis-calore.js`	17 kB	Heat and diffusion: 1D and 2D heat equation, Newton cooling, walls.
`bsi-fis-ising.js`	26 kB	Ising model: Metropolis, phase transition, hysteresis.
`bsi-fis-particelle.js`	59 kB	Particle physics: Standard Model, conservation, invariant mass, lifetimes.
`bsi-fis-analitica.js`	40 kB	Analytical mechanics: phase space, double pendulum, Noether, Liouville.

File	Size	Role
`bsi-fis-laser.js`	34 kB	Lasers and quantum optics: threshold, cavity modes, Rabi, Mach–Zehnder.
`bsi-fis-clima.js`	34 kB	Climate physics: radiative balance, CO2 forcing, ice-albedo, Coriolis.
`bsi-chi-fasi.js`	38 kB	Phase diagrams: water and CO2 (IAPWS, Span-Wagner), eutectic, Clapeyron.
`bsi-chi-pes.js`	33 kB	LEPS potential energy surface for H + H2 and trajectories.
`bsi-articoli-nobel-fisica.js`	118 kB	Topic explanations of the Nobel Prizes in Physics (no data).
`bsi-articoli-premi-matematica.js`	123 kB	Topic explanations of the Abel Prize and Fields Medal (no data).
`bsi-materie.js`	11 kB	Subject switch buttons: chemistry, physics, mathematics.
`bsi-fis-astro.js`	36 kB	Astrophysics: H-R diagram, black body, solar system, exoplanets, cosmology.
`bsi-fis-relativita.js`	30 kB	Special relativity: Minkowski, twins, dynamics.
`bsi-materie-app.js`	15 kB	Three apps in one: the chosen subject adapts the ✨ menu, Spectra, Study and Utility.
`bsi-attrezzi.js`	23 kB	Quick tools of the ✨ menu for physics and mathematics: calculator, derivative, graph, constant, units, formula, glossary, quiz.
`bsi-studio.js`	66 kB	Study-kit engine: summaries, quizzes, Leitner flashcards, formula sheet, glossary.
`bsi-studio-fisica.js`	306 kB	Physics study-kit data (no code).
`bsi-studio-matematica.js`	290 kB	Mathematics study-kit data (no code).
`bsi-risorse-materie.js`	149 kB	Web resources, bibliography and citations for the three subjects; CODATA constants and units; mathematical tables with their proofs.
`bsi-astro-sistemasolare.js`	110 kB	Solar system 3D at any date: Standish JPL elements, Kepler, Earth, Moon and eclipses, Jupiter and Saturn.
`bsi-astro-orbite.js`	69 kB	Orbital mechanics 3D: elements and state, manoeuvres, satellites, Lambert.
`bsi-astro-matcelesti.js`	60 kB	Celestial mathematics: Kepler equation, rotations and quaternions, N-body integrators, periodograms.
`bsi-astro-stelle.js`	63 kB	Life of stars 3D: Lane-Emden, evolution, nucleosynthesis.
`bsi-astro-buchineri.js`	72 kB	Black holes and general relativity: Schwarzschild ray tracing, geodesics, curvature, Hawking.

File	Size	Role
`bsi-astro-ondegrav.js`	55 kB	Gravitational waves: inspiral, chirp, detector, Hulse-Taylor pulsar.
`bsi-astro-galassie.js`	70 kB	Galaxies 3D: Hubble types, rotation curves, collision, Local Group.
`bsi-astro-cosmo.js`	70 kB	Cosmology 3D: Friedmann-Lemaître, distances, CMB, thermal history.
`bsi-astro-cielo.js`	75 kB	Planetarium 3D: the sky from any place and time, coordinates, precession, analemma.
`bsi-astro-esopianeti.js`	57 kB	Exoplanets 3D: habitable zone, transit, radial velocity, direct imaging.
`bsi-astro-chimica.js`	28 kB	Chemistry of an interstellar cloud (formation and photodissociation of H_2 and CO) and the Saha equation with the spectral classes.
`bsi-chem-mappe.js`	47 kB	Chemical space map (exact t-SNE on Tanimoto distances) and peak deconvolution (Levenberg–Marquardt).
`bsi-fis-spin.js`	55 kB	Spin, qubits and entanglement: Stern-Gerlach, Bloch sphere, CHSH, BB84.
`bsi-fis-statistica.js`	47 kB	Statistical mechanics: Schottky, Maxwell-Boltzmann, Fermi and Bose, Einstein and Debye.
`bsi-fis-nuclei.js`	49 kB	Nuclei and reactions: valley of stability, Q values, tunnelling, 3D nucleus, reactor.
`bsi-fis-materiali.js`	48 kB	Elasticity and materials: tension, beams, Euler, sound in solids.
`bsi-fis-elettronica.js`	48 kB	Electronics: rectifiers, diodes and Zener, common emitter, op-amps, ADC and DAC.
`bsi-mat-edp.js`	75 kB	Partial differential equations: string, membrane, heat, Laplace, Burgers.
`bsi-mat-analisicomplessa.js`	70 kB	Complex analysis: domain colouring, conformal maps, Joukowski, residues.
`bsi-mat-markov.js`	65 kB	Markov chains and stochastic processes: PageRank, gambler's ruin, walks, queues.
`bsi-mat-svd.js`	58 kB	SVD and applied linear algebra: 2×2 and 3×3 SVD, compression, PCA, least squares.
`bsi-mat-codici.js`	60 kB	Codes and elliptic curves: Hamming, CRC, curves over ℝ and F_p, ECDH.
`bsi-spectra-live.js`	114 kB	Spectra Live: voice console, local commands, screen reading, nine agent tools.
`bsi-spectra-androide.js`	45 kB	Spectra's face on the «Noir» model: animated head and mouth, 17 emotions from 11 expressive fields.
`bsi-spectra-mente.js`	17 kB	Spectra's mind: local memory, internal state, thoughts (a simulation of behaviour).

File	Size	Role
`bsi-spectra-robot.js`	20 kB	Fallback face: android head built in three.js.
`bsi-spectra-volto.js`	16 kB	Last-resort face: SVG vector portrait.
`bsi-lezione.js`	33 kB	Guided lesson with Spectra for every section: chapters from the tabs, controls demonstrated, spotlight, check question from the study kit.
`bsi-spectra-voce.js`	12 kB	Spectra's voice: TTS audio (with an OpenAI key) through a Web Audio chain (bass, metal, room) and lip-sync from the audio; falls back to browser synthesis.
`bsi-spectra-corpo.js`	49 kB	Body driver for any humanoid avatar (bones, ARKit blendshapes, black hair, gaze, gestures, walking, sitting) and a rigged test mannequin.
`bsi-spectra-olointerattivo.js`	74 kB	Spectra's interactive hologram: IR/^{1}H/^{13}C/mass/UV spectra of reference molecules or of a written trace with each signal assigned; universe with NASA data, Kepler and vis-viva; maths and physics with the assistant's working; every step explained aloud and in writing.
`bsi-spectra-esplora.js`	42 kB	«⚡ Explore with Spectra»: one section per subject in the style of explorable lessons, ten chapters from high school to advanced PhD with prediction, a recomputing slider, live view in the hologram and full explanation.
`bsi-spectra-spiega.js`	36 kB	«🧠 Explain»: the complete, exhaustive explanation of chemistry, physics and mathematics from high school to advanced PhD, including what is on screen (text, values, tables and an image of spectra, graphs and 3D scenes); without a model it shows what the app knows on its own.
`bsi-spectra-figura.js`	55 kB	Spectra with a body in one module: the body controller (embedded in the app and packed, rig, aligned «Noir» face) used by the studio, and the figure in the AI sections that thinks, speaks and gestures with the answers.
`bsi-spectra-rig.js`	35 kB	In-browser auto-rig for humanoid figures without a skeleton: measuring, Mixamo-named skeleton, skin weights (distance, normal, geodesic), head removed keeping the hood, night-blue recolouring.
`bsi-spectra-studio-lezioni.js`	34 kB	Spectra Studio lessons (predict → test → prove → reveal): the fall, the shape of water, the Basel problem, with verified computations.
`bsi-spectra-creatore.js`	11 kB	Spectra recognises its creator: a secret-phrase check compared as a SHA-256 fingerprint (a deterrent, not access control).
`bsi-aggiorna.js`	5 kB	«Check for updates»: version in use from the service worker, published version from sw.js.
`bsi-fis-fluidi.js`	31 kB	Fluids: Venturi and Bernoulli, viscosity and Stokes, capillarity.
`bsi-mat-vettoriale.js`	43 kB	Vector calculus: fields, Green, gradient.
`bsi-chi-cinetica.js`	31 kB	Complex kinetics: composite and oscillating reactions, Turing patterns.

File	Size	Role
`bsi-fis-meccanica.js`	37 kB	Mechanics in motion: projectile with air, collisions, real pendulum, coupled oscillators.
`bsi-fis-onde.js`	33 kB	Waves and optics: superposition, lenses and mirrors, diffraction.
`bsi-fis-circuiti.js`	17 kB	Circuits: RLC transients, alternating current, filters and Bode.
`bsi-fis-termo.js`	44 kB	Thermodynamics in motion: cycles, particle gas, Brownian motion.
`bsi-mat-numerici.js`	24 kB	Numerical methods: roots, integration, ODEs, interpolation with measured orders.
`bsi-mat-geometria.js`	24 kB	Analytic geometry and vectors: vectors, lines and planes in 3D, conics.
`bsi-chi-equilibrio.js`	27 kB	Chemical equilibrium and Le Chatelier: kinetic simulator and K(T).
`bsi-lab3d.js`	63 kB	3D Laboratory: surfaces,
`bsi-vista-molecole.js`	8 kB	Quality settings and control bar for every 3Dmol molecular viewer.
`bsi-assistente.js`	71 kB	The three macro-areas Chemistry, Physics and Mathematics: natural-language assistant with full working and inverse problem, tool catalogue, levels.
`bsi-chi-esame.js`	98 kB	Chemistry exam from basic to expert: 40 problem models, solved papers.
`bsi-fis-elettrodi.js`	14 kB	Electrode kinetics (Butler-Volmer, Tafel) and transition-state theory (Eyring).
`bsi-fis-superfici.js`	18 kB	Photophysics (Jablonski, Stern-Volmer), adsorption isotherms (Langmuir, Freundlich, BET), polymers.
`bsi-mat-statistica.js`	27 kB	Laboratory statistics: t, F, Grubbs, Dixon, uncertainty propagation, calibration line with LOD and LOQ.
`bsi-nmr2d.js`	21 kB	Two-dimensional NMR: COSY, edited HSQC and HMBC, built by walking the graph over the two per-atom predictions.
`bsi-moldraw.js`	44 kB	Interactive NMR spectrum: zoom, assignment table, peak↔atom link, SMILES/InChI history, CSV/JCAMP-DX/PNG exports.
`bsi-documento.js`	23 kB	Opening of any document — PDF, images, text, Word, OpenDocument, PowerPoint, Excel — with an explicit list of what could NOT be read.
`bsi-quesito.js`	19 kB	Recognition of spectroscopic data in a problem's text, with each datum's provenance, and step-by-step working through bsi-elucida.js.
`bsi-spettrolettore.js`	104 kB	Spectrum reader: JCAMP-DX with ASDF compression, prominence-based peak finding, IR band assignment, neutral losses.
`bsi-mol3d.js`	42 kB	3D molecule viewer: formation animation, atom selection, measurement of angles, lengths and dihedrals.

File	Size	Role
`bsi-lingue.js`	262 kB	Internationalisation engine: fourteen languages, writing direction included.
`bsi-molingue.js`	54 kB	Chemical languages: twenty-five outputs from SMILES, with IUPAC naming over a declared class.
`bsi-pannelli-lingua.js`	66 kB	Panels for languages and chemical languages.
`sw.js`	9 kB	Service Worker: offline operation and network strategy.
`rdkit_lab.html`	260 kB	Cheminformatics laboratory.
`astro.html`	2660 kB	Astrochemistry module.
`chimorga.html`	195 kB	Organic chemistry module.
`proxy/spectra-proxy.js`	10 kB	Optional proxy (Cloudflare Worker) for the AI calls.

Digests

File	SHA-256
`index.html`	`1ae3cd538a35a2b8397d9dc967b60ff2852057d848cc5e78cf4291afae520a20`
`bsi-ai-hub.js`	`c2db3c146724e49d9694cb3b3476301274475200290 8c289c79bab3259265f2b`
`bsi-spettri.js`	`36911b64fd173917f9151c8ef82fc1c6f903eb93205fc851f20534a46d958921`
`bsi-cheminfo.js`	`59a3111b3b9436add5376fde1be6f90ded3dc08c095cebdc04da50c7f8268bbc`
`bsi-pretsch.js`	`27f727240363579fba3dba50ab3b567fbcdb26a8a9a06c0e43968b195c479db3`
`bsi-nmr.js`	`881917dbb141dfcbab9a25b9c518eaa4a8c1c52dc1020cbac90985a1cf859b72`
`bsi-geom3d.js`	`a616b91efa8f790388c99f22cc9f2d48630cb760d4f7fa87f92b99b9db106054`
`bsi-fisica-comune.js`	`5fe08ef43a14a93e09caef439a07f0742221bdf1d395ce1c50dfed9e90cc73bb`
`bsi-fis-schrodinger.js`	`9954f7b14d9c84a2539480c2e61b468f2c37c72d03b3a6147d414c9870480d9b`
`bsi-fis-hartreefock.js`	`541b5340edbb392f5f85d473f430a586282a848e0dc432b752ab898407582c84`
`bsi-fis-rotovib.js`	`0ce3ad61451a86506e9e653776d73b9e2cd21986c84a00e65db0fb6290ebbdde`
`bsi-fis-termostat.js`	`0bb9f04d7d5b144791764fa4efd9f8e658be5b5d78b40f94e18752c2366589a9`
`bsi-fis-dinamica.js`	`0b1ab11071eab8a1af5def46919bc73aa2dbd9dd58c77cea30aa6dee020f4755`
`bsi-fis-matematica.js`	`616a2d4f338fe1d371f5ba3fd4fe569e94b2d94626a8d323bba8ae58f8550f62`
`bsi-fis-cristalli.js`	`02e2224e77713d071720d6394452a21eb8e9e1c9103871f41193e82f346fe805`
`bsi-fis-nucleare.js`	`8ae148a2f37a40f6902faa4148e44468b4d659b6c7d1230378de92c2703a5874`
`bsi-mat-cas.js`	`eea3a1eef74d31d3fde78b7e4a2795169e8af0f357edc10050eb98d8eaf386ac`
`bsi-mat-basi.js`	`db30e16c58bde3638f13de5c3311a00598628676b7b418dc023b175efa77c214`

File	SHA-256
bsi-mat-funzioni.js	68f8780d4fb4fcda6f028e4749adef333253efbf270d0caab47def5bdf30c66c
bsi-mat-calcolo.js	3d5cff0520cf385dfe35af1b3f916a66c5afb4ff60d880f04123ac5fe73a9279
bsi-mat-lineare.js	a84ca3fb6720da50f474633152b84e03e1d2287e3271cdadca790b735c69d686
bsi-mat-ode.js	bfbb1528e5d025d7492c69503115dc0dfb1c50fbbd6735133306ad1327830163
bsi-mat-multi.js	5a025543a12c66994e8ac08a2705da5761eac8bbcc5a5be471536372762b2e37
bsi-mat-complessi.js	8ad8fff97934e7224d97523decec29bb3b4bc4631dc20cc6ea78a0caec69320b
bsi-mat-risolutore.js	709349e1d81f03506c02b9b05a060d8d3801dc37183c0e3894cc93c0aafe9f0c
bsi-mat-esame.js	de4af253244bd7d610b1a3607297d1678a529504afa23d7123aebe042e1f2c27
bsi-fis-esame.js	c9ffb883e94a07b7cda3777b34d6141bd4c47bb3c9857218a9a37f65f8728947
bsi-percorsi.js	950c686202f2934009827bc300c787063b325528b7d5ab5385a8ab3b6706345c
bsi-mat-probabilita.js	adc628789ec90772c14ce0e61aed74981cf60b261ac15ff6076d1d4b3e8d0bec
bsi-mat-visiva.js	cb624f3e1e6c6df94c287f961d0f21a9d8ec852e8245ff5aeed09a083f2c5b14
bsi-fis-em.js	1c45fb608ccda1e500ab3038998cee4cb9c27b091c0b1d1b4f7af0137bf9a6ab
bsi-fis-suono.js	fead9e5dde961fc56c3a649cce6e8eed23046f327c8392dbfd8de5024cf0a1bf
bsi-fis-solido.js	e185ce2e4f5cb4e1101bbb5d398fa3893a017c8df8fa4a26246291fda60ca2ad
bsi-chi-distillazione.js	8d13592cf7523c1dd159d3e954947a4956e44539308312dc97cd3588207fcfc0
bsi-chi-voltammetria.js	4db1a71e0ac3569b821f6cc5280470453f3a4df5c62b9695a3339367c5aa3e41
bsi-mat-numeri.js	f30d041cd9963342aa476e128e4b29d543e113d3483253377ea486de3d46484f
bsi-mat-grafi.js	4670e67bd25b97d4c7d2b8e72b03ed8ed4f6e6aa5c1c43dd74c56b437c1c8c33
bsi-mat-serie.js	722b1a1b7397b7710689af555d1fd60de73ee405f5061819d9e59d7f4631d060
bsi-mat-variazioni.js	ccb7379618067a0412d27fe95cb19d5823e0c0b739a371afc7cc69f162a5ca7a
bsi-mat-topologia.js	87a8df768b8fa968608b52462df4340484ccaeb7ed3e6a987987a62603ceb8fa
bsi-mat-algebra.js	59d66b5c6be431f49e5808e5c78758a7ca3cc33fba6155e56457f13dec14e237
bsi-mat-giochi.js	d1ae8048acab5a518f1ca9c9ef31e4c9346459c7a0d83ec935733a57465c4b72
bsi-mat-logica.js	26cfc591156a6c232dd40938252ce238374a6a678f08557d3bcfa899a5d0785e
bsi-mat-inferenza.js	4dadc036537e3833e396ab5dfa9968f0ccab398ccc18959ab474e6652d27e43c
bsi-mat-noneuclidea.js	a687dbe43e91cacc5b2287af6e135188363e49644cd870b48db9b4a66d555119
bsi-fis-gravita.js	195d35280eb3484f0374b9e51801e96634c797ba21209fce1bdb7b8aed867e33
bsi-fis-calore.js	81b9c1c60c7ba5a022f2f95e36ac17274ed1e2027d0ba36bcc24f8d6243ad3b8
bsi-fis-ising.js	8ccd29082a4388ce63bdc1e39d9aa5fa92322e378384df0951d98bde6fe79b8c
bsi-fis-particelle.js	4dce551cb464150e0faae0bcddcea834059692187310450 3bd6a2dd10b41140c
bsi-fis-analitica.js	01b4178998372c6db7d18a312c5a5ed1c8c26267bd0b72ae63ad6fbf92ba2895
bsi-fis-laser.js	8f876f1552b50b22643526a2d16a4a8b0c0bbf5b8d9161d325f38efd16c917fa
bsi-fis-clima.js	8e136c02f7559cd22d57ccd857d06f8d6a6a89b2f38790d2782e6a45a7f2c822

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bsi-chi-fasi.js	99854760b11db89cf853bb4c2be08d00f018fb510fd6a56aff1c3006e68ed1af
bsi-chi-pes.js	c4e74b909e7987275930b05c0c25cdc0968bcf338ea78a144bc7ff949a821e36
bsi-articoli-nobel-fisica.js	afc22b073e64168605b3d792cd83bcdf7d838e0c80548a0a58caae98456d7d30
bsi-articoli-premi-matematica.js	d62f01d66783d7fb558684959abd1cda2f81ee6885c8cdc0879559f2b64cbf43
bsi-materie.js	763063fb3d3c1e43da8066c6febab6309f889a6550e7da1f45ce1d001c27c314
bsi-fis-astro.js	64901258a641c54ee99a862990a7e5d86acc5c2367408bb38db9371a39c7363e
bsi-fis-relativita.js	d90cb347a69e668fde69df4805815938f8da8b9a5f288ce7a16914aa0fec32bd
bsi-materie-app.js	a9d57912a69bf7c599a6c4d16be815be4e654ab424daeca0e12f02c00904911f
bsi-attrezzi.js	dedf9939e96c9544239f5769b68f0547604aff78e3ee3f47906afddcf1e20a38
bsi-studio.js	aa1c97df67465affd4f7d0c96240b0d0ae6dd9d8c8dd6bf3c93abe9cef49e377
bsi-studio-fisica.js	973bab09dad47980f0d411aef35ff0378f2266030c3fcb14e1cefe64eb63531b
bsi-studio-matematica.js	3a2d88a7f9cea52bc5c46bf86018904b2d032320726988759b3e0fb84110c5c7
bsi-risorse-materie.js	03e2fa89af1157985cdab1a08c60153ed57d8c648a5f7b67cd3f5d910c43e417
bsi-astro-sistemasolare.js	d59c52b7135d020d67a8163e9bc4d0cecea2446df93aa66389f0477a2a908ef7
bsi-astro-orbite.js	32c8123ac5a7b39a9a4a816713d0305a9d90ddb8fb714614fcbba3eb3e834c4b
bsi-astro-matcelesti.js	b2ee40c8a53fb25baf42020693539d91dc5278a14436115c69643470d57cda56
bsi-astro-stelle.js	84064a5bdf8fdc1111011201cf97a11a47396acbd3c7442d52e2c5ba74a388a3
bsi-astro-buchineri.js	c75a84cba9be45c5fec1b4f77f4ca79f9d898067006a3c309d951147ddde083c
bsi-astro-ondegrav.js	dcbc8ea2e45f55a463ea344c26e0de8e0aa6a14f6adc73d6b2a2e43aee9736db
bsi-astro-galassie.js	893eee603ec4af64ceaf440f3a22da3dab77a55333911e80e06212459e5f7125
bsi-astro-cosmo.js	356152c5251593c68e8dd66b1182f5ee3f7c5f4f1cb26c722bdd384f8fe66d47
bsi-astro-cielo.js	26b3e33cefe7d1606d0abf8d47be6d7d9b412891ab193d386835d4e40f9fdcf6
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bsi-astro-chimica.js	7b56d2acb2d7b1805a9873397043006d5b419f0278c78ccbde286c6130ad1c13
bsi-chem-mappe.js	594d53b5a6675791d66df42d0d3dead84e670b4672f962392f9241c5ce8b8add
bsi-fis-spin.js	a24dc05385fbff4826a02f95315ae1d8c57c8ee555da868a08620f24127fe3ad
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bsi-fis-nuclei.js	3f393cf853d4171beb53153fa2d03da2f526ba40af2b9ad1ce4c4e426678d4db
bsi-fis-materiali.js	b2869b83fa17200da5f3900abf6959214df0dd6f88c4d8d02be88d5a6d832269
bsi-fis-elettronica.js	8620e8ba116f1d3edb3a487a7bf2aab869fbf1afc2230462e251062e7a2bf61d
bsi-mat-edp.js	525fae4f04fcc0ec7884ebba30e47d013c972a5ff77b1088ab4bca41bd04a50b
bsi-mat-analisicomplessa.js	0eb6e5181471af7e85eb3ee42012daddbe1fdac932c0c4c28cf67135875302d8
bsi-mat-markov.js	0e122fac691f1faa2e464214a7d24c9a687b16491c5964b760e0df191b13a35b
bsi-mat-svd.js	ce6d77e921c469c532f9f6de2689f7b2a8c8de656f18ee32b1fc4e1977538931

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bsi-mat-codici.js	9aaf44beb1922a95239285067941899e6ae4424024546c8b38dc2663e5f091b2
bsi-spectra-live.js	c36ef98cfaf7abde37fda9cffb7c76c2730372ec97037c3ee63b5109c10f6c22
bsi-spectra-androide.js	36bd9ad900e9da2860ddf95469c48ff81387d6da6ca71672bb52975cebb6bd08
bsi-spectra-mente.js	29a9666fb7278adcd8747060927f29403b53b47ab00e5d2b5dc6e9ccff311436
bsi-spectra-robot.js	227944042a092f360a6f27e2d8221686101fa6f29514c4e56eb4401d7de5b1b7
bsi-spectra-volto.js	7061605f2d7e124d2a80ec2f19531b7fa32fed843b60ca3d07843f4926105bf6
bsi-lezione.js	9d254c1cfaaf0c9ea5ec33086e71c665941f352378a14c874e27b3cb7e8f861a
bsi-spectra-voce.js	fa6f3440ffadc9a9a4c2b57821141ea3d2d4ef36dca9f85d92dc21e652dad9a7
bsi-spectra-corpo.js	ae8e5ea95d2a8946b7318d5ac4f4e3646297a4ebeeb30924d8f8c1bd0110a04f
bsi-spectra-olointerattivo.js	19cd15b7c5e74fd97992aba0c13defbd533dbcd386aaf996f9a5621a53bf04b8
bsi-spectra-esplora.js	71b0362cbc8c12442ba5fe9b804bb982b95911afce317684a4cdf418586f7c91
bsi-spectra-spiega.js	789b3560c1372da180a86a486e9ce26f1d471768d39bd39fe1c6f55b85dc8b95
bsi-spectra-figura.js	1c072d7b2f9de5fdf627dc985e200366c1ef383ed41ff088d9380f4fbea6ec98
bsi-spectra-rig.js	fcdd67c468f543a6e98f682d49c45b26173bb987c1aa529cb63d4e3e2a40918b
bsi-spectra-studio-lezioni.js	a7344a2b7a62986827dff100a44676d10daa841d76af97b056b796657a991a9c
bsi-spectra-creatore.js	73954030d3fac2a5b12fd26c92f61a1c8bb986813941d33fb6b154e8cf89bee2
bsi-aggiorna.js	5950d90103a7d603175e4b577e2394bb9a6d006a4ff22e48d4a1cc70661f1b8e
bsi-fis-fluidi.js	33b66d5c555b28b0b65224aef0d23b1f113fdc402ef58e00de4e2c630e38fc7f
bsi-mat-vettoriale.js	ced5e1917ee286455c37ca008c9154b74118ef40021c7df6b97561437c544fdb
bsi-chi-cinetica.js	e1a42d19404deb16f0c9cf9ac19de6b0970c955ec7b4427c4838ca97705bf3b4
bsi-fis-meccanica.js	a73522108a4fdcd7b99719157b87111af8520f219c3a086d6d73eee6fefed569
bsi-fis-onde.js	83ac1416df79f6fde4057382143fce24f9c6bb178f10fd44d44c96c0ffe6660a
bsi-fis-circuiti.js	098679cdbb52241b6035e23076c82a25fe3d9a72b6f5bf7d4f880f887223f375
bsi-fis-termo.js	61a151f01860e31d7c5470d6ace6e6ee8d007c4797f316cf6b005d8de21d8224
bsi-mat-numerici.js	8656bdf657dcc3468f78951ed3d8e1056d953b147e04595c6bfaad647fa8ca93
bsi-mat-geometria.js	609bf6c951d438aa0b0639fd2e736a52b340a363294ee564acfe9510966334be
bsi-chi-equilibrio.js	15d9d99c63c1db49dc3834b7b1db83f5ad2e03d1b9cc1b2f69cdb8ba6cffcefd
bsi-lab3d.js	9b1424f3ad744cc2abd42be04f9db0907578452ecb66156e2c809a5c774a1604
bsi-vista-molecole.js	4a7eb7cd7d8a0033ad0381d936d8e36b2b2e6b75af04988ecc5e9f4e8022cbc4
bsi-assistente.js	103375186ee638e89e324a0d737be266c8464df20d763bb85edafe0e6af4daef
bsi-chi-esame.js	ddb5d73fcf887e03770b3f0794fecaa2f8c918f4fae445bf57ff7a637f4643e7
bsi-fis-elettrodi.js	249bb92ac39b78e916eaa0ecf80a2b1aac5d10249bb6b51628f61839c4b64240
bsi-fis-superfici.js	3daec64d2c4755e70279b8ef2f433e4c98d6e90ff7d8c7d9c0689a24f080abd8
bsi-mat-statistica.js	0b0c82afa13173b84308589cf16fbaa8b30c5e575fd8fdee00034b70c34d17fc

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`bsi-nmr2d.js`	`a0b1db87f27ffa8c124d617e4903f61e35b89c30c35162527ad023325c03fa78`
`bsi-moldraw.js`	`45cacdca20c8091cbfb32e256c179de8c32c71068a5d1008fbef9be005b5df0e`
`bsi-documento.js`	`0e9ad6043a396652549b30879764cc401f8b7bd25689b5bb5f37cf26f7d7d8d9`
`bsi-quesito.js`	`6f686abcfdc3959d8c865a3e6ee7129224de52992ab1ba3da07372a986769cdf`
`bsi-spettrolettore.js`	`14daf41305b0a7271c6ff5f7c9078a4e5f0e0149f32460a9608ed1996f810256`
`bsi-mol3d.js`	`93fea4a6811ba4263de450db4e7ae1801d4fde196cf7683d0100846c51751893`
`bsi-lingue.js`	`e31add1c339356b8c5ad24c9575acbae6841c6b1da1cc2ece8fbf7d56f7f698e`
`bsi-molingue.js`	`e45fec1ee20e3e7b9a7bea3d6e20aa4e33e74ec43578bfa7bfb94646b7bc53fb`
`bsi-pannelli-lingua.js`	`9437192c6a3995425b3ef305ab86c70685011d40d789c2cc45736880a7369186`
`sw.js`	`5f6e67c31c26c9643649de91ff8d6a49f31d201b60cce3b6a5985a129a03ba28`
`rdkit_lab.html`	`066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad`
`astro.html`	`3376b2b7f26ed483f2d698c6692b31f21c9f245f87d1fb1884216fb8b1da489d`
`chimorga.html`	`cf0a850b5780649ce496d673645f9e8829110e9bf9722a43d2d0020dade17c54`
`proxy/spectra-proxy.js`	`808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962`

3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

Package	Version	Licence
`playwright-core`	1.62.1	Apache-2.0

4. Runtime dependencies

None. The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

Service	Purpose	If unreachable
PubChem PUG-REST	IUPAC name, CAS, GHS pictograms, 3D conformers	the app uses local data and says so
NASA / ESA	astronomical images and data	the module shows the tabulated data
AI providers (user's choice)	assistant answers	the assistant reports it and offers alternatives

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

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5. Licence compatibility

All licences of the distributed components (MIT, BSD-3-Clause, CC0, CC BY 4.0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

> **The application's licence is distinct from those of its components.** The code written for BioSpecInfo is **proprietary** — *All rights reserved* (see `LICENSE`): readable and open to assessment, but copying, reuse and commercial use require the Author's written permission. The permissive licences listed above cover **only** the bundled third-party libraries, and do not extend to the application.

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Traceability Matrix — BioSpecInfo

Field	Value
Software	BioSpecInfo
Author	Samuele Pio Provenzano
Version described	`bsi-v250`
Purpose	Connect every declared requirement to the implementation that realises it and to the bench that verifies it.

> **How to read this matrix.** Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.
>
> The outcomes of the benches cited here are reported in full, with their complete output, in `../evidence/RAPPORTO-VERIFICA.md`, which is generated automatically.

1. Scientific requirements

ID	Requirement	Implementation	Verification bench
SCI-01	A drug structure must come from an external source and match that source, not merely weigh correctly	`FARM_DATA` in `index.html` — **263 entries**, 36 declaring a ChEMBL identifier and formula	`audit_farmaci` — the formula is **rebuilt by counting the graph's atoms** and compared with ChEMBL's: 36 of 36. Weight alone was not enough: three pre-existing entries (lenalidomide, palbociclib, aripiprazole) were **isomers** — same weight, different molecule — and were replaced. Two ratchets: verified provenance ≥ 36, readable structures ≥ 240
SCI-02	Functional groups must be recognised on the structure, not on the text of the SMILES	`bsi-spettri.js` §1, SMARTS patterns via RDKit	`test_spettri` — 8 reference molecules, additive recognition
SCI-03	A molecule with several functional groups must show them all	`bandeDaGruppi()`, additive construction	`test_spettri` §1 — acetylsalicylic acid: ester **and** acid

ID	Requirement	Implementation	Verification bench
SCI-04	Spectra must respect the conventions of their technique	`svgIR()`, `makeNMRsvg()`, `drawUVSpectrum()`, `drawMSSpectrum()`	`test_assi` (SVG), `test_assi_canvas` (canvas)
SCI-05	A computed spectrum must be reproducible	every `Math.random()` removed from the traces	`test_assi` §3, `test_spettri` §8 — byte-for-byte comparison
SCI-06	The height of ^{1}H peaks must represent the integration, or not be asserted at all	`makeNMRsvg()`; `_nmrSvg()` draws bands of uniform height	`test_assi` §4
SCI-07	Conjugation must act on the individual carbonyl	SMARTS `acidoConiug`, `estereConiug`, `chetoneConiug`, `aldeideConiug`	`test_spettri` §1 — aspirin 1760/1690; acetophenone 1690
SCI-08	An ambiguous physical constant must not be resolved arbitrarily	`costante_fisica` in `bsi-ai-hub.js` — collects every candidate	`test_costanti` — 45 checks, including the refusals for ambiguity
SCI-09	The limits of the predictors must be declared to the user	chart headers; `docs/06` §3.5	`verifica_guida` — consistency between promises and code
SCI-10	Molecular symmetry must determine the selection rules correctly	`ssimm` section in `index.html`	`test_simmetria` — 26 checks: normal modes, centre of inversion, mutual exclusion
SCI-12	A chromatogram must follow from the parameters, not be drawn by hand	`scroma` section: $\sigma = t_R/\sqrt{N}$, R_s from distance and widths	`audit_stabilita` (opening and drawing), inspection of the values against the definitions
SCI-13	A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads	`staratura` section: ordinary least squares, F-test line⟷parabola, deleted residual	Verified on the three example sets included: linear ($F = 0.3$), non-linear ($F = 186.6 > 7.71$), with an outlier (1 detected)
SCI-14	A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set	`bsi-cheminfo.js` §9 — `divisionePerScaffold()` alongside `divisioneCasuale()`	`test_cheminfo` — the scaffold split leaves no scaffold shared between training and test

ID	Requirement	Implementation	Verification bench
SCI-15	A model score must be compared against chance, not presented on its own	`bsi-cheminfo.js` §9 — retraining on scrambled labels, eight repetitions	`test_cheminfo` — verified **in both directions**: on a learnable signal R^2 0.394 beats all eight twins (margin 0.287); on random labels R^2 −0.199 does **not** beat them
SCI-16	Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split	`bsi-cheminfo.js` §1 — deduplication on the canonical SMILES after the largest fragment	`test_cheminfo` — `OC(=O)C` and `CC(=O)O` collapse into one entry; salts lose the counterion
SCI-17	Similarity between molecules must be computed, not estimated	`bsi-cheminfo.js` §3 — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints	`test_cheminfo` — compared against hand-computable values, including the 0/0 = 1 convention
SCI-18	A validation on a single split, over a small set, is a noisy number: it must come with its dispersion	`bsi-cheminfo.js` §9-bis — `validazioneIncrociata()` over **scaffold-grouped** folds	`test_cheminfo` — the folds cover every molecule once, **no scaffold sits in two folds**, and the same seed gives the same folds
SCI-19	A result that changes on every run is verifiable by nobody	seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report	`test_cheminfo` — two calls with the same seed give identical folds; the method report declares the seed
SCI-20	Results must be able to leave the tool, and the method must leave with them	`bsi-cheminfo.js` §9-ter — `esportaCsv()` and `rapportoMetodo()`, both formed in memory in the browser	`test_cheminfo` — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits
SCI-21	A number presented as structural similarity must come from a fingerprint, and one implementation must compute it	`rdkit_lab.html` loads `bsi-cheminfo.js` and uses its `tanimoto()` and `fingerprint()`; the eight-bit similarity has been removed	`test_cheminfo` — page and engine give the same Tanimoto over all **28 pairs**, deviation **exactly 0**; the fake-similarity functions no longer exist; every reference structure is valid and carries the **monoisotopic mass** of the drug it names
SCI-22	Conversions between molecular languages must be checked against an external source or against hand calculation, and	`bsi-molingue.js` — **seventeen outputs**, from SMILES to CML	`test_lingue` — 7 **literature InChI keys**, full SMILES → molfile → SMILES round-trip, 26 hand-written names, **7 required refusals**; mass composition against hand calculation (alanine C

ID	Requirement	Implementation	Verification bench
	coordinates written to a file must be real and declared		40.44%, aspirin C 60.00%); XYZ and PDB with non-null coordinates **and** declared 2D
SCI-23	Angles, lengths and dihedrals are measured on the coordinates, and on flat coordinates they are not shown	dot product for angles, cross product for dihedrals	`test_mol3d` — tetrahedron **109.4712°**, water 104.47°, CO_2 180°, BF_3 120°, ammonia 106.13°; dihedrals 0°, 60°, 90°, 180°; ethanol C–C 1.509 Å; and the refusal tested in both directions
SCI-24	A spectrum read from a file must be decoded according to the format, and the peaks found must be the ones that are there — no more, no fewer	`bsi-spettrolettore.js` — JCAMP-DX with full ASDF (PMAI, DIF, DUP) and the integrity check the format prescribes; peaks by **prominence**, noise from MAD × 1.4826	`test_spettrolettore` — **61 checks**. On three Gaussians built at 1715, 2950, 3400 cm^{-1} under noise it finds three: 1716, 2952, 3402. **Both ways**: on pure noise it finds **zero**. ASDF verified symbol by symbol (`abcdefghi` = −1…−9; `n` = −5, not −4)
SCI-25	A band must be declared *compatible with*, never assigned to a single possibility	`BANDE_IR` — 24 ranges; `assegnaIR()` returns **all** compatible assignments	`test_spettrolettore` — at 1715 cm^{-1} both a ketone C=O and a carboxylic acid C=O appear; the section states that it does not deduce the structure
SCI-26	A published model must be checked against **its own** declared error, and a learned model against chance	`bsi-cheminfo.js` §10 — ESOL (Delaney 2004) with its uncertainty, four drug-likeness filters reported separately, random forest with out-of-bag error	`test_spettrolettore` §4-5 — ESOL against four experimental values, mean deviation **0.68**, inside the declared ~1 log unit; forest out-of-bag R^2 **0.90** on a non-linear relationship and **−0.16** on pure noise; deterministic for a given seed and different for another
SCI-27	A structure without a source must not get in, not even one that is known	collected from ChEMBL, `tools/dati/farmaci_v187.json` carries each entry's identifier	**Two refusals**: ivermectin (`structure_type NONE` , a mixture of homologues) and semaglutide (a protein, `SEQ`). And two entries kept under the **record's** name rather than the query's: «morphine» → apomorphine, «levothyroxine» → liothyronine
SCI-28	A predicted spectrum must say WHICH atoms produce each signal, and the deviation must be measured on	`bsi-pretsch.js` — the estimation tables transcribed in full from the source (Pretsch 4th ed.: 91 ^{13}C and 66 ^{1}H rows for benzenes, 42 for ethylenes, 31 for alkanes, 24 for ^{13}C aliphatics, 4×4 steric corrections); `bsi-nmr.js` — the reasoning that applies them: ring position obtained by walking the cycle,	`test_nmr` — **56 checks**. Two separate sets: tuning **0.71 ppm** over 9 molecules, **validation 0.92 ppm** over 22 never used for tuning, worst case **4.9 ppm**, ^{1}H **0.06 ppm** (0.03 on the aromatics alone). The bench requires validation to stay **worse** than

ID	Requirement	Implementation	Verification bench
	molecules that did not choose the parameters	additive scheme with a cyclic reference compound, chemical equivalence by shell environment code	tuning: were they equal, someone would have moved a molecule between the two sets
SCI-29	Chemically equivalent atoms must give ONE signal, not one per atom	environment code built as concentric shells by distance (the HOSE-code principle), with aromatic bonds written as such and not in Kekulé form	`test_nmr` — eight molecules whose signal count is known: benzene **1** (not six), toluene 5, p-xylene 3, naphthalene 3, biphenyl 4, aspirin 9. All correct
SCI-30	From an image one can recover the SHAPE of a spectrum, not its calibration	`bsi-digitalizza.js` — `bsi-spettrolettore.js` delegates `daImmagine()` to it, a single implementation. The axis frame is found first, the trace is extracted at **sub-pixel** resolution (darkness-weighted centroid), the run is chosen by continuity **and** by darkness, both axes are calibrated linearly or logarithmically, %T is converted to absorbance with $A = -\log_{10} T$	`test_digitalizza` — **45 checks** over seven constructed figures: mean deviation **2.2 cm⁻¹**, worst **7.3** (one column is worth ~4 cm^{-1}). **Both ways**: from a blank sheet and from the app icon no curve comes out, and the crude fallback on the same figure with frame and labels finds **one** peak instead of three. `test_spettrolettore` for the routing
SCI-31	A two-dimensional NMR map must say WHICH atoms produce each correlation, and must not connect what the topology does not connect	`bsi-nmr2d.js` — one-bond HSQC with inverted sign on CH_2, three-bond COSY between non-equivalent protons, two- and three-bond HMBC; exchangeables excluded from off-diagonal spots	`test_nmr2d` — **40 checks**, in both directions: hexafluorobenzene gives no HSQC spots, benzene **no** off-diagonal COSY spots (its six protons are equivalent), methane **no** HMBC spots despite having four protons
SCI-32	A program-built 3D geometry must carry THE SAME indices as the predictor, otherwise the peak↔atom link lights up the wrong atom	`bsi-geom3d.js` — distance geometry over the `bsi-nmr.js` graph, hydrogens appended at the end, generator seeded from the SMILES	`test_geom3d` — **29 checks**: every heavy atom has the same index and element across five molecules; bonds and angles within tolerance of literature values; **in both directions** benzene comes out planar and cyclohexane does not
SCI-33	A document must be opened in full, and whatever could NOT be read must be declared — a reader that returns little without saying so makes the document look nearly empty	`bsi-documento.js` — PDF page by page with its text layer (PDF.js), images, text, Office archives opened with `DecompressionStream` and no libraries	`test_documento` — **47 checks**: PDF, Word and text genuinely opened and their content verified; **in both directions** an image declares there is no optical character recognition and produces no working, an unknown format is refused rather than guessed, a binary file named `.txt` is recognised as binary

ID	Requirement	Implementation	Verification bench
SCI-34	Spectroscopic data recognised in a text must carry their PROVENANCE, and the working must declare what it is missing	`bsi-quesito.js` — recognition by section (IR, MS, ^{1}H, ^{13}C) with the boundaries between sections, then `bsi-elucida.js` for the step-by-step working	`test_documento` — formula, bands, masses with intensities, ^{1}H signals with integral and multiplicity, ^{13}C signals; **in both directions**: $CDCl_3$ does not become the compound, a band outside 400-4000 is not a band, without an MS label there are no masses, a new paragraph closes the section, a text with no spectra produces no data
SCI-35	A figure with a frame, labels and a grid must not be digitised as if it were only the curve: the axes and the numbers are dark pixels just like the trace	`bsi-digitalizza.js` — the frame is sought among the CONTINUOUS lines in the margins, and a border is told from a gridline by its **irregular spacing** relative to the family **or** by its thicker, darker stroke; the crop is moved past the line thickness; vertical gridline columns are marked as not measured and reconstructed from their neighbours	`test_digitalizza` §1-2 — on a figure with a frame **and** a grid (11 vertical families, 7 horizontal) it finds all four sides and the peaks stay within 2.2 cm^{-1}; **both ways**, on a grid-only figure it does **not** invent a frame and says so, and a **flat baseline** is not mistaken for an axis
SCI-36	A digitised curve must be able to leave the application, and the exported file must be valid	`bsi-digitalizza.js` — CSV output with the units in the header, and JCAMP-DX 4.24 in (XY..XY) form with FIRSTX/LASTX/NPOINTS and the provenance written inside the file	`test_digitalizza` §4 — the **full round trip**: the exported JCAMP is read back by `bsi-spettrolettore.js`'s `leggi()` and returns with the same number of points, the same end points and the same units
SCI-38	A quantum solver must reproduce the exact solutions that exist, not merely draw plausible curves	`bsi-fis-schrodinger.js` — finite differences with Sturm bisection and inverse iteration; Crank-Nicolson in time; Morse with literature parameters	`test_fisica` — harmonic oscillator, infinite well, numerical Morse = exact, norm conserved to 10^{-9}, exact transmission
SCI-39	An ab initio calculation must reproduce the published energies for the same basis and geometry	`bsi-fis-hartreefock.js` — RHF/STO-3G, McMurchie-Davidson integrals, Boys function, SCF with DIIS	`test_fisica` — Szabo-Ostlund H_2 (−1.116714 Eh) and HeH^+ (−2.860662), Crawford H_2O (−74.942079928) to 10^{-6}, nuclear repulsion, Mulliken charges summing to zero
SCI-40	Computed physical-chemistry quantities (entropies, spectra, diffraction, decays) must match independent reference values	`bsi-fis-termostat.js`, `bsi-fis-rotovib.js`, `bsi-fis-cristalli.js`, `bsi-fis-nucleare.js`, `bsi-fis-dinamica.js`, `bsi-fis-matematica.js`	`test_fisica` — NIST-JANAF S° of eight gases, HCl r_e and B from combination differences, 2θ of the NaCl/Si/Cu/Fe reference patterns and absences, ^{99m}Tc peak, ^{14}C, ^{56}Fe B/A, energy conserved in molecular dynamics, Hückel, FFT, RK4, N_2O_5 E_a

ID	Requirement	Implementation	Verification bench
SCI-41	A symbolic calculation must be exact or stay silent: a wrong antiderivative or derivative teaches the error with the face of a result	`bsi-mat-cas.js` — canonical form over exact fractions, derivatives with a log of the rules, antiderivatives returned **only** if their derivative matches the function at six points, Taylor series by truncated power-series arithmetic, limits by series and Richardson extrapolation	`test_matematica` — table derivatives and a complicated derivative against finite differences, antiderivatives by parts and by substitution, and **both ways**: e^(−x²) does **not** get an antiderivative; ∫e^(−x²) = √π, Taylor coefficients 1/120 and 17/315, standard limits,
SCI-42	Linear algebra must give exact results and the steps that produce them; balancing a reaction must come from conservation of atoms and charge	`bsi-mat-lineare.js` — Gauss-Jordan in BigInt fractions, Faddeev-LeVerrier characteristic polynomial, exact rational roots before Durand-Kerner; null space of the composition matrix	`test_matematica` — integer inverse of the Hilbert matrix and det 1/2160, a 4×4 det recomputed separately, eigenvalues of tridiag(−1, 2, −1), a defective matrix, permanganate, redox with charges, Wöhler; **both ways** $H_2 + O_2 \rightarrow H_2O + H_2O_2$ declared not unique
SCI-43	Laboratory statistics must reproduce the reference tables and examples, and uncertainty propagation must be checkable	`bsi-mat-statistica.js` — t and F from the incomplete beta function, Grubbs with the computed critical value, Rorabacher's Dixon, regression with s_y/x, LOD, LOQ and sample uncertainty; propagation with symbolic derivatives and Monte Carlo	`test_matematica` — t(0.975; 9), t(0.975; 4), F(0.975; 4, 4), Grubbs n = 5 and 10, the Miller & Miller calibration example (b, a, s_y/x, r, x_0 and $s(x_0)$), propagation matching a 20 000-draw Monte Carlo
SCI-44	An exam solver must produce the correct working, not just the result, and state when a step is numerical	`bsi-mat-risolutore.js` (domains, limits via Taylor, partial fractions, improper integrals, series with tests, two variables, extrema on compact sets, double integrals, differential forms, ODEs with Cauchy) and `bsi-fis-esame.js` (48 General Physics I and II models)	`test_esami` — results checked against SymPy, hand calculation or a different formula (range, Cramer, conservation laws, geostationary orbit); **every** solved-paper exercise produces a working; **both ways**: e^(−x²) has no antiderivative, ∫₁^∞ 1/x diverges, Σ 1/(n ln n) diverges
SCI-45	Electrode kinetics, transition-state, photophysics, adsorption and polymer quantities must recover the parameters the data were generated from	`bsi-fis-elettrodi.js`, `bsi-fis-superfici.js`	`test_esami` — α and i_0 from Tafel, ΔH‡ and ΔS‡ from Eyring, n_m and c from BET, n_m and K from Langmuir, Mn and Mw by hand, quantum yields summing to 1
SCI-46	Numerical chemistry problems, from basic to expert,	`bsi-chi-esame.js` — 40 models on four levels (General and Inorganic, Analytical, Physical Chemistry, statistical thermodynamics and	`test_esami` — $Ca_3(PO_4)_2$, glucose from composition, limiting reagent, ICE, acetic acid, 10^{-8} M HCl at pH 6.98 (**both ways**: not 8),

ID	Requirement	Implementation	Verification bench
	must give textbook values and balance reactions by themselves	activities), IUPAC atomic masses, reactions balanced via the composition-matrix null space, exact pH from the charge balance	AgCl and Ag_2CrO_4, Daniell, Faraday, Hα, Hess, titration, H_3PO_4, $BaSO_4$, HCl B, argon Sackur-Tetrode, Pourbaix, ionic strength; every solved paper
SCI-47	Astrophysics, nuclear physics, quantum physics and PhD-level problems must reproduce independent known values	`bsi-fis-esame.js` (astrophysics, nuclear, quantum, variational, perturbation), `bsi-chi-esame.js` (Marcus, isotope effect, Wigner, organic, spectroscopy, biochemistry), `bsi-mat-risolutore.js` (residues, heat equation), `bsi-percorsi.js`	`test_esami` — Sirius, Proxima, Kepler for Earth and Jupiter, Schwarzschild, ^{56}Fe and ^{235}U, D-T, ^{131}I, ^{14}C, 2p orbital, Larmor at 9.4 T, variational helium, perturbation = exact, Marcus, KIE, aspirin DBE, lysine pI, Cl_2 pattern, $\int 1/(x^4+1) = \pi/\sqrt{2}$, heat equation against the exact solution
SCI-48	An assistant reading natural-language questions must say what it understood and where each datum comes from, and the result must be the one computed another way	`bsi-assistente.js` — rules for mathematics (limits, derivatives, integrals, ODEs brought to the form G(x, y, y′, y″) = 0, series, matrices), IDF keyword scoring and re-ranking by data used for physics and chemistry, unit conversion, inverse problem by bisection; `bsi-mat-cas.js` (trigonometric integrals, e^(c·ln u), polynomial GCD)	`test_assistente` — 80 questions: model understood, data read (0.1 M is not 0.1 m; «carbon 14» is not a datum; «brakes», «falls», «an electron»), result against the independent formula; **both ways**: an arbitrary sentence does not produce an invented model; Spectra's tools with a question and with an explicit model
SCI-49	What is shown must be right: an animated or three-dimensional figure must not show a shape theory rules out, and an animation must end exactly on the picture the module would have drawn	`bsi-lab3d.js` — hydrogen-like orbitals sampled exactly from	ψ
SCI-50	A simulation must recover the exact answer where one exists, and declare its approximations elsewhere	`bsi-fis-meccanica.js` (projectile with quadratic drag by RK4, collisions with restitution, pendulum with the elliptic integral, normal modes), `bsi-fis-onde.js` (conjugate points, N-slit diffraction), `bsi-fis-circuiti.js` (closed-form RLC transients, impedance, Bode), `bsi-fis-termo.js` (cycles with per-process balances, hard-disc gas with elastic collisions, random walks), `bsi-mat-numerici.js`, `bsi-mat-geometria.js`, `bsi-chi-equilibrio.js`	`test_laboratori` — range and flight time in vacuum, T/T_0 = 1.18034 at 90°, energy lost $\frac{1}{2}\mu(1-e^2)v_{rel}^2$, Kirchhoff in the three RLC regimes, −3.01 dB and −45° at cut-off, ΔU = ΔS = 0 around cycles and efficiencies equal to the formulas, Maxwell-Boltzmann mean speed and hard-disc pressure, numerical method orders, Runge's phenomenon, conic foci, skew lines, kinetic equilibrium = computed equilibrium, Le Chatelier **both**

ID	Requirement	Implementation	Verification bench
		(mass-action kinetics against the computed equilibrium, van 't Hoff with ΔH° and ΔS° declared constant)	**ways** (exothermic/endothermic) and in the page. **Found and fixed**: with a low hot temperature and high r the Otto cycle «burned» by cooling (efficiency −130 %)
SCI-51	Astrophysics, relativity, fluids, vector calculus and complex kinetics must reproduce known values and theorems	`bsi-fis-astro.js`, `bsi-fis-relativita.js`, `bsi-fis-fluidi.js`, `bsi-mat-vettoriale.js` (symbolic curl and divergence with a numeric zero check), `bsi-chi-cinetica.js`	`test_scienze` — ΛCDM age 13.8 Gyr and Einstein-de Sitter 2/(3H_0), integrated Planck = σT^4, Earth-Mars Hohmann 259 days, Earth transit 84 ppm, HD 209458 b K, Kepler on the planets; twins and invariant interval; Poiseuille R^4, Stokes, Jurin; Green in both forms, the vortex with zero curl and circulation 2π **both ways** (with and without the origin inside); consecutive t_max, Hopf threshold and Brusselator period
SCI-52	Probability, visual mathematics, electromagnetism, sound, semiconductors, distillation and voltammetry must reproduce known results, and models must declare where they deviate from measurement	`bsi-mat-probabilita.js`, `bsi-mat-visiva.js`, `bsi-fis-em.js`, `bsi-fis-suono.js`, `bsi-fis-solido.js` (Varshni, N_c and N_v ∝ T^(3/2)), `bsi-chi-distillazione.js` (Antoine, van Laar, McCabe-Thiele), `bsi-chi-voltammetria.js` (finite differences with a second-order implicit boundary, Butler-Volmer)	`test_esplorazioni` — Bayes 1/6, central limit σ/√n, exact Fourier, logistic map periods, closed-form dipole flux = integral, n_i of Si/Ge/GaAs, V_bi, Antoine boiling points, ethanol-water azeotrope against measurement, McCabe against Fenske, reversible peak = Randles-Ševčík within 1 %, ΔE_p 58 mV, irreversible peak = 0.4958 nFAC√(αnFvD/RT). **Found and fixed**: with a first-order boundary gradient the simulated peak was 95.6 % of theory and ΔE_p 63.5 mV
SCI-53	The nineteen new sections (numbers and cryptography, graphs and optimisation, sequences and series, calculus of variations, topology, algebraic structures, game theory, logic and computability, statistical inference, non-Euclidean geometries;	`bsi-mat-numeri.js`, `bsi-mat-grafi.js`, `bsi-mat-serie.js`, `bsi-mat-variazioni.js`, `bsi-mat-topologia.js`, `bsi-mat-algebra.js`, `bsi-mat-giochi.js`, `bsi-mat-logica.js`, `bsi-mat-inferenza.js`, `bsi-mat-noneuclidea.js`, `bsi-fis-gravita.js`, `bsi-fis-calore.js`, `bsi-fis-ising.js`, `bsi-fis-particelle.js`, `bsi-fis-analitica.js`, `bsi-fis-laser.js`, `bsi-fis-clima.js`, `bsi-chi-fasi.js` (IAPWS, Span–Wagner), `bsi-chi-pes.js` (Sato LEPS)	`test_approfondimenti` — π(10^5) = 9592, RSA d = 2753, Dijkstra = Floyd–Warshall, Kruskal = Prim, figure-eight, Earth–Moon Lagrange points, Onsager, boiling at 373.124 K, H_3 saddle at 0.93 Å and 0.42 eV. `test_pareggio` — 462 checks in twelve scripts in `verifiche/`: $π^2/6$, the cycloid faster than any other curve, χ of the solids from their vertices, 549 946 tic-tac-toe nodes, busy beavers, table t quantiles, Girard, pion momentum 29.79 MeV/c, exact pendulum period, Rabi against Schrödinger, 255 K and 288 K. **Found and fixed**: the

ID	Requirement	Implementation	Verification bench
	gravitation, heat, Ising, particles, analytical mechanics, lasers, climate; phase diagrams, potential energy surface) must reproduce known results or an independent second method		first LEPS had its asymptote at −6.40 eV instead of −D_e; the hyperbolic tiling side length had p and q swapped; the Noether demo potential let the particle escape
SCI-54	The Articles section must have the **same level** for chemistry, physics and mathematics: same tabs, same rules, no year or name written by hand	`bsi-articoli.js` (three subjects), `bsi-articoli-nobel.js` (135 explanations), `bsi-articoli-nobel-fisica.js` (140), `bsi-articoli-premi-matematica.js` (140; Abel Prize and Fields Medal from Wikidata)	`test_articoli` — for each of the three subjects at least 130 explanations, none under 450 characters, no year, no duplicate expression, eight article searches; fake answers injected for the Physics Nobel and for Wikidata: the names received appear and no others
UI-18	Each subject must have its own bar: whoever studies physics or mathematics must not scroll through the chemistry groups	`bsi-materie.js` — two round buttons above Spectra and ✨, always the two subjects you are not in (flask, atom, sigma); the bar shows the subject's groups plus Study and Utility; opening a section of another subject switches subject by itself; the choice is remembered	`test_approfondimenti` — visible groups for each subject, two different icons, accessible name, position above Spectra, subject remembered after reload, no section unreachable
SCI-55	The ten sections of the Astrophysics group (solar system, orbital mechanics, celestial mathematics, stars, black holes, gravitational waves, galaxies, cosmology, planetarium, exoplanets) must reproduce published values or an independent second method, and declare illustrative data	`bsi-astro-sistemasolare.js` (Standish JPL elements, Meeus), `bsi-astro-orbite.js`, `bsi-astro-matcelesti.js`, `bsi-astro-stelle.js` (Lane–Emden, AME2020), `bsi-astro-buchineri.js` (Schwarzschild), `bsi-astro-ondegrav.js`, `bsi-astro-galassie.js` (Freeman, NFW), `bsi-astro-cosmo.js` (Planck 2018), `bsi-astro-cielo.js`, `bsi-astro-esopianeti.js`	`test_v237` — ten scripts in `verifiche/`, 314 checks: Meeus Julian date 2436116.31, Curtis ex. 4.3, elements ↔ state over 2000 orbits ($2.8\cdot10^{-14}$), Laplace limit 0.6627434, $\xi_1 = 6.89685$ for n = 3, Mercury's perihelion 42.982″, deflection 1.7512″, GW150914 chirp mass 28.096 $M_\odot$, Freeman peak at 2.150 R_d, ΛCDM age 13.791 Gyr, GMST of Meeus ex. 12.a, Earth's T_eq 254.8 K

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SCI-56	The ten new physics laboratories (spin and qubits, statistical mechanics, nuclei, materials, electronics) and mathematics laboratories (PDEs, complex analysis, Markov, SVD, codes) must reproduce known results or a second method	`bsi-fis-spin.js`, `bsi-fis-statistica.js`, `bsi-fis-nuclei.js` (AME masses), `bsi-fis-materiali.js`, `bsi-fis-elettronica.js`, `bsi-mat-edp.js`, `bsi-mat-analisicomplessa.js`, `bsi-mat-markov.js`, `bsi-mat-svd.js`, `bsi-mat-codici.js`	`test_v237` — ten scripts in `verifiche/`, 391 checks: HZH = X, Schottky peak (0.41678 Δ; 0.43923 k), B/A(^{56}Fe) 8.7903 MeV, Q of D + T 17.589 MeV, I-section moment with Huygens–Steiner, diode with Newton = Lambert W, Bessel zeros (j_{01} = 2.404825557695773), Cauchy–Riemann on nine functions, stationary distribution by powers = Cramer, σ of [[1,1], [0,1]] = φ and 1/φ, CRC-32 = 0xCBF43926, 112 single errors corrected by Hamming(7,4)
SCI-57	Study material must be checked: minimum quantities, valid schema, quiz answers computed where possible, tables with a numeric proof, citations in the declared formats	`bsi-studio.js` (`valida`, `MINIMI`), `bsi-studio-fisica.js`, `bsi-studio-matematica.js`, `bsi-risorse-materie.js` (citations, tables with `prova`)	`test_v237` — `verifica-studio` (167 checks on the engine: generator, shuffle, Leitner, score, HTML cleaning), `verifica-studio-fisica` and `verifica-studio-matematica` (minimums and schema; **264 mathematics answers recomputed** out of 346 questions), `verifica-risorse` (87 checks: APA 7, IEEE, ISO 690, BibTeX on hand-built records, table proofs)
UI-19	Every tab of the twenty new sections must open full, without warnings, with its canvases drawn and an accessible name, on a large screen and on a phone, without blocking the page	group 9 «🔭 Astrophysics» in `index.html`; sections built with `bsi-fisica-comune.js` and, for 3D, `bsi-lab3d.js`	`test_v237` — at 1280 and 390 px every tab is full, has no ⚠, every canvas draws (for 3D the scene is sampled) and has a name; **2D tabs within 250 ms, 3D tabs within 6 s** (in the bench WebGL runs in SwiftShader, with no GPU); the Astrophysics group has at least 12 sections; nothing overflows at 390 px; no JavaScript errors; the section names in English
UI-20	«Check for updates» must always give an answer: up to date, new version, site unreachable	`bsi-aggiorna.js` (service worker version via `BSI_VERSIONE`, `sw.js` from the same site with no cache); in `index.html` the 🔄 Updates window, with GitHub as fallback; in `sw.js` the answer to `BSI_VERSIONE`. **Found and fixed**: the window looked for a `CACHE_VERSION` line that `sw.js` no longer has, and on GitHub: the published version was never read	`test_controllaagg` — 12 checks: the «Check» button actually pressed in the three cases (same `sw.js`; `sw.js` with a higher version (v999); request aborted), «Update now» only in the second, and `BSIAggiorna.controlla()` with its pop-up notice in the same three cases

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SCI-58	The 3D tabs added to 29 chemistry, physics and mathematics sections must show quantities computed from the section's model and compared with known values, a closed form or a second method	`bsi-chi-pes.js`, `bsi-chi-fasi.js`, `bsi-chi-cinetica.js`, `bsi-chi-equilibrio.js`, `bsi-fis-cristalli.js`, `bsi-fis-schrodinger.js`, `bsi-fis-hartreefock.js`, `bsi-fis-rotovib.js`, `bsi-fis-dinamica.js`, `bsi-fis-meccanica.js`, `bsi-fis-em.js`, `bsi-fis-onde.js`, `bsi-fis-gravita.js`, `bsi-fis-relativita.js`, `bsi-fis-fluidi.js`, `bsi-fis-termo.js`, `bsi-fis-solido.js`, `bsi-fis-ising.js`, `bsi-fis-particelle.js`, `bsi-mat-multi.js`, `bsi-mat-vettoriale.js`, `bsi-mat-calcolo.js`, `bsi-mat-lineare.js`, `bsi-mat-ode.js`, `bsi-mat-probabilita.js`, `bsi-mat-topologia.js`, `bsi-mat-noneuclidea.js`, `bsi-mat-complessi.js`, `bsi-mat-variazioni.js` (scene through `bsi-lab3d.js`). **Found and fixed before publication**: `applicaMobius` did not recognise the pole for an angle of exactly π; in molecular collisions the normal was taken from spheres that had already overlapped (fraction above threshold 0.345 against $e^{-1} = 0.368$, now 0.370)	`test_3d238` — runs the 29 scripts `tools/banchi/verifiche/verifica-<id>-3d.js` (for `smattopo` and `smatnoneucl` the section's own script): 311 checks, among them the LEPS saddle with a single imaginary frequency, h*/R = 1.32549 for the soap film, for 3D Ising
SCI-59	The Spectra Studio lessons must show numbers computed from a model and compared with the exact solution, an experimental value or a second method	`bsi-spectra-studio-lezioni.js` — `calcolo.caduta` (semi-implicit Euler, optional quadratic drag), `calcolo.vsepr` (minimum of the energy of domains on the sphere, lone pairs with w = 1.25), `parziale`, `coda`, `fourierX2`, `coprimi`	`test_spectra_studio` — runs `verifiche/verifica-studio-spectra.js` (23 checks): fall time $\sqrt{(2h/g)} = 2.0193$ s for h = 20 m and the two spheres together; range 12.116 m; with air the thrown one lands later; tetrahedron $\arccos(-1/3) = 109.471°$, trigonal 120°, linear 180°; water 104.40° (experimental 104.5°, w calibrated); ammonia 106.69° with the same w (prediction); S(100 000) within the tail of $\pi^2/6$; tail between 1/(N+1) and 1/N for N = 10, 100, 1000; Fourier series of x^2 at $\pi = \pi^2$; harmonic series as $\ln N + \gamma$; coprimes → $6/\pi^2$ within 3σ; each lesson with exactly one right answer, every option explained, at least three proof steps, exercises on four levels
SCI-60	The Saha equation and the Boltzmann distribution must	`bsi-astro-chimica.js` — `saha(T, P_e, χ, Z_I, Z_II)`, `ionizzatoH`, `boltzmann`, `balmer`, `piccoBalmer`, `soleCalcio`,	`test_potenzia250` — at P_e = 20 Pa half ionisation at 9556 K (textbook ~9600) and Balmer maximum at

ID	Requirement	Implementation	Verification bench
	give the textbook values: hydrogen ionisation, the Balmer-line maximum, the calcium–hydrogen comparison in the Sun	`classeDi`; section `ssaha`	9870 K (~9900); $N_2 = N_1$ at 85,400 K; in the Sun 7.70×10^{-5} ionised, 5.05×10^{-9} in n = 2, 394 Ca II ions per absorbing H atom (~400). **Limit**: local thermodynamic equilibrium, pure-hydrogen gas
SCI-61	Interstellar-cloud chemistry must respect the exact solutions of the simple cases and the layered structure of a photodissociation region	`bsi-astro-chimica.js` — `evolvi` (implicit Euler on geometric steps), `gammaH2` with Draine and Bertoldi self-shielding, `tassiCO` (reduced scheme after Nelson and Langer), `profilo`; section `snubeism`	`test_potenzia250` — H_2 formation time 5.28 million years at n = 100 cm^{-3} and 100 K; numerical equilibrium = 2kn/(2kn + Γ) within 10^{-6}; f(1/2kn) = 1 − 1/e; fractions monotonic with depth; H → H_2 before C^+ → CO; a field a hundred times stronger pushes CO deeper. **Limit**: zero-dimensional teaching model, no CO self-shielding: orders of magnitude and structure, not real-cloud abundances
SCI-62	Peak deconvolution must recover the parameters of a spectrum whose truth is known, and the areas must be those of the closed form	`bsi-chem-mappe.js` — `profilo`, `area`, `stime` (second-derivative minima), `deconvolvi` (Levenberg–Marquardt, constraints on height, width, η); section `sdeconv`	`test_potenzia250` — three synthetic spectra with fixed-seed noise (five Gaussians of amide I, two Lorentzians, pseudo-Voigt with a shoulder): centres within 0.5, widths and heights within 6 %, $R^2 > 0.999$; η recovered; closed-form areas against the numerical integral; data pasted by the user. **Limit stated in the section**: a deconvolution is not unique, a high R^2 does not prove the components are real
SCI-63	The chemical-space map (t-SNE) must keep neighbourhoods, be repeatable and say how far to trust it	`bsi-chem-mappe.js` — `tsne` (exact: perplexity by bisection, early exaggeration, momentum, adaptive gains, fixed seed), `conservazione`, `purezza`, `pca2`; Morgan fingerprints and Tanimoto distances from `bsi-cheminfo.js`; section `sspaziochim`	`test_potenzia250` — on four Gaussian groups in ten dimensions: neighbourhoods kept better than PCA, groups separated, KL decreasing, same seed same map, P symmetric and normalised; in the browser, with RDKit, 55 molecules of seven families: neighbourhoods kept 81 %, same-family neighbour 95 %. **Limit stated in the section**: the axes mean nothing, nor do distances between far-apart groups

ID	Requirement	Implementation	Verification bench
UI-23	Every tab of the 29 sections with the new 3D tab must open full, without warnings, with its canvases drawn and an accessible name, and the last tab must be a 3D scene that draws; a chart without a name must take the tab's	`bsi-fisica-comune.js` — `prepara()` and `mostra()` give canvases without a name (or with a one-letter name) «Chart: <tab> — <section>»; the 3D scene is built after the click, outside the handler	`test_3d238` — at 1280 and 390 px, for every section: all tabs full, no ⚠, every canvas draws and has a name of at least three characters; the last tab is a scene with `Scena.campiona() > 0`; **synchronous work on click under 250 ms** (WebGL in SwiftShader); nothing overflows at 390 px; no JavaScript error
UI-22	The Spectra Live console must stay usable on a phone with Android's status bar and navigation keys over the page, with no overlapping elements, and every section must be reachable from it	`bsi-spectra-live.js` — `env(safe-area-inset-*)` margins with minimums of 34 and 46 px (class `.bsiJV-bordi`) when the window covers the screen on a touch device or the app is installed; layout computed in the drawing loop; 🧭 navigator per subject; context shortcuts; 💬 button; round Spectra Live button (since v239 Spectra's only button, UI-25)	`test_spectra238` — on a simulated phone (412×915, touch): safe margins active, first button ≥ 34 px from the top edge, controls ≥ 46 px from the bottom edge, telemetry in two columns not covering the face; on a large screen no forced margins; on both, header, face, state, dialogue, controls and attribution without overlaps; the navigator lists, for each of the three subjects, exactly the sections of the bar, the quick tools and the phrases; opening a section of another subject switches subject and leaves the Spectra Live button
UI-24	Every section must have a guided lesson built from the section itself: one chapter per tab, the controls shown at work and put back as they were, a check question with its explanation, a summary; during the lesson no floating button over the panel, and at the end everything as it was	`bsi-lezione.js` — `BSILezione.costruisci()` reads title, subtitle, tabs, explanations, canvases, sliders and selects; check question from the subject's study kit (most words in common with the section); player `#bsiLZ` with spotlight, ◀ ▶, pause and voice; «🎓 Guided lesson with Spectra» button in every section; demonstrated controls return to their starting value even when skipping ahead or closing; floating buttons hidden and restored; «reduce motion» respected. `bsi-spectra-live.js` — «give me a lesson» («fammi lezione») command and 🎓 shortcut	`test_lezione` — 30 checks: going through all **194** sections of the bar in the three subjects, each has its lesson (opening … summary) and the button, and each of the **99** with tabs has one chapter per tab; at least 60 lessons with a check question (**79** at the time of writing); at 390 and 1280 px: the lesson starts from the opening, floating buttons are hidden, a slider really changes value under the spotlight and returns to its starting value when skipping ahead half-way, a select goes through its options and returns, the check shows the right answer and the explanation, the summary lists the chapters, nothing overflows, on closing the panel is

ID	Requirement	Implementation	Verification bench
			gone and the buttons are visible again; «fammi lezione» starts the lesson; no JavaScript errors. The voice is not tested by the bench
UI-25	Spectra must have a single visible button, its face in a circle, from which both console and chat are reached	`bsi-spectra-live.js` — `#bsiJV-fab` (58 px, round) in place of the `#bsi-spectra-fab` pill, which stays in the document hidden by the `bsiJV-solo-volto` class because other modules and the benches use it; the circle opens the console, 💬 opens the chat; vector face at once, portrait after `load`	`test_spectra238` — after loading the circle is visible, the pill is not, and the circle has `border-radius: 50%`
UI-26	Spectra Studio must lead every lesson through the four phases predict → test → prove → reveal on a large screen and on a phone, remember the predictions, recommend the exercise level and remain usable without WebGL	`spectra-studio.html` — three.js scene with Spectra from the shared controller of `bsi-spectra-figura.js` (embedded body `models/spectra-corpo.bsi` since v241, rigged, and «Noir» face on the collar; while it arrives the `#invito` notice «Spectra sta arrivando…», if it does not arrive «↻ Riprova»; no substitute, no « 🧍 Corpo» button); panel `#pan` with all the text, `aria-live`, lesson tabs with `role=tab`; memory `bsi-spectra-studio-v1`; voice switch 🔊/🔇 remembered in `bsi-spectra-studio-voce`; without WebGL no scene, panel and measurements remain. `bsi-spectra-live.js` — «Spectra Studio» entry in the 🧭 navigator	`test_spectra_studio` — at 1280 and 390 px the three lessons go through the four phases with the measurements in the scene (fall, water, Basel: S_{1000} = 1.64393, coprimes 0.6079), the real body is always there without loading it and the notice is gone, no « 🧍 Corpo» button and no file choice, the published file is a «BSIC» block and not a `.glb`, an invalid block rejected without errors, the face rests on the real body's collar; for every lesson four predictions, measurements in the table and formulas in the proof, the reveal marks the right answer with four exercises of which one recommended, the memory remembers the prediction and the exercise opened; at 390 px nothing overflows; without WebGL the lesson gives the measurements (2.019 s); no JavaScript errors
UI-27	In every AI section (Spectra AI and the three areas with the assistant) Spectra must appear with the body chosen by the author, **always present without having to load it** (since v241), under the title, and react to what the app	`bsi-spectra-figura.js` — `Spectra()` (`carica()`: the embedded body `models/spectra-corpo.bsi`, reopened by `apri`; an old body in IndexedDB `bsi-spectra-modelli` is deleted; in the app, with `soloDispositivo`, no requests for the development files, no 404s; «Spectra sta arrivando…» and «↻ Riprova»), `monta()` (lab-style frame, «SPECTRA · IN ASCOLTO / PENSA / PARLA», «🎙️ Parla con me», «🧠 Spiega», «🎓 Studio»), `sezioni()` (one figure moved into the open section);	`test_spectra_figura` — 111 checks at 412 and 1280 px (28 of them for «🧠 Spiega», AI-18, 14 for the hologram and 10 for the catalogue, UI-28, and 6 for the Spectra Live console): button face only, pill hidden; in the four sections the figure inside the section under the title (≥ 280 px) and only one in the whole app; `#main` at 0 px above the sections outside it; Spectra's body is always there, without loading it (embedded, bones

ID	Requirement	Implementation	Verification bench
	does; the icon must be the face only; no substitute; the sections outside `#main` must have no empty space above them	events `bsi-spectra-stato`, `bsi-spectra-risposta`, `bsi-spectra-dice` (from `bsi-spectra-live.js` and `bsi-assistente.js`); `index.html` — class `bsi-sez-fuori`. **Since v242 also in the Spectra Live console**: with the console open, the figure moves into `#bsiJV-figura` in place of the bust `#bsiJV-volto` (hidden) and the console's state drives it through `bsi-spectra-stato` (thinking → concentrated, hand on chin; listening → curious; the console's event carries the `daLive` signature and the console ignores it, so no recursion); with the console closed, it returns to the section	20/23); no « 🧍 Corpo» button and no file choice; on the real body face on the collar (0.0 cm), only the face visible; thinks, explains with lip sync on body and face, doubt on error, speaks with Spectra Live, explains the local assistant's working; in the Spectra Live console the figure with the body in place of the bust, the console's state → hand on chin, with the console closed the figure returns to the section; on the phone the buttons do not cover the floating ones; no JavaScript errors. `test_spectra_studio` — in the studio the embedded body is always there and the published file is a packed block. **Not tested**: the visual rendering of the body, which has to be judged by looking
UI-28	To Spectra's right a hologram must show live scenes of the sections (computed 2D and 3D animations, or the open section live), follow the conversation's subject and lead to the section with one tap; **every section** of chemistry, physics and mathematics must be showable and explainable there («non deve mancare nulla in questo ologramma»), including those without graphs; and **Spectra's figure must not disappear**: showing a section must not take the user away from the	`bsi-spectra-figura.js` — `Ologramma` (glass screen with base and beam, on the viewer's right), `SCENE_OLO`: planets (Kepler's equation M = E − e sin E solved by Newton every frame, T ∝ a^1.5), molecule (benzene with the π cloud), spectrum (IR with Gaussian bands and a cursor), surface (z = sin(r − t)/r), orbital (3d_z² with phases), wave (Gaussian packet with free Schrödinger dispersion), screen (the section behind the window, re-rendered with `BSISpectraLive.cattura` 5 times a second); `sceneDi(text)` picks the scene from the words; event `bsi-spectra-tema` from `bsi-spectra-spiega.js`; «… · apri ›» label that opens the window on «🧠 Spiega» and the section behind it (`BSISpiega.mostraSezione`), without leaving the AI section; on closing the window the return to the starting AI section (`sai`, `sareachim`, `sareafis`, `sareamat`) and the tour stopped; «indica» gesture before «spiega»; button bar under the scene; in «schermo» a section without graphs is projected as a card (title, tab, up to 6 controls with values, visible text). `bsi-spectra-spiega.js` — `catalogo()` (every	`test_spectra_figura` — 12 checks per width (412 and 1280 px): the catalogue has every section of the bar (201/201 since v250: chemistry 78, physics 63, mathematics 31, shared 29); a question that names a topic finds its section (titolazioni → `stitr`, Schrödinger → `sschrod`, integrali per parti → `smatcalc`, spettro di massa → `sms`, spettro infrarosso → `sirvis`, meccanica orbitale → `sastroorbite`, «ciao come stai» → none); asking about titrations Spectra opens the section behind the window, shows it in the hologram and explains it with its data; the Periodic Table (no graphs) is projected as a card; «🎞 Mostrami tutto» runs through the subject's sections; the hologram is to Spectra's right (screen x); six live scenes; perihelion/aphelion speed ratio = (1+e)/(1−e) = 1.94 within 8 %; the subject follows the conversation (IR answer → spectrum, orbit question → planets); the label leads to the section; a tap on the label opens «Sistema solare 3D» **behind**

ID	Requirement	Implementation	Verification bench
	AI section, and closing the window must return to where the figure is	`.nav-group-panel [data-s]` button of the bar with group and subject from `BSIMaterie.materiaDi`, not written by hand), the «🗂️ Tutte le sezioni nell'ologramma» card with search and subject filter, `mostraSezione(id)` (`BSISpectraLive.calcolo.apriSezione`), `giro(materia)` / `giroFerma()` («🎞️ Mostrami tutto», one section every 7 s), `sezioneDi(testo)` (the section the question names: words with stems, weighted acronyms, synonyms)	the «🧠 Spiega» window: Spectra stays visible in front and the hologram shows the section live; once the window is closed, the app returns to Spectra's section and the figure is there; also, with the 3D scene open behind «🧠 Spiega», the hologram shows it live, and on answering Spectra points at the hologram and then explains with the hands. **Not tested**: the visual rendering of the scenes, which has to be judged by looking
UI-29	For each subject (chemistry, physics, mathematics) a «⚡ Explore with Spectra» section in the style of explorable lessons (every-joule): a page to scroll with **ten chapters from high school to advanced PhD**, Spectra with the body **on stage while scrolling** and the hologram following the chapter; in every chapter predict (four answers, the reason for each), try it with **computed, not hand-written, numbers**, watch the linked section live, get the full explanation; predictions remembered and erasable	`bsi-spectra-esplora.js` — `CAPITOLI` (10 per subject, levels 0→4; prediction with four answers, the right one and a reason for each; linked section and hologram scene), `calcola(id, v)` (19 sliders: pH with autoprotolysis, Henderson–Hasselbalch, t½, Marcus, fall, range, pendulum, Carnot, box, tunnelling, muon γ, derivative, e, ln, Basel, harmonic, eigenvalues, `gibbsMax(N)`, heat), `costruisci(materia, pannello)` (the «active» chapter = the one crossing the middle of the free space under Spectra; `bsi-spectra-dice` and `olo.mostra(scena)`; «📺 Guardalo dal vivo» with `bsiOpenAIHub('spiega')` and `BSISpiega.mostraSezione`; «🧠 Spiegamelo tutto» with `BSISpiega.chiedi` at the chapter's level; «🔊 Ascolta Spectra» with `BSISpectraLive.parla`), three sections `sesplchim`, `sesplfis`, `sesplmat` registered with `BSIFisica.sezione`; memory `bsi-esplora-v1` (erased by «Cancella dati → Memoria persistente», `bsi-ai-hub.js`). `bsi-spectra-figura.js` — class `bsiEX-palco` (`position: sticky` under the app header; shorter on phones); `bsi-spectra-spiega.js` — the Explore sections count as AI sections	`test_spectra_esplora` — 75 checks. In node: per subject ten chapters, levels 0→4 in order, well-formed predictions, unique ids; 22 slider numbers against the formulas (pH of HCl 10^{-8} M = 6.978, t½ = 6.931 s, Marcus λ/4 at −2λ, fall 2.019 s, range 40.775 m, pendulum 2.006 s, box 0.376 eV, γ = 31.63, harmonic 7.4855, Gibbs ≈ 1.179…). At 412 and 1280 px: the three sections in their subject's bar and every linked section exists; per subject ten chapters and Spectra in the section, sticky, in view after scrolling; the chapter in front active and its scene on the hologram; the prediction marks right and wrong with the reason, Spectra reacts (surprise); memory and bar 1/10; the slider recomputes; nothing overflows at 412 px; «📺 Guardalo dal vivo» opens the section behind the Spectra window with the hologram live and closing returns to the Explore section; no JavaScript errors. **Not tested**: the voice of «🔊 Ascolta Spectra», a real model's explanation, the didactic quality of the texts
UI-30	The hologram must be selectable and become **interactive**: in	`bsi-spectra-olointerattivo.js` — «⛶ Interagisci» button on the hologram (`bsi-spectra-figura.js`); window `#bsi-olo-ov` (the floating buttons hide	`test_spectra_olo` — 69 checks (since v248: «🗂️ Tutta l'app» with all 201 sections and the 15 ✨ menu entries chosen by the user,

ID	Requirement	Implementation	Verification bench
	spectra, change type and see them change with the written trace and data; for planets and the universe everything about them; in mathematics and physics the hologram changing in real time with the question; **every step** explained in writing and aloud by Spectra; numbers computed, not hand-written	through `_bsiSyncFabs`) with the shared figure in `#bsiOI-figura`; `SPETTRI` (six molecules, IR, ¹H, ¹³C, MS, UV, typical literature values), `spettro(tipo, molecola|traccia, { MHz })` (multiplets in Hz converted to ppm with the field; inset of the selected multiplet), `leggiTraccia` (assignments from group-frequency, chemical-shift and neutral-loss tables; band shape matters), `massa`, `insaturazione`; `PIANETI` (NASA fact sheet), `pianeta(nome|{a, e})` (third law, perihelion/aphelion, vis-viva, gravity, escape), 3D scene with Kepler's equation; mathematics and physics from `bsi-assistente.js` and `bsi-mat-cas.js` (computed steps); «🔊 Spectra spiega i passaggi» with the Spectra Live voice; `scegli(i)`; «🧠 Approfondisci» → `BSISpiega`; event `bsi-olo-interattivo`. Since v242 (request: «it won't let me select it and I can't hear the voice»): a tap on the hologram itself opens it in interactive mode; at each new computation Spectra says the steps on her own (`voceAutomatica`); «🧠 Spiega» says its answers aloud and without a key answers with the local working or the named section.	those with graphs live and those without hosted in the frame; on the phone the whole window scrolls; «🔔 Prova audio»; since v242 also: automatic voice on the written trace, the tap on the hologram that opens it, «🧠 Spiega» without a key answering and speaking). In node: six molecules × five spectra; acetone 58.08 g/mol; unsaturation 4|1|0; triplet 2J/ν_0 at 400 MHz, 10 times wider at 60 than at 600 MHz; IR trace (narrow 2950 is not acid O–H) and MS (M^+, M − 15, acylium); Earth T = 1 yr, vis-viva 30.29/29.29 km/s; Mars (1+e)/(1−e) and 3.71 m/s²; escape 11.19 km/s; a = 4 AU → T = 8 yr. At 412 and 1280 px: the button opens in «spettri», the figure moves in, the floating buttons disappear; type and trace redraw and redo the steps; MS trace with two steps; the tapped peak 43 selects its step and Spectra says it; Mars with six steps in 3D; zeros −√3, 0, √3 and stationary ±1 with ≥ 8 steps; projectile 35.31 m; pendulum 2.837 s; the voice says every step; nothing overflows at 412 px; «🧠 Approfondisci» opens Spiega; Spectra returns to the section; no JavaScript errors. **Not tested**: the audible voice, a real model's explanation; the reference spectra are typical literature values, not measured, and the trace gives candidate assignments, not a structure
UI-31	The chemistry, physics and mathematics assistant must live **inside the round button with Spectra's face**, not in the top bar	`bsi-spectra-live.js` — `apriAssistente()` (exposed as `BSISpectraLive.apriAssistente`): tapping `#bsiJV-fab` opens the area of the current subject (`BSIMaterie.corrente()` → `sareachim` / `sareafis` / `sareamat`) and scrolls to the top; if the area is missing it opens the console (`apriHud`); a CSS rule hides `.nav-btn[data-s="sareachim	sareafis

ID	Requirement	Implementation	Verification bench
UI-32	Spectra's voice must be heard on the phone too: a speech engine slow to start must not get the sentence cancelled, and whoever cannot hear it must be able to find out why	`bsi-spectra-live.js` — `parla()`: cancels only if something is in progress and resumes synthesis; while the browser says it is speaking or has the sentence queued it waits up to ~8 s (it used to cancel after 1.2 s: the cause of the silence); «onend» moves on even without «onstart»; lip-sync only if synthesis does not start; unlock on first tap (empty utterance at zero volume, quality-voice audio context resumed, `bsi-spectra-voce.js` `sblocca()`; a context still suspended falls back to the system voice); voices re-chosen when they arrive; command «prova la voce» / «non ti sento» with `statoVoce()` (how many voices, which one, what to do on Android)	`test_impronta` — a fake synthesis starting after 2.5 s without «onstart»: no sentence cancelled, 2/2 reach the end; a synthesis that never starts: lip-sync carries the speech forward; «prova la voce» answers; `statoVoce()` readable. **Not tested**: audible sound on a real phone
UI-33	Every Spectra answer in the Chat must be listenable with a male robotic voice, low and cold, using only the Web Speech API (offline too): pitch 0.58, rate 0.88, volume 1; an Italian male voice if available; a 🔊 button on every answer; «Leggi automaticamente» remembered; stop when the chat is closed or the section changes; no button without synthesis	`bsi-ai-hub.js` — `speakRobotic(text, {pulsante})` (whole answer, sentence by sentence, text cleaned by `BSISpectraLive.calcolo.perLaVoce`; cancels only if something is in progress), `stopSpeaking()`, `toggleAutoSpeak()` (`localStorage` `spectraAutoSpeak`), `createSpeakButton()` («🔊 Ascolta» ↔ «■ Ferma», `aria-pressed`), voice choice Italian male → Italian → male → first, re-chosen on `voiceschanged`; switch in the Chat bar; automatic reading at the end of an answer (not if the console is open); `closeHub()` and the section buttons stop the voice; console (`bsi-spectra-live.js`) and fallback (`bsi-spectra-voce.js`) with the same parameters	`test_voce` — 21 checks with a fake synthesis (since v248 also on Android: no voice chosen by name, no empty unlock utterance, 160 ms after a cancel): voice «Luca» (Italian male, not the female nor the English one); 0.58/0.88/1 on every sentence; a 1228-character answer read entirely in 28 sentences, without markdown; no needless cancel; in the Chat the switch, remembered on and off, automatic reading of Gemini's answer (fake network); a button on every answer, «■ Ferma» and stop; stop on closing and on section change; console at the same parameters; no button without synthesis. **Not tested**: the real sound
UI-14	One function must not have two entry points. The spectra lived in **nine sections** across two menus; Spectra in three places; and the ✨ panel showed **two "Data Science", two "File**	`bsi-centrospettri.js` — the panels are **moved** (`appendChild` moves, it does not copy) into a single tabbed section; the old buttons stay hidden because they are the target of `goSection()`, and a tab **clicks the button** instead of showing the panel, so lazy initialisation goes through the one road that already existed. `bsi-menu.js` — the "Utility" category is in the **main bar** with all its entries, and the rule is: if something	`test_articoli` §5-6 — the **ten** tabs are inside the centre, and each identifier exists in the document **exactly once** (the proof that it was moved and not copied); `goSection()` on each opens the centre on the right tab and shows **only** one. `test_menu` — Utility visible in the bar with all **fifteen** entries plus the two language ones, **no** menu entry repeating a

ID	Requirement	Implementation	Verification bench
	Manager", two "Articles", two "Updates", because the panel has a hardwired list that opened the very sections the menu module was inserting — and nobody compared the two lists	has a **reachable** button in the bar it is not in the ✨ menu, if it has none it is only in the menu. "Reachable" is not measured with `offsetParent` — the bar shows one group at a time, and a perfectly reachable button has a null `offsetParent`	bar section, **no** duplicate label in the panel, across **two viewports**: large screen and phone. **Both ways**: no repeated id, no duplicate button, Spectra and the reader out of the bar but with their button still in the document because `goSection()` uses it
UI-17	Anything that moves must be pausable, slowable and restartable, must not consume when not visible, and must honour «reduce motion» (WCAG 2.2.2)	`bsi-fisica-comune.js` — `F.anima`: bar with ⏸/▶, ↺, speed ×¼…×4 and time; step with real dt capped at 1/20 s; stops off-screen (IntersectionObserver), with the page hidden and when the canvas is removed; with «reduce motion» it starts paused with the first frame drawn. Every laboratory uses it	`test_scienze` — time advances, stops when paused, runs faster at ×4 than at ×¼, stops off-screen and resumes when visible; with «reduce motion» it starts paused and ▶ starts it; every bar has accessible names
UI-15	A data table must not exist twice: two copies diverge, and the reader gets a different answer depending on where they came in	The J coupling constants existed in two independent tables with **different** numbers (aromatic ortho ^{3}J 6–9 Hz against 7–10 Hz). The one inside the Spectra centre remains, which sits in the section with its own reference list; `_apriCostantiJ()` is the single road and leads there	`test_articoli` §6 — asking for the J constants opens the centre on the reference-tables tab, on the right sub-tab, and the table has its rows
ING-02	A hand-written list of articles is stale the next day, and a prize year written from memory is an error nobody can see	`bsi-articoli.js` reads **live** from Crossref and from the Nobel Foundation; `bsi-articoli-nobel.js` holds **explanations only**, matched to the **topic of the official motivation** and not to the year, so no attribution comes from the application. The addresses are not written: they are **pure functions** of the identifier (DOI, year)	`test_articoli` §1-4 — the links tested as functions (stable for the same datum, different for a different one); the explanations matched against nine test motivations and **none** invented when the topic is not recognised; and the proof that counts: a response is **injected** with a name that exists nowhere, and the screen must show **that** name. **Three distinct failures tested**: failed request, empty answer, unrecognised shape — and they must be stated **differently**
UI-16	Changing language must cost a gesture, not a navigation	`bsi-sellingua.js` — a **round** button top right, at the title's height, with the flag in use; it opens into a disc with the fourteen flags on **two shells** (six inside, like a benzene ring) and, at the centre, the language name and **how much it is**	`test_sellingua` across **two viewports** — round, top, right, and "at the title's height" **measured** as the overlap of the two boxes (with `align-self:center` it lined up on a large screen and not on a phone,

ID	Requirement	Implementation	Verification bench
		translated, measured by `coperturaPerLingua()`. The actual switch stays in `bsi-lingue.js`: here there is only the way to call it, and the "Language" section remains because it INFORMS while this one ACTS	because there the flex aligns to the top and the logo block is two lines tall); an accessible name that says what it does rather than "flag of Italy"; the disc **inside the screen**; and **both ways**: the language really changes and going back to Italian the text is identical
ING-01	One function must not have two implementations: the React/FastAPI stack must run THE SAME files as the application	`stack/api/worker/motore.mjs` — a Node process loads the repository modules with RDKit WebAssembly; no Python engine	`test_stack` — **15 checks**: compares the SHA-256 fingerprints declared by `GET /salute` with those of the files on disk, verifies the numbers match the browser's, and that no `.ts` / `.tsx` file contains a shift table

2. AI agent requirements

ID	Requirement	Implementation	Verification bench
AI-01	A model retired by the provider must not block the assistant	`modelloSuggeritoDaErrore()`, `bsi_modelli_ko` blacklist	`test_404` — 36 checks
AI-02	A provider unreachable from the browser must be remembered and demoted	`bsi_prov_ko`, 24 h TTL, `providerUtilizzabili()`	`test_ko` (35), `browser_ko` (37)
AI-03	A temporary overload must produce a retry, not an error	`erroreTemporaneo()`, `TENTATIVI_TEMPORANEO`; since v243 a 503/529 on one model moves **at once** to another candidate model of the same provider, with the same key (`segnaSovraccarico`, `altroModello`: the overloaded model is set aside for 10 minutes, not rejected); only when the candidates run out does it wait and then fall back to another provider	`test_503` — 40 checks (three different models, no waiting; always down → fallback to another provider); `test_impronta` — Gemini 503 on the first model → answer from the second in under 5 s
AI-04	An imposed wait longer than 10 s must switch provider if a ready one exists	`ATTESA_TROPPO_LUNGA_MS`	`test_attesalunga` — 14 checks

ID	Requirement	Implementation	Verification bench
AI-05	The provider's token ceiling must be learned from the error, not hard-coded	`tettoDaErrore()`, `MARGINE_TETTO`	`test_tetto` — 32 checks
AI-06	Opaque reasoning signatures must be returned to the provider	capture and replay in `appendAgentTurn()`	`test_firma` — 31 checks
AI-07	Two overlapping submissions must not corrupt the history	`_turnoInCorso` in `send()`	`test_doppioinvio` — 7 checks
AI-08	The user must be able to measure which providers they can actually reach	`provaTuttiIFornitori()`	`browser_prova` — 29 checks
AI-09	Paid providers must not be offered as free ones	separation by `PROVIDERS[id].free`	`browser_prova` §D2
AI-10	A documented route must exist to remove CORS and rate limits	`proxy/spectra-proxy.js` + in-app guide	`test_guidaproxy` (12), `browser_proxy` (15), `browser_proxyui` (17)
AI-11	Spectra's face must really move with what he says and hears: emotions recognised in sentences, a face that changes, eyes in the theme's colour	`bsi-spectra-androide.js` — «Noir» model (Skyfox13, CC BY 4.0, modified); 17 emotions from 11 expressive fields blended on the CPU into one attribute (WebGL allows at most 16 per vertex); mouth from the spoken text; fallbacks `bsi-spectra-robot.js` and `bsi-spectra-volto.js`	`test_v237` — 17 emotions and 11 fields; nine sentences give the expected emotion; eyes at rest `#1a3050`; more than 1000 animated vertices; face still in neutral and different between joy and empathy; the set emotion persists; a compliment produces the warm smile; more than 1000 pixels drawn at the centre; no JavaScript errors. The no-WebGL fallback is not exercised by the bench
AI-12	Spectra's mind must stay in the browser and declare itself for what it is: a state that responds to events, not consciousness	`bsi-spectra-mente.js` — key `bsi-spectra-mente-v1`, state of curiosity, attention, care, mood, bond; `coscienza()` passes a summary to the model; «forget everything»	`test_v237` — a compliment raises the mood, «how are you» answers from the internal state, the summary for the model contains the mood. «Forget everything» and the persistence of the key are not exercised by the bench

ID	Requirement	Implementation	Verification bench
AI-14	The face must show how demanding the question is and stay alive without looking conscious: concentration from the section's level and from the question, eyelids and eye tint per emotion, sleep when nobody is there	`bsi-spectra-live.js` — `difficolta(testo)` and `livelloSezione(id)` (first level of the path in `bsi-percorsi.js`); `bsi-spectra-androide.js` 3.0 — eyelids as a deformation of each eye's vertices around its centre, tint per emotion in the `#1a3050` family, `concentra()`, damped springs, micro-expressions, sleep after 120 s without contact	`test_spectra238` — an advanced-course question weighs more than a sum, «in simple words» lowers it, the same question weighs more in a PhD-level section than in a school-level one and the level is the path's; joy narrows the eyes, surprise widens them, doubt makes them asymmetric, `#1a3050` at rest, four emotions give four tints, concentration narrows and brightens the eyes, the blink closes the eyelids, after two minutes it sleeps and wakes at the first touch
AI-15	Spectra's body must work with any humanoid avatar: recognise bones, blendshapes and hair, report what is missing, and drive gaze, gestures, emotions, lip sync, eyelids, walking and sitting; the hair must turn black	`bsi-spectra-corpo.js` — recognition by name (Mixamo, Ready Player Me, VRM, Unreal, generic), `rapporto()`, `orienta()` (poses as world directions), emotions as ARKit weights, `capelliNeri()` (colour `#0a0a0c` and texture desaturated in the shader), `fai`, `cammina`, `siedi`, `audio`; test `manichino()`	`test_spectra_studio` — on the mannequin (11 checks): bones 22/23 (no jaw), blendshapes 14/14 and hair found; red hair turned black (RGB 0.039 0.039 0.047); arms down from the T-pose; pointing with the forearm on the target (1.7° off); thinking with the hand near the chin; the gaze turns; walking moves the body 0.96 m moving the legs; sitting lowers the hips by 42 cm; joy → smile, worry → mouth corners down; lip sync from audio opens the jaw; the blink closes the eyelids. The fallback to the «cyberpunk» model (`models/spectra-cyberpunk.glb`) was removed in v241: the body is the embedded one (AI-17)
AI-16	The quality voice must go through a measurable effects chain, give the lip sync from the audio and fall back on the browser's synthesis without a key	`bsi-spectra-voce.js` — `catena()` (bass, mud, presence, ring, comb, saturation, reverb from `impulso()`, compressor), `leggiBocca()` (energy and centroid), `qualita()`, `TIMBRO`; `bsi-spectra-live.js` and `bsi-lezione.js` use it when there is an OpenAI key	`test_spectra_studio` — in an offline audio context (3 checks): at 80 Hz the bass comes out boosted (6.9 dB), the reverb leaves a tail after the sound ends, without a key `qualita()` is `'sistema'`. **Not tested**: the call to OpenAI and listening to the voice
AI-17	A humanoid body without a skeleton must become drivable by the body driver without external tools: a	`bsi-spectra-rig.js` — `misura()` (height, axis, legs, hands, shoulders; human canon for the points that cannot be measured), skeleton with Mixamo names (20 weighted bones and 7 tips), weights ∝ $1/d^4$ with a per-region gate, normal, geodesic-distance-from-the-hand and	`test_spectra_rig` — 30 checks on a synthetic A-pose humanoid: 27 bones with Mixamo names recognised by the driver, locomotion; weights summing to 1, the hand to the hand and forearm, the flank next to the hanging hand with zero arm weight,

ID	Requirement	Implementation	Verification bench
	recognised skeleton, weights that do not tear the torso, a hood rigid with the head with an opening for the «Noir» face, the app's colours, a definition suited to high-density screens	head-axis filters, three smoothing passes; `rigga()` with `cappuccio`, `gonfia`, `viso` (elliptical opening in the shader), `suddividi` (Phong tessellation) and the options `visiera: false` (helmet as matte shadow) and `testa` (scale), which the studio does not use; bump map from the texture's luminance; `ricolora()` (night blue, turquoise → emissive `#00c9b7`). `spectra-studio.html` and the figure use it on the **embedded** body `models/spectra-corpo.bsi` (since v241; packed by `tools/impacchetta-corpo.js`, reopened by `apri` in `bsi-spectra-figura.js`); only in the studio, in development, if the block is missing, `models/spectra-corpo.glb` (glossy black visor kept, pixel ratio up to 3); `bsi-spectra-androide.js` — «solidale» face	legs never crossing; in the «spiega» gesture the hand moves 0.52 m and the flank 1 mm; walking moves the body; with the face only no triangle removed, ellipse in the shader, double side, hood rigid with the head (480/480 vertices), with `visiera: false` the helmet in shadow without removing vertices; night blue and teal light, grey emits nothing; tessellation ×4 with weights normalised on the new vertices and the skin closed (open border edges only double); no JavaScript errors. `test_spectra_studio`, on the **real embedded body**: face–hood sync in 24 poses (threshold 1 mm, measured 0.0 mm), of the «Noir» model only mask, eyes and details visible, «solidale» face; at 1280 and 390 px the body is always there without loading it, no «🧍 Corpo» button and no file choice, the published file is a «BSIC» block and not a reusable `.glb`, a block that is not a model rejected without errors. **Not tested**: the visual rendering of the folds and the hood on the real body, which has to be judged by looking
AI-18	«🧠 Spiega»: exhaustive explanation of chemistry, physics and mathematics from high school to advanced PhD, including of the screen (spectra, graphs, 3D scenes)	`bsi-spectra-spiega.js` — `BSISpiega.sistema(level, subject)` (five levels with `PROFONDITA`, «do not simplify below this level and do not leave out anything relevant», nine-part mandatory structure with the quantitative reading of NMR/IR/MS/UV-Vis spectra, 3D scenes and mathematical graphs), `schermo()` (section, tab, controls with value and range, tables up to 4 × 40 rows, visible text), `immagine()` (visible canvases, `BSILab3D` scenes re-rendered, SVGs rasterised; JPEG at most 1400 px wide), `chiedi()` (image in the first turn only, conversation in memory; a box unchecked by the user, `ST.schermoNo`, is respected even when a named or catalogue section is opened: only the question); without a model it says so and shows what the app knows on its own; `bsi-ai-hub.js` — «🧠 Spiega» tab	`test_spectra_figura` — 14 checks per width (412 and 1280 px): second tab, «Lezione» absent; figure in the tab; with «Spettri Ref.» open the screen is read (≥ 1 graph, control values, image); five levels; the model receives screen text, question and image; exhaustive, non-simplified request with δ, multiplicity, J, integral and 3D scenes; mandatory structure; formatted answer and Spectra thinking; follow-up without re-sending the image; advanced PhD with frontier, open problems and primary literature; without a model it does not pretend; SVG spectra read as an image; 3D scene of «Meccanica orbitale 3D» (2 canvases); privacy: with «Includi lo schermo» unchecked, the named section opens but the model receives only the

ID	Requirement	Implementation	Verification bench
		second after «💬 Chat», `window.bsiLLM`, per-provider image conversion; `bsi-spectra-figura.js` — the figure in the tab, «🧠 Spiega» button	question, with no screen text or image. **Not tested**: the quality of a real model's explanation — the model's answers are simulated, the bench checks what is sent and how it is shown

3. Stability requirements

ID	Requirement	Implementation	Verification bench
STA-01	A long session must accumulate neither DOM nodes nor timers	section lifecycle management	`audit_stabilita` §1 — 194 sections × 5 rounds
STA-02	Exhausting `localStorage` must not make the app unusable	guarded writes; persistent warning in the File Manager	`audit_quota` (10), `test_filemanager` (15)
STA-03	No rejected promise may remain unhandled	systematic `.catch()`	`audit_promesse` — 22 checks, healthy network and dead network
STA-04	Corrupted saved data must not prevent startup	`loadJSON()` with fallback	`audit_stabilita` §4
STA-05	A slow network must not be treated as an absent one	`ATTESA_RETE_MS = 3500` in `sw.js`	`test_sw` §3 and §3-bis — measured 3,507 ms
STA-06	An update must not destroy work in progress	`BSI_SW_UPDATED` message, decision taken page-side	`test_sw` §4
STA-07	Browsing molecules must not exhaust the WebGL contexts	reuse of the 3D viewer at all 5 creation points	`test_visore3d` — from 7 constructions down to 1
STA-08	An unavailable IndexedDB store must be declared, not fatal	`dbStore()` over a promise; persistent warning	`test_filemanager` §2
STA-09	Start-up must not load the RDKit chemistry engine (6.7 MB) on its own while the user	`bsi-pannelli-lingua.js` — the name archive is built when the converter is opened (`agganciaConvertitore`), no longer 3 s after start-up; `bsi-spettri.js` — RDKit warms up only in Chemistry,	`test_avvio` — 9 behaviour checks: it does not start while in use; not in Physics; not with «Data saver»; in Chemistry when still it warms up; the archive is absent at start-

ID	Requirement	Implementation	Verification bench
	is using the app, nor for those not studying chemistry	only after the app has been still for 4 s, never with «Data saver»; on-demand `bsiLoadRDKit` is unchanged. Measured on a simulated slow phone (CPU ×4), in Chemistry: longest start-up block from ~1960 ms to ~250 ms, total blocking time from ~2560 ms to ~640–815 ms	up and is built on opening «Linguaggi delle molecole» (283 molecules); on demand it still arrives. **Not tested**: the milliseconds on a real phone
STA-10	Every section of the app must open when asked for, not be empty, not make the page overflow and not show missing values; the same for the separate pages	`bsi-spectra-live.js` — `apriSezione()`: the exact identifier comes before the similarity search (before, «smo» opened «smol», and so did five others); `index.html` — `bsiNormRXN()` brings to the bank's schema the 36 reactions written with another schema, which printed «undefined»; `chimorga.html`, `guidaret.html`, `accademia.html`, `sr_completo.html`, `sr_essenziale.html` — rules for narrow screens	`test_giro` — 24 checks: all catalogue sections (201) opened one by one at 412 and 1280 px: the one asked for is the one opened, visible, not empty, no overflow, no «undefined»/«NaN», no JavaScript errors; 14 separate pages with no overflow, suspicious text or errors. **Not tested**: the correctness of each section's content, which belongs to the scientific benches

4. Interface and accessibility requirements

ID	Requirement	Implementation	Verification bench
UI-01	Every page must open without JavaScript errors	—	`audit_stabilita` §5 — 14 pages
UI-02	Charts must be sharp on high-density screens	`bsiNitido()` on the canvas contexts	`audit_grafici` — 40 canvases
UI-03	The app must work at 390 px width, and content that does not fit must stay **reachable**	`auto-fit` grids, no fixed column, and wide tables inside a container with `overflow-x:auto`	`audit_mobile` — **two distinct measurements** across all 194 sections at 390 px: the document's `scrollWidth` (the horizontal bar the user sees) **and** elements wider than their section **with no scrollable ancestor**, that is content clipped and unreachable — a written column that cannot be read and gives no sign of existing. **Both ways**: a wide table inside a scrolling container is **not** reported, because it can be read by scrolling. `audit_stabilita` for errors in a phone viewport
UI-04	The user must be able to know which version they are running and force the update	"Updates" entry in the ✨ panel	`test_aggiorna` — 9 checks

ID	Requirement	Implementation	Verification bench
UI-05	Data deletion must be selective and reversible in its choices	`bsiCancellaDati()` by group	`browser_reset` — 24 checks
UI-08	Surfaces that appear only after an action must also meet WCAG AA contrast	the six panels of the Cheminformatics section	`test_cheminfo` — the analysis is run, the panels opened in turn, **460 text elements measured, 0 defects**, with the WCAG formulas rewritten inside the bench
UI-09	Opening a section must not block the page	3Dmol's first `render()` moved out of the click; the 296 synthesis figures **and the 263 drug cards** drawn in 8 ms slices	`test_fluidita` — **194 sections traversed**, median switch 16 ms, worst 87 ms, **0 sections above 100 ms** (there were 3 with the new drugs, at 159 ms); plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute. For the viewer the claim is **structural, not temporal**: immediately after the click handler returns, the WebGL context must not exist; shortly after it must — because a millisecond threshold was measuring SwiftShader rather than the code (14…192 ms on identical code)
UI-10	The interface must switch among many languages, declaring how much each is translated, and carrying the writing direction with it	`bsi-lingue.js` with **fourteen languages**, a searchable selector, `dir="rtl"` for Arabic	`test_lingue` — **170 skeleton elements out of 170 in each of the thirteen languages**, coverage computed by applying the dictionary and counting; writing direction tested in both directions; navigation categories and the search placeholder included
UI-11	The molecule must show how it holds together, and the viewer must choose the angle	`bsi-mol3d.js`: formation from dust with staggered arrival, and atom selection by click	`test_mol3d` — the canvas **pixels** at 130 ms and 2.3 s, the arrival order (skeleton before hydrogens) measured on each atom's progress, and the selection of 1, 2, 3 and 4 atoms
UI-12	Whoever has a file must be able to open it: a picker listing extensions does not narrow the choice on a phone, it **hides** it	`bsi-spettrolettore.js` — a single `<input type="file">` **with no `accept`**, a single button, and `apriQualunque()` routing by type, by the extension the format declares and **by content**, announcing the decision in `#bsiSP-deciso`	`test_spettrolettore` §7 — one visible button, an **empty** `accept` (a non-empty `accept` here *is* the defect), and routing verified on PDF, `.docx`, `.txt`, `.jdx`, two `.csv` columns and an image; **both ways**, a binary disguised as `.txt` is not poured out as text but declared binary

ID	Requirement	Implementation	Verification bench
UI-13	The version shown in the header must be the one actually running	`index.html` — the badge is filled from `window.BSI_APP_VERSION`, no longer a hand-written string	`test_spettrolettore` §8 — the badge text is compared with `BSI_APP_VERSION` measured in the running page. It was needed: for about eighty versions the header read `v13l` when the application was already past version 190, and no check looked at it

5. Security and privacy requirements

ID	Requirement	Implementation	Verification bench
SEC-01	No API key may be present in the repository	keys only in `localStorage` or in the Worker's secrets	`tools/verifica-sicurezza.js` — 616 tracked files, 8 credential shapes
SEC-02	No personal data may leave the device without an explicit action	local-first architecture, telemetry disabled	`audit_rete` — **direct verification**: a canary value seeded into 71 stores of the user's data, the application used across 194 sections on 6 pages, and the URL, headers and body of every request inspected. Plus `verifica-sicurezza` on the two exit mechanisms
SEC-03	Passwords must not appear in clear text in the source	SHA-256 in `file_manager.html`	`tools/verifica-sicurezza.js`
SEC-07	No credential may EVER have entered the repository, not even in a commit later fixed	no key has ever been committed; keys live in `localStorage` or in the Worker's secrets	`audit_storia` — **1,551 distinct versions of text files across 390 commits**, 9 patterns tested in both directions; fails on a shallow clone, because it would measure less surface
SEC-04	No conflict marker may reach publication	—	`verifica_guida` §12 — 58 text files
SEC-08	The phrase with which the creator makes himself known to Spectra must not appear in the code nor be stored, and must not reach	`bsi-spectra-creatore.js` — SHA-256 fingerprint compared with the File Manager's; in proof mode the console field is a password field and sending skips `elabora()`; three wrong attempts, then a one-minute wait; in `localStorage` only `bsi-`	`test_spectra238` — «I am your creator» asks for proof and the field becomes a password field; a wrong phrase is rejected and the field goes back to normal; the typed phrase remains neither in the dialogue, nor in the page, nor in memory; after

ID	Requirement	Implementation	Verification bench
	dialogue, memory, voice or model	`spectra-creatore-v1` with `ok`, `dal`, `ultima`, erasable through «Clear data → Persistent memory»	three attempts the wait; for the model the guest is not recognised, with the instruction never to ask for the phrase in chat; recognised: greeting by name, telemetry, context for the model; «forget that I am your creator» removes the recognition. `tools/verifica-sicurezza.js` — no plaintext password in the code
SEC-09	The text Spectra speaks must leave the device only if the user has saved an OpenAI key, and the flow must be declared	`bsi-spectra-voce.js` — `parla()` calls `api.openai.com/v1/audio/speech` only if `chiaveOpenAI()` finds a key in `bsi_api_keys` and Web Audio is available; otherwise `speechSynthesis`. In Spectra Studio the 🔊/🔇 switch turns the voice off and the choice is kept in `bsi-spectra-studio-voce`: with the voice off no text goes to OpenAI. Declared in `03-Security-Privacy-Compliance.md` §1.2 and in `09-Release-Conformance-Statement.md`	`test_spectra_studio` — without a key `qualita()` is `'sistema'`, i.e. no request to OpenAI. **Not tested by the bench**: that with a key the text sent is only the text to be spoken (the bench does not reach OpenAI) and the studio's voice switch
SEC-10	The creator must be able to make himself known with the phone's fingerprint (or face), **only him**: the fingerprint is bound only after the secret phrase, never leaves the phone, and once bound the browser's memory alone is no longer enough	`bsi-spectra-creatore.js` — WebAuthn: `attivaImpronta()` (only if already recognised with the phrase; platform credential, user verification required, no attestation; only the credential id and the date are stored), `conImpronta()` (the bound credential, UV bit of the authenticator data), `riconosciuto()` which, with a bound fingerprint, holds only in the session where the sensor confirmed it, `dimentica()` which also removes the fingerprint; `bsi-spectra-live.js` — commands «attiva l'impronta», «riconoscimi», buttons «👆 Attiva l'impronta» after the phrase and «👆 Sono io» on opening. **Honesty**: without a server the check is in the browser (a lock against passers-by); the sensor accepts any fingerprint enrolled on the phone	`test_impronta` — also from the 💬 Chat tab (the phrase opens the console, it does not go to the model); with Chromium's virtual authenticator: no fingerprint without the phrase; wrong phrase rejected; bound after the phrase, only `id` and date stored; new session → fingerprint needed; right fingerprint with verified user → recognised and the model knows; unverified user or another sensor → not recognised; in the console «attiva l'impronta», «👆 Sono io» and the greeting by name; «dimentica che sono il tuo creatore» removes the fingerprint. **Not tested**: a real phone sensor

> **SEC-03 — declared limitation.** GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v188`. The old one remains in the history and no longer opens anything.

> **SEC-08 — declared limitation.** The File Manager's limitation applies: the fingerprint is in the code and the recognition mark is in `localStorage`, where whoever holds the device can write it by hand. It is a deterrent, not an access control, and it protects no data: it tells Spectra whom it is talking to.

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6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

ID	Requirement	Current coverage
UI-06	Full WCAG 2.1 AA conformance	the mechanical part is automated across **13 pages and 194 sections** (`tools/verifica-accessibilita.js`, 38,407 text elements: it was 19,751 before text over gradients entered the measurement, and 37,409 before the «Spectrum reader» section). **Of the 1,069 contrast defects that emerged, none remain: 0 measured** over those same 194 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are **zero**: see the box in `docs/09` §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09)
PERF-01	First-paint time on a low-end device	manual cross-device testing (`docs/02` §4)
SCI-11	Structures of 2 entries that are not single molecules (it was 6)	**not representable**: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see `docs/06` §2.4
PERF-02	Code coverage of the verification benches	**partially measured**: 49.89 % of statements over the widest path (`audit_copertura`). Whole-battery coverage and branch coverage remain unmeasured; see §8
UI-07	Behaviour on Firefox and WebKit	**not verified** — the battery runs on Chromium only; see §8
UI-21	Three apps in one: the ✨ menu, Spectra, Study and Utility must	**not automated** — `test_approfondimenti` checks only the bar's groups per subject (UI-18). The per-subject ✨ menu and the quick tools of `bsi-attrezzi.js` are tested by hand; their computations use `BSIMatCAS`, verified by `test_matematica` (SCI-41)

ID	Requirement	Current coverage
	change with the subject, and the quick tools must work	
AI-13	Spectra Live: local commands without network, voice conversation, tools that read the tab and act on its controls	**not automated** — `bsi-spectra-live.js` is tested by hand. Speech recognition and synthesis are not available in the battery's headless Chromium

7. Overall coverage

Category	Requirements	Verified by a bench	Coverage
Scientific (SCI-01…63)	61	61	100 %
AI agent (AI-01…18)	17	17	100 %
Stability (STA-01…10)	10	10	100 %
Interface (UI-01…33)	30	30	100 %
Security (SEC-01…10)	8	8	100 %
Architecture (ING-01…02)	2	2	100 %
Automated total	**128**	**128**	**100 %**
Not automated (§6)	7	0	0 %
Declared total	**135**	**128**	**95 %**

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

> **Why two totals.** Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **118 requirements verified out of 125 declared** — and the seven that are missing are listed by name in §6, not summarised into a percentage.
>
> This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

Gap	What it entails	Why it is so
Partial code coverage	Measured: **49.89 %** of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches	Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (c8, istanbul), not to the problem. Value recorded; `audit_copertura` fails if it falls
Chromium only	Behaviour on Firefox and WebKit is verified by hand, not by a bench	The battery uses `playwright-core`, which downloads a single engine. In the verification environment the CDN for the other engines answers **403** to the network policy: they cannot be installed there
No visual regression	An unintended graphical change would not be caught	Spectra are deterministic and comparable byte for byte (SCI-05): the comparison exists on the traces, not on the whole page
2 entries without a structure	Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six	See `06-Scientific-Accuracy-Data-Provenance.md` §2.4
Performance on slow devices	First-paint time is not measured on low-end hardware	It requires physical devices; the test is manual

Document updated to version `bsi-v250`.

Release Conformance Statement — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version	`bsi-v250`
Author and release owner	Samuele Pio Provenzano
Repository	`github.com/samupropio1-ship-it/BioSpecInfo-v11`
Distribution	GitHub Pages — `samupropio1-ship-it.github.io/BioSpecInfo-v11/`
Nature of the software	Client-side Progressive Web App, no backend

1. Subject of this statement

The undersigned declares that version `bsi-v250` of BioSpecInfo has been put through the verification procedure described in `02-Verification-Validation-Report.md` and that the outcomes are those reported, without selection, in `../evidence/RAPPORTO-VERIFICA.md` — a document generated automatically by `tools/genera-evidenza.js`, which transcribes the benches' output with no manual intervention.

This statement does **not** attest the absence of defects. It attests that:

1. an automated verification procedure exists and can be reproduced by third

parties;

1. the outcome reported is the one actually produced, failures included;
2. the known limitations are declared explicitly and not presented as features.

2. Intended use and limitations

2.1 Declared purpose

BioSpecInfo is a **teaching and study-support** tool for chemistry, biochemistry, pharmacology and astrochemistry. It is designed for university students and for exam preparation.

2.2 What the software is NOT intended for

These exclusions are an integral part of this statement.

Domain	Exclusion
Clinical use	It is not a medical device within the meaning of Regulation (EU) 2017/745. It does not support diagnosis, prescription, dosing or therapeutic decisions. The pharmacological data are for teaching purposes only, and the application says so at

Domain	Exclusion
	the head of the section.
Certified analytical use	The spectra are **predictions** from functional-group tables, not measurements nor quantum-mechanical simulations. They do not replace an experimental database (SDBS, NIST) nor an instrumental determination, and cannot be used for the formal identification of substances.
Regulated GxP environment	It is not validated for use in a GLP/GMP environment. The alignment with GAMP 5 described in `docs/03` is methodological, not a formal qualification.
Access control	The File Manager protects its contents with a client-side deterrent. On a static site there is no effective access control: whoever reads the source bypasses it. It must not be used for confidential material.

3. Summary of verification outcomes

The full outcomes, with each bench's textual output, the execution environment, the commit and the SHA-256 digests, are in `../evidence/RAPPORTO-VERIFICA.md`.

Family	What it demonstrates
Scientific data	The chemical data are verified against an independent source
AI agent	The assistant stays usable when the external provider fails
Stability	The app withstands long sessions, exhausted storage, degraded networks
Interface	The panels respond as documented
Documentation/code consistency	What the documentation promises exists in the code

4. Known deviations at the date of release

Listed in full. None has been removed from verification in order to make it pass.

#	Deviation	Impact	Treatment
D-01	**2 entries** (it was 6) that are not single molecules: Ivermectin, a mixture of B1a/B1b homologues, and Coartem, a combination of two active ingredients	For these two no 2D/3D representation and no spectral prediction is available	Entries left **without SMILES**: the pharmacological data remain, the structure is not shown. For Digoxin, Vincristine and Tacrolimus (topical and systemic) the structure was taken from **ChEMBL** and passes the comparison with the declared molecular weight: those four left the registry. The two that remain have no structure to show, not a missing one.

#	Deviation	Impact	Treatment
D-02	19 entries with no SMILES by their very nature (monoclonal antibodies, peptides)	None: for those molecules SMILES notation is not the appropriate representation	Not a substantive deviation; listed for completeness
D-03	~~SEC-02 verified only by proxy~~ — **remedied**: see S-10	—	`audit_rete` observes the traffic during real use, with a canary value seeded into the user's own data
D-04	~~The File Manager password in the git history~~ — **remedied**: see S-12	—	The password was **changed** at version `bsi-v188`. The old one is still readable in the history and no longer opens anything. At the author's request it was changed **again** at `bsi-v245` (only its SHA-256 fingerprint is in the code)
D-06	Code coverage **measured but partial**: 49.89 % of statements, not branches	It is the coverage of the widest path a bench walks (194 sections plus the other pages), not of the whole battery; and an `if` entered from one side only counts as covered	`audit_copertura` measures it with Chromium's profiler, with no build and without rewriting the source. The value is **recorded**: if it falls, the battery fails
D-07	Verification on Chromium only	Firefox and WebKit are tested by hand	Declared in `docs/08` §8. In the verification environment the reason is checkable: the CDN from which `playwright-core` downloads the other engines answers **403** to the network policy, so Firefox and WebKit cannot be installed there. **Re-verified on 2026-10-08**: `npx playwright install firefox` fails with «Download failure», and `cdn.playwright.dev` still answers 403 to the tunnel. It is not a choice, it is a constraint of the environment, and it stands
D-08	~~Partial English translation~~ — **remedied**: see S-13	—	All **16 documents** are in English. `verifica-documenti` and `verifica-affermazioni` read `docs/en/` just like the Italian set: a disagreement between the two languages fails the battery

#	Deviation	Impact	Treatment
D-09	Full WCAG 2.1 AA conformance not verifiable entirely by automation	What stays outside is text inside SVGs (8,888 elements), text over a real background **image** (108) and everything requiring human judgement. Text over a **gradient** has entered the measurement: 591 elements, judged against the worst stop of the gradient	The skipped elements are **counted** and reported on every run
D-10	The `bsi-v181` tag points at the wrong commit	The release workflow, started without filling in the version field, used its own default — `bsi-v181` — and, being rerunnable by design, **moved that tag** from commit `cb0cf03` to the `bsi-v188` one, also attaching the v187 packages to it. The foreign assets were removed. **The tag cannot be moved back by an app**: GitHub refuses to let an app write a ref pointing at a commit whose `.github/workflows/*` differs from the current one, and at `cb0cf03` that file differs by 319 lines. Three routes were tried, all blocked by the same rule: `git push --force` («refusing to allow a GitHub App to create or update workflow»), `PATCH /git/refs` with `force` (403), and delete-and-recreate — where the **403 already lands on creating a throwaway tag** at that commit, so the real tag was left untouched. `GITHUB_TOKEN` cannot be granted the `workflows` permission. **Re-verified on 2026-10-08** with the session's own GitHub client, which uses a different credential: `PATCH /git/refs/tags/bsi-v181` answers **403 «Write access to this GitHub API path is not permitted through this proxy»**. It is a second block, independent of the first: even without the workflow-file rule, writing to git refs does not get through. The tag points at `64e8c316`; it should point at `cb0cf03e`	The three packages attached to the release are the correct ones; the «Source code (zip)» archive must be ignored, and the `bsi-v181` code is downloadable from commit `cb0cf03`. Declared in the CHANGELOG's `bsi-v181` entry and in the release body, with the command that restores the tag from personal credentials. The default was removed and the workflow now **refuses** a version that does not match `sw.js` and the package filenames — verified both ways
D-11	Part of Spectra Live, the per-subject ✨ menu and the quick tools have no bench	Voice, continuous listening, local commands other than the creator's, the tools that act on the page and the physics and mathematics ✨ menu are tested by hand: a failure in them would not fail the battery	Declared in `docs/08` §6 as **UI-21** and **AI-13**. Since v238 the console layout, the navigator, the level, the face and the creator commands are covered by `test_spectra238` (UI-22, AI-14, SEC-08); since v239 the guided lesson and the «fammi lezione» command by `test_lezione` (UI-24) and the single button by `test_spectra238` (UI-25), while the lesson's voice is still tested by hand. Speech recognition and synthesis do not exist in the battery's headless Chromium;

#	Deviation	Impact	Treatment
			the quick tools' computations rest on `BSIMatCAS`, verified by `test_matematica`
D-12	Two Spectra local keys stay outside selective erasure	`bsi-spectra-ritratto-v4` (the icon derived from the face) and `bsi-spectra-sentinella` (continuous-listening mark) are not in the «Clear data» groups. The mind, the creator recognition, the subject and the study kits are (Persistent memory and Progress groups)	Neither holds user data: the first is regenerated, the second is removed when listening stops. Declared in `docs/11` §6
D-13	Spectra's creator recognition is not an access control	The SHA-256 fingerprint of the secret phrase is in the code, and the recognition mark (`bsi-spectra-creatore-v1`) is in `localStorage`, where whoever holds the device can write it by hand. An unsalted fingerprint of a weak phrase can also be searched by trial outside the app	Like the File Manager password (SEC-03), it is a deterrent: it protects no data, it tells Spectra whom it is talking to. The phrase is neither in the code nor stored (SEC-08, `test_spectra238`)
D-14	Spectra with the body: the body is included in the app, but **packed**, not as a `.glb` (since v241)	The body chosen by the author (the «Bounty Hunter», free on CGTrader) is under CGTrader's Royalty Free licence: it may be used inside a product, not offered as a standalone file. Up to v240 it was not in the app and each user had to load it; since v241 it is always there, for everyone, as the packed block `models/spectra-corpo.bsi` (4.40 MB, «BSIC» format: the `.glb` bytes mixed with an xorshift32 stream from a fixed seed) that only the app's code reopens. **It is not absolute protection**: someone reading the code can rebuild the `.glb`, as with any model shown on a screen	It is how games and apps embed purchased models: the published file is not a downloadable, reusable `.glb`; the original `.glb` stays out of git (`.gitignore`). The block is regenerated with `tools/impacchetta-corpo.js`, which checks that it reopens identical. Rig, figure and reactions are tested on the **real embedded body** (`test_spectra_studio`, `test_spectra_figura`: AI-17, UI-26, UI-27); the visual rendering has to be judged by looking

Contrast, from 1,069 defects to zero — how it surfaced and how it was closed

Up to version `bsi-v168` this document declared WCAG conformance "verified by sampling" and the bench reported **0 defects**. Those were two statements at once true and meaningless: the bench inspected only the **visible** elements, and `index.html` shows one section at a time. Out of 19,751 text elements it was looking at **41**.

Walking all 87 sections then present, **1,069** contrast defects surfaced. They had not appeared: they had always been there.

Today they are **zero**, measured with the same bench over every section — 88 of them now. Not by wholesale substitution — that road, attempted once, introduced 997 new defects — but by going back to the **common cause** every time and re-measuring after each change. The **40 unlabelled fields** have likewise fallen to **zero**.

The gravest causes were not wrong colours but **theme tokens whose value equalled the background**:

Structural defect	Measured effect
`--g800` is `#0d1522`, **identical to** `--bg` — and the rule `a { color: var(--g800) }` applied it to every link	Every link without a colour of its own was at contrast **1:1**: invisible
`--g700` is `#16263d`, nearly identical to `--white` (`#16273e`)	Text at **1.01:1**
Buttons that changed their **background** without changing their **text colour**	The selected state (or the deselected one, depending on the panel) became unreadable: **1.39:1**
`lightenColor(hex,pct){ return hex; }`	A function called "lighten" that returned the colour **unchanged**: an unfinished stub, in all likelihood written for this very problem
The quiz answer panel: light background, light text	The student **could not read the correct answer**

The structural remedy was an explicit token — `--testo-forte` — for prominent text on a dark surface, applied only to the uses as a *text colour*: `--g800` and `--g700` stay where they serve as background or border.

Cause	Defects	Remedy
A dark-theme safety net with higher specificity than the components: text meant for the **white** cards was forced to `#d4dce6`, contrast 1.38:1	159	Rewritten with `:where()` (zero specificity): it protects text that has no colour of its own and stops fighting text that has one
A per-category accent used as **text** on a dark header, with `#0d1522` (near-black) as fallback: 1.13:1	~400	`bsiAccentoLeggibile()` preserves the hue and raises the lightness just enough. Applied at the 14 points where the accent is chosen — not in the data, so it holds for future entries too
Two hues used only on dark backgrounds, just under threshold (3.75:1 and 3.04:1)	139	Replaced after **verifying** that they never appeared on a light background
Table zebra striping white/near-black with always-light text: the white rows were unreadable	57	Two dark tones, consistent with the theme
Periodic-table cells just under threshold (4.46:1 and 3.88:1)	43	Text at `#eceff4`, verified across all nine categories
`"#var(--text)"` — one `#` too many makes the colour invalid: in SVG it falls back to black, on canvas the assignment is **ignored** and the previous colour persists	8 places	Replaced with the real value

The direction of the correction depends on the surface, not on the colour. The first attempt always lightened: it removed thirty defects on the dark header and added eighteen on the light formula box that uses the same accent. The function now darkens on light backgrounds and lightens on dark ones.

The final tail: 80 → 0. The last eighty were scattered over 33 distinct causes and looked like finishing work. They were not: three more structural causes lay underneath.

Final cause	Remedy
The "automatic contrast corrector" was producing defects instead of removing them. It used *perceived* brightness instead of WCAG luminance, so its thresholds corresponded to no real ratio; it **ignored alpha**, so `rgba(255,180,84,.08)` over a dark surface looked to it like a "light background" and it darkened the text to `#16273e` — **creating** a defect at 1.21:1 on top of a source that was already correct; and it looked only at elements carrying a background of **their own**, that is, never the commonest case	Rewritten: it walks the ancestors **compositing alpha** to obtain the effective background, measures the true WCAG ratio, and delegates the correction to `bsiAccentoLeggibile()`
`--g900` is `#0d1522` — **the same as `--bg`**, exactly like `--g800`: the legacy of a greyscale born for the *light* theme, where `--g900` was "the darkest text". With the theme inverted, that role no longer exists	The uses as a **text colour** (30 rules today) move to `--testo-forte`; the 16 uses as a **background** stay where they are
A colour chosen from data used as a **background** with the text hard-coded to `#fff`: it works while the hue is dark and stops as soon as someone adds `#e65100`	`bsiEtichettaLeggibile()` — the dual of the previous function: it picks the text by looking at the background and, if neither extreme suffices, **darkens the background while preserving the hue**, because a label must stay recognisable by colour

The steps as measured with the official bench, not estimated: **80 → 27 → 26 → 10 → 4 → 0**. Each step is a full re-measurement over the 87 sections after a group of changes; none of the figures above is attributed to a single cause, because the causes were not measured one by one.

The number remains **recorded as a baseline** in `evidence/accessibilita-riferimento.json`. Zero is not an achievement to file away: it is the value the bench defends. If a future change reintroduces even one defect, the battery **fails**.

What stays outside automation is what automation cannot judge: text inside SVGs and text over gradient backgrounds (D-09). Those elements are **counted** and reported on every run, not passed over in silence.

4-bis. Deviations remedied in this version

They are listed because they concerned **the verification apparatus itself**, and a conformance statement that says nothing about the defects of its own evidence is not worth the paper it is written on.

#	Deviation remedied	Why it mattered
S-01	32 of the 39 benches were not in the repository: they lived in the working area of the session in which they had been written	It made **false** the sentence this whole document rests on — "anyone can re-run them". Whoever cloned the repository found 7 out of 39. They now live in `tools/banchi/`
S-02	A missing bench was counted but did not make the battery fail	A battery missing *all* its benches would have declared "CONFORMANT" with zero failures. A missing bench now makes the outcome **NON-CONFORMANT**
S-03	`package.json` and `package-lock.json` were excluded from the repository	`npm install`, the documented command for reproducing the evidence, installed nothing: the battery failed at the first bench with *Cannot find module*
S-04	The accessibility check inspected only the visible section	On `index.html` that meant one section out of 87. The "0 defects" held for the Dashboard, not for the application
S-05	`SCI-10` was defined twice in the matrix, and the coverage totals were wrong (36 declared out of 37 defined)	The first number an assessor reads was wrong. A bench now verifies it
S-06	50 broken links out of 127 in the delivery packages	Documentation handed to third parties with references that led nowhere
S-07	15 attachable PDFs stuck at documents 00-05 and at an earlier version	An obsolete attachment states false things with an air of authority
S-08	The numeric-claims bench looked at **the Italian documents only**	The English set drifted undisturbed: `docs/en/05` declared "84 sections" while the app has 87, and stayed stuck at version `bsi-v146` while the code was at 171. The English set is now compared against the measurement just like the Italian one, and a disagreement between the two languages fails the bench
S-13	The English translation stopped at 10 documents out of 16	A non-Italian-speaking assessor read less than two thirds of the documentation, and the missing part was the operational one: data model, interfaces, deploy, functional specifications, manual. All translated: the English set is now as complete as the Italian one, and the two consistency benches check it the same way
S-12	The File Manager password had been left in clear text in the git history	Taking it out of the files does not take it out of the history: `git log -p` hands it to anyone, and as long as that password is in use the deterrent deters nobody. The only remedy is to **change it**, and that has been done: the source carries only the new SHA-256 digest, the old password remains in the history and no longer opens anything. Verified in the browser: the new one opens, the old one does not

#	Deviation remedied	Why it mattered
S-10	SEC-02 — "no personal data leaves the device" — was verified by checking the two *mechanisms* of exit, not the exit	A check on mechanisms says "I cannot see how it would leave", which is not "it did not leave". `audit_rete` seeds an unrepeatable value into 71 stores of the user's data, then **uses** the application and inspects the URL, headers and body of every request. Proved in reverse with `BSI_PROVA_FUGA=1`, which causes a leak on purpose: the bench must fail, and it does
S-11	The deviations registry had no converse guard: an entry declared that **no longer deviates** simply stayed	An exemption nobody revokes is a permission left switched on, and tomorrow it would silently cover a wrong structure put in its place. The bench now fails for that too — verified by putting a remedied entry back
S-09	The "automatic contrast corrector" **created** the defects it was meant to remove	It ignored alpha: a `rgba(255,180,84,.08)` background over a dark surface looked light to it, and it darkened with `!important` a text that was already correct in the source. A remedy tool that worsens the thing it remedies is the hardest kind of defect to see, because it presents itself as the solution

5. Acceptance criteria applied

A version is published only if **all** the following criteria are satisfied.

- ☑ No non-network JavaScript error on any page
- ☑ `tools/verifica-farmaci.js` — no defect; every deviation recorded
- ☑ `verifica_guida` — no promise in the documentation without a counterpart in the code
- ☑ No conflict marker in the tracked files
- ☑ Service Worker cache version bumped and aligned with `BSI_APP_VERSION`
- ☑ Known deviations listed in §4 of this document
- ☑ Evidence report regenerated against the published commit
- ☑ **No missing bench** — a bench that is not there is not a bench that passed
- ☑ `tools/genera-pacchetti.js` closed with **0 broken links**
- ☑ `tools/genera-pdf.js` re-run: the attachments describe this version

6. Change management

Aspect	Procedure
Traceability	Every change goes through a pull request describing the defect, its cause and its verification
Versioning	The application version coincides with the Service Worker cache version (`bsi-vNNN`), incremented at every publication
Distribution	Automatic from `main` via GitHub Pages; the Service Worker signals the new version to open sessions without interrupting them

Aspect	Procedure
Regression	The full battery is re-run before every merge into `main`
Reversibility	Every version corresponds to a commit; rollback is a `git revert`

7. Intellectual property and licences

All third-party components are distributed under **permissive** licences (MIT, BSD-3-Clause, Apache-2.0, CC0, Public Domain), mutually compatible and free of strong copyleft constraints. The «Noir» 3D model used as Spectra's face (`models/spectra-noir.glb`) is by Skyfox13 (<https://sketchfab.com/podderkuntal25>), under the CC BY 4.0 licence, and has been modified; the attribution appears in the Spectra Live console, in the README, in `THIRD_PARTY_NOTICES.md` and in `07-SBOM.md`. The full bill of materials, with digests and residual obligations, is in `07-SBOM.md` and in CycloneDX format at `../evidence/sbom.cdx.json`.

The code written for this project is the author's own work.

8. Data handling

The application is **local-first**: there is no backend, no personal data is collected and no persistent identifier is present. Data generated by the user (notes, chats, API keys, uploaded files) stay in the device's `localStorage` and IndexedDB and can be deleted selectively from the interface itself.

Spectra's mind (`bsi-spectra-mente.js`) keeps conversation episodes, study time and an internal state in `localStorage`: it is a simulation of behaviour, not consciousness. When a question goes to the artificial-intelligence provider the user chose, a summary of this memory and of the open tab goes with it; if the question is about what is on screen, an image of its canvases too.

The creator recognition (`bsi-spectra-creatore.js`) keeps only a mark (`bsi-spectra-creatore-v1`: recognised yes, date of first and last access). The secret phrase is compared as a SHA-256 fingerprint and enters neither the dialogue, nor memory, nor voice, nor the requests to the provider; the model receives only whether the user has been recognised.

The guided lesson (`bsi-lezione.js`) is built from the open page: it does not call the model, has no keys of its own in `localStorage` and goes online only for the voice, as below; at the end it notes an emotion («fiera», proud) in Spectra's mind, which ends up in `bsi-spectra-mente-v1` like the others.

Spectra's quality voice (`bsi-spectra-voce.js`, since v239): if the user has saved an OpenAI key, the text Spectra speaks in the console, in the guided lesson and in Spectra Studio is sent to OpenAI (`gpt-4o-mini-tts`) to receive the audio, even when the chat uses another provider and without going through the proxy. Without an OpenAI key the browser's speech synthesis is used and the text does not leave. Spectra Studio (`spectra-studio.html`) keeps in

`localStorage` the predictions and the exercises opened for each lesson (`bsi-spectra-studio-v1`) and the voice choice (`bsi-spectra-studio-voce`), and also sends each prediction to the mind. Both keys are covered by «Clear data → Persistent memory». With the voice off in the studio (🔇) no text goes to OpenAI. Since v241 Spectra's body is embedded in the app (`models/spectra-corpo.bsi`): the user no longer loads any file and nothing is written to IndexedDB; an old body stored by a previous version (`bsi-spectra-modelli`) is deleted on start, and «Clear data → Progress in the rest of the app» deletes that database.

Network calls are optional and go to documented public services (§4 of `07-SBOM.md`). If the user configures the proxy, the API keys reside on a Worker they own and not on the device.

Details in `03-Security-Privacy-Compliance.md`.

---

9. Signature

This document is drawn up by the author of the software, who takes responsibility for it. It does not constitute certification by a third-party body, and is not presented as such.

Anyone can verify what is declared here by re-running the procedure in §5 of `../evidence/RAPPORTO-VERIFICA.md` against the commit indicated, and comparing the SHA-256 digests of the files.

Samuele Pio Provenzano *Version* `bsi-v250`.

Interface Reference — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v250`
Purpose	Document the project's HTTP interfaces and the external APIs the application consumes.

0. A necessary premise

BioSpecInfo exposes no API. It is an entirely client-side Progressive Web App, served as static files: there is no backend, no application server, no REST endpoints belonging to the product.

This document would therefore be empty, were it not for two interfaces that do exist and that an assessor is entitled to see documented:

	What it is	Documented in
Spectra proxy	An **optional** Cloudflare Worker, owned by the user, that forwards calls to the AI providers. It is the only HTTP interface written in this project.	§1, §2
External APIs consumed	Public third-party services the application queries, always optionally.	§3
Internal JavaScript APIs	The extension points of the code, for whoever has to work on it.	§4

> An OpenAPI specification of non-existent endpoints would be plausible and false documentation. Only what exists is described here.

1. Spectra proxy — overview

The proxy solves two problems that have no other solution from the browser:

1. **CORS.** Several AI providers do not send the headers that allow a web page

to call them directly. From the Worker the call originates from a server and the constraint does not apply.

1. **Custody of the keys.** With the proxy the API key lives among the Worker's

secrets and not on the user's device.

It is **optional**: without a proxy the application works using the keys the user enters locally.

Deploy

```
cd proxy
npx wrangler login
npx wrangler deploy
npx wrangler secret put GROQ_KEYS     # one or more keys, comma-separated
```

Configuration

Variable	Type	Default	Description
`ORIGINI`	secret/variable	*(empty = all)*	List of permitted origins, comma-separated. **Leaving it empty makes the proxy open to anyone:** not advisable.
`LIMITE_IP`	variable	`20`	Requests per single IP within the time window
`TETTO_GIORNO`	variable	`2000`	Total daily requests
`GROQ_KEYS`, `GEMINI_KEYS`, `ANTHROPIC_KEYS`, `OPENAI_KEYS`, `XAI_KEYS`, `NVIDIA_KEYS`, `ZAI_KEYS`, `DEEPSEEK_KEYS`	secrets	—	API keys. Several keys comma-separated: when the first one's quota is exhausted (429), the proxy moves to the next **within the same request**.

> The variable names are kept in Italian because that is what they are called in the Worker: translating them here would document a configuration that does not exist.

Endpoints

Method	Path	Description
`OPTIONS`	`/*`	CORS preflight
`GET`	`/stato`	Which providers have a key configured
`GET` `POST`	`/{provider}/{path...}`	Forwarding to the provider

Recognised providers: `anthropic` · `gemini` · `groq` · `xai` · `nvidia` · `zai` · `openai` · `deepseek`

Status codes

Code	Meaning
`200`	The provider's answer, forwarded as it is
`204`	CORS preflight accepted
`400`	Request body unreadable
`403`	Origin not permitted by the `ORIGINI` list

Code	Meaning
`404`	Unknown provider
`405`	Method other than GET, POST or OPTIONS
`429`	Per-IP limit or daily ceiling exceeded
`503`	No key configured for that provider

Security choices

- **No header is forwarded blindly.** Only `anthropic-version` and

`anthropic-beta` pass through; any credential sent by the client stays out. A proxy that forwards everything is an open relay.

- **The `key` parameter is not propagated.** If a client puts it in the query

string, the proxy discards it: the key is the server's.

- **The body is read once and kept**, because retrying with a different key

requires being able to send it again — and a stream is consumed.

---

2. OpenAPI 3.0 specification

> The `description` fields are translated here; the **error messages in the examples are left in Italian**, because that is what the proxy actually returns to a user of the Italian application. Translating them would document responses that do not exist.

```
openapi: 3.0.3
info:
  title: Spectra Proxy
  version: "1.0.0"
  description: |
    Optional proxy (Cloudflare Worker) for calls to language-model providers.
    It removes CORS blocks and keeps the API keys on the server rather than on
    the user's device.

    It is not part of the BioSpecInfo application: it is published by the user
    on their own Cloudflare account, and the application uses it only if it
    knows its address.
  license:
    name: Proprietary — All rights reserved (Samuele Pio Provenzano)
servers:
  - url: https://{workername}.{account}.workers.dev
    description: The user's instance
    variables:
      workername: { default: spectra-proxy }
      account:    { default: example }

tags:
  - name: Status
    description: Querying the configuration without revealing secrets
```

```
  - name: Forwarding
    description: Forwarding requests to the model providers

paths:
  /stato:
    get:
      tags: [Status]
      summary: Configured providers
      description: |
        Lists the providers for which the proxy holds at least one key.
        It reveals no key. The application uses it to show only the models that
        actually work, instead of letting them fail one by one.
      responses:
        "200":
          description: List of active providers
          content:
            application/json:
              schema:
                type: object
                required: [fornitori]
                properties:
                  fornitori:
                    type: array
                    items:
                      type: string
                      enum: [anthropic, gemini, groq, xai, nvidia, zai, openai, deepseek]
              examples:
                twoKeys:
                  summary: Two providers configured
                  value: { fornitori: [groq, gemini] }
                none:
                  summary: Worker published but with no secrets
                  value: { fornitori: [] }
        "403":
          $ref: "#/components/responses/OrigineNonAmmessa"

  /{fornitore}/{percorso}:
    parameters:
      - name: fornitore
        in: path
        required: true
        description: Destination provider
        schema:
          type: string
          enum: [anthropic, gemini, groq, xai, nvidia, zai, openai, deepseek]
      - name: percorso
        in: path
        required: true
        description: |
          Path of the endpoint at the provider, reassembled as it is.
          Examples: `openai/v1/chat/completions` for Groq,
          `v1beta/models/gemini-3.6-flash:streamGenerateContent` for Gemini.
        schema: { type: string }
        example: openai/v1/chat/completions
    post:
      tags: [Forwarding]
      summary: Forward a request to the provider
      description: |
        The body is transmitted unchanged. The proxy adds the key in the way
```

```
        that provider requires (an `Authorization: Bearer` header, `x-api-key`,
        or a query parameter) and, if the call fails for exhausted quota,
        retries with the next key within the same request.
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              description: Body expected by the provider's API, transmitted unchanged
            examples:
              groq:
                summary: Streaming completion on Groq
                value:
                  model: llama-3.3-70b-versatile
                  stream: true
                  messages:
                    - role: user
                      content: Which IR bands identify a carboxylic acid?
      responses:
        "200":
          description: |
            The provider's answer, forwarded without modification (typically
            `text/event-stream` when `stream: true`).
          content:
            application/json:
              schema: { type: object }
            text/event-stream:
              schema: { type: string }
        "400": { $ref: "#/components/responses/CorpoIlleggibile" }
        "403": { $ref: "#/components/responses/OrigineNonAmmessa" }
        "404": { $ref: "#/components/responses/FornitoreSconosciuto" }
        "429": { $ref: "#/components/responses/TroppeRichieste" }
        "503": { $ref: "#/components/responses/ChiaveAssente" }
    get:
      tags: [Forwarding]
      summary: Forward a read request to the provider
      description: |
        It serves, among other things, the resolution of the Gemini model, which
        queries `ListModels` with GET and could not pass without a key in the
        browser.
      responses:
        "200":
          description: The provider's answer
          content:
            application/json:
              schema: { type: object }
        "403": { $ref: "#/components/responses/OrigineNonAmmessa" }
        "404": { $ref: "#/components/responses/FornitoreSconosciuto" }
        "429": { $ref: "#/components/responses/TroppeRichieste" }
        "503": { $ref: "#/components/responses/ChiaveAssente" }

components:
  schemas:
    Errore:
      type: object
      required: [errore]
      properties:
        errore:
```

```
          type: string
          description: Message in Italian, meant to be shown to the user
  responses:
    OrigineNonAmmessa:
      description: The request's origin is not in the `ORIGINI` list
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Origine non ammessa da questo proxy." }
    FornitoreSconosciuto:
      description: The first path segment does not match a known provider
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Fornitore sconosciuto: pippo" }
    CorpoIlleggibile:
      description: The request body could not be read
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Corpo della richiesta illeggibile." }
    TroppeRichieste:
      description: The per-IP limit or the daily ceiling has been exceeded
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example: { errore: "Troppe richieste da questo indirizzo: riprova fra un minuto."
}
    ChiaveAssente:
      description: The proxy has no key configured for that provider
      content:
        application/json:
          schema: { $ref: "#/components/schemas/Errore" }
          example:
            errore: "Il proxy non ha una chiave per groq. Impostala con: wrangler secret
put GROQ_KEYS"

security: []
# CLIENT authentication is deliberately absent: the access control is the
# ORIGINI list plus the per-IP limits. Adding a client-side token would mean
# writing it into a static page, that is, making it public.
```

3. External APIs consumed by the application

All **optional**: with no network the application stays fully functional on local data, and says so to the user.

Service	Endpoints used	What it is for	If it does not answer
PubChem PUG-REST	`/compound/name/{name}/property/...` `/compound/smiles/{smiles}/SDF?record_type=3d` `/compound/cid/{cid}/PNG`	IUPAC name, formula, CAS, GHS pictograms, 3D conformers, 2D images	The app uses the local tabulated data and says so

Service	Endpoints used	What it is for	If it does not answer
RCSB PDB	`https://files.rcsb.org/download/{pdb}.pdb`	Protein structures for the 3D viewer	Explicit message in the viewer
NASA / ESA	Public images and data	Astrochemistry module	The module shows the tabulated data
AI providers	At the user's choice, or via the proxy	Assistant answers	The assistant declares it and offers a reachable alternative
Crossref REST	`https://api.crossref.org/works/{doi}`	A work's record for citations (APA 7, IEEE, ISO 690, BibTeX) in the Resources and bibliography sections	Says so, with the request address; the hand-filled form remains
The app's own site	`./sw.js?controllo=<time>` with `cache: 'no-store'`	«Check for updates»: the `var CACHE = 'bsi-vNNN'` line is the published version	The GitHub fallback is tried
GitHub (fallback)	`https://raw.githubusercontent.com/samupropio1-ship-it/BioSpecInfo-v11/main/sw.js`	Same reading, only if the own site does not answer	«Cannot check now», the app keeps working offline

None of these calls transmits personal data. There are no persistent identifiers and no active telemetry (`BSI_TELEMETRY_URL` is empty).

> This is not asserted from the architecture alone: `audit_rete` seeds a canary value into the user's own data, uses the application across 6 pages and 87 sections, and inspects the URL, headers and body of every outgoing request.

---

4. Internal JavaScript APIs

Extension points for whoever has to work on the code. They are globals on `window` because the application is a set of classic scripts, not ES modules.

`window.BSISpettri` — spectral engine

Function	Signature	Description
`gruppi`	`(smiles, callback)`	Recognises functional groups via SMARTS/RDKit. Calls back with `null` if RDKit is unavailable: the caller decides whether to fall back or declare that it cannot compute.
`gruppiSync`	`(smiles) → object | null`	Synchronous version: answers only if the molecule is already cached
`bandeDaGruppi`	`(groups) → band[]`	Table of IR bands `{n, pk, w, i, f}`
`svgIR`	`(bands, options) → string`	IR spectrum in transmittance, SVG, deterministic
`legendaIR`	`(bands) → string`	Assignments in HTML
`quandoPronto`	`(callback)`	Notifies when the analysis of a molecule is available, so it can be redrawn
`precarica`	`()`	Warms up RDKit while nobody is waiting

`window.bsiProxy` — proxy state

Function	Description
`url()`	Configured address, empty string if absent
`stato()`	Promise: queries `/stato` and records which providers are covered
`copre(providerId)`	Whether that provider is covered by the proxy
`ricarica()`	Re-reads the configuration

Other notable points

Symbol	Description
`window.bsiLoadRDKit(cb)`	Loads RDKit once only and calls back with the instance
`window.irBandList(smiles)`	IR bands — replaced by the new engine, keeps the original signature
`window.makeIRsvg(bands, colour)`	IR drawing — likewise
`window.nmrPeakList(smiles)`	^{1}H peaks with integration
`window.bsiCancellaDati(groups)`	Selective deletion of the saved data
`window.bsiOpenUpdate()`	Opens the updates window
`window.bsiAccentoLeggibile(colour, background, threshold)`	Adjusts an accent so it reaches the WCAG threshold on that surface, preserving the hue
`window.bsiEtichettaLeggibile(background, threshold)`	The dual: picks the text colour from the background and, if neither extreme suffices, darkens the background while preserving the hue

`window.BSIAggiorna` — update check (`bsi-aggiorna.js`)

Function	Signature	Description
`controlla`	`() → Promise<'aggiornata' | 'nuova' | 'offline'>`	Compares the version in use with the published one and says so in a pop-up notice (`#bsi-agg-esito`). If the version differs it asks the service worker to update (`registration.update()`). It always answers
`versioneInUso`	`() → Promise<string | null>`	Asks the active service worker for its cache name with the `{ type: 'BSI_VERSIONE' }` message over a `MessageChannel`; `null` without a service worker or after 1.5 s without an answer
`versionePubblicata`	`() → Promise<string | null>`	Downloads `./sw.js?controllo=<time>` with no cache and reads its `var CACHE = '…'`

The service worker answers the `BSI_VERSIONE` message with `{ cache: CACHE }` on the port it received. The 🔄 Updates window uses these functions and, if the own site does not answer, reads the copy of `sw.js` on GitHub.

`window.BSIAttrezzi` — quick tools of the ✨ menu (`bsi-attrezzi.js`)

Symbol	Signature	Description
`apri`	`(key, subject) → boolean`	Opens the tool in a window above the page. Keys: `calcolatrice`, `derivata`, `grafico`, `costante`, `unita`, `formula`, `glossario`, `quiz`. `false` if the key does not exist
`chiudi`	`()`	Closes the window
`STRUMENTI`	object	Icon, name, description and opening function of each tool
`PER_MATERIA`	object	Which tools the ✨ menu shows for `fisica` and `matematica` (none for `chimica`)
`calcolo.calcola`	`(text) → { esatto, html, valore } | { errore }`	Exact computation with `BSIMatCAS`
`calcolo.derivaTesto`, `calcolo.integraTesto`	`(text, x?) → { testo, html } | { errore }`	Symbolic derivative and antiderivative
`calcolo.campioni`	`(text, a, b, n)`	Points of a function of x for the graph; `null` without the symbolic engine
`calcolo.cerca`	`(list, q, fields)`	Accent-insensitive search
`calcolo.quizLampo`	`(subject, n, seed)`	Questions drawn from the study kit with a reproducible generator; `null` without a kit

`window.BSIStudio` — study-kit engine (`bsi-studio.js`)

Symbol	Signature	Description
`registra`	`(subject, data) → boolean`	Merges a subject's data (`argomenti` , `riassunti` , `quiz` , `flash` , `formule` , `glossario`) and creates the five sections `s<pre>riass` , `s<pre>quiz` , `s<pre>flash` , `s<pre>form` , `s<pre>gloss` (`pre` : `fis` , `mat` , `chi`). A data file loaded before the engine queues itself in `window.BSIStudioCoda`
`dati`	`(subject) → object | null`	The subject's merged data
`materie`	`() → string[]`	The registered subjects
`sezioni`	`(subject) → string[]`	The ids of the five sections
`valida`	`(data, opts) → string[]`	The problems found, in Italian; an empty list means «fine». With `{ quantita: true }` it also checks the minimums in `MINIMI`
`MINIMI`	object	`{ riassunti: 60, quiz: 300, flash: 250, formule: 200, glossario: 250 }`
`pulisci`	`(html) → string`	Keeps only `p, b, i, ul, li, sub, sup, br` , without attributes
`norm`	`(s) → string`	Text without tags and accents, lower case
`calcolo.mulberry32`	`(seed) → () → number`	Reproducible generator in [0, 1)
`calcolo.mescola`	`(arr, rnd?) → array`	Fisher–Yates on a copy
`calcolo.leitner`	`(state, id, correct) → state`	Correct: one box up (at most 5); wrong: box 1. Returns a new object
`calcolo.punteggio`	`(answers) → { giuste, totale, perc, perArg }`	Overall and per-topic score

`window.BSISpectraAndroide` — Spectra's face (`bsi-spectra-androide.js`)

Symbol	Signature	Description
`disponibile`	`() → boolean`	Whether the browser offers WebGL
`crea`	`(container) → Promise<Androide>`	Loads three.js, `GLTFLoader` and `models/spectra-noir.glb` , and draws the face in the container
`foto`	`(px) → Promise<string>`	A PNG image of the face (data URL), for the icons
`emozioneDi`	`(sentence) → string`	A sentence's emotion from keywords; `'serena'` with no signal
`EMOZIONI`	object	The 17 emotions, each as weights of the expressive fields plus eye light and head pose
`CAMPI`	string[]	The 11 expressive fields

Symbol	Signature	Description
`COLORI`	object	Eye colour per state; `attesa: 0x1a3050`
`URL`, `BOCCA`, `versione`	—	Model path, mouth geometry, module version

The object returned by `crea` has, among others, `stato(s)` (`'attesa'`, `'ascolta'`, `'pensa'`, `'parla'`), `dici(text)` (the mouth follows the text), `allinea(i)`, `taci()`, `espressione(name, seconds, intensity) → boolean`, `emozione() → string`, `sente(sentence)` (reacts to a user sentence and passes it on to the mind), `passo(dt)`, `ferma()`, `libera()`. Since version 3.0 of the module (v238) also `concentra(c)` (target concentration between 0 and 1), `concentrazione() → number` (the current one), `addormentato() → number` (0 awake, 1 asleep), `aperture() → [left, right]` (visible opening of the two eyelids, 1 = at rest) and the `coloreOcchi` property (current eye tint, `#rrggbb`).

`window.BSISpectraRobot` (`bsi-spectra-robot.js`) exposes `disponibile`, `crea` and `foto` with the same signatures: it uses the android and, if the model does not arrive, the head built in three.js. `window.BSISpectraVolto` (`bsi-spectra-volto.js`) is the last-resort SVG portrait.

`window.BSISpectraMente` — Spectra's mind (`bsi-spectra-mente.js`)

Function	Signature	Description
`ricorda`	`(question, outcome)`	Records an episode (time, subject, section, question cut to 160 characters) and updates curiosity, attention and bond
`sente`	`(emotion, who)`	Moves the mood (and the care) according to the emotion detected in a sentence
`stato`	`() → { curiosita, attenzione, premura, umore, legame }`	The internal state, values between 0 and 1
`fondo`	`() → string`	The face's background emotion when nobody is speaking
`profilo`	`() → object`	Favourite sections, minutes per subject, favourite hour, neglected subject, quiz mistakes, recent topics
`pensieri`	`() → [{ peso, testo }]`	Suggestions ordered by weight
`saluto`	`() → string`	Greeting that takes the time and the last visit into account
`coscienza`	`() → string`	The summary added to the question sent to the model; empty without memory. Also exposed as `window._bsiMente`
`cosaRicordi`, `comeStai`	`() → string`	The answers to the «what do you remember» and «how are you» commands
`dimentica`	`()`	Clears the memory and removes the `bsi-spectra-mente-v1` key

It is a simulation of behaviour: the functions read and write numbers and text in `localStorage`, nothing else.

`window.BSISpectraLive` — the console (`bsi-spectra-live.js`), v238 additions

Function	Signature	Description
`difficolta`	`(text) → { valore, livello, nome }`	Difficulty of the question between 0.05 and 1, level 0–3 and its name (high school … PhD)
`livelloSezione`	`(id) → number`	First level of the `bsi-percorsi.js` path in which the section appears; −1 if none
`navigatore`	`(subject?)`	Opens (or closes) the 🧭 navigator; with a subject, shows that one
`provaCreatore`	`(active)`	Puts the console field in proof mode (password type for one minute) or takes it out
`verificaCreatore`	`(phrase)`	Checks the phrase with `BSISpectraCreatore.verifica` and answers in the dialogue, without recording the phrase

`window.BSISpectraCreatore` — the creator (`bsi-spectra-creatore.js`)

Symbol	Signature	Description
`NOME`, `NOME_BREVE`	string	`'Samuele Pio Provenzano'`, `'Samuele'`
`riconosciuto`	`() → boolean`	Whether the creator has been recognised on this device
`verifica`	`(phrase) → Promise<{ ok, rimasti?, attendi? }>`	SHA-256 fingerprint of the phrase compared with the File Manager's; after three errors `attendi` (seconds) for one minute. The phrase is not stored
`dimentica`	`()`	Removes `bsi-spectra-creatore-v1`
`visita`	`()`	Updates the last-visit date of the recognised creator
`saluto`, `chiTiHaCreato`	`() → string`	The greeting by name and the answer to «who created you?»
`contesto`	`() → string`	The `[CREATORE]` line added to the model's context

`window.BSILezione` — the guided lesson (`bsi-lezione.js`, v239)

Symbol	Signature	Description
`avvia`	`(section?, { muta? }?) → boolean`	Builds the lesson for the given section (or the visible one) and starts it in the player; `muta: true` starts with the voice off. `false` if no section is visible. A lesson already running is closed
`chiudi`	`()`	Stops the lesson, puts the control being demonstrated back as it was, removes panel and spotlight, makes the floating buttons visible again
`costruisci`	`(section?) → { sezione, titolo, capitoli, passi }`	Construction only, without the player. `capitoli`: «Introduzione», one name per tab (or «Esploriamo»), «Verifica» if there is a question, «Riepilogo» (English labels when the interface is in English). `passi`:

Symbol	Signature	Description
	`| null`	objects with `tipo` among `titolo`, `scheda`, `leggi` (placeholder replaced by the tab's steps when the tab is opened), `testo`, `tela`, `cursore`, `scelta`, `quiz`, `fine`, and `cap` (chapter index)
`vai`	`(i)`	Goes to step `i` like a tap on ◀ ▶ (interrupts the voice)
`stato`	`() → { i, passi, tipo, capitoli, auto } | null`	Current step, number of steps, step type, chapters, whether auto-advance is on; `null` with no lesson
`versione`	string	`'1.0'`

Spectra Live's local command «fammi lezione» («give me a lesson») in `BSISpectraLive.comando` calls `BSILezione.avvia()` after closing the console.

`window.BSISpectraVoce` — the voice (`bsi-spectra-voce.js`, v239)

Symbol	Signature	Description
`qualita`	`() → 'audio' | 'sistema' | 'testo'`	`'audio'` with an OpenAI key in `bsi_api_keys` and Web Audio; `'sistema'` with `speechSynthesis` only; `'testo'` without voice
`parla`	`(text, { suBocca?, timbro?, soloSistema? }?) → Promise<{ ok, motore, durata? } | { annullato }>`	In audio mode sends the text (up to 3,800 characters) to `api.openai.com/v1/audio/speech` (`gpt-4o-mini-tts`, voice «onyx»; `tts-1` as fallback), runs it through the chain and calls `suBocca({ energia, centroide, apertura, rotondo, largo, frazione })` every frame; if the request fails, or with `soloSistema`, uses `speechSynthesis`. Interrupts the previous sentence
`ferma`	`()`	Stops the audio and the browser's synthesis
`catena`	`(audioContext, timbre?) → { ingresso, uscita, analizzatore, ferma }`	The effects chain of `TIMBRO`
`impulso`	`(audioContext, seconds) → AudioBuffer`	Synthetic reverb impulse response (stereo)
`leggiBocca`	`(analyser, data) → { energia, centroide, apertura, rotondo, largo }`	Lip sync from audio: RMS energy and spectral centroid
`TIMBRO`	object	`bassiHz` 140, `bassiDb` 5, `fangoHz` 350, `fangoDb` −3, `presenzaHz` 2800, `presenzaDb` 3, `anelloHz` 55, `anello` 0.12, `pettineMs` 4.3, `pettineRetro` 0.35, `pettine` 0.18, `calore` 1.6, `stanzaS` 1.4, `stanza` 0.16, `volume` 0.95

`window.BSISpectraCorpo` — the body (`bsi-spectra-corpo.js`, v239)

Symbol	Signature	Description
`crea`	`(root, { animazioni?, capelliNeri? }?) → Corpo`	Attaches the driver to a three.js object with a skeleton; with `animazioni` (the GLB's clips) it recognises `idle`, `walk`, `sit`, `stand`, `wave`, `point`, `talk`; the hair turns black unless `capelliNeri: false`
`manichino`	`({ neon? }?) → THREE.Group`	The test mannequin (since v240 only for the driver's bench, never as Spectra): bones with Mixamo names in T-pose, head with 26 ARKit blendshapes, separate eyes, red hair cap
`EMOZIONI`, `OSSA`, `normaNome`	objects, function	ARKit weights per emotion; regular expressions for bone names; name normalisation

Methods of a `Corpo`:

Method	Signature	Description
`rapporto`	`() → { ossa, mancanti, blendshape, capelli, clip, locomozione, viso }`	What the model can do (the requirements check of `docs/spectra/01`)
`capelliNeri`	`(colour?)`	Colour (default `#0a0a0c`), roughness, and texture desaturated in the shader
`orienta`	`(bone, direction, weight)`	Brings the bone → child direction onto a world direction
`emozione`	`(name, seconds?, intensity?) → boolean`	One of the emotions in `EMOZIONI` for a few seconds
`dici`, `audio`, `taci`	`(text)`, `(mouthState)`, `()`	Lip sync from text, from audio (`suBocca` of `BSISpectraVoce`), stop
`guarda`	`(point)`	Eyes, neck and head towards a point
`fai`	`(gesture, target?, seconds?)`	`'indica'` (point), `'spiega'` (explain), `'pensa'` (think), `'attende'` (wait), `'conclude'`
`cammina`, `siedi`	`(point)`, `(yes?)`	Walk to a point; sit (`false` to stand up)
`peso`, `imposta`	`(name) → number | null`, `(name, value) → boolean`	Reads and writes a blendshape weight
`passo`	`(dt)`	Advances the behaviour by `dt` seconds; to be called every frame

`window.BSISpectraFigura` — Spectra with the body (`bsi-spectra-figura.js`, v240; embedded body since v241)

Symbol	Signature	Description
`Spectra`	`new (T, scene, renderer,`	Spectra's body in an existing scene. Since v241 the body is **embedded**: `carica()` downloads `models/spectra-corpo.bsi`, reopens it (internal function `apri`: «BSIC» signature, version, length, bytes mixed with an xorshift32 stream from a fixed seed) and hands it to `daBuffer`; first it deletes an old

Symbol	Signature	Description
	`opz?) → controller`	body from IndexedDB `bsi-spectra-modelli`, if any. If the block does not arrive: with `opz.soloDispositivo` (enabled for the figure from `monta`) no further attempt and `caricato = false`; without it (the studio, in development) it tries `models/spectra-corpo.glb`. The `models/spectra-cyberpunk.glb` fallback is gone. No substitute body (the mannequin only with `opz.manichino === true`, for the driver's bench). Rig with `bsi-spectra-rig.js` (`RIG`), «Noir» face on the collar, «solidale»
`monta`	`(container, opz?) → figure`	Lab-style frame with Spectra standing, HUD «SPECTRA · IN ASCOLTO / PENSA / PARLA», buttons «🎙️ Parla con me», «🧠 Spiega», «🎓 Studio» (since v241 no «🧍 Corpo» and no file input); while the body arrives «Spectra sta arrivando…», if it does not arrive «Spectra non è arrivato: serve la rete la prima volta» with «↻ Riprova»; listens to `bsi-spectra-stato`, `bsi-spectra-risposta`, `bsi-spectra-dice`
`sezioni`	`(id[]) → figure`	One shared figure (one WebGL context) moved into the open AI section (`sai`, `sareachim`, `sareafis`, `sareamat`; since v242 also `sesplchim`, `sesplfis`, `sesplmat`, where the figure takes the `bsiEX-palco` class and stays on stage while scrolling); the 3D files load on first opening. **Since v242 the open Spectra Live console takes precedence over everything**: the figure goes into `#bsiJV-figura` (created if missing) and the console takes the `con-figura` class, which hides the bust `#bsiJV-volto`; with the console closed, the figure returns to the section
`prepara`	`() → Promise`	Loads three.js, the GLB loader, the body driver and the rig if they are missing
`dbCorpo`	`('leggi' | 'metti' | 'togli', buffer?) → Promise`	The IndexedDB of the old user-loaded body (`bsi-spectra-modelli`, store `file`, key `corpo`); since v241 `carica()` uses it only to read and **remove** an old body, it no longer writes to it
`emozioneDi`	`(text) → string`	The emotion suggested by an answer's content
`Ologramma`	`new (T, scene) → hologram`	The holographic screen to Spectra's right; methods `costruisci(name)`, `mostra(name)`, `passo(dt)`; scenes computed every frame, plus `'schermo'` (the section behind the window, re-rendered with `BSISpectraLive.cattura` 5 times a second; a section without graphs is projected as a card: title, tab, up to 6 controls with values, visible text)
`SCENE_OLO`	object	The six scenes (`pianeti`, `molecola`, `spettro`, `superficie`, `orbitale`, `onda`): for each `nome` (it, en), `sez` (the id of the section the label opens) and `parole` (the regular expression that selects it)
`sceneDi`	`(text) → string | null`	The scene matching the words of a text, or `null`
`LIMITE`, `RIG`, `figura`	number, object, figure	60 MB; the rig parameters for the author's body; the sections' shared figure (or `null`)

Controller methods: `carica() → Promise<boolean>` (the embedded body; sets `caricato`), `daBuffer(buffer)`, `usaScena(scene)` (for benches), `passo(dt)`, `emozione(name, seconds?)`, `dici(text)`, `taci()`, `fai(gesture, target?, seconds?)`, `presente() → boolean`. `daFile(file)` was removed in v241 together with the «🧍 Corpo» button.

New event: **`bsi-spectra-dice`** (`CustomEvent` on `document`, `detail: { testo }`), dispatched by `parla()` in `bsi-spectra-live.js` and by `bsi-assistente.js` when it solves an exercise; the figure says it with lip sync and a gesture.

New event: **`bsi-spectra-tema`** (`CustomEvent` on `document`, `detail: { testo, schermo }`), dispatched by `bsi-spectra-spiega.js` with the question (and when the tab opens with a section behind); the hologram picks the scene with `sceneDi(text)`, or shows the screen live if `schermo` is true. When answering, the figure first points at the hologram («indica» gesture, about 2.5 s) and then accompanies with «spiega».

`window.BSISpiega` — «🧠 Spiega» (`bsi-spectra-spiega.js`, v240)

Symbol	Signature	Description
`costruisci`	`(panel)`	Builds the «🧠 Spiega» tab in the given panel (called by `bsi-ai-hub.js`): question, level, «📺 Includi lo schermo» checkbox with the screen card, quick prompts, «↻ Nuova spiegazione», the «🗂️ Tutte le sezioni nell'ologramma» card (search, subject filter, «🎞️ Mostrami tutto»)
`catalogo`	`() → [{ id, nome, gruppo, g, materia }]`	Every section of the app bar (`.nav-group-panel [data-s]`), with its group label and subject (`BSIMaterie.materiaDi`; `'comune'` if none): 201 sections since v242 (chemistry 78, physics 63, mathematics 31, shared 29). Read from the bar, not written by hand
`sezioneDi`	`(text) → catalogue entry | null`	The section a question names: meaningful words of the names (with stems), acronyms (MS, IR, UV, NMR, MO…) weighted more, synonyms («spettro di massa» → MS, «infrarosso» → IR, «risonanza magnetica» → NMR, «orbitali molecolari» → MO); ties go to the shorter name. E.g. «titolazioni» → `stitr`, «integrali per parti» → `smatcalc`, «ciao come stai» → `null`
`mostraSezione`	`(id) → Promise<boolean>`	Opens the section behind the Spectra window (`BSISpectraLive.calcolo.apriSezione`, which switches subject if needed) and refreshes the screen, so the hologram shows it live
`giro`	`(subject?)`	«🎞️ Mostrami tutto»: every section of the subject (`'chimica'`, `'fisica'`, `'matematica'`; all if empty) in the hologram, one every 7 s, with the counter «n/N · name»
`giroFerma`	`()`	Stops the tour
`chiedi`	`(question, { schermo?, livello? }?) → Promise`	Sends the question to the chat's model; with `schermo` it adds the screen text and, in the first turn, the image; without `schermo`, if the question names a section (`sezioneDi`) other than the open one, it first opens it (`mostraSezione`) and then asks with the screen, unless the user unchecked the box (`ST.schermoNo`): then only the question; follow-up questions continue the conversation. Without a model it shows what the app knows on its own

Symbol	Signature	Description
`schermo`	`() → { testo, tela, svg, grafici, titolo, materia }`	Reads the open section: title, tab, controls with value and range, visible tables (up to 4 × 40 rows), visible text, visible canvases and SVGs
`immagine`	`(S) → Promise<dataURL>`	Composes into a JPEG (at most 1400 px wide) the visible canvases (`BSILab3D` scenes re-rendered first) and the visible SVGs rasterised
`sistema`	`(level, subject) → string`	The system prompt: level, `PROFONDITA`, mandatory structure, rigour
`aggiornaSchermo`	`()`	Refreshes the screen card (title, number of graphs and 3D scenes, thumbnail)
`LIVELLI`, `PROFONDITA`	array	The five levels (High school to Advanced PhD · frontier) and what is asked at each
`stato`	`() → { turni, livello, inCorso }`	The state of the conversation

The **`bsi-spectra-stato`** event, since v242 also from the console: `bsi-spectra-live.js` emits it at every state change with `detail: { stato, dettaglio, daLive: true }` (`stato`: `'attesa'`, `'ascolta'`, `'pensa'`, `'parla'`); the figure reacts (`'pensa'` → concentrated, «pensa» gesture, hand on chin; `'ascolta'` → curious, «LISTENING TO YOU»). The console ignores events carrying `daLive`: that signature is what stops the console from calling itself forever.

`window.BSIEsplora` — «⚡ Explore with Spectra» (`bsi-spectra-esplora.js`, v242)

Symbol	Signature	Description
`CAPITOLI`	`{ chimica, fisica, matematica } →` array of 10 chapters	Each chapter: `id` (`c-…`, `f-…`, `m-…`), `liv` (0 high school … 4 advanced PhD · frontier), `tit`, `sez` (the id of the linked app section), `scena` (the hologram scene: `orbitale`, `molecola`, `spettro`, `onda`, `pianeti`, `superficie`), `testo`, `pred: { d, o[4], g, p[4] }` (question, four answers, index of the right one, a reason for each), `prova?: { et, min, max, passo, val }` (the slider, in 19 chapters out of 30)
`calcola`	`(id, value) → { v, testo } | null`	A chapter's slider quantity: `v` the number, `testo` the line shown. E.g. `calcola('c-ph', 8).v` ≈ 6.978 (HCl 10^{-8} M, with autoprotolysis), `calcola('f-pendolo', 1).v` ≈ 2.006 s, `calcola('m-armonica', 3).v` ≈ 7.4855; `null` for chapters without a slider
`costruisci`	`(materia, pannello) → element`	Builds the subject's page in the panel: header with the «n/10» bar, the ten chapters (`section.cap`, id `bsiEX-<id>`) with prediction, try and the «📺 Guardalo dal vivo», «🧠 Spiegamelo tutto», «🔊 Ascolta Spectra» buttons; on scroll the chapter crossing the middle of the free space under the figure takes the `attivo` class, emits `bsi-spectra-dice` and brings the hologram to its scene; sets the CSS variable `--bsiEX-top` (the height of the app's fixed header)
`gibbsMax`	`(N) → number`	The maximum, computed numerically over 4000 points in (0, π/2], of the partial sum of the square wave (4/π) Σ sin(kx)/k over odd k ≤ N (≈ 1.179 with N = 201)

Symbol	Signature	Description
`SEZIONI`	`['sesplchim', 'sesplfis', 'sesplmat']`	The three sections, registered with `BSIFisica.sezione` (one tab, «⚡ Esplora»)
`MEM`	`'bsi-esplora-v1'`	The `localStorage` key of the predictions: `{ <chapter id>: { giusta, quando } }`

The module also loads in node (`require`), without `document`: `CAPITOLI`, `calcola` and `gibbsMax` are pure, which is how `test_spectra_esplora` compares them with the formulas. «🧠 Spiegamelo tutto» uses `BSISpiega.chiedi(question, { schermo: true, livello, nonAprire: true })` after `BSISpiega.mostraSezione(sez)`.

`window.BSIOloInterattivo` — the interactive hologram (`bsi-spectra-olointerattivo.js`, v242)

Symbol	Signature	Description
`apri`	`(modo?, testo?)`	Opens the `#bsi-olo-ov` window (built on first opening; calls `_bsiSyncFabs`, so the floating buttons hide) in the given mode — `'spettri'`, `'universo'`, `'matematica'`, `'fisica'` — or in the last one used; `testo` pre-fills the trace, the orbit or the question. Emits `bsi-olo-interattivo` with `{ aperto: true }`
`chiudi`	`()`	Stops the voice, closes the window and emits `bsi-olo-interattivo` with `{ aperto: false }`; the shared figure returns to the section
`modo`	`(modo, testo?)`	Switches to another mode (tab at the top) and redoes hologram and steps
`stato`	`() → { aperto, modo, tipo, chi, MHz, passi, sel, dati }`	The state: spectrum type, molecule or trace (`chi`), NMR field, step titles, selected step, the data drawn
`imposta`	`(testo)`	Writes into the mode's field (trace, question or orbit) and redraws; for mathematics and physics it also starts solving
`scegli`	`(i, parla?)`	Selects step `i`, highlighting it in the text and on the hologram; with `parla` Spectra says it
`SPETTRI`	object	The six reference molecules (`acetone`, `etanolo`, `acetico`, `benzene`, `toluene`, `etile`): `nome`, `formula` and the `ir`, `h1`, `c13`, `ms`, `uv` signals — typical literature values, not measured spectra
`PIANETI`	array	The eight planets (NASA fact sheet): name, semi-major axis (AU), eccentricity, period, mass, radius, moons (as of 2024), colour
`spettro`	`(tipo, molecola | traccia, { MHz }?) → { tipo, assi, curva, passi, segnali, righe, MHz }`	The spectrum of a molecule or of a written trace: the curve to draw and one step per signal (position, shape, assignment, why); for ^{1}H the multiplets in Hz converted to ppm with the field (400 MHz if not given)

Symbol	Signature	Description
`pianeta`	`(nome | { a, e }) → { nome, a, e, T, q, Q, vq, vQ, passi }`	Period from the third law (years), perihelion and aphelion (AU), vis-viva speeds at perihelion and aphelion (km/s), and the steps (with surface gravity and escape velocity for real planets)
`leggiTraccia`	`(tipo, testo) → signals`	Interprets a written trace: IR «1715 s, 3300 br» (s/m/w, br = broad), ¹H «2.17 s 6H; 1.2 t 3H J=7», ¹³C a list of δ, MS «m/z intensity», UV «λ ε»; candidate assignments from tables
`massa`, `insaturazione`	`(formula) → number`	Molar mass (C, H, O, N) and degree of unsaturation C − H/2 + N/2 + 1

New event: **`bsi-olo-interattivo`** (`CustomEvent` on `document`, `detail: { aperto }`), on opening and closing the window. `bsi-spectra-figura.js` adds the «⛶ Interagisci» button on the hologram and, with the window open, moves the shared figure into `#bsiOI-figura`.

`window.bsiLLM` — the chat's provider (`bsi-ai-hub.js`, v240)

Symbol	Signature	Description
`provider`	`() → string`	The provider chosen in the Spectra window (or the saved one)
`chiave`	`(provider) → string | null`	The key to use for that provider, as the chat does; used by «🧠 Spiega»

`window.BSIStudioLezioni` and `window.SpectraStudio` — the studio (v239)

`BSIStudioLezioni` (`bsi-spectra-studio-lezioni.js`, also usable with `require` in node) exposes `LEZIONI` (for each lesson `id`, `materia`, `predici`, `testa`, `dimostra`, `rivela`), `SCENE` (each lesson's 3D objects) and `calcolo`: `caduta(h, v0, g?, dt?, k?)`, `vsepr(nBonds, nLonePairs, wLP?, iter?)`, `parziale(N)`, `armonica(N)`, `coda(N)`, `fourierX2(x, N)`, `mcd(a, b)`, `coprimi(M, max, seed?)`.

`SpectraStudio`, defined inside `spectra-studio.html`, serves the benches: `apri(id)`, `scegli(i)`, `testa()`, `dimostra(k)`, `rivela()`, `stato() → { lez, fase, misure, previsione, scena, volto }`.

`window.BSISpectraAndroide.crea` — the face in someone else's scene (v239)

`crea(parent, { scena, renderer, senzaBusto }?)`: with `scena` the face lives in an existing three.js scene, without its own renderer, lights, camera or drawing loop (the host calls `passo(dt)` and attaches `radice` to the body); with `senzaBusto` the model's neck and shoulders are hidden. The object also exposes `parti` (the model's meshes by name).

Document updated to version `bsi-v250`.

Data Model — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v250`
Purpose	Document where the data live, with what structure, and what lifecycle they have.

0. A necessary premise

BioSpecInfo has no server database. There is no backend, so there are no centrally managed relational tables, no migrations, and no users stored anywhere.

There is, however, a real data model, and an assessor is entitled to see it documented. It is organised across three levels:

Level	Technology	What it holds	Where it lives
Preferences and progress	`localStorage`	Settings, API keys, AI history, study progress	The user's browser
Document archive	IndexedDB	Photos, documents, HTML sections, notes, activity log	The user's browser
Spectra Studio body	IndexedDB (`bsi-spectra-modelli`)	No longer written since v241: the body is embedded in the app; the database is read only to delete an old body	The user's browser
Teaching datasets	SQLite via `sql.js`	Exercise tables, created in memory at each session	Memory, not persisted

All data stay on the device. They are not transmitted, there is no server-side user identifier, and the user can delete them selectively from the interface.

1. Entity diagram

```
erDiagram
    DISPOSITIVO ||--|| PREFERENZE : "possiede"
    DISPOSITIVO ||--|| ARCHIVIO : "possiede"
    DISPOSITIVO ||--o{ SESSIONE_SQL : "apre"

    PREFERENZE {
        string bsi_ai_provider "fornitore AI scelto"
        json   bsi_api_keys "chiavi per fornitore"
        string bsi_proxy_url "indirizzo del proxy"
        json   bsi_ai_threads "conversazioni salvate"
        json   bsi_ai_memory "memoria dell-agente"
        json   bsi_prov_ko "fornitori irraggiungibili"
```

```
        json   bsi_modelli_ko "modelli ritirati"
        json   bsi_tetti "tetti di token appresi"
        json   bsi_srs "ripetizione spaziata"
        json   bsi_quiz_progress "avanzamento quiz"
        string bsi_section "ultima sezione aperta"
    }

    ARCHIVIO ||--o{ PHOTOS : contiene
    ARCHIVIO ||--o{ DOCS : contiene
    ARCHIVIO ||--o{ SECTIONS : contiene
    ARCHIVIO ||--o{ NOTES : contiene
    ARCHIVIO ||--o{ ACTIVITY : registra

    PHOTOS {
        int    id PK "autoIncrement"
        string name
        string type "MIME"
        int    size "byte"
        string data "data URL"
        int    added "epoch ms"
    }
    DOCS {
        int    id PK "autoIncrement"
        string name
        string type
        int    size
        string data "data URL"
        int    date "epoch ms"
    }
    SECTIONS {
        int    id PK "autoIncrement"
        string name
        string icon
        string html "contenuto"
        int    size "caratteri"
        int    added "epoch ms"
    }
    NOTES {
        string key PK "sempre main"
        string text
        int    updated "epoch ms"
    }
    ACTIVITY {
        int    id PK "autoIncrement"
        string icon
        string msg
        string time "data locale"
    }
```

> The diagram keeps the entity and field names as they appear in the code, in Italian: renaming them in the translation would document a model that does not exist.

2. Preferences and state — `localStorage`

2.1 AI assistant

Key	Type	Content	Lifecycle
`bsi_ai_provider`	string	Identifier of the selected provider	Until the user changes it
`bsi_api_keys`	JSON	`{ providerId: key }`	Until explicitly deleted
`bsi_api_key`	string	Single key, legacy format	Migrated to `bsi_api_keys`
`bsi_proxy_url`	string	Proxy address	Until "Disconnect"
`bsi_ai_threads`	JSON	`{ threads: [...], activeId }` — max **30 chats**, **100 messages** each	Pruned automatically
`bsi_ai_memory`	JSON	Facts the agent remembers between sessions	Deletable
`bsi_ai_history`	JSON	Condensed history	Deletable
`bsi_nucleo`	string	"Core" mode active	—

2.2 What the assistant learns about providers

These three keys are the reason the assistant improves with use instead of repeating the same mistakes. They have an expiry because today's failure must not be a permanent sentence.

Key	Structure	Expiry	What it is for
`bsi_prov_ko`	`{ providerId: { t: epoch, n: count } }`	**24 hours**	Providers that do not answer from this device. Demoted in the selection, never removed.
`bsi_modelli_ko`	`{ digest: { modelli: [...], t } }`	**7 days**	Models retired by the provider, so as not to offer them again. Indexed on the digest of the key.
`bsi_tetti`	`{ providerId: tokenCeiling }`	Permanent	Token ceiling learned from the error message rather than hard-coded

2.3 Study and progress

Key	Content
`bsi_srs`	Spaced repetition: cards, intervals, next review
`bsi_sm2`	Parameters of the SM-2 algorithm
`bsi_quiz_progress`, `bsi_quiz_history`	Quiz progress and history
`bsi_studypath`	The study path chosen
`bsi_notes`, `bsi_note_<section>`	Personal notes, global and per section
`bsi_ds_prog`, `bsi_ds_proj`	Progress in the data-science module
`bsi_section`	Last section opened, to resume from there

Key	Content
`bsi_guide_v3`	Introductory guide already seen
`bsi-materia`	Chosen subject: `chimica`, `fisica` or `matematica`
`bsi-studio:<subject>:studiati`	Study kit: summaries marked as studied
`bsi-studio:<subject>:leitner`	Study kit: Leitner box (1–5) of each flashcard, `{ id: box }`
`bsi-studio:<subject>:quiz`	Study kit: quiz history and mistakes to review

`<subject>` is `fisica`, `matematica` or `chimica`. Every read and write of the kit is guarded: in private browsing the storage may be missing or full, and the kit stays usable without remembering progress.

2.3-bis Spectra Live and Spectra's mind

Key	Content	Lifecycle
`bsi-spectra-mente-v1`	JSON: `episodi` (at most 300: time, subject, section, question cut to 160 characters, outcome), `sezioni` (time and visits per section), `ore` (24 counters), `materie` (seconds per subject), `sessioni`, `ultimaVisita`, `stato` (`curiosita`, `attenzione`, `premura`, `umore`, `legame`, between 0 and 1), `emozioni` (at most 60)	Until the user says «forget everything», which resets the object and removes the key
`bsi-spectra-ritratto-v4`	PNG image (data URL) of the face, derived once from the 3D model and used as Spectra's icon	Replaces `bsi-spectra-ritratto-v3`, which is removed
`bsi-spectra-sentinella`	`'1'` while the console's continuous listening is on	Removed when listening stops
`bsi-spectra-creatore-v1`	JSON `{ ok: true, dal, ultima }` (milliseconds): the creator has been recognised on this device. **It contains neither the secret phrase nor its fingerprint**	Written when the proof succeeds; removed by «forget that I am your creator» or by «Clear data → Persistent memory»
`bsi-spectra-studio-v1`	JSON per Spectra Studio lesson (`caduta`, `acqua`, `basilea`): `previsioni` (list of `{ quando, scelta, giusta }`) and `esercizi` (exercise index → time it was opened)	Written by `spectra-studio.html` at every prediction and every exercise opened; removed by «Clear data → Persistent memory»
`bsi-spectra-studio-voce`	`'0'` if Spectra Studio's voice has been turned off with 🔇; absent or other than `'0'` = on	Written by the studio's voice switch; removed by «Clear data → Persistent memory»
`bsi-esplora-v1`	JSON per «⚡ Explore with Spectra» chapter (v242): `{ <chapter id>: { giusta, quando } }` — the outcome of the last prediction (right or wrong) and when it was made (milliseconds); the «n/10» bar of each subject comes from here	Written by `bsi-spectra-esplora.js` at every prediction (reviewing a chapter already predicted does not rewrite it); removed by «Clear data → Persistent memory»

The mind is a simulation of behaviour: the key holds numbers and text, nothing else. Its summary (`BSISpectraMente.coscienza()`) is added to the questions that go to the artificial-intelligence provider the user chose; without questions to the model it does not leave the device.

The phrase with which the creator makes himself known (`bsi-spectra-creatore.js`) is not stored data: it is typed into a password field, its SHA-256 fingerprint is computed, compared with the File Manager's and discarded. The model receives only one line of context: whether the user has been recognised or not.

«🧠 Spiega» (`bsi-spectra-spiega.js`, v240) stores nothing: the conversation is kept in memory for the session and «↺ Nuova spiegazione» clears it. With «📺 Includi lo schermo» checked, the screen text and an image of the graphs and 3D scenes go to the configured AI provider together with the question (see `03-Security-Privacy-Compliance.md`).

The guided lesson (`bsi-lezione.js`, v239) has no keys of its own: the lesson is built from the page at every start and lives in memory while it is open. The controls it shows at work return to the value they had. At the end of a lesson the mind records an emotion («fiera», proud) in `bsi-spectra-mente-v1`.

Spectra Studio (v239) derives from its key the recommended exercise level (right predictions plus exercises opened, up to 3) and also sends each prediction to the mind (`BSISpectraMente.ricorda`). The studio's body and scenes store nothing.

«⚡ Explore with Spectra» (`bsi-spectra-esplora.js`, v242) keeps only the predictions, in `bsi-esplora-v1`; slider positions and the active chapter are not saved. It sends nothing to anyone: «🧠 Spiegamelo tutto» goes through «🧠 Spiega», with its rules.

The quality voice (`bsi-spectra-voce.js`) reads the OpenAI key from `bsi_api_keys` and writes nothing: the audio received stays in an in-memory cache for the session. The text to be spoken, however, leaves the device for OpenAI when the key is there and the voice is on (`03-Security-Privacy-Compliance.md` §1.2). In Spectra Studio the choice to turn it off is kept in `bsi-spectra-studio-voce`.

2.4 Identifiers and licence

Key	Content	Privacy note
`bsi_device_id`	Random local identifier	**Not transmitted.** It only distinguishes installations within the same browser.
`bsi_user_email`	Email, if entered by the user	Entered voluntarily, not transmitted
`bsi_pro_license`, `bsi_trial_start`	State of Pro mode	Local
`bsi_rdkit_lib2`	Molecule library of the laboratory	Local

> That none of these ever leaves the device is verified directly by `audit_rete`, which seeds a canary value into these very keys and then inspects every outgoing request.

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3. Document archive — IndexedDB

Database name	`bsi_filemanager_v1`
Schema version	`2`
Object stores	`photos`, `docs`, `sections`, `notes`, `activity`

3.1 Structure of the stores

Store	Key	Fields	Notes
`photos`	`id` (autoIncrement)	`name`, `type`, `size`, `data`, `added`	Images as data URLs. Per-file limit: **20 MB**
`docs`	`id` (autoIncrement)	`name`, `type`, `size`, `data`, `date`	Generic documents
`sections`	`id` (autoIncrement)	`name`, `icon`, `html`, `size`, `added`	HTML pages uploaded by the user
`notes`	`key`	`key` (`"main"`), `text`, `updated`	A single entry
`activity`	`id` (autoIncrement)	`icon`, `msg`, `time`	Log of the most recent operations

> The `files` store, present in earlier schema versions, is no longer created from version 2.

3.2 Opening and unavailability

Opening IndexedDB is **asynchronous**, and in some configurations (Safari in private browsing, site data blocked) it **never happens**. The code awaits a promise rather than reading a variable that might not yet be set, and distinguishes two cases:

- **reading** with the store absent → returns an empty set, the page shows

"nothing yet";

- **writing** with the store absent → the error reaches whoever pressed the

button, and a persistent warning appears at the top of the page saying nothing will be kept.

If another tab requests a version upgrade, the connection is closed (`onversionchange`) so as not to block both.

3.3 The Spectra Studio body — `bsi-spectra-modelli`

Database name	`bsi-spectra-modelli`
Object store	`file`
Key	`corpo`
Content	Up to v240: the `.glb` file (as an `ArrayBuffer`) the user had loaded with the « 🧍 Corpo» button. **Since v241 nothing is written any more**
Who writes it	Nobody, since v241 (the « 🧍 Corpo» button and the file input were removed)
Who reads it	`Spectra.carica()` in `bsi-spectra-figura.js`, only to **delete** an old body, if any, and free the space
Erasure	On start, by `Spectra.carica()`; and «Clear data → Progress in the rest of the app» (`DATI_CANCELLABILI.app` in `bsi-ai-hub.js`) deletes the whole database

Since v241 Spectra's body is **embedded in the app** as the packed block `models/spectra-corpo.bsi` (not a `.glb`: CGTrader's Royalty Free licence lets the model be used inside the product, not offered as a standalone file). It is not user data: it is downloaded from the app's own site and kept in the service worker's cache, so after the first start it works without a network.

---

4. Teaching datasets — SQLite in memory

The data-exercise module creates an SQLite database **in memory** through `sql.js` (SQLite compiled to WebAssembly). The tables are generated at each session from the datasets bundled in the repository and are **not persisted**: closing the tab makes them vanish.

The choice is deliberate: the exercise must start from a known state every time, and no user data end up in a database they did not create.

---

5. Reference data — constants in the code

The scientific data are not in a database but in constant structures inside `index.html`, loaded with the application. They are read-only and versioned together with the code.

Structure	Content	Verification
`FARM_DATA`	**263 drugs**: name, category, SMILES, molecular weight, mechanism of action, indications, adverse effects, class; 36 entries also carry `chembl` and the record's `formula`	`audit_farmaci` — structure ⟷ weight, duplicates, and the formula rebuilt from the graph against ChEMBL's
`UV_DATA`	Chromophores, λmax, extinction coefficients, Woodward's rules	`audit_dati`
`MS_DATA`	Characteristic fragmentations by class	`audit_dati`

Structure	Content	Verification
`MOLECOLE_ESAME`	Tabulated exam molecules	`test_spettri`
`PHYS_SIMBOLI`	Physical constants with symbol and unit	`test_costanti`
`SMARTS`, `SMARTS_H`	Patterns for functional-group recognition	`test_spettri`

6. Lifecycle and deletion

The user can delete data **by group**, from the interface, without wiping everything. The groups are defined in `bsiCancellaDati()`.

Group	What it removes
`chat`	Saved conversations and question history
`chiavi`	API keys (including the old format), chosen provider, proxy address, memory of unreachable providers and models
`memoria`	The assistant's persistent memory, Spectra's mind (`bsi-spectra-mente-v1`) the creator recognition (`bsi-spectra-creatore-v1`) and Spectra Studio (`bsi-spectra-studio-v1`, `bsi-spectra-studio-voce`) and the «⚡ Explore with Spectra» predictions (`bsi-esplora-v1`)
`ripasso`	Spaced repetition
`app`	Quizzes, notes, study path, data-science projects, chosen subject (`bsi-materia`), study-kit progress (`bsi-studio:*`) and the IndexedDB database `bsi-spectra-modelli` (the Spectra Studio body loaded from the device)

Left outside the groups: `bsi-spectra-ritratto-v4` (the icon derived from the face, which is regenerated) and `bsi-spectra-sentinella` (which removes itself).

Deleting the archive proceeds even if a single entry fails, and reports how many were not removed: a "delete everything" operation that stops halfway without saying so would be worse than the problem it solves.

7. Retention and space limits

Constraint	Value	Behaviour when exceeded
`localStorage`	5–10 MB depending on the browser	Writes fail. The app stays operational: chats are pruned (30 chats, 100 messages), and every write is guarded.
IndexedDB	Quota granted by the browser	Error reported to whoever pressed, with a persistent warning
Single image	20 MB	Refused with an explicit message; the upload of the others continues

Behaviour with space exhausted is verified by the `audit_quota` and `test_filemanager` benches: ten pages out of ten stay usable with `localStorage.setItem` forced to fail.

Document updated to version `bsi-v250`.

Deploy Guide — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v250`
Purpose	Operational procedure for publishing, updating and rolling back the application.

0. What gets published, in practice

BioSpecInfo is a set of **static files**. There is nothing to compile, no process to keep running, no database to provision.

Build	None. The repository files *are* the application.
Server runtime	None
Environment variables	None for the application. Only the optional proxy has any (§4).
Database	None server-side
Server requirement	Serve static files over HTTPS with the correct MIME types

The one non-obvious requirement: the server must return `application/wasm` for `.wasm` files, otherwise RDKit and SQLite do not start. GitHub Pages, Netlify, Vercel and Cloudflare Pages already do.

1. Environments

Environment	How it is obtained	What it is for
Development	`python3 -m http.server 8899` from the root	Daily work and verification benches
Preview	A branch published on Pages, or `npx serve`	Checking before merging
Production	GitHub Pages from the `main` branch	`samupropio1-ship-it.github.io/BioSpecInfo-v11/`

> **Why a server is needed even in development.** Opening `index.html` with a double click (`file://`) does not work: RDKit and SQLite are WebAssembly modules and require an HTTP context. It is the reason the verification benches also run against a local server.

2. Publishing to production

2.1 Procedure

```
# 1. from the working branch, the full battery
python3 -m http.server 8899 &
node tools/genera-evidenza.js          # must close with 0 failures

# 2. bump the version — with the tool, NEVER by hand
node tools/porta-versione.js bsi-v<NNN+1> --prova   # look first
node tools/porta-versione.js bsi-v<NNN+1>           # then write

# 3. regenerate SBOM, evidence, PDFs and packages — IN THIS ORDER
#    The packages bundle the PDFs and the evidence: built first, they carry
#    the typeset documents and the report of the PREVIOUS version.
#    genera-pacchetti now notices and stops, but better not to get there.
node tools/genera-sbom.js
node tools/genera-evidenza.js       # the battery; needs a server on :8899
node tools/genera-pdf.js            # 36 PDFs; every footer carries version and commit
node tools/genera-pacchetti.js      # the three packages; fails on broken links, stale PDFs
or stale evidence

# 4. merge into main
git add -A && git commit -m "…"
git push -u origin <branch>
#    then pull request and squash merge
```

GitHub Pages publishes automatically from `main`: the *pages build and deployment* workflow takes about a minute.

2.1-bis Publishing the version's release

The application is online as soon as `main` updates; the **release** serves a different purpose: giving an assessor a fixed point to download — a tag, the version notes and the delivery packages attached.

It is published from the repository, not from outside:

> **Actions → release → Run workflow**, field `versione`: `bsi-v<NNN>`.

The workflow creates the tag on the current commit, opens the release and attaches the three packages from `consegna/`. The full archive — `models/` included, about 124 MB — is generated by GitHub itself as *Source code (zip)*.

Why from the repository and not from the command line. The credentials a remote session uses on the repository are authorised to write to branches and **not to tags**: the `POST` to `git-receive-pack` for `refs/tags/…` answers `403 Forbidden`, while a new branch goes through. The Actions token, by contrast, writes to the repository that hosts it.

What the workflow **refuses** to do:

Condition	Outcome
`CHANGELOG.md` has no `## [bsi-v<NNN>]` entry with content	fails: a release without notes looks like a version and does not say what changed
fewer than three packages in `consegna/`	fails: a missing attachment would make the release silently incomplete
the tag does not point at the commit the packages came from	fails: a release must have a single provenance

That last row exists because of a real defect. On re-running the workflow to refresh the packages, `gh release edit` updated notes and assets but **did not move the tag**: the *Source code (zip)* archive, which GitHub generates from the tag, stayed at the first publication's commit while the packages came from a later one. Two provenances, and nothing saying so. The tag is now moved, and a following step does not take that on trust: it asks GitHub where the tag points and compares.

The workflow is **re-runnable**: if the release already exists it updates notes, assets and tag instead of failing halfway. Moving an already-published tag is not free — whoever downloaded it already has different content — so it applies to a release just published, not an old one: for that, publish a new version.

2.1-ter Why the version bump is not done with a find-and-replace

It looks like the most harmless operation in the world: search `bsi-v<NNN>`, write `bsi-v<NNN+1>`. **For three releases in a row it was done that way, and every time it rewrote historical sentences** — sentences recording a fact that happened at a past version. What came out of it sat inside a signed conformance statement:

the document said	the truth
the password had been **changed at the current version**	it was changed at version `bsi-v188`
the tag had been **moved to the current version's commit**	the incident happened at version `bsi-v188`
the thirty-six entries were **added in the current version**	they were added in `bsi-v188`
"bump `CACHE` (e.g. `bsi-vN` → `bsi-vN`)"	an example equal to itself

No check could catch it: every line named a version that **exists**, in the document describing that version.

> This table uses placeholders rather than the real numbers because the two checks described below **would reject it** — which is the proof that they work.

`tools/porta-versione.js` touches **only** the lines that DECLARE the version — headers, footers, the README badge, package names, the two code lines — and **lists** everything else so it gets read. With `--prova` it writes nothing.

Two checks in `tools/verifica-documenti.js` now close the door: no past-tense sentence may name the current version, and no upgrade example may show the same version on both sides.

> **Name the thing, not the number.** "From the Pretsch tables" rather than "from version such-and-such": it reads better and does not age.

2.2 The two version lines

`CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` **must match**.

- `CACHE` determines whether the Service Worker considers the cache stale:

without a bump, devices keep serving the old files and the publication reaches nobody.

- `BSI_APP_VERSION` is what the user sees in the "Updates" window. «Check for

updates» compares the active Service Worker's version (it answers the `BSI_VERSIONE` message with its `CACHE`) with the `var CACHE = '…'` line of the published `sw.js`, read from the same site with no cache and, only if that does not answer, from GitHub. `BSI_APP_VERSION` enters the comparison only when no Service Worker answers.

> **The shape of the line matters.** The check reads `sw.js` with the expression `var CACHE = '…'` . Up to version `bsi-v236` it looked for a `CACHE_VERSION` line that `sw.js` no longer contained, and the published version was never read. If the declaration in `sw.js` changes shape, `bsi-aggiorna.js` must change too; `test_controllaagg` notices.

They drifted apart once in the past — the second had been stuck for months — and the result was that the update check was comparing the wrong number.

2.2-bis Large files outside the precache

`models/spectra-noir.glb` (Spectra's face, about 8 MB) is not in the Service Worker's precache, like the other `models/*.glb`: it is downloaded the first time the app derives Spectra's icon or opens the Spectra Live console. It must be published with the rest, with its CC BY 4.0 attribution (`07-SBOM.md`); if it is missing, the console uses the fallback face.

2.2-ter One secret phrase, two fingerprints

Since version `bsi-v238` the same secret phrase serves the File Manager and the recognition of Spectra's creator. Its SHA-256 fingerprint is written in two places: `file_manager.html` and `bsi-spectra-creatore.js` (constant `IMPRONTA`). If the phrase changes, **both** must change in the same release; otherwise the File Manager opens with the new phrase and Spectra recognises only the old one. A recognition mark already written in a browser (`bsi-spectra-creatore-v1`) is not revoked by the change: it is removed with «forget that I am your creator» or with «Clear data». The plaintext phrase must never be written to a file in the repository.

2.2-quater A new module: three places to keep aligned

A new application `.js` file, such as `bsi-lezione.js` (the guided lesson), must be declared in `index.html` (with `defer`; `bsi-lezione.js` comes before `bsi-spectra-live.js`, which uses its `BSILezione` object when it receives «fammi lezione»), in the `PRECACHE` list of `sw.js`, so that it works offline, and in the module list of `tools/genera-sbom.js`, so that it appears in the SBOM. A module missing from the precache works while there is a network and disappears offline.

The latest module added this way is `bsi-spectra-esplora.js` (v242, «⚡ Explore with Spectra»): in `index.html` with `defer`, **after** `bsi-fisica-comune.js`, because it registers its three sections (`sesplchim`, `sesplfis`, `sesplmat`) with `BSIFisica.sezione` as soon as it runs; in the `PRECACHE` of `sw.js`, next to `bsi-spectra-spiega.js`; in the list of `tools/genera-sbom.js`. The three containers `<div id="sesplchim" class="section">`… and the three bar buttons are in `index.html`. The module brings no 3D files of its own: it uses the shared figure of `bsi-spectra-figura.js`.

In the same version `bsi-spectra-olointerattivo.js` (v242, the «⛶ Interagisci» interactive hologram) follows the same three rules: in `index.html` with `defer`, **right after** `bsi-spectra-figura.js`, which creates the button on the hologram and moves the figure into its window; in the `PRECACHE` of `sw.js`; in the list of `tools/genera-sbom.js`. It uses three.js and the figure already loaded by `bsi-spectra-figura.js` and the local assistant (`bsi-assistente.js`, `bsi-mat-cas.js`), and brings no data files of its own: molecules and planets are in the code.

2.2-quinquies Spectra Studio, a page of its own

`spectra-studio.html` (v239) does not go through `index.html`: it loads by itself `three.min.js`, `gltf_loader.js`, `bsi-spectra-mente.js`, `bsi-spectra-creatore.js`, `bsi-spectra-androide.js`, `bsi-spectra-corpo.js`, `bsi-spectra-rig.js`, `bsi-spectra-figura.js`, `bsi-spectra-voce.js` and `bsi-spectra-studio-lezioni.js`. The page and these modules are in the `PRECACHE` of `sw.js`; `bsi-spectra-figura.js` (v240) is also in `index.html`, with `defer`, because it shows Spectra with the body in the AI sections: it loads `three.min.js`, `gltf_loader.js` and the other 3D modules only the first time one of those sections is opened; `bsi-spectra-spiega.js` (v240, the «🧠 Spiega» tab of the Spectra window) is also in `index.html` with `defer` and in the precache; `bsi-spectra-voce.js` is also in `index.html`, because the console and the guided lesson use it. The studio's «Noir» face is the same `models/spectra-noir.glb` as the console's, outside the precache (§2.2-bis): if it does not arrive, the body stays without a face.

Spectra's body (v241): `models/spectra-corpo.bsi` must be deployed. The «Bounty Hunter» body (free on CGTrader, Royalty Free licence: used inside the product, not offered as a standalone file) is embedded in the app as the packed block `models/spectra-corpo.bsi` (4.40 MB), which is in the repository and must be deployed with the app: without it the figure and the studio stay on «Spectra non è arrivato: serve la rete la prima volta» with «↻ Riprova». The block is not a `.glb`: only `bsi-spectra-figura.js` (function `apri`) reopens it. Like the `models/*.glb` files it is not in the precache: the service worker (network first) caches it on the first request, and from then on it works offline. The original `.glb` (`models/spectra-corpo.glb`) stays out of git (`.gitignore`); the studio tries it only in development, if the block is missing. The `models/spectra-cyberpunk.glb` fallback no longer exists.

Regenerating the block (for example after changing the model):

```
node tools/impacchetta-corpo.js path/to/body.glb models/spectra-corpo.bsi
# ✓ models/spectra-corpo.bsi (4.40 MB), verificato: si riapre identico al .glb
```

The tool checks that the input is a `.glb`, writes the block («BSIC» · version · length · bytes mixed with an xorshift32 stream from the fixed seed, the same as in `bsi-spectra-figura.js`) and checks that it reopens identical. Then the new `models/spectra-corpo.bsi` is deployed; clients' caches refresh with the `sw.js` version change. A change of model must also be reflected in `models/CREDITS.md` and `THIRD_PARTY_NOTICES.md`. The quality voice calls `api.openai.com` directly from the browser: any `Content-Security-Policy` (`03-Security-Privacy-Compliance.md` §2.4) must allow it in `connect-src`, otherwise the voice falls back on the browser's synthesis.

2.3 How the update reaches users

1. The browser detects a different `sw.js` and installs the new Service Worker.
2. `skipWaiting()` activates it at once; the old caches are deleted.
3. The Service Worker **notifies** the open tabs (`BSI_SW_UPDATED`) — it does not

reload them on its own authority.

1. The page decides: if no work is in progress it reloads itself; if the

assistant is answering or there is unsaved text, it shows "Update now / Later".

Whoever opens the app at that moment already receives the new version.

---

3. Publishing alternatives

The application does not depend on GitHub Pages. Any static file host will do, provided it is over HTTPS (the Service Worker requires it).

Platform	Command / configuration	Notes
GitHub Pages	Settings → Pages → branch `main`, folder `/`	In use. Free, custom domain possible
Netlify	`netlify deploy --prod --dir=.`	No build command
Vercel	`vercel --prod`	Static project, no framework
Cloudflare Pages	`npx wrangler pages deploy .`	Convenient if the proxy is already in use
VPS with nginx	Copy the files to `/var/www/biospecinfo`	See the configuration below

nginx configuration

```
server {
    listen 443 ssl http2;
    server_name biospecinfo.example.org;
    root /var/www/biospecinfo;

    # Without this MIME type, RDKit and SQLite do not start
    types { application/wasm wasm; }

    # sw.js must not be cached by the CDN: it is what announces new versions
    location = /sw.js {
        add_header Cache-Control "no-cache, no-store, must-revalidate";
    }

    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

A rule that holds everywhere: `sw.js` must not be cached

If a CDN keeps `sw.js`, the browser keeps seeing the old version and **no update ever arrives**, however much one publishes. This holds on every platform.

---

4. Optional proxy — a separate deployment

The proxy is a **distinct** component, owned by the user, published on their own Cloudflare account. It is not required in order to use the application.

```
cd proxy
npx wrangler login
npx wrangler deploy
# the address is printed: https://spectra-proxy.<account>.workers.dev

npx wrangler secret put GROQ_KEYS      # one or more keys, comma-separated
npx wrangler secret put GEMINI_KEYS    # optional
```

The address is then pasted into the assistant's "Proxy" box, inside the app. The same four steps are available **inside the application**, with the commands ready to copy.

Recommended configuration

```
# Without this variable the proxy accepts requests from any origin
npx wrangler secret put ORIGINI
# value: https://samupropio1-ship-it.github.io
```

Variable	Default	Why change it
`ORIGINI`	*(empty = all)*	**Always set it.** An open proxy is usable by anyone, at the expense of your quota.

Variable	Default	Why change it
`LIMITE_IP`	`20`	Requests per IP within the window
`TETTO_GIORNO`	`2000`	Overall daily ceiling

> **Keys must never go into `wrangler.toml`.** That file is versioned: a key written there ends up on GitHub, and the bots that scan it find it within hours. Use `wrangler secret put`, which keeps them out of the repository.

---

5. Domain and certificate

Aspect	Situation
HTTPS	Mandatory: without it the Service Worker does not register and the app does not work offline
Certificate	Provided and renewed automatically by GitHub Pages
Custom domain	Settings → Pages → Custom domain, plus a `CNAME` record pointing to `<user>.github.io`

---

6. Monitoring

There is no server to monitor: there are no processes that could fall over, nor resources that could run out.

What to watch	How
Outcome of the publication	GitHub → Actions → *pages build and deployment*
Availability	An HTTP check on `index.html` (UptimeRobot or similar)
Client-side errors	The browser console. Remote telemetry exists but is **disabled** (`BSI_TELEMETRY_URL` empty): enabling it requires a deliberate decision about data handling.
Proxy consumption	Cloudflare → Workers → Analytics

---

7. Rollback and emergency procedures

7.1 Going back to the previous version

```
git revert <commit>
git push origin main
```

Pages republishes in about a minute. Devices receive the restored version as a normal update — **provided `CACHE` changes**: a revert that also takes the version number backwards is fine, because what counts is that the value be *different*, not that it be greater.

7.2 "I published but I do not see the changes"

In order:

1. **Was the version bumped?** It is the cause in ninety per cent of cases.

Without it, the Service Worker serves the cache.

1. **Has Pages finished?** GitHub → Actions.
2. **Is `sw.js` cached by a CDN?** It must answer `no-cache`.
3. From the app: ✨ menu → 🔄 **Updates** → **Check for updates**. It tells

which version is served and which is published; if they differ it forces a `registration.update()`. If it answers «up to date» with the old number, the `sw.js` the site serves is still the old one (points 2 and 3).

1. As a last resort: developer tools → Application → Service Workers →

Unregister, then reload.

7.3 "The assistant has stopped answering"

This is not a publishing problem: it concerns the AI provider. In the app, the 🔌 **Test** button measures in a few seconds which providers the device can actually reach, without needing any key.

7.4 Restoring the user's data

The data live on the device and **there are no copies on a server**: they cannot be restored remotely. This is a direct consequence of the local-first architecture, and it is declared to the user. Exports (notes, documents) remain the only form of backup.

---

8. Pre-publication checklist

- ☐ `node tools/genera-evidenza.js` closes with **0 failures**
- ☐ `CACHE` in `sw.js` and `BSI_APP_VERSION` in `index.html` bumped and **equal**
- ☐ `node tools/genera-sbom.js` re-run
- ☐ `node tools/genera-pdf.js` re-run — otherwise the attachments describe the previous version
- ☐ `node tools/genera-pacchetti.js` re-run and closed with **0 broken links**
- ☐ Evidence report regenerated against the commit being published
- ☐ Any new deviations noted in `09-Release-Conformance-Statement.md` §4
- ☐ No conflict markers (`verifica_guida` checks this across 58 files)
- ☐ No API key in the versioned files

- ☐ The accessibility debt **has not grown** — if it has fallen, regenerate the

baseline with `node tools/verifica-accessibilita.js --aggiorna-riferimento` and record it in the commit

- ☐ Code coverage **has not fallen** — `audit_copertura` fails if it has

Document updated to version `bsi-v250`.

Functional Specifications — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v250`
Purpose	Describe what the product does, for whom, under which rules and with which limits.

1. Product description

BioSpecInfo is an **interactive cheminformatics** platform for university study of chemistry, biochemistry, pharmacology and astrochemistry.

The whole scientific logic — interpretation of molecular structures, spectral prediction, calculations, 2D and 3D visualisation — runs **in the student's browser**. There is no server: the application installs as an app, works with no network, and transmits no data.

1.1 Objectives

Objective	How it is pursued
Total accessibility	No registration, no cost, no server. Whoever opens the link is already using it.
Independence from the network	Everything needed is bundled and cached: it works in a lecture hall, on a train, in a library with no signal.
Verifiable correctness	Scientific data are put through automatic checks; the limits of the models are declared rather than hidden.
Privacy by construction	There is no backend that could collect data: the absence of collection is not a promise, it is a consequence of the architecture.

1.2 What the product is not

Not a medical device	It does not support diagnosis, prescription or dosing (Reg. EU 2017/745)
Not a certified analytical instrument	The spectra are predictions, not measurements
Not a collaborative platform	No accounts, no sharing, no synchronisation between devices

2. Actors

Actor	Description	What they can do
Student	Primary user. Studying for a university exam.	Every function: consultation, calculation, quizzes, spaced repetition, AI assistant
Teacher	Uses the app as teaching support	The same functions; uploading their own material into the File Manager
Technical assessor	A company or committee examining the work	Reading the documentation and re-running the verification benches

There are no technical roles: there is no authentication, so there is no distinction of permissions. The File Manager has a local protection that is a **deterrent**, not an access control (see §6).

3. Functional areas

The application has **201 sections**, grouped by area of study.

3.1 Spectroscopy

Function	Description
Spectroscopy centre	A molecule is entered (SMILES, name, formula) and one obtains IR, $^{1}H/^{13}C$ NMR, MS, UV-Vis, Raman, 2D and 3D structure, functional groups
Visual IR spectra	Characteristic bands by class of compound
Mass spectrum	Fragmentation patterns and diagnostic rules
UV-Vis	Chromophores, λmax, Woodward's rules
Per-atom NMR prediction	Draw or type a structure and get ^{1}H and ^{13}C with WHICH atoms produce each signal; interrogable spectrum (zoom, intensity, click a peak), assignment table, SMILES/InChI history, CSV/JCAMP-DX/PNG exports
Two-dimensional NMR	COSY, edited HSQC and HMBC built by walking the graph; clicking a spot highlights the atoms producing it. It is not a simulation of the experiment, and the panel says so
Open any file, and the program decides	One entry point, with no extension filter: open what you have and the program **states what it decided the file is**. A document (PDF, photograph, Word, OpenDocument, PowerPoint, Excel, text) opens, **everything** it contains is shown (every page drawn, all the text), the recognised spectroscopic data appear in a table **with their provenance**, and the working proceeds step by step; a JCAMP-DX file or two columns of numbers become a measured spectrum; an image does both. Whatever could not be read is declared
Header language switcher	A round button top right, at the title's height, with the flag of the language in use; it opens into a disc with the fourteen flags on two shells and, at the centre, the language name and **how much it is translated, measured**. The actual switch stays in the languages module: this is only the way to call it

Function	Description
Spectrum from a figure (digitisation)	From a photograph of a chart recorder, a crop from a paper or the instrument screen, the trace is extracted from the **pixels**: the axis frame is recognised and only its inside is digitised, the line position is the **darkness-weighted centroid** (sub-pixel), the grid is worked around by continuity and darkness, both axes are calibrated linearly or logarithmically, %T is converted to absorbance with $A = -\log_{10}T$. The chart, the peaks and the assignments appear by themselves and are redone at every change of the calibration; a diagnostics line says what was decided and why. CSV and JCAMP-DX outputs. Measured deviation: mean 2.2 cm^{-1}, worst 7.3 over seven constructed figures
3D model from the drawn structure	Coordinates built from the graph, carrying the predictor's own indices: a clicked peak lights up the atom in three dimensions too
⚛️ **Physics: eight computed sections**	**Schrödinger laboratory** (stationary states of wells, oscillators, Morse and double well; wave packet and tunnelling with Crank-Nicolson; vibrations of HCl, H_2, CO, N_2 in cm^{-1} compared with experiment); **live Hartree-Fock** (RHF/STO-3G in the browser, eight molecules, orbitals drawn, dissociation curves); **rovibrational spectroscopy** (P and R branches with temperature and isotopes, Raman with spin statistics, bond length from the lines); **statistical thermodynamics** (partition functions, S° and Cp against NIST-JANAF, populations); live Lennard-Jones **molecular dynamics**; **mathematics for chemistry** (FID → spectrum by FFT, Hückel, kinetics by Runge-Kutta, Arrhenius least squares); **crystallography** (14 Bravais lattices, Miller indices, powder pattern with absences); **radiochemistry** (Bateman, dating, Weizsäcker). Every number is checked by `test_fisica` against the literature
📐 **Mathematics: eight sections on a symbolic engine**	A computer-algebra engine written for the app (`bsi-mat-cas.js`: exact fractions, canonical form, rule-based derivatives, antiderivatives verified by differentiating them, Taylor series by truncated power-series arithmetic, limits) underpinning: **basics** (significant figures and measurement rules, logarithms and pH, equations from linear to polynomial and the fractional ones of equilibrium, practice with marking); **function analysis** (domain, zeros, extrema, inflections, asymptotes, tangent; families and transformations); **derivatives, integrals and series** (rules used, Riemann, adaptive Simpson, Taylor, left and right limits); **linear algebra** in exact fractions with Gauss-Jordan steps, eigenvalues, plane transformations and reaction balancing as a null space; **differential equations** (direction field, order of methods, phase plane with equilibria classified from the Jacobian); **several variables** (3D surfaces, critical points, Maxwell relations, exact differentials); **complex numbers and Fourier**; **laboratory statistics** (t, F, Grubbs, Dixon, uncertainty propagation, calibration line with LOD/LOQ)
🎓 **UniBA exams and advanced physical chemistry**	**Mathematics exam**: Istituzioni di Matematiche I and II syllabi explained, step-by-step solver for 16 exercise types, 16 solved papers (64 exercises), timed simulation. **Physics exam**: General Physics I and II syllabi, 48 standard problems solved with the user's data (laws, substitutions, units), 12 solved papers. **Electrodes and transition state** (Butler-Volmer, Tafel, Eyring) and **photochemistry, surfaces and polymers** (Jablonski, Stern-Volmer, Langmuir/Freundlich/BET, Mn/Mw, Carothers). **Chemistry exam** (🔬 category): 40 standard problems on four levels — basic (General and Inorganic), intermediate (Analytical), advanced (Physical Chemistry), expert (Sackur-Tetrode, Debye-Hückel, Pourbaix) — and 7 solved papers. Mathematics and Physics also gain an **advanced and expert** level (eigenvalues, ODE systems, Fourier,

Function	Description
	Laplace, triple integrals; driven oscillations, waves, Doppler, photoelectric, Compton, relativity, blackbody, particle in a box, tunnelling). The traces are built on the syllabi, they are not the official papers
🧭 **From high school to PhD**	A map of chemistry, physics and mathematics on four levels (high school, bachelor's, master's, PhD) with links to the tools. New solvers: **astrophysics** (10 models), **nuclear physics** (9), **quantum physics** (11, with variational and perturbation methods), **organic chemistry, spectroscopy and biochemistry** (10), **PhD** (Marcus, isotope effect, Wigner; residues, heat equation)
🔐 🕸️ ♾️ 📈 🍩 🔣 ♟️ 🧠 📊 🌐 **Mathematics: deeper topics**	**Numbers and cryptography** (sieve, Euclid and Bézout, modular clock, real RSA, π(x) against Li(x)); **Graphs and optimisation** (Dijkstra, Kruskal, gradient descent, linear programming); **Sequences and series** (ε–N limit, convergence tests, rearrangement, radius of convergence); **Calculus of variations** (brachistochrone, catenary, catenoid, Fermat); **Topology** (V − E + F, surfaces and orientability, fundamental polygons, knots and colourings); **Algebraic structures** (dihedral groups, permutations, rings and finite fields, Lagrange); **Game theory** (Nash equilibria, iterated dilemma and replicator, tic-tac-toe with minimax, Nim); **Logic and computability** (truth tables and normal forms, adder circuits, Turing machine, sets); **Statistical inference** (confidence intervals, tests and power, regression with Anscombe's quartet, bootstrap); **Non-Euclidean geometries** (sphere and Girard, Poincaré disk and tilings, curvature and parallel transport)
🪐 🔥 🧲 ⚛️ 🎡 🔦 🌍 **Physics: deeper topics**	**Gravitation** (N bodies, Lagrange points, slingshot); **Heat and diffusion**; **Ising model**; **Particles** (Standard Model, conservation laws, Z invariant mass, muon lifetimes); **Analytical mechanics** (phase space and elliptic integral, chaotic double pendulum, bead on a hoop and Noether, Liouville); **Lasers and quantum optics** (rate equations and threshold, cavity modes and mode locking, Rabi and Bloch, photons in a Mach–Zehnder); **Climate physics** (radiative balance and layers, CO_2 forcing, ice–albedo with hysteresis, Coriolis)
🧊 🏔️ **Physical chemistry: deeper topics**	**Phase diagrams** (water and CO_2 with the IAPWS and Span–Wagner curves, eutectic and lever rule, Clapeyron and Gibbs); **Potential energy surface** (LEPS for H + H_2, trajectories, reaction path, Polanyi)
🧪 ⚛️ 📐 **Subjects and Articles**	Two buttons above Spectra and ✨ switch subject: the bar shows only the chemistry, physics or mathematics groups, plus Study and Utility. The **Articles** section has the same two tabs for all three subjects: latest articles (Crossref) and prizes with a topic explanation — Nobel in Chemistry and in Physics, Abel Prize and Fields Medal for mathematics (Wikidata)
🔭 **Astrophysics: a group of twelve sections**	The bar's 🔭 **Astrophysics** group gathers 🔭 Astrophysics and 🪐 Gravitation (previously in the physics group) and ten 3D sections: **Solar system 3D** (positions at any date from Standish's JPL Keplerian elements, valid 1800–2050; Earth, Moon and eclipses with Meeus's truncated theory; Jupiter and Saturn); **Orbital mechanics 3D** (elements ↔ state, Hohmann and bi-elliptic manoeuvres, Earth satellites, Lambert and porkchop); **Celestial mathematics** (Kepler's equation with three methods, celestial sphere and quaternions, N-body integrators, Lomb–Scargle and BLS periodograms); **Life of stars 3D** (a star in cross-section, Lane–Emden, evolution, nucleosynthesis); **Black holes and general relativity** (Schwarzschild ray tracing in the shader, geodesics, curvature, deflection and GPS, Hawking radiation); **Gravitational waves** (inspiral, chirp, detector, Hulse–Taylor pulsar); **Galaxies 3D** (Hubble types, rotation curves, collision, Local Group); **Cosmology 3D** (Friedmann–Lemaître expansion, distances, CMB, thermal history); **Planetarium 3D** (about 170

Function	Description
	stars and 28 constellations from any place and time, coordinates, precession, analemma); **Exoplanets 3D** (habitable zone, transit, radial velocity, direct imaging and population)
🌀 🎲 ☢️ 🧱 🔌 **Physics: five laboratories**	**Spin, qubits and entanglement** (Stern–Gerlach, Bloch sphere, Bell and CHSH, BB84); **Statistical mechanics** (levels and the Schottky anomaly, Maxwell–Boltzmann, Fermi and Bose, Einstein and Debye solids); **Nuclei and reactions** (valley of stability, Q values, barrier and tunnelling, 3D nucleus, reactor); **Elasticity and materials** (tension test, beams, Euler buckling, sound in solids); **Electronics** (rectifiers, diodes and Zener, common emitter, op-amps, ADC, DAC and counters)
🌊 🌀 🔗 🧩 📡 **Mathematics: five laboratories**	**Partial differential equations** (string, 3D membrane, heat in 1D and 2D, 3D Laplace, transport and Burgers); **Complex analysis** (domain colouring, 3D landscape of |f|, conformal maps, Joukowski, residues, real integrals); **Markov chains and stochastic processes** (chain, PageRank, gambler's ruin, walks in 1, 2 and 3 dimensions, Poisson and queues); **SVD and applied linear algebra** (2×2 and 3×3 SVD, image compression, 3D PCA, least squares); **Codes and elliptic curves** (Hamming(7,4), CRC and check digits, curves over $\mathbb{R}$ and over F_p, ECDH)
🧪 ⚛️ 📐 **Three apps in one**	The chosen subject also decides the ✨ **menu**, **Spectra**, **Study** and **Utility** (`bsi-materie-app.js`). In physics and mathematics the ✨ menu's chemistry block gives way to the **quick tools** (`bsi-attrezzi.js`), opened in a window above the page: 🧮 exact calculator, ∂ derivative and ∫ antiderivative, 📈 graph, 🔢 CODATA constant, 🔁 unit converter, 📐 formula, 📖 glossary and ⚡ lightning quiz taken from the subject's study kit. Spectra receives the subject's system instruction, suggestions and greeting
🧊 **3D tabs in the sections (v238)**	Twenty-nine sections (4 chemistry, 15 physics, 10 mathematics) have, after their other tabs, a 3D tab built on the section's model: LEPS surface with the trajectory, P–V–T surface with the dome, molecular collisions in a box, equilibrium molecule by molecule, unit cells with Miller planes, the $\|\psi\|^2$ cloud, RHF orbitals, a molecule rotating and vibrating, a Lennard-Jones fluid, a launch with wind, fields and the Lorentz force, a ripple tank, a potential well with the Lagrange points, the light cone, flow past an obstacle, the p-V-T surface, the Brillouin zone, 3D Ising, a particle detector; tangent plane, field and divergence, solids of revolution, space transformations, chaos in phase space, the bivariate normal, surfaces with χ and orientability, sphere and hyperboloid, Riemann sphere, soap film. Text and controls are built on click, the scene right after. Every tab has a verification script (`15-Test-Documentation.md` §3.1-ter). Canvases without a name take the tab's and the section's
🎓 **Guided lesson in every section (v239)**	Under the title of every section the «🎓 Guided lesson with Spectra» button starts a lesson in chapters built from the section itself (`bsi-lezione.js`): an opening with title and subtitle; one chapter per tab, in which the tab opens, the explanation is read in pieces, canvases and 3D scenes are highlighted, every slider runs from minimum to maximum and returns where it was, every select goes through its options and returns as it was; a check question from the subject's study kit with the explanation after the answer; a summary. Sections without tabs are read by headings and paragraphs. The player sits on the right on a large screen (the page shifts left) and at the bottom on a phone; it has ◀ ▶, pause, Spectra's voice and a spotlight that dims the rest of the page. During the lesson the floating buttons are hidden. At the time of writing: 194 sections with a lesson, 99 with tabs, 79 with a check question
🌌 **Spectra Studio (v239)**	A page of its own (`spectra-studio.html`), opened from the console's 🧭 navigator: a 3D laboratory with Spectra standing next to a holographic bench. Three lessons with the predict → test → prove → reveal method: 🪂 the fall (dropped or thrown, they land

Function	Description
	together: 2.019 s from 20 m, range 12.12 m), 💧 the shape of water (VSEPR as an energy minimum: tetrahedron 109.47°, water 104.4°, ammonia predicted 106.7°), ∑ the Basel problem (partial sums, tail, Fourier series of x², coprimes → 6/π²). Every option is explained; exercises on four levels with the one recommended by the memory of predictions (`bsi-spectra-studio-v1`). The body is the one chosen by the author, **always present** (since v241 embedded in the app, `models/spectra-corpo.bsi`), rigged, with the «Noir» face on the collar; while it arrives «Spectra sta arrivando…», no substitute; the voice is turned off with 🔊/🚫 and the choice is remembered; the panel holds all the text and the lesson can be followed without WebGL too. Design in `docs/spectra/`
🎲 🎞️ ⚡ 🔊 💡 🌡️ 🔋 **Explorations**	**Probability in motion** (Galton, central limit, Monte Carlo, Bayes); **Visual mathematics** (animated linear maps, Fourier epicycles including from a hand drawing, Mandelbrot and Julia with zoom, logistic map and bifurcations); **Electromagnetism in motion** (draggable charges with equipotentials and field lines, induction, generator, EM waves and polarisation); **Sound and acoustics** with audio (harmonics, timbre, beats, Doppler and Mach cone, decibels); **Semiconductors** (Fermi-Dirac, n_i, doping, p-n junction, LEDs); **Distillation** (T-x-y with Antoine and van Laar, azeotrope, McCabe-Thiele, Rayleigh); simulated **Cyclic voltammetry** (diffusion and Butler-Volmer, Randles-Ševčík, scan-rate analysis)
🔭 ⏱️ 🌊 ∇ 🌀 **Astrophysics, relativity, fluids, vector calculus, complex kinetics**	**Astrophysics**: H-R diagram with real stars and the main sequence by mass, black body and star colours, animated solar system with Kepler and Hohmann, exoplanet transits with light curve and radial velocity, Friedmann cosmology with the age of the universe and receding galaxies. **Special relativity**: Minkowski, twins, dynamics and Doppler. **Fluids**: Venturi, Poiseuille, Reynolds, Stokes and centrifuges, capillarity. **Vector calculus**: fields with particles, divergence and curl, potential, Green's theorem verified, gradient. **Complex kinetics**: consecutive, parallel and reversible reactions, Brusselator and Hopf bifurcation, Turing patterns. Every animation in the app has pause, restart and speed, stops off-screen and honours «reduce motion»
🎯 〰️ 🔌 🔥 🧮 📐 ⚖️ **Laboratories in motion**	Seven sections: **Mechanics in motion** (projectile with air drag and optimal angle, collisions with a coefficient of restitution, the real pendulum with its phase space, coupled oscillators and normal modes); **Waves and optics** (beats, standing waves, group velocity; lenses and mirrors with principal rays; N-slit diffraction in the colour of light); **Circuits** (RC, RL, RLC transients with the three regimes; alternating current with phasors and resonance; filters with the Bode plot); **Thermodynamics in motion** (Carnot, Otto, Diesel and Stirling cycles in the p-V and T-S planes with balances; a gas of particles reaching Maxwell-Boltzmann; Brownian motion); **Numerical methods** (roots, integration, ODEs, interpolation, with measured convergence orders); **Analytic geometry and vectors** (vectors and lines and planes in 3D, conics with the invariants); **Equilibrium and Le Chatelier** (a reaction shifting when disturbed, K as a function of T)
🧊 **3D Laboratory and animated charts**	New section in the Physics category: **surfaces** z = f(x, y) with contour lines and critical points, hydrogen-like **orbitals** as a
⚗️ ⚛️ 📐 **The three macro-areas**	At the head of the Chemistry, Physics and Mathematics categories. Each has an **assistant** to which you write the problem in natural language (English too): it recognises the kind of exercise or the model, reads numbers with units, formulas, reactions and lists, says what it understood and where each datum comes from, also

Function	Description
	solves the **inverse problem** and shows the working; the **catalogue** of every tool of the subject with search; the **levels** from high school to PhD. Spectra has the same solvers as tools
NOE and 2D NMR (tables)	Reference values for spatial correlations
Reference tables	Tabulated values for quick consultation
Spectrum quiz	Recognising the functional group from a spectrum

3.2 General and physical chemistry

Interactive periodic table · chemical calculator · reaction balancing · degrees of unsaturation · distribution diagrams · VSEPR · molecular orbitals · advanced thermodynamics · electrochemistry (Nernst) · pKa and buffers · laboratory calculator · constants and formulae.

3.2-bis Separation — chromatography

Chromatography section: a chromatogram computed from the parameters (theoretical plates, retention factor, selectivity, dead time) with an overview and an expanded view of the peak pair; a **van Deemter** curve with the three terms separated and the minimum computed in closed form; a **Kováts retention index** calculator; a table of the quantities with their working ranges; GC/HPLC comparison; a guide to detectors; diagnostics for peak-shape defects.

> **Rule applied.** The chromatogram is not an illustration: the Gaussians have $\sigma = t_R/\sqrt{N}$ and the resolution shown is the one that would be measured on those peaks. The section explicitly declares what it does **not** do — predict a molecule's retention time from its structure, which would require experimental parameters of the stationary phase that cannot be derived from the SMILES alone.

3.2-ter Calibration and uncertainty

Calibration section: ordinary least-squares regression on data pasted from a spreadsheet, with slope and intercept **accompanied by their standard deviation**; a plot of the line with a 95 % confidence band and a **residual plot**; the unknown's concentration with a full confidence interval (the three sources 1/M, 1/N and distance from the centroid are shown separately); **LOD and LOQ** per ICH Q2(R1); uncertainty propagation indicating which contribution dominates; significant-figure rules.

Two diagnostics that an r^2 does not provide:

Diagnostic	Method	Why r^2 is not enough
Non-linearity	F-test between line and parabola on the same data	An r^2 of 0.984 accompanies an obvious saturation curve: r^2 measures how much variance the line explains, not whether the line is the right model

Diagnostic	Method	Why r² is not enough
Outlier	**Deleted** residual: the judging line is rebuilt without the point being judged	The naive criterion `

> **The second row is a trap met while writing the section**, not a textbook case: the 3 σ criterion was already written, and the example meant to demonstrate it was not being flagged. The check passed without seeing anything.

3.2-quater Cheminformatics and QSAR

Cheminformatics section: a workbench that goes from the raw set to a verdict on the model, entirely in the browser. SMILES are pasted in — with a name and an activity, where there are any — and six panels return what a cheminformatics group produces in a day.

Panel	What it does
Descriptors	Standardisation (largest fragment, canonical SMILES, deduplication), 43 RDKit descriptors, Lipinski's rule of five, Veber filter, QED recomputed — MinimalLib does not expose it
Similarity and groups	Morgan/ECFP4, RDKit, pattern and MACCS fingerprints; Tanimoto and Dice; similarity matrix; nearest neighbours; **Butina clustering** and **MaxMin** selection for diversity
Chemical space	PCA on the standardised descriptors, with explained variance per component and the **loadings** that say which descriptor moves which axis; **Bemis–Murcko** scaffolds with the molecule count per scaffold
QSAR model	Kernel ridge (Tanimoto kernel), kNN over Tanimoto or logistic regression; **scaffold split**; **grouped cross-validation** over 3, 5 or 10 folds; **model comparison on identical folds** with the trivial reference at the top; hyperparameters chosen by **nested validation**; **conformal prediction intervals**; **learning curve**; applicability domain
Search	Substructure search across the whole set with RDKit's `SubstructLibrary`, SMARTS or SMILES query, occurrence counts, eight ready queries
SAR table	R-group decomposition around a core: one row per molecule, one column per substitution position, with each substituent's median activity. Molecules without the core are **excluded**, not shown with empty cells
Matched pairs	Pairs differing by a **single** substitution, grouped by transformation with median, range and **number of concordant pairs**
Activity cliffs	Structurally close pairs with distant activities, ranked by **SALI**: these are the pairs every model gets wrong, and it is honest to show them
Structural alerts	PAINS and Brenk, with the matched fragment highlighted and the reason it is flagged

Three choices that separate a workbench from a demonstration:

Choice	Method	Why
Scaffold split	The test set contains **only scaffolds never seen** in training	A random split puts close analogues on both sides: the model recognises rather than predicts, and the R^2 that comes out does not survive the first new molecule
Null model by scrambling	The same model is retrained on **shuffled labels**, eight times; the verdict is the comparison	An R^2 of 0.4 can arise by chance when the molecules are few. If the random twins reach the same score, the model has learnt nothing — and the panel **says so**, instead of showing only the good number
Deduplication on the canonical form	`OC(=O)C` and `CC(=O)O` are the same molecule	The same molecule in training and in test is an information leak, and the written string is not enough to see it

> **The verdict is written out in words, not left to the reader.** The QSAR panel does not stop at the metrics: it states whether the model beat all its random twins, and by how much. Measured on the 28-inhibitor example: R^2 0.861 against a maximum of 0.574 among the eight twins — margin 0.287. On the same molecules with random labels the model does **not** beat the twins, and the panel says so in the same way.

The results can be taken away

Two exports, both formed in memory by the browser: there is no network request, which is why the note beside the buttons can say that no data leaves the device without asking anyone to take it on trust.

Export	Content
CSV table	One row per molecule: name, canonical and original SMILES, activity, scaffold, nine descriptors, QED, Lipinski violations, Veber outcome, alerts found. With cross-validation active, **predicted** and **residual** are added — and only then, because cross-validation is the one scheme that gives every molecule a prediction from a model that had not seen it
SDF	The structures with **generated 2D coordinates**, the name on each block's first line and the data fields attached: it is the format in which sets are exchanged between groups and programs. Verified by round-trip
Method report (.md)	What is needed to **redo** the same analysis: RDKit version, fingerprint, split type, number of molecules and scaffolds, **the seed**, the metrics, the null-model verdict, the folds one by one, and the declared limits

> **Why the seed is in the report.** Without it, "random split" and "scrambled labels" mean nothing: two runs would give different numbers and nobody could say which was the right one. The generator is a seeded mulberry32, and the same seed gives the same folds.

A measurement no other bench could make

`verifica-accessibilita` walks the 194 sections and measures what is **visible**. The six panels of this section live inside a container that stays `display:none` until the analysis has been run: that bench never saw them, and its "contrast 0" was not speaking about them.

`test_cheminfo` runs the analysis for real, opens the six panels in turn and measures WCAG contrast on every piece of text that appears there — **460 elements, 0 defects**. The formulas are rewritten inside the bench rather than taken from the application: a bench that asked the code under test how good its own contrast was would be measuring nothing.

The engine lives in `bsi-cheminfo.js`, is exposed as `window.BSIChem` and is verified by the `test_cheminfo` bench (189 checks) against hand-computable values.

3.3 Organic chemistry

Retrosynthesis (98 exercises) · synthesis and mechanisms with electron arrows · animated mechanisms · structure editor · *Organic Chemistry 1+2* module with 25 modules.

3.4 Biochemistry

Amino acids · biomolecules · animated metabolic pathways · enzyme kinetics (Michaelis-Menten) · 3D macromolecules · interactive biological structures · biochemistry guide · exam biochemistry.

The **biochemistry guide** is a document of its own (`Biochimica_Guida_Definitiva.html`, 8.2 MB) loaded into a frame by the section: 12 chapters plus the index, 78 subsections, 98 embedded figures, and two chapters devoted to the exam — "Oral exam guide" and "Real exam questions and correct answers". It contains **no scripts and makes no network requests**: it cannot raise JavaScript errors, and the alternative text of the 98 figures comes from the captions the author wrote, not from invented descriptions (see `docs/06` §3-ter).

The document closes with a **self-test of 12 multiple-choice questions**, recovered from the previous guide — where it was a JavaScript quiz — and rewritten **without a line of code**: the four options are in plain sight, the solution sits in a native `<details>` that opens on a click. In print the `<details>` disappears and a dedicated copy of the solution appears, so the printed copy does not lose the explanations.

3.5 Pharmacology

Function	Description
3D Atlas	One chooses a body region, a disease and a drug: the model highlights the target organ
Database	**263 drugs** with structure, molecular weight, mechanism, indications, adverse effects, class. For **36** the entry carries the ChEMBL identifier and the record's formula, and the formula is rebuilt from the graph and compared
Interactions	Checking interactions between active ingredients
Pharmacokinetics	Calculator for half-life, clearance, volume of distribution

Function	Description
Clinical cases	Diagnosis-therapy paths for teaching purposes

3.6 Astrochemistry

Molecules of the interstellar medium · exoplanets and JWST data · nebulae · stellar spectra · nucleosynthesis · cometary chemistry · quizzes.

3.6-bis Spectrum reader

You load a file from your own instrument, or paste a text, and it is read.

Function	Description
Formats	**JCAMP-DX** (`.jdx` , `.dx`) with full **ASDF** compression — PMAI for digits, DIF for differences, DUP for repeats — or two columns of numbers separated by spaces, commas or tabs
Integrity check	The JCAMP format prescribes that, after a run of differences, the next value written in full must equal the computed one. That check is performed here
Peak finding	By **prominence**, with noise estimated from the median absolute deviation (MAD × 1.4826) rather than the standard deviation, which one tall peak inflates
IR assignment	For each band, **all** compatible assignments among 24 ranges are listed, not the first
Mass spectra	A peak list is recognised as such; differences between ions are matched against 17 **neutral losses**
NMR	Integration of areas by chemical-shift region
Drawing	IR with **decreasing** wavenumbers, the way it is read

What it does not do, stated in the section itself: it does not deduce the structure. A set of bands is compatible with many molecules, and a program that produced a name from three peaks would make a claim the data cannot support.

3.6-ter Languages and chemical languages

Function	Description
Interface language	**14 languages** with a selector and a search field: Italian, English, Spanish, French, German, Portuguese, Dutch, Polish, Romanian, Greek, Russian, Chinese, Japanese, Arabic. Each one's coverage is **measured**, not declared
Writing direction	Arabic carries `dir="rtl"` ; every other language `ltr` , including on the way back
Chemical languages	**25 outputs** from one SMILES: formula and mass composition, canonical/flat/explicit-hydrogen SMILES, CXSMILES, InChI and InChIKey, the key's skeleton block, Murcko scaffold, CIP stereo tags, SMARTS and CXSMARTS, molfile V2000 (also aromatic) and V3000, XYZ, PDB, CML, JSON, and the **six fingerprints** (Morgan, MACCS, RDKit, atom pairs, topological torsions, pattern)

Function	Description
IUPAC naming	Over a **declared class** (acyclic, neutral, C H O N F Cl Br I, one kind of principal group, linear substituents). Outside that class it **refuses** instead of guessing

3.7 Study tools

Function	Description
Spaced repetition	SM-2 algorithm on cards generated from the contents
Quizzes	By subject, with score and history
Study path	A guided sequence of topics
Notes	Global and per section
Pomodoro	Study timer
File Manager	Personal archive of photos, documents and HTML pages
Physics and mathematics study kits	Five sections per subject (`bsi-studio.js`): 🗺️ summaries with «mark as studied» and progress; ❓ quizzes by topic and level, with shuffled answers, review of mistakes and history; 🃏 flashcards with the five-box Leitner method; 📐 formula sheet; 📖 glossary. Physics: 69 summaries, 325 questions, 273 flashcards, 227 formulas, 273 entries. Mathematics: 66 summaries, 346 questions, 266 flashcards, 224 formulas, 279 entries
Resources and bibliography	For physics, mathematics and chemistry (`bsi-risorse-materie.js`): web resources by category, a bibliography by level with no years or ISBNs (search links are computed from title and author), citations from a DOI read live from Crossref in APA 7, IEEE, ISO 690 and BibTeX; without network, a form to fill in by hand
🔢 **Constants and units**, 📚 **Mathematical tables**	CODATA 2018 constants with the exact SI ones marked; a converter that checks dimensions. Tables of derivatives, integrals, series, identities, Laplace and Fourier transforms and constants: every numeric entry carries a proof function that the bench runs

3.8 The "Spectra" assistant

An AI agent with **47 tools** that can interrogate the application itself: compute descriptors, generate spectra, look up constants, balance reactions, open sections, read the open tab and act on its controls. It supports ten providers, and stays usable when one of them fails (§5.2).

Spectra Live (`bsi-spectra-live.js`) is its full-screen voice console: hands-free conversation with the browser's speech recognition and synthesis, local commands that work without network and without a key (open sections, switch subject, compute, differentiate, convert, look up constants and definitions, plot, quiz, move sliders, change tab), and the screen context added to the questions that go to the model. In the console Spectra has the **face** of the «Noir» android (model by Skyfox13, CC BY 4.0), with a mouth that follows the spoken text and 17 emotions; and a **mind** (`bsi-spectra-mente.js`) that locally remembers episodes and study time and keeps an internal state of five quantities. It is a simulation of behaviour, not consciousness: see `05-AI-Agent-Architecture.md` §4.10–4.12.

The console is the main entry point. Since v239 Spectra has **a single button**, the face in a circle, present on every screen in place of the «Spectra» pill: it opens the console, and from the console 💬 opens the chat. The circle shows a drawn face at once and, once the page has loaded, the portrait. On a full-screen phone it respects the margins of the status bar and of Android's keys (at least 34 and 46 px when the system does not report them) and lays out face, state, dialogue and controls without overlaps. The 🧭 navigator lists every section per subject, the quick tools and the recognised phrases; the context shortcuts (Explain this section, Guide me, What am I looking at?, Quiz me, Review, and since v239 🎓 Lesson) change with the screen; 💬 opens the chat. «Give me a lesson» («fammi lezione») starts the guided lesson of the open section. The **difficulty** of a question starts from the study level of the open section in the paths (high school, bachelor, master, PhD) and grows with advanced-course words, symbols and length; it becomes the face's **concentration** (eyes half closed and brighter). The face (v3.0) has eyelids, an eye tint per emotion, muscle springs, micro-expressions and sleep after two minutes without contact. With the console silent for more than a minute Spectra may voice an important thought from the mind unprompted.

The creator (`bsi-spectra-creatore.js`). Spectra knows it was created by Samuele Pio Provenzano. Recognition requires the File Manager's secret phrase, compared as a SHA-256 fingerprint and never stored; once recognised, the author is greeted by name and the model knows it. It is a deterrent, not an access control (§6).

The quality voice (`bsi-spectra-voce.js`, v239). With an OpenAI key saved in the app, Spectra speaks in the console, in the guided lesson and in Spectra Studio with a deep, metallic voice: the text goes to OpenAI (`gpt-4o-mini-tts`), the audio comes back and goes through an effects chain, and the mouth follows the audio. Without a key the browser's speech synthesis remains. The spoken text leaves the device only in this case (`03-Security-Privacy-Compliance.md` §1.2).

The body (`bsi-spectra-corpo.js`, v239). A generic driver for humanoid avatars (bones, ARKit blendshapes, black hair, gaze, gestures, walking, sitting), used by `bsi-spectra-figura.js` (v240) in Spectra Studio and in the AI sections. The body chosen by the author (the «Bounty Hunter», free on CGTrader, Royalty Free licence) is, since v241, **always there, for everyone, without loading it**: it is embedded in the app as a packed block (`models/spectra-corpo.bsi`) that only the app's code reopens, because the licence allows using the model inside the product but not offering it as a standalone `.glb` file. While it arrives «Spectra sta arrivando…» («Spectra is on the way…») appears; if there is no network at the first start, «Spectra non è arrivato: serve la rete la prima volta» with «↻ Riprova»; after the first load it stays in the cache and works offline. The «🧍 Corpo» button and the invitation to load the body are gone; no substitute (§6).

Spectra with the body in the AI sections (`bsi-spectra-figura.js`, v240). In Spectra AI and in the three areas with the assistant, under the title, a lab-style frame shows Spectra standing: it thinks while the model works, explains with its hands and lip sync when the answer arrives, speaks with Spectra Live and explains the working when the local assistant solves an exercise; buttons «🎙️ Parla con me», «🧠 Spiega», «🎓 Studio». One figure for the whole app. Spectra's icon is the face only. The buttons sit in a bar under the scene.

Spectra with the body in the Spectra Live console (`bsi-spectra-figura.js`, v242). The author's request, looking at the console: «it still comes out like this (the bust)». While the console is open, the face-only bust is replaced by Spectra with the body and the hologram (the same figure, moved into the console, between the telemetry and the dialogue; on a large screen on the right). The console's state drives the figure: while listening it is curious, while working it is concentrated with its hand on its chin, then it speaks. With the console closed, the figure returns to the section it came from.

The hologram (`bsi-spectra-figura.js`, v240). To Spectra's right (the viewer's right) a holographic screen shows live scenes, computed rather than recorded: four planets on Kepler orbits (Kepler's equation solved every frame, faster at perihelion), benzene with its π cloud, an IR spectrum tracing itself, the surface $z = \sin(r - t)/r$, the 3d_z² orbital rotating, a Schrödinger wave packet spreading; or the section open behind the Spectra window, live. The scene follows the subject of the question and the answer (orbits → planets, spectrum → IR, orbital → 3d_z²…); with «📺 Includi lo schermo» it shows the open section; at rest the scenes alternate every 14 s. The label at top right («🪐 Orbite di Keplero · apri ›»…) opens the matching section with one tap **without removing Spectra's figure**: the AI section is not left; the Spectra window opens on «🧠 Spiega» (if it is not open already) with the section behind it, so Spectra stays visible in front and the section is seen live in the hologram. When the window is closed, if Spectra opened other sections (the catalogue, «🎞️ Mostrami tutto», a question naming a topic, the label), the app returns to the AI section the user started from, where the figure is, and the tour stops. When answering, Spectra first points at the hologram and then explains with the hands. **Every section is showable**: a section without graphs (text, tables, controls) is projected as a card drawn on the hologram (title, open tab, up to six controls with their values, visible text) instead of falling back to the planets.

«🧠 Spiega» (`bsi-spectra-spiega.js`, v240). In the Spectra window the **«🧠 Spiega»** tab is the second one, after «💬 Chat» (then Esame Orale, Ripassa Oggi, Genera Guida). It gives an **exhaustive explanation** of any chemistry, physics or mathematics question, at one of five levels (High school, Bachelor, Master, PhD, Advanced PhD · frontier), without simplifying below the chosen level and without leaving out anything relevant. The answer has a mandatory structure: in short; what you are looking at (with the screen: the quantitative reading of the graph — for an NMR spectrum every signal with δ, multiplicity, J, integral and assignment, for IR, MS and UV-Vis bands, fragments and transitions; for a 3D scene geometry, symmetry, nodes, critical points and isosurfaces; for a mathematical graph domain, asymptotes, extrema, inflections, convergence); theoretical foundations; formalism and full derivation; computed numerical examples; limiting cases, approximations and common errors; links between the three subjects; applications and research frontier; three self-check questions with answers. With **«📺 Includi lo schermo»** the explanation is about what is open: the model receives the section's text (controls with values, tables, text) and an image of the graphs and 3D scenes. Quick prompts: «📈 Spiegami questo grafico», «🧪 Lo spettro, picco per picco», «🧊 La scena 3D», «∴ Dimostrazione completa», «🔭 Frontiera della ricerca»; «↺ Nuova spiegazione» starts over. A configured AI provider is needed: without one, the tab says so and shows what the app knows on its own (the local assistant for nu-

meric exercises, the chapters of the section's guided lesson, the screen data). Spectra with the body moves into the tab, thinks while the model writes and then explains with voice and gestures.

The catalogue: every section in the hologram (`bsi-spectra-spiega.js`, v240). The author's request: «ogni sezione della chimica, della fisica e della matematica deve esserci, non deve mancare nulla in questo ologramma: deve esserci tutto spiegabile e mostrabile» (every section of chemistry, physics and mathematics must be there, nothing may be missing from this hologram: everything must be explainable and showable). The catalogue is not written by hand: it is read from the app bar (every section button, with its group — Chimica, Spettroscopia, Biochimica, Farmacologia, Chimica Fisica, Studio, Utility, Fisica, Matematica, Astrofisica — and its subject): 201 sections (chemistry 78, physics 63, mathematics 31, shared 29, since v242); a new section appears by itself. In the «🧠 Spiega» tab the collapsible card «🗂️ **Tutte le sezioni nell'ologramma**» lists them by group, with a search box and a subject filter; one tap opens the section behind the Spectra window (switching subject if needed), the hologram shows it live and «📺 Includi lo schermo» is checked, so the explanation uses its data. «🎞️ **Mostrami tutto**» runs every section of the chosen subject (or all of them) through the hologram, one every 7 s, with the counter «n/N · name»; it can be stopped. **A question that names a topic finds its section** (the meaningful words of the section names looked for in the question, also by stem; the acronyms MS, IR, UV, NMR, MO… weigh more; synonyms such as «spettro di massa» → MS, «infrarosso» → IR, «risonanza magnetica» → NMR, «orbitali molecolari» → MO; ties go to the shorter name): asking in «🧠 Spiega» without the screen included, the named section opens behind the window, is shown in the hologram and the explanation includes its screen data. **A box unchecked by the user is respected**: if the user unchecked «📺 Includi lo schermo», the section named or chosen from the catalogue still opens and is shown, but the box stays unchecked and the model receives only the question, with no screen text or image; the quick prompts about the screen check it again (explicit choice). Examples: titolazioni → Titolazione, Schrödinger → the Schrödinger equation, integrali per parti → «Derivate, integrali e serie», spettro di massa → Spettro MS, spettro infrarosso → IR Visivi, meccanica orbitale → 3D orbital mechanics; «ciao come stai» → none.

The interactive hologram, «⛶ Interagisci» (`bsi-spectra-olointerattivo.js`, v242). The author's request: select the hologram and interact with it, with every step explained aloud by Spectra and in writing. The **«⛶ Interagisci»** (interact) button sits on the hologram wherever the figure is (AI sections, Explore sections, the «🧠 Spiega» window, the Spectra Live console) and opens a full-screen window with Spectra with the body, the big hologram and the steps panel; the app's floating buttons hide. The mode starts from what the hologram was showing:

Mode	What you do	What the steps explain
🧪 Spectra	IR, ^{1}H-NMR (field 60–600 MHz, with the selected multiplet zoomed in), ^{13}C-NMR, mass (EI), UV-Vis; one of six reference molecules (acetone, ethanol, acetic acid, benzene, toluene, ethyl acetate) or a	For each signal position, shape, assignment and why (group frequencies, chemical shifts, neutral losses; a narrow band is

Mode	What you do	What the steps explain
	written trace («1715 s, 2950 m, 3300 br», «2.17 s 6H; 1.2 t 3H J=7», «58 30, 43 100», «254 200»), redrawn while you type	never a carboxylic O–H); for molecules formula, molar mass, degree of unsaturation
🪐 Universe	The eight planets in 3D (NASA data, Kepler orbits, distances compressed as √a); tap a planet, or write an orbit («a=2 e=0,6»)	Kepler's third law, perihelion and aphelion, vis-viva speeds and the (1+e)/(1−e) ratio, surface gravity, escape velocity
📐 Mathematics, ⚛️ Physics and chemistry	Write the question; the app's own assistant (`bsi-assistente.js`, `bsi-mat-cas.js`) solves it and the graph follows: function and derivative with zeros and stationary points; projectile trajectory; swinging pendulum	The steps computed by the assistant, not written by a model

«🔊 **Spectra spiega i passaggi**» reads them one by one, highlighting each in the text and on the hologram; tapping a peak, a planet or a step jumps there and Spectra says it; «🧠 **Approfondisci**» opens «🧠 Spiega» with the hologram's data. With the window closed, Spectra returns to the section. The reference spectra are typical literature values for teaching, not measured spectra; from a trace the interpreter proposes candidate assignments taken from tables, it does not determine a structure.

⚡ **Explore with Spectra** (`bsi-spectra-esplora.js`, v242). The author's request: «there is no section for each subject like every-joule». Three sections, one per subject, in their subject's bar right after the «Area … · assistente» button: «⚡ **Esplora la chimica con Spectra**», «⚡ **Esplora la fisica con Spectra**», «⚡ **Esplora la matematica con Spectra**». Each is a page to scroll, in the style of explorable lessons: Spectra with the body stays on stage at the top while scrolling (right under the app header; shorter on a phone) and the hologram changes with the chapter in front. Ten chapters per subject, from high school to advanced PhD · frontier:

Subject	Chapters
Chemistry	the atom and ionisation energy · the shape of molecules (VSEPR) · the pH of a very dilute strong acid · buffers and Henderson–Hasselbalch · first-order kinetics · IR spectroscopy · ^{1}H-NMR and spin–spin coupling · molecular orbitals: why O_2 is magnetic · Marcus theory and the inverted region · the frontier: the delocalisation error of DFT
Physics	falling bodies · projectile motion · Kepler's laws · the pendulum · Carnot efficiency · two-slit interference · the particle in a box · tunnelling · time dilation and muons · the frontier: gravitational waves (GW150914)
Mathematics	the derivative as a slope · the number e · the integral and the logarithm · the Basel series · the harmonic series · eigenvalues: the rotation of the plane · Fourier series and the Gibbs phenomenon · complex analysis and residues · the heat equation · the frontier: the Riemann hypothesis

Each chapter: the phenomenon in a few exact lines; 🔮 **Prevedi** (predict), four answers including the most common misconceptions, then the reveal (right or wrong, and why each answer is right or wrong) and Spectra reacts (joy or surprise); 🧪 **Prova** (try), in 19 chapters out of 30, a slider that recomputes the real quantity (pH with water autoprotolysis,

Henderson–Hasselbalch, t½, Marcus ΔG‡ and relative rate, fall time, range, pendulum period, Carnot, the box's E_1, tunnelling transmission, muon γ and mean path, derivative, $(1+1/n)^n$, ln b, Basel sums with the tail, H_N against ln N + γ, rotation eigenvalues, the computed Gibbs maximum, the decay of a heat mode); «📺 **Guardalo dal vivo»** (watch it live) opens the linked section behind the Spectra window («🧠 Spiega» tab) and the hologram shows it live, and closing the window returns to the Explore section; «🧠 **Spiegamelo tutto»** (explain it all) asks «🧠 Spiega» for the explanation at the chapter's level, with the section's screen (an AI provider is needed); «🔊 **Ascolta Spectra»** reads the chapter with the Spectra Live voice. While scrolling, the chapter crossing the middle of the free space under Spectra becomes «active» (even when taller than the screen): Spectra tells it, with lip-sync and gestures, and the hologram takes its scene. Predictions stay on the device (`bsi-esplora-v1`) with the «n/10» progress bar, and are erased by «Cancella dati → Memoria persistente».

Body, face and mind are a simulation of behaviour, not consciousness.

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4. Main flows

4.1 Analysing a molecule

```
1. Open the Spectroscopy centre
2. Enter the molecule:
     · SMILES                → used directly
     · name or formula       → resolution via PubChem (needs network)
     · drawing in the editor → SMILES generated
3. The application shows: 2D structure, functional groups, descriptors,
   predicted spectra (IR, NMR, MS, UV-Vis), drug-likeness
4. The 3D structure loads if there is a network (conformer from PubChem)
```

With no network: point 3 stays complete for SMILES entered directly; resolution by name and the 3D structure are unavailable, and the application says so instead of hanging.

4.2 Asking the assistant

```
1. Tap Spectra's face (the circle at the bottom right), then 💬
   for the chat
2. On first use: choose a provider and enter an API key,
   or connect a proxy (and then no key is needed)
3. Type the question
4. The assistant answers, using its tools when needed:
   every tool used is shown, not hidden
```

If the provider does not answer: the assistant notes it, demotes it in the selection for 24 hours and moves to another configured provider. The 🔌 **Test** button measures in a few seconds which providers the device actually reaches, **without needing any key**.

4.3 Studying with spaced repetition

1. Open the study section
2. Generate a deck from the contents
3. For each card, rate how well it was remembered
4. The SM-2 algorithm computes when to show it again
5. On reopening, only the cards that are due appear

4.4 Archiving your own material

1. File Manager → local password
2. Upload photos, documents or HTML pages
3. The material stays ON THE DEVICE (IndexedDB), it is not transmitted

4.5 Checking for updates

1. ✨ menu → 🔄 Updates → «Check for updates»
2. The app asks the active service worker which version it is serving (BSI_VERSIONE message) and downloads sw.js from the same site, with no cache; if that fails, it tries the copy on GitHub
3. It always answers, in one of three ways:
   - same version → «You already have the latest published version», with the number
   - new version → announces it with both versions and offers «Update now»
   - site unreachable → says so; the app keeps working offline

4.6 Following a guided lesson

1. Open a section and tap «🎓 Guided lesson with Spectra» (or say or type «give me a lesson» to Spectra)
2. The lesson advances by itself, chapter by chapter: it opens the tabs, highlights the charts, moves the sliders and selects and puts them back
3. ◄ ► move one step, pause stops the advance, 🔊/🔇 the voice, × or Esc close
4. At the check question the lesson waits for the answer, then shows the explanation
5. The summary lists the chapters and offers «Replay the lesson» and «Ask Spectra»

4.7 Learning in Spectra Studio

1. Spectra's console → 🧭 → «Spectra Studio» (at the top)
2. Choose a lesson: 🪂 The fall, 💧 The shape of water, ∑ The Basel problem
3. PREDICT — choose an answer; the prediction is recorded
4. TEST — the simulation runs on the holographic bench, with measurements
5. PROVE — the derivation, one step at a time (◄ and «Passo successivo ►»)
6. REVEAL — why each option is right or wrong, the connections, exercises on four levels (one recommended)
7. Spectra speaks, looks at the scene, points and gestures; dragging turns the camera. Without WebGL the panel with all the text remains

4.8 Exploring a subject with Spectra (v242)

1. Subject bar → «⚡ Esplora la chimica / la fisica / la matematica con Spectra» (right after the area with the assistant)
2. Scroll: Spectra stays at the top, the chapter in front becomes active, Spectra tells it and the hologram takes its scene
3. PREDICT — tap one of the four answers; the reveal explains each one, Spectra reacts, the «n/10» bar advances
4. TRY — move the slider: the quantity is recomputed
5. «📺 Guardalo dal vivo» — the linked section behind the Spectra window, live on the hologram; closing returns here
6. «🧠 Spiegamelo tutto» — the full explanation at the chapter's level; «🔊 Ascolta Spectra» — the chapter read aloud

4.9 Interacting with the hologram (v242)

1. On Spectra's hologram → «⛶ Interagisci»: full-screen window
2. Choose the mode: 🧪 Spectra · 🪐 Universe · 📐 Mathematics · ⚛️ Physics and chemistry
3. Change the spectrum type or molecule, write a trace, tap a planet or write the question: the hologram is redrawn and the steps redone
4. «🔊 Spectra spiega i passaggi» — one by one, highlighted; tapping a peak, a planet or a step jumps there
5. «🧠 Approfondisci» — «🧠 Spiega» with the hologram's data
6. With the window closed, Spectra returns to the section

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5. Business rules

5.1 Scientific data

Rule	Rationale
A molecular structure enters the database only if the recomputed weight agrees with the literature one (±0.6 u)	A plausible but wrong datum is indistinguishable from a correct one for the user
If a structure does not pass verification, the entry stays **without a structure**	A wrong datum replaced by another wrong one is no progress
Every absence must be **recorded with its reason**	An unjustified exception becomes a shortcut
Spectra are deterministic	Two spectra of the same molecule must be comparable
An ambiguous physical constant is not resolved at random	A wrong value given with confidence is worse than a request for clarification

5.2 AI assistant

Rule	Behaviour
A provider that does not answer is remembered for **24 hours**	Then it is retried: today's failure is not a sentence

Rule	Behaviour
A retired model is learned from the error message	Hand-coded lists grow stale
An imposed wait longer than **10 seconds** switches provider, if a ready one exists	Beyond that threshold the user believes the app has frozen
One turn at a time	Two overlapping submissions corrupt the history
Paid providers are not offered as free ones	Without active billing one hits the limit at the first question

5.3 User data

Rule	Behaviour
At most 30 conversations, 100 messages each	Beyond that, `localStorage` fills up and **every** subsequent save fails
Images up to 20 MB	Beyond that, explicit refusal; the others proceed
Selective deletion by group	Wiping everything to free space is an avoidable loss
A failed write **must** be declared	"✅ Saved" when it is not true is worse than an error

6. Current limitations

Declared, not hidden. The detail is in `09-Release-Conformance-Statement.md` §4.

#	Limitation	Practical consequence
1	**2 entries with no structure to show** (it was 6)	Ivermectin is a mixture of B1a/B1b homologues and Coartem a combination of two active ingredients: a single SMILES notation does not represent them, and for these two the 2D/3D and predicted spectra are missing. Digoxin, vincristine and tacrolimus (topical and systemic) now have their structure, taken from ChEMBL and verified against the molecular weight
2	**Predicted, not measured, spectra**	Useful for recognising functional groups; not for formal identification
3	**No synchronisation**	Data do not pass from one device to another, and there are no copies on a server
4	**File Manager: deterrent, not security**	On a static site, whoever reads the source bypasses any page-side control. The source carries only the SHA-256 digest of the password, never the password
5	**Functions that need a network**	IUPAC name, CAS, GHS, 3D conformers, AI assistant

#	Limitation	Practical consequence
6	**Accessibility automated, not complete**	Contrast, accessible names, labels, heading hierarchy across 13 pages and all 194 sections: **0 defects**, including text over gradients (judged against the worst stop). What stays outside is text inside SVGs and text over a real background image, counted on every run, and everything requiring human judgement
7	**Chromatography: no retention prediction**	The section computes resolution, efficiency and indices **given** k, α and N; it does not predict k from a structure, which would require experimental parameters of the stationary phase
8	**Partial code coverage**	Measured: **49.89 %** of statements over the widest path a bench walks. It is not whole-battery coverage, and it is statements, not branches; see `08-Traceability-Matrix.md` §8
9	**Verification on Chromium only**	Firefox and WebKit are tested by hand, not by a bench
10	**Spectra's mind is a simulation**	Memory and internal state are data and rules in the browser, not understanding or feelings. The mind's summary and the screen context go to the provider with the question, when the question goes to the model
11	**Some astrophysics data are illustrative**	The exoplanet population is not a catalogue; the rotation-curve «observations» are a model with noise; the CMB map is synthesised from an illustrative power spectrum. The sections say so
12	**Spectra Live and the quick tools have no bench**	Since v238 `test_spectra238` tests layout, navigator, level, face and creator commands, since v239 also the single button, and `test_lezione` the guided lesson and «fammi lezione»; voice, the other local commands, the tools that act on the page and the per-subject ✨ menu are tested by hand (`08-Traceability-Matrix.md` §6, UI-21 and AI-13)
13	**Spectra's face weighs about 8 MB**	`models/spectra-noir.glb` is not in the service worker's precache: it is downloaded the first time the app derives Spectra's icon (a few seconds after the first start, if WebGL is available) or opens the console. If it does not arrive, the console uses the fallback face
14	**Creator recognition is not an access control**	The SHA-256 fingerprint is in the code and the recognition mark (`bsi-spectra-creatore-v1`) is in `localStorage` : whoever holds the device can write it by hand. Like the File Manager password, it is a deterrent and protects no data
15	**The guided lesson comes from the page, not from an author**	The lesson reads the section's explanations, canvases and controls: where a tab has little text, the chapter says little. The check question is chosen by words in common with the section and may be only close to the topic; 115 sections out of 194 have none. The voice is not tested by the bench. The model it is inspired by (Every Joule's interactive lesson) was not reachable during development: the lesson is an interpretation of it
16	**Spectra Studio is a first prototype; the body needs the network at first start**	Three lessons only. The body chosen by the author (CGTrader Royalty Free licence) is embedded in the app as a packed block (`models/spectra-corpo.bsi` , 4.40 MB) and is downloaded the first time: with no network at the first start «Spectra non è arrivato: serve la rete la prima volta» appears with «↻ Riprova»; no substitute. The packing is not absolute protection: the published file is not a reusable `.glb` , but someone reading

#	Limitation	Practical consequence
		the code can rebuild it. Black hair is guaranteed by the driver but tested only on the mannequin of its bench. The studio's interface texts are in Italian
17	**With an OpenAI key and the voice on, the spoken text goes to OpenAI**	The quality voice sends OpenAI the text Spectra speaks, even when the chat uses another provider and without going through the proxy. It is avoided by turning the voice off (🔇) in the console, in the guided lesson or in Spectra Studio, where the choice is remembered, or by not saving an OpenAI key

7. Possible developments

Not commitments: directions consistent with the architecture.

Area	Proposal	Why
Data	The 2 remaining entries have no structure to show	It is not a gap to close: ivermectin is a mixture, Coartem a combination. At most one could show the **components**, declared as such
Spectra	Comparison with experimental databases (SDBS/NIST) inside the app	It would make the distance between prediction and measurement visible
Accessibility	Add `axe-core` alongside, for the rules the bench does not implement	ARIA roles, tab order and focus management stay outside the current measurement
Export	Full export of the user's data	The only possible remedy for the absence of backups
Assistant	A preconfigured proxy for those who do not want to publish one	It would remove the last barrier to AI access

Document updated to version `bsi-v250`.

A guide for the people who use it — BioSpecInfo

This guide is written for people who study, not for people who program. You do not need to know anything about computing.

What it is

BioSpecInfo is an application for studying chemistry, biochemistry, pharmacology and astrochemistry. It opens like a website, but you can **install** it on your phone or computer and it works **without internet** too.

Three things to know straight away:

- **No sign-up needed.** No account, no email, no password.
- **It costs nothing.**
- **Your data stay yours.** Notes, files and conversations never leave your

device: there is no server they could reach.

👉 **samupropio1-ship-it.github.io/BioSpecInfo-v11**

Installing it (optional, but worth it)

Installing gives you an icon on your screen, full-screen opening and offline operation.

Device	How to do it
Android / Chrome	⋮ menu → *Install app* (or *Add to Home screen*)
iPhone / iPad	Share button → *Add to Home Screen*
Computer	Install icon in the address bar, on the right

After installing, open the app once with the network on: it will download everything it needs. From then on it works with no signal too.

Finding your way around

Element	Where it is	What it is for
🔍 **Search**	At the top	Searches the whole app. Shortcut: `Ctrl+K`
Section menu	Navigation bar	The 201 sections, grouped by subject
✨	Bottom right	Tools, settings and updates

Element	Where it is	What it is for
Spectra	Bottom right, the face in a circle	The assistant that answers questions: a tap opens the console, and from there 💬 the chat
🎓 Guided lesson	Under the title of every section	A lesson in chapters on the section, led by Spectra

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The things you will do most often

Analysing a molecule

1. Open the **Spectroscopy centre**
2. Type the molecule into the box. You can use:

- the **name** — `aspirin`, `caffeine`
- the **SMILES formula** — `CC(=O)Oc1ccccc1C(=O)O`
- or **draw it** with the editor

1. Press Enter

You get the 2D and 3D structure, the functional groups, the molecular weight, the predicted spectra (IR, NMR, mass, UV-Vis) and the pharmacological properties.

> **If you have no network:** searching by *name* does not work, because it has to consult an external database. The SMILES formula, on the other hand, always works.

Reading an IR spectrum

The spectrum is drawn as on a real instrument: the scale at the bottom runs from 4000 to 400 cm⁻¹ (**from left to right the numbers go down**) and the bands point **downwards**.

The coloured area below 1500 cm⁻¹ is the **fingerprint region**: that is where the specific molecule is recognised, not just its class. Below the chart you will find the list of bands with their assignment.

> The spectra are **predicted** from the structure, not measured in a laboratory. They are there to help you recognise functional groups and to understand the link between structure and spectrum. For experimental data, the app sends you to the official databases.

Everything about spectra, in one place

What you do with a spectrum used to live in **nine sections** across two different menus. Now there is 🔬 **Spectra centre**, a single section with ten tabs:

tab	what it is for
Read your spectrum	a file, a photo, a PDF
Predict NMR	^{1}H and ^{13}C assigned per atom
2D NMR and NOE	COSY, HSQC, HMBC
Mass	fragmentation and neutral losses
IR	bands and functional groups
UV-Vis	chromophores and λmax
Reference tables	^{1}H, ^{13}C, IR, J constants
Everything on one molecule	editor, spectra, groups, comparison
Elucidate a structure	from the data to a proposal
Practise	recognise the group from the spectrum

The panels are the same ones as before: they were **moved**, not copied — there is no second copy of anything. The old links still work: if something in the application used to take you to "Mass spectrum", it now opens the Spectra centre on the **Mass** tab.

> **Two tables of J constants, with different numbers.** There were two: one gave the aromatic ortho ^{3}J as 6–9 Hz, the other as 7–10. Whoever looked that number up got a different answer depending on where they came in. **One** remains, the one inside the Spectra centre, which sits in the section with its own reference list.

Changing language: the round button, top right

At the title's height, at the far right, there is a **round button** with the flag of the language in use. It opens into a **disc**: the fourteen flags on two shells — six inside, like a benzene ring, and eight outside — and at the centre appears the name of the language you are moving to and **how much it is translated**. That percentage is not hand-written: it is measured by applying the dictionary and counting what stays in Italian.

Touch a flag and the interface changes at once. `Esc` closes the disc.

> The 🌍 **Language** section remains, and does something else: it explains the languages, shows each one's coverage and says that Arabic carries the writing direction with it. That one **informs**, this one **acts**.

⚛️ Physics: eight computed sections

In the ⚛️ **Physics** category of the bar there are eight sections where the physics underneath chemistry is **computed**, not narrated. Every number you see is compared by the bench with an independent literature value.

section	what you do
🌊 **Schrödinger**	pick a potential (well, harmonic oscillator, Morse, double well) and see levels and wavefunctions with the exact solution beside them; launch a wave packet at a barrier (tunnelling); see the vibrations of HCl, H_2, CO and N_2 in cm^{-1}
🧮 **Hartree-Fock**	a real ab initio calculation (RHF/STO-3G) on eight molecules: energy, orbitals with their energies, Mulliken charges, a map of the chosen orbital, the dissociation curve
🎻 **Rovibrational**	the IR spectrum of a gaseous diatomic with its P and R branches: change temperature, resolution, isotope; rotational Raman of N_2, O_2, H_2, CO_2; from the lines to the bond length
🎲 **Statistical thermodynamics**	entropy and heat capacity of nine gases computed from molecular constants, contribution by contribution, and rotational and vibrational populations
🎱 **Molecular dynamics**	live Lennard-Jones particles: heat, compress and watch gas, liquid and crystal, with the speed distribution and g(r)
∑ **Mathematics for chemistry**	from FID to spectrum by Fourier transform; Hückel on twelve π systems; kinetics by Runge-Kutta; the Arrhenius line with errors from your data
💎 **Crystallography**	the 14 Bravais lattices to rotate, Miller planes, the powder pattern of eleven crystals with absences
☢️ **Radiochemistry**	decay chains (the hospitals' technetium generator), dating with ^{14}C, K-Ar and U-Pb, nuclear binding energy

📐 Mathematics: from basics to advanced

The 📐 **Mathematics** category holds eight sections that run on a real computer-algebra engine: type expressions as you would by hand (`2x² + sin(3x)`, `√(x+1)`, `e^(-k*t)`, `ln(x)/x`) and the results are **exact** — 1/3 stays 1/3. Antiderivatives appear only if differentiating them gives the original function back.

section	what you do
🔢 **Mathematics basics**	count significant figures and apply the measurement rules; logarithms and pH; solve equations with the working, from linear to polynomials (complex roots too) and the fractional equations of equilibrium (weak acid); practise with ever-new marked exercises
📈 **Function analysis**	type f(x): domain, symmetry, zeros, exact derivatives, maxima, minima, inflections, vertical, horizontal and oblique asymptotes, a moving tangent; and function families with a·f(b(x−c))+d
∫ **Derivatives, integrals and series**	the derivative with the list of rules used; the antiderivative and the definite integral with Riemann sums; the degree-n Taylor polynomial; left and right limits
🔲 **Linear algebra**	systems, determinant, inverse, rank, null space, eigenvalues, in exact fractions with every Gauss-Jordan step; plane transformations; balancing a reaction (redox with charges too) as the null space of the composition matrix
🌀 **Differential equations**	the direction field of $y' = f(x, y)$ with solutions starting where you tap; Euler, Heun and RK4 compared; the phase plane of predator-prey, pendulum, Van der Pol, Brusselator, SIR, with classified equilibria

section	what you do
🗻 **Functions of several variables**	3D surfaces to rotate and contour maps, gradient, Hessian, minima, maxima and saddles (the transition state), gradient descent; state functions, the Maxwell relation, exact differentials
ℂ **Complex numbers and Fourier**	complex numbers on the Argand plane, n-th roots, Euler's formula, Fourier series with the Gibbs phenomenon
📏 **Laboratory statistics**	mean, standard deviation, confidence interval; Grubbs and Dixon for outliers; t and F tests with p-values; uncertainty propagation; calibration line with LOD, LOQ and the sample uncertainty

🎓 UniBA exams: Chemistry, Mathematics and Physics, from basic to expert

🎓 **Chemistry exam** (🔬 Chemistry category) has 40 standard problems on four levels: basic (moles, stoichiometry, gases, solutions, thermochemistry, kinetics, equilibria, acids and bases, solubility, electrochemistry, the atom), intermediate (analytical: titrations, Beer-Lambert, gravimetry, EDTA, polyprotic acids), advanced (physical chemistry: van ’t Hoff, Clausius-Clapeyron, polyenes, rotor and oscillator, Michaelis-Menten) and expert (Sackur-Tetrode, Debye-Hückel, Pourbaix). Reactions balance themselves. In Mathematics and Physics level **A** goes beyond the two courses: eigenvalues, ODE systems, Fourier, Laplace, triple integrals; waves, modern and quantum physics.

🎓 **Mathematics exam** (📐 category) and 🎓 **Physics exam** (⚛️ category) follow the syllabi of the Chemistry degree at the University of Bari. Each has the **syllabus** explained topic by topic with the method; the **solver**, which writes the step-by-step working with your data; **solved papers** in the style of the written exam (built on the syllabi: not the official papers). Mathematics also has a timed **exam simulation**. ⚛️ Physics also gains ⚡ **Electrodes and transition state** and 🌟 **Photochemistry, surfaces and polymers**.

🧪 ⚛️ 📐 Switching subject

Above **Spectra** and ✨ there are two round buttons: the two subjects you are not in — the **flask** for chemistry, the **atom** for physics, the **Σ** for mathematics. Tapping one, the bar shows only that subject's groups (plus 📚 Study and 🛠️ Utility, which serve all) and opens its Area. The choice is kept when you reopen the app; if you open a section of another subject, for instance from search or from Spectra, the bar follows by itself.

The subject does not change only the bar: it changes the whole app. In 📚 **Study** and 🛠️ **Utility** the common entries stay and the subject's own appear (for physics, for instance, the physics summaries, quizzes and constants); Spectra gets the subject's instructions, suggestions and greeting; in physics and mathematics the ✨ menu shows the **quick tools**, which open in a window above the page without losing your place:

Tool	What it does
🧮 **Calculator**	Exact computation: fractions and roots stay as they are, then the decimal

Tool	What it does
∂ **Derivative** and ∫ **antiderivative**	With the app's symbolic engine
📈 **Graph**	One or more functions of x
🔢 **Constant** and 🔁 **units**	CODATA constants and unit conversion with a dimension check
📐 **Formula**, 📖 **glossary**, ⚡ **lightning quiz**	Taken from the subject's study kit

If a piece of data a tool needs is not loaded, the tool says so instead of making something up.

🔭 The Astrophysics group

The bar has an 🔭 **Astrophysics** group with twelve sections: 🔭 **Astrophysics** and 🪐 **Gravitation**, which used to be in ⚛️ Physics, and ten 3D sections — 🪐 Solar system 3D (pick a date and see where the planets are), 🛰️ Orbital mechanics 3D, 📐 Celestial mathematics, ⭐ Life of stars 3D, 🕳️ Black holes and general relativity, 〰️ Gravitational waves, 🌌 Galaxies 3D, 🌠 Cosmology 3D, 🔭 Planetarium 3D (the sky from Bari or a place you choose, at any time; a click on a star gives its coordinates and times) and 🌍 Exoplanets 3D. Scenes rotate by dragging and zoom with two fingers or the wheel. On a slow phone the 3D scenes may take a moment to open. Some data are illustrative and the section says so: the exoplanet population is not a catalogue, and the cosmic microwave background map is synthesised.

🌀 The new physics and mathematics laboratories

In ⚛️ **Physics**: 🌀 Spin, qubits and entanglement, 🎲 Statistical mechanics, ☢️ Nuclei and reactions, 🧱 Elasticity and materials, 🔌 Electronics. In 📐 **Mathematics**: 🌊 Partial differential equations, 🌀 Complex analysis, 🔗 Markov chains and stochastic processes, 🧩 SVD and applied linear algebra, 📡 Codes and elliptic curves. As in the other laboratories, the sliders sit next to the simulation and the note below explains what you see.

🗺️ Physics and mathematics study kits and resources

With physics or mathematics chosen, 📚 **Study** has five sections: 🗺️ **Summaries** (mark the ones you have studied and watch your progress), ❓ **Quiz** (choose topic, level and number of questions; keys 1–4 and Enter; at the end the score by topic and a review of mistakes), 🃏 **Flashcards** (five-box Leitner method: a card you miss goes back to the first box), 📐 **Formula sheet** (with «copy») and 📖 **Glossary**. 🛠️ **Utility** has 🌐 **Resources and bibliography** (links by category, books by level, and the **Citations** tab: paste a DOI and get the citation in APA 7, IEEE, ISO 690 or BibTeX), plus 🔢 **Constants and units** for physics and 📚 **Mathematical tables** for mathematics. Chemistry has its own 📚 Bibliography and citations section. Your progress stays on this device only.

📰 Articles for the three subjects

In 📰 **Articles** choose 🧪 Chemistry, ⚛️ Physics or 📐 Mathematics at the top. Each subject has the **latest articles** (eight topics) and its **prizes**: the Nobel for chemistry and physics, the **Abel Prize** and the **Fields Medal** for mathematics, which has no Nobel. Years, names and motivations arrive live from the Nobel Foundation and Wikidata; every prize has an explanation written for its topic.

🔐 The deeper topics

In 📐 **Mathematics**: Numbers and cryptography (in the RSA tab you type a message and watch it encrypted and decrypted), Graphs and optimisation, Sequences and series, Calculus of variations, Topology (solids rotate when dragged), Algebraic structures, Game theory (play tic-tac-toe against a perfect opponent, and Nim), Logic and computability (type a formula and get its truth table), Statistical inference (regression points can be dragged) and Non-Euclidean geometries. In ⚛️ **Physics**: Gravitation, Heat and diffusion (on the plate a tap moves the source), Ising model, Particle physics, Analytical mechanics, Lasers and quantum optics and Climate physics. In ⚗️ **Physical Chemistry**: Phase diagrams and Potential energy surface.

🎲 The explorations

In 📐 **Mathematics**: 🎲 **Probability in motion** and 🎞️ **Visual mathematics** (in the Fourier epicycles you can choose «draw it yourself» and trace a shape with your finger; in the fractals a tap zooms in). In ⚛️ **Physics**: ⚡ **Electromagnetism in motion** (charges can be dragged), 🔊 **Sound and acoustics** (the 🔊 Listen button plays the sound) and 💡 **Semiconductors**. In ⚗️ **Physical Chemistry**: 🌡️ **Distillation and mixtures** and 🔋 **Cyclic voltammetry**.

🔭 Astrophysics, relativity, fluids, vector calculus, complex kinetics

The 🔭 **Astrophysics** group has 🔭 **Astrophysics**; in ⚛️ **Physics** there are ⏱️ **Special relativity** and 🌊 **Fluids**; in 📐 **Mathematics ∇ Vector calculus**; in ⚗️ **Physical Chemistry** 🌀 **Complex kinetics**. Under every animation there is a bar: ⏸/▶ pauses and resumes, ↺ restarts, **×¼ … ×4** changes speed, on the right the elapsed time. Animations stop by themselves when they leave the screen; with «reduce motion» on in the system they start paused and run with ▶.

🎯 The laboratories in motion

In the ⚛️ **Physics** category there are 🎯 **Mechanics in motion**, 〰️ **Waves and optics**, 🔌 **Circuits** and 🔥 **Thermodynamics in motion**; in 📐 **Mathematics** there are 🧮 **Numerical methods** and 📐 **Analytic geometry and vectors**; in ⚗️ **Physical Chemistry** there is ⚖️ **Equilibrium and Le Chatelier**. Each tab has sliders on the left and the simulation on the right, with the numbers describing it under the controls. In the equilibrium simulator the buttons add or remove a species, change volume or temperature: concentrations move until Q equals K again, and the note says which way the reaction goes.

🧊 The sections' 3D tabs

Twenty-nine sections have, after their other tabs, a 3D tab built on the section's own model: for instance the potential energy surface with the trajectory running over it, crystal cells with Miller planes, field lines and the motion of a charge, the light cone, the Brillouin zone, the Ising spin cube, the particle detector, the tangent plane, the flux through a sphere, Lorenz chaos, the Riemann sphere, the soap film between two rings. They rotate and zoom like the other 3D scenes. Text and controls appear at once; the scene is built a moment later, and on a slow phone it may take a little while.

🧊 3D Laboratory, charts and molecular views

In the ⚛️ **Physics** category the 🧊 **3D Laboratory** shows surfaces, atomic orbitals, motion in space, crystal structures and field lines in three dimensions. Rotate by dragging with a finger or the mouse and zoom with two fingers or the wheel. The buttons at the top right start auto-rotation (⟳), go back to the initial view (⌖), save an image (📷) and switch to full screen (⛶). Moving a finger or the mouse over a chart in any section shows a crosshair with the value of every curve at that point. Molecular 3D views have the same bar, plus the ✦ button choosing quality: automatic, high (with ambient occlusion), medium or low for slow phones. If «reduce motion» is on in the operating system there are no animations.

⚗️ ⚛️ 📐 The three macro-areas and the assistant

At the head of the 🔬 **Chemistry**, ⚛️ **Physics** and 📐 **Mathematics** categories is the area of that subject. In the 🤖 **Assistant** tab, write the problem as you would write it to your teacher, for example «a car at 90 km/h brakes at 5 m/s²», «how many grams of NaCl for 250 mL of 0.5 M solution» or «y'' + 4y = sin(x), y(0)=0, y'(0)=1», and press **Solve and explain**. The answer says what was understood; the **Data used** table shows in green the data read from the text, in yellow those inferred and in red those assumed, to be checked. If the chosen model is not the right one, the **Did you mean…?** buttons offer the others, and **Edit the data in the solver** opens the problem with its fields. The 🗂️ **All tools** tab gathers the subject's sections, with search; 🧭 **Levels** orders them from high school to PhD.

🧭 From high school to PhD

In the 📚 **Study** category, 🧭 **From high school to PhD** tells you, for chemistry, physics and mathematics, what you should be able to do at each level and opens the right tools with one click. The **Physics exam** now covers astrophysics, nuclear and quantum physics (up to variational and perturbation methods); the **Chemistry exam** organic chemistry, spectroscopy, biochemistry and a PhD level (Marcus, isotope effect, tunnelling); the **Mathematics exam** residues and the heat equation.

Where things are

The sections live in the **navigation bar**, grouped by subject, with 🛠️ **Utility** among them: laboratory, GHS, concept map, molecule comparison, molecule search, atlas, resources, materials, notes, file manager, statistics, data science, pomodoro, lab calculator, **articles**, and

the two language ones.

The ✨ menu holds only what is *not* a section: the editors, the overlay labs, and actions such as exporting a figure, printing, sharing, updating. The rule is simple and always holds: **if something is in the bar, it is not also in the menu**. It used not to be, and the menu showed two "Data Science", two "File Manager", two "Articles" and two "Updates".

Two exceptions, and they have a reason:

- **Spectra** is not in the bar: it has its own dedicated button, the pulsing

one at the bottom. Three roads to the copilot were too many.

- The **spectrum reader** is not in the bar: it is a tab of the Spectra centre.

> **Two File Managers, and they are two different things.** In the bar, 📁 **File Manager** is for uploading and organising your own notes (PDF, images, Word). In the ✨ menu, 🔒 **Protected File Manager** is a separate page with the password, for the gallery and the custom sections. They had the same name and looked like a duplicate: now they are named for what they do.

The articles and the Nobel Prizes

📰 **Articles** (from the ✨ menu) has two tabs.

Latest articles shows what came out in the last two months, by topic: spectroscopy, synthesis, catalysis, drugs, materials, biochemistry, computational. It is not a hand-written list — that would be stale tomorrow: it comes from **Crossref**, the DOI registry, at the moment you open the tab. Each title leads to its **DOI**, which is a permanent address, and beside it there is a link to **read it in Italian**.

Nobel Prize in Chemistry lists every prize. Years, laureates and motivations come from the **Nobel Foundation**, live: they are not written inside the application. The **explanation** in Italian is — and it is matched to the *topic* of the official motivation, not to the year, so no attribution comes from here. For each prize you get the official page, the Foundation's scientific and popular material, the Italian translated view and an Italian Wikipedia search — all addresses **computed** from the year, not written.

> **If a prize has no explanation, I say so.** I would rather write "for this prize I have no explanation" and leave you the official motivation and the Foundation's material, which is the source, than make something up.

> **It needs the network.** Without it you do not see a stale list passed off as current: you see what I asked for, from whom, and why it did not arrive — with the address of the request, which you can open by hand.

Reading a spectrum you measured yourself

The **Spectrum reader** section does the opposite of the one above: it does not predict a spectrum from a structure, it reads one you already have.

There are **two buttons**, and they are not duplicates.

`📂 Open a file` filters nothing: whatever you have in your folder, open it. `📷 Photo or image of a spectrum` instead opens the **phone gallery** and also lets you take the picture there and then — on a phone these are two different system pickers, not two filters of the same one, and without the second there is no way to hand over a photograph.

A line under the buttons then **tells you what it decided the file is**:

What you open	What happens
a **PDF**, a **Word** file, an OpenDocument, a spreadsheet	it opens, shows you all of it and **works through** what it contains
a **photograph** (`.png` , `.jpg` , …)	both things: the **trace** is extracted and the image is also shown as a page, because it may be a photo of a spectrum or of a sheet
a **JCAMP-DX** (`.jdx` , `.dx`), compressed form included	it is a spectrum: the format itself says so
a **text** file or a `.csv`	decided **from the content**: if the numbers form a curve it is a spectrum, otherwise it is **problem text** and gets worked through

If you prefer, **paste** the data or the problem text into the box and press **Read and work it out**: the same reasoning applies.

> There used to be two cards, and the spectrum one allowed only `.jdx` and `.csv` . On a phone that filter did not narrow, it **hid**: anyone arriving with the PDF of an exam saw an empty folder. Now there is one door and it filters nothing.

If you have no file at hand, the three buttons **IR example**, **MS example** and **NMR example** load a sample spectrum, and **example: an exam problem** loads a complete problem.

Inside the section there is 📖 **How to use it — full guide**: seven short chapters covering the two buttons step by step, the calibration, what the program does by itself, what it cannot do, how to recognise a wrong result from the symptom you see, and how to take the numbers away.

A spectrum you only have as a figure

The trace is extracted from the **pixels**, and the chart appears at once. Then it must be **calibrated**, because the axis numbers are not in the image:

field	what you write in it
x at left / x at right	the values at the two edges of the plot — for an IR typically 4000 and 400 **in that order**, because an IR is written with decreasing wavenumbers
y at bottom / y at top	the values of the vertical axis; for a figure in %T, 0 at the bottom and 100 at the top

field	what you write in it
y unit	`%T`, `T`, `A` or intensity. Picking `%T` derives the absorbance with **$A = -\log_{10} T$**, which is the definition
x/y logarithmic	if an axis is logarithmic, say so: applying a linear map to it does not raise an error, it produces wrong numbers that look right
frame	normally I find it; "the whole figure" is for when the image is already cropped to the plot only

Change a field and the chart redraws by itself: there is nothing to confirm. Below it the **diagnostics** appear — how many columns, whether the frame was found and on which sides, background and threshold, how many columns were grid — because a number without its how cannot be challenged.

What I do by myself: I find the **axis frame** and digitise only what is inside it (the numbers written on the axes are dark pixels just like the trace); I **survive the grid**, picking in each column the run near where the trace was a pixel earlier and dark enough to be the trace; I read at **sub-pixel**, because the centre of a two-pixel line sits in between.

> **How much I get wrong.** Measured over seven constructed figures whose band centres are known: **mean 2.2 cm⁻¹** deviation, **worst 7.3**. One column of a figure 900 pixels wide over 3600 cm⁻¹ is worth about 4 cm⁻¹: below that there is nothing to gain, it is the limit of the figure.

> **If the calibration is wrong**, the peaks come out at wrong numbers while sitting in the right place on the figure: the shape is recovered from the pixels, the calibration is not. And a photo taken at an angle carries a distortion I do not straighten — frame the figure head-on.

With **download CSV** and **download JCAMP-DX** the numbers leave this place: any spectroscopy program opens the JCAMP, and this very reader reads it back identical. Inside it is written that the calibration was given by hand.

Below the chart you will find the peaks found with their signal-to-noise ratio, and for each band **all** compatible assignments. If you read two rows at 1715 cm⁻¹, that is not a defect: at that wavenumber a ketone and a carboxylic acid are both possible, and naming only one would be inventing.

> The reader **does not tell you which molecule it is**. A set of bands is compatible with many different molecules: the step from spectrum to structure is yours to take, with everything else you know about the sample.

Opening a problem and having it worked through

The same button, `📂 Open a file`. It takes the PDF of the exam, a photograph of the sheet, a Word file, plain text — and shows you **everything** inside: every page drawn (click to enlarge) and the full text.

Then, if the text contains spectroscopic data, it recognises them and puts them in a table **with where it took each one from**. That table sits above the working on purpose: look at it first. If it mistook a mass for an IR band, the working that follows will be impeccable and the answer wrong, and you catch it there.

The working lists the steps with the **evidence** for each and how certain it is: degrees of unsaturation are arithmetic, the carbon count from the M+1 peak is an estimate with a margin, an IR band is *compatible with* several things.

> **It does not give you the structure.** A set of spectroscopic data is compatible with more than one molecule, and a program producing a single one would offer a certainty the data do not contain. What you can do is the opposite: write a structure yourself and the program **compares** it, signal by signal, saying where it fails.

> **If the document is a scan**, the text cannot be read: a scanned page contains pixels, not letters, and there is no optical character recognition. The app tells you instead of pretending the sheet was empty. The pages stay visible, and if one is a photograph of a spectrum the trace can be extracted with **Open an image**.

Choosing the language

The **Languages** section holds **fourteen** languages, with a search field to find them quickly: Italian, English, Spanish, French, German, Portuguese, Dutch, Polish, Romanian, Greek, Russian, Chinese, Japanese and Arabic. Choosing Arabic also flips the page's **direction**, right to left.

Next to it is **Chemical languages**, which is a different thing: it converts a molecule among the twenty-five notations chemistry writes it in — SMILES, InChI, SMARTS, molfile, XYZ, PDB, CML, the fingerprints — and, for a declared class of simple molecules, writes its **IUPAC name**. Outside that class it says so, rather than attempting a wrong name.

Studying with cards

1. Open the study section
2. Generate a deck of cards
3. For each card, after answering, say how well you remembered it

The app works out **when** to show it to you again: the things you know well come back rarely, the uncertain ones come back soon. When you reopen, you only find the cards that are due.

Looking up a drug

Open **Pharmacology**. For each of the 263 drugs you will find the structure, molecular weight, mechanism of action, indications, adverse effects and class.

> ⚕ **For teaching purposes only.** This information does not replace the judgement of a doctor or a pharmacist, and is not there to decide a therapy.

Spectra, the assistant

Spectra answers chemistry questions and can use the app's own tools: calculate, generate spectra, look up constants, balance reactions.

Getting it working

Spectra relies on an external artificial-intelligence service, and you need a free **key**. Two routes:

The simplest — a key of your own

1. Go to console.groq.com and create a key

(free, it takes two minutes)

1. Open Spectra (the face at the bottom right, then 💬), paste the key into

the 🔑 box

1. Done

The definitive one — a proxy

If services often fail to answer for you, or you would rather not keep the key on your phone, you can publish a *proxy*: a small free program that keeps the key safe and makes even the services that are blocked from the browser work.

In the assistant's **Proxy** box you will find **"How to set it up, in 4 steps"**, with the commands ready to copy.

When Spectra does not answer

Almost always it is not the app's fault but the external service's. Press 🔌 **Test** in Spectra's panel: in a few seconds it tells you **which services your device can actually reach** — and you need no key to find out.

If a service stops answering, Spectra remembers it for 24 hours and moves it to the bottom of the list, under *"Did not answer from this device"*. It does not delete it: tomorrow it might work again.

Spectra Live: talking to him

Spectra has **a single button**: his face in a circle, at the bottom right, on every screen. A tap opens the console; from the console the 💬 button opens the written chat. Right after the app opens the circle shows a drawn face; once the page has loaded it becomes the portrait. The ◈ **Live** button in Spectra's header and `Ctrl+Shift+J` work too. The console fills the screen; on a phone, with the app installed, it leaves the status bar at the top and Android's keys at the bottom free. Press the microphone and speak: Spectra listens, answers aloud and then listens again, so you can carry on without touching anything. Say «stop» or tap the screen to interrupt. If the browser has no speech recognition or synthesis, the console says so and you can type.

In the console there is **Spectra with the body**, standing, with the hologram on his right (no longer the face-only bust): on a phone between the telemetry and the dialogue, on a large screen on the right. He follows what the console is doing: while listening to you he is curious, while working he puts his hand to his chin, then he speaks and gestures. When you close the console, Spectra goes back to the section where you left him.

Some requests Spectra carries out by himself, **without network and without a key**: «switch to physics», «calculate 3/4 + 1/6», «derivative of x^2 sin(x)», «convert 5 km to m», «constant planck», «plot sin(x)», «quiz me», «pause», «where am i». Everything else goes to the artificial-intelligence service you chose, together with a short summary of what is on your screen. If you ask about «this chart» or press 👁, Spectra also attaches an image of the open tab's charts.

Above the controls are the **shortcuts**: with a section open «Explain this section», «Guide me», «What am I looking at?», «Quiz me», «Review» and 🎓 **Lesson**, which starts the section's guided lesson (see below). At the top, the 🧭 button opens the navigator: for chemistry, physics and mathematics every section of the bar, the quick tools and the phrases Spectra understands even offline, with a search box. A tap on a section opens it, even if it belongs to another subject. The 💬 button switches to the full chat.

On the left, the telemetry shows among other things who you are to Spectra (**USER**), the **LEVEL** of the question and the **FOCUS**. The level starts from that of the open section in the study paths (high school, bachelor, master, PhD) and rises with advanced-course questions; «in simple words» lowers it. If the console stays open and silent for more than a minute, Spectra may tell you something important unprompted, for instance that it is time for a break.

🎓 The guided lesson

In every section, under the title, there is the «🎓 **Guided lesson with Spectra**» button. Tap it, or say or type «give me a lesson» to Spectra: the section becomes a lesson in chapters, with the real simulation in front of you.

- It starts from the section's title. Then **one chapter per tab**: the tab

opens by itself, the explanation appears one piece at a time, word by word, and a **spotlight** dims the rest of the page and lights up the chart or control being discussed.

- The **sliders** move by themselves, from minimum to maximum, while the chart

or the 3D scene changes; the **selects** go through their options. At the end every control is back as it was, even if you skip ahead or close half-way.

- Many sections have a **check question** taken from the subject's study kit:

the lesson waits for your answer, then shows the right one and the explanation.

- The **summary** lists the chapters covered; from there you can replay the

lesson or ask Spectra to go deeper.

The lesson panel sits on the right on a large screen (the page shifts left to make room) and at the bottom on a phone. ◀ ▶ move one step, the middle key pauses, 🔊/🔇 turns Spectra's voice on and off, ✕ (or Esc) closes. During the lesson the app's round buttons are hidden and they come back at the end. If «reduce motion» is on in your system, sliders jump to the value instead of sliding and the text appears all at once.

The lesson is built by reading the section: where a tab has little text, the chapter is short; and in sections with no suitable question in the study kit there is no check.

🌌 Spectra Studio: Spectra's laboratory

In Spectra's console tap 🧭 and, at the top of the list, **Spectra Studio**. A separate page opens: a 3D laboratory where Spectra stands next to a holographic bench and teaches you in four steps.

1. **Predict** — Spectra stages a phenomenon and asks what will happen. Choose

an answer: it is recorded.

1. **Test** — the simulation runs on the bench, with the measured numbers.
2. **Prove** — the rigorous explanation, one step at a time.
3. **Reveal** — why the right answer is right and the others are not, what it

connects to, and four exercises from high school to research. The one highlighted is the level Spectra recommends, based on how your predictions went and on the exercises you have already opened.

For now there are three lessons: 🪂 **The fall** (a sphere dropped and one thrown: which lands first?), 💧 **The shape of water** (why is the angle 104.5°?) and ∑ **The Basel problem** (what is 1 + 1/4 + 1/9 + …?). Dragging turns the camera, the wheel zooms in. If the device has no 3D graphics, the whole lesson stays in the panel, as text. Your predictions stay on this device; they are erased with «Clear data → Persistent memory». In the studio's button bar there is 🔊/🔇: it turns Spectra's voice off and on, and the studio remembers your choice next time too. The studio's own buttons and messages are in Italian.

Spectra's body is always there. You do not have to load anything: the body chosen by the author is already inside the app, in the studio and in the AI sections. The first time a network is needed: while it arrives you see «Spectra sta arrivando…» («Spectra is on the way…»); without a network «Spectra non è arrivato: serve la rete la prima volta» appears —

press **«↻ Riprova»** («↻ Retry») once you are connected. After the first start Spectra stays in the app and appears offline too. The «🧍 Corpo» button is gone; if in a previous version you had loaded the body yourself, that copy is deleted automatically to free space.

Spectra in the AI sections. In Spectra AI and in the areas with the assistant for Chemistry, Physics and Mathematics, under the title, Spectra stands in a lab-style frame: while the model works it brings its hand to its chin, when the answer arrives it explains with its hands and moves its lips. **«🎙️ Parla con me»** opens Spectra Live, **«🧠 Spiega»** opens the explanation, **«🎓 Studio»** opens Spectra Studio. Spectra's round button, on every screen, is its face.

The hologram. To Spectra's right there is a holographic screen with scenes that really move: orbiting planets, a benzene molecule, an IR spectrum drawing itself, a wave, an orbital, a surface. It changes by itself with the subject you are talking about (ask about orbits and the planets appear; talk about a spectrum and the spectrum appears) and, if you have a section open behind the Spectra window, it shows it live. **Tap the label at the top right of the hologram** («🪐 Orbite di Keplero · apri ›»…) to open the section where that scene is really studied: it opens behind the Spectra window, on the «🧠 Spiega» tab, so Spectra stays in front of you and the section is seen live in the hologram. When you close the window you return to the AI section you started from, with Spectra there (this also applies to sections opened from the catalogue, from «🎞️ Mostrami tutto», which stops, or from a question). Every section of the app can be shown in the hologram (below, «Every section in the hologram»). When answering, Spectra points at the hologram and then explains with the hands. The buttons are in the bar under the scene.

Explain everything: «🧠 Spiega». In the Spectra window open the second tab, **«🧠 Spiega»** (or press «🧠 Spiega» in Spectra's frame). Write the question — anything in chemistry, physics or mathematics — and choose the level, from high school to advanced PhD, at the research frontier: Spectra does not simplify below the chosen level. The answer is complete and always has the same structure: the summary, the theory, the step-by-step derivation, worked numerical examples, limiting cases and common errors, the links between the three subjects, applications and research, and three questions to check you understood. Then you can go on: «go deeper», «prove step 3», «and if I double the temperature?».

What you are looking at. If behind the window you have a section open with a graph, a spectrum or a 3D scene, leave **«📺 Includi lo schermo»** checked (the card shows which section and a thumbnail) or use a quick prompt: **«📈 Spiegami questo grafico»**, **«🧪 Lo spettro, picco per picco»**, **«🧊 La scena 3D»**. Spectra reads the control values and the tables and looks at the image of the graphs: for an NMR spectrum it explains every signal (chemical shift, multiplicity, coupling constant, integral) and which atom it belongs to. **Note**: with the box checked the screen text and the image of the graphs go to the AI provider you configured, together with the question; uncheck it if you do not want that, and your choice stays: even when Spectra opens a section you named or chose from the catalogue below, the model receives only the question. The quick prompts about the screen («📈 Spiegami questo grafico»…) check it again. A configured AI provider is needed, as for the

chat; without one, the tab tells you so and shows what the app knows on its own (the assistant's working for a numeric exercise, the chapters of the section's guided lesson). **«↺ Nuova spiegazione»** starts over; the conversation is not saved.

Every section in the hologram. In the «🧠 Spiega» tab open the **«🗂️ Tutte le sezioni nell'ologramma»** card (in English «Every section in the hologram»): it lists every section of the app (201: 78 chemistry, 63 physics, 31 mathematics and 29 shared), grouped as in the bar. Search for one in the box (for example «titration», «Schrödinger», «integrals») or filter by subject; tap it and Spectra opens it behind the window, shows it live in the hologram and checks «📺 Includi lo schermo», so the explanation uses its data (if you had unchecked it, it stays unchecked). Even a section without graphs (text, tables, controls, such as the Periodic Table) appears in the hologram, as a card with the title, the controls and the text. **«🎞️ Mostrami tutto»** («Show me everything») runs through every section of the chosen subject (or all of them) in the hologram, one every 7 seconds, with a counter («12/60 · …»); **«■ Ferma il giro»** stops it. You do not even need to search: if your question names a topic (in Italian, for example «spiegami le titolazioni», «lo spettro di massa», «la meccanica orbitale»), Spectra finds the section about it, opens it behind the window, shows it in the hologram and explains it with its screen data. New sections appear by themselves: the list is read from the app bar.

⛶ Interacting with the hologram

On the hologram, on Spectra's right, there is the **«⛶ Interagisci»** (interact) button (in the AI sections, in the Explore sections, in the «🧠 Spiega» tab and in the Spectra Live console). Tap it: the hologram opens full screen, with Spectra beside it and the steps panel at the side. It starts from what the hologram was showing; you can change mode at the top.

- 🧪 **Spectra.** Choose the type (IR, ^{1}H-NMR, ^{13}C-NMR, mass, UV-Vis) and a

molecule (acetone, ethanol, acetic acid, benzene, toluene, ethyl acetate), or **write your own trace** and the spectrum is redrawn as you type. Examples: IR «1715 s, 2950 m, 3300 br» (wavenumber and intensity: s strong, m medium, br broad); ^{1}H-NMR «2.17 s 6H; 1.2 t 3H J=7»; mass «58 30, 43 100» (m/z and intensity); UV-Vis «254 200» (λ and ε). Each signal becomes a step: where it is, its shape, what it is assigned to and why. In ^{1}H-NMR choose the instrument field (60 to 600 MHz): multiplets really narrow or widen, and an inset zooms into the selected one. For the molecules you also see formula, molar mass and degree of unsaturation.

- 🪐 **Universe.** The eight planets in 3D on their orbits. Tap a planet (or

pick it from the list, or write an orbit such as «a=2 e=0,6»): period, perihelion and aphelion, speeds, surface gravity and escape velocity, computed step by step.

- 📐 **Mathematics** and ⚛️ **Physics and chemistry.** Write the question, for

example «studia f(x)=x^3-3x» or a projectile or a pendulum: the app's own assistant solves it, offline, and the graph changes with the question (the function with its derivative, zeros and extrema; the trajectory; the swinging pendulum).

Press «🔊 **Spectra spiega i passaggi**»: Spectra reads them aloud one by one, and each lights up in the text and on the hologram. Tap a peak, a planet or a step to jump there. «🧠 **Approfondisci**» opens «🧠 Spiega» with the hologram's data (an AI provider is needed). When you close, Spectra returns to the section.

The spectra of the six molecules are typical literature values, for study: they are not measured spectra. From your trace the app proposes possible assignments according to tables; it does not tell you which molecule it is.

⚡ Explore with Spectra: a subject to scroll

In the chemistry, physics and mathematics bars, right after the «Area … · assistente» button, there is «⚡ **Esplora la chimica con Spectra**» (and physics, and mathematics; in English «⚡ Explore chemistry with Spectra»…). It is a page to scroll, like an explorable lesson: **ten chapters**, from high school to advanced PhD, at the frontier of research.

- **Chemistry**: ionisation energy, the shape of molecules (VSEPR), the pH of a

very dilute strong acid, buffers and Henderson–Hasselbalch, first-order kinetics, IR spectroscopy, ^{1}H-NMR and coupling, the molecular orbitals of O_2, Marcus theory and the inverted region, the delocalisation error of DFT.

- **Physics**: falling bodies, projectile motion, Kepler's laws, the pendulum,

Carnot efficiency, two-slit interference, the particle in a box, tunnelling, time dilation and muons, gravitational waves (GW150914).

- **Mathematics**: the derivative as a slope, the number e, the integral and

the logarithm, the Basel series, the harmonic series, the eigenvalues of a rotation, Fourier and the Gibbs phenomenon, residues, the heat equation, the Riemann hypothesis.

While you scroll, **Spectra stays at the top**, under the app header (on a phone in a shorter frame, to leave room for the text). The chapter in front of you lights up: Spectra tells it to you, with his lips and gestures, and the hologram on his right changes scene. In every chapter:

1. 🔮 **Prevedi** (predict) — tap one of the four answers. You see at once

whether it was right and **why each answer is right or wrong**; Spectra is pleased or surprised.

1. 🧪 **Prova** (try) — in many chapters a slider: move it and the number is

really recomputed (the pH, the fall time, the pendulum period, the Gibbs phenomenon…).

1. 📺 **Guardalo dal vivo** (watch it live) — the app section that shows the

phenomenon opens behind the Spectra window and the hologram projects it. When you close the window you are back at the chapter.

1. 🧠 **Spiegamelo tutto** (explain it all) — the full «🧠 Spiega»

explanation, at the chapter's level, with the section's data (an AI provider must be configured). 🔊 **Ascolta Spectra** reads it aloud.

At the top, the **«Il tuo percorso»** (your path) bar counts the chapters you have already predicted (for example 3/10); a chapter already done shows «Già previsto ✓» or «×» and you can review it by tapping an answer. Predictions stay on this device; **«Cancella dati → Memoria persistente»** erases them.

Body, face and voice imitate a behaviour: **they are not a consciousness**. The design is described (in Italian) in `docs/spectra/`.

Spectra's voice

Without setting anything, Spectra speaks with your device's voice, the deepest one it finds. If you have saved an **OpenAI key** among Spectra's keys, in the console, in the guided lesson and in Spectra Studio he speaks with a deeper, metallic voice, and the mouth follows the voice. In that case **the text Spectra speaks is sent to OpenAI** to turn it into audio, even if you use another service for the chat. If you do not want this, turn the voice off with 🔇: in the console, in the guided-lesson panel and in Spectra Studio's button bar (which remembers the choice). Or do not save an OpenAI key.

Spectra's face and mind

Spectra has a face: a 3D android (the «Noir» model by Skyfox13, CC BY 4.0 licence, modified; the credit is at the bottom of the console). In the console, since v242, you see it on Spectra's body, which takes the place of the bust. The mouth moves on the text he speaks, the head follows your gaze, and the expression changes with what he says and what you tell him: 17 expressions, from a smile to doubt. The eyes have the app's blue and change shade with the emotion; the eyelids narrow in a smile, open wide in surprise and half close when the question is hard. The eyes follow your finger or the mouse. If nobody touches anything for two minutes, Spectra falls asleep; the first touch wakes it. Without 3D graphics you see a simpler face.

Spectra remembers conversations and the time you spend in the sections, and has an internal state (curiosity, attention, care, mood, bond) that changes the tone of the greeting and the suggestions, for instance a break after an hour and a half of study. **It is a simulation of behaviour, not consciousness**: numbers and rules, not feelings. Everything stays on this device; say «what do you remember» to see it and «forget everything» to erase it. When a question goes to the artificial-intelligence service, a summary of this memory goes with it.

Spectra and its creator

Ask «who created you?» and Spectra answers: Samuele Pio Provenzano, the author of BioSpecInfo. If you say «I am your creator», Spectra asks for proof: the author's secret phrase, to be typed into the field, which becomes hidden like a password (do not say it aloud). The phrase is not stored and does not enter the conversation; after three wrong attempts you must wait a minute. Whoever passes is greeted by name on that device, until they say «forget that I am your creator» or erase the memory from «Clear data».

Like the File Manager password, this recognition is a **deterrent**: it tells Spectra whom it is talking to, it protects nothing.

The File Manager

A personal archive for photos, documents and HTML pages, protected by a password.

> ⚠️ **Something you need to know.** This protection is a **deterrent**, not real security: whoever knows where to look can get around it. Do not put confidential documents in it.

The files stay on your device. **There is no copy anywhere else:** if you clear your browser data or change phone, they are gone. Use the export for the things you care about.

Frequently asked questions

Do I have to pay for anything? No. The app is free. If you use Spectra, the AI service's free key has daily limits, but nothing is paid.

Does it really work without internet? Yes, after the first opening. The only things left out are those that by their nature have to ask someone else: searching molecules by name, 3D structures downloaded on the spot, and Spectra.

Are my notes saved somewhere? Only on your device. There is no server: nobody can read them, but nobody can recover them for you either if you lose them.

How do I delete my data? ✨ menu → Settings → delete data. You can choose **what**: only the keys, only the conversations, only the study data, or everything.

I updated but I do not see anything new. ✨ menu → 🔄 **Updates** (*Aggiornamenti*) → **Check for updates** (*Controlla aggiornamenti*). There you can also see which version you are running, and the check always answers in one of three ways: *you already have the latest version* (with the number), *new version available* (with the **Update now** button), or *cannot check now*, if the site does not answer — the app keeps working offline anyway.

Can I trust the spectra for a report? For recognising functional groups and understanding the structure-spectrum relationship, yes. For a figure to be cited as experimental, no: they are predictions. The app sends you to the official databases (NIST, SDBS) with direct links.

I found an error in a chemical datum. Report it: open an *issue* on the GitHub repository. The data go through automatic checks, but no check finds everything.

If something goes wrong

Problem	What to try
The app does not open	Reload it. If it persists, ✨ menu → Updates → Check
I do not see the new features	The version is in the ✨ menu → Updates
Spectra does not answer	The 🔌 **Test** button: it tells you which services you reach
"Out of space"	✨ menu → Settings → delete the old conversations
A section stays empty	Check the network: some sections load data from the internet
The charts are blurred	Reload the page; the charts are redrawn at your screen's resolution

Contacts

Author: Samuele Pio Provenzano **Repository and reports:** github.com/samupropio1-ship-it/BioSpecInfo-v11

Guide updated to version `bsi-v250`.

Test Documentation — BioSpecInfo

Field	Value
Software	BioSpecInfo
Version described	`bsi-v250`
Purpose	Describe how the tests are organised, how to run them, what they cover and where they leave gaps.

1. Kinds of test present

Kind	Present	Notes
End-to-end (E2E)	✅	Playwright on headless Chromium against a local HTTP server. It is the prevailing form.
Data verification	✅	Scientific data compared against an independent source
Stability tests	✅	Long sessions, exhausted storage, degraded network
Documentation/code consistency	✅	Verifies that what the documentation promises really exists
Isolated unit tests	❌	See §6
Instrumented code coverage	✅	`audit_copertura`, with Chromium's profiler: no build, no source rewritten. See §6

Why E2E and not unit tests

The application is made of classic scripts operating directly on the DOM, with no modules and no dependency injection: there are no units that could be isolated without a deep re-write. The defects that actually occurred — a blurred canvas, a button unreachable because its container is collapsed, an AI provider that stops answering — **would not have surfaced** from a unit test anyway: they live in the integration between code, DOM and browser.

The choice is therefore reasoned, not endured. Its limits remain, and are declared in §6.

2. How to run the tests

Requirements

```
npm install                      # playwright-core only
python3 -m http.server 8899 &    # RDKit WASM requires an HTTP context
```

> Without a server every test fails: opening the files over `file://` prevents the WebAssembly modules from loading.

The whole battery, with a report

```
node tools/genera-evidenza.js
```

It runs every bench, transcribes its full output and produces `docs/evidence/RAPPORTO-VERIFICA.md` with environment, commit and SHA-256 digests. It exits with code `0` only if no bench fails.

```
node tools/genera-evidenza.js --veloce   # skips the benches that need a browser
```

The `--veloce` option is useful for a quick check but **is not sufficient for an attestation**: the benches it skips are precisely the ones measuring real behaviour.

A single bench

```
node tools/verifica-farmaci.js           # a stand-alone tool
node tools/verifica-farmaci.js --json    # machine-readable output

cd tools/banchi && node test_spettri.js  # one E2E bench
```

The E2E benches live in **`tools/banchi/`**, versioned like the rest of the repository; the `BSI_BANCHI` variable allows another folder to be pointed at.

> **It has not always been so, and that is worth saying.** Up to version `bsi-v168` the default folder was the working area of the session in which the benches had been written: **32 of the 39 benches were not in the repository**. Whoever cloned it and ran the documented command found seven, and received no warning — because a missing bench was counted but did not make the battery fail, which closed with "CONFORMANT" and zero failures.
>
> Both things have been corrected: the benches are versioned, and **a missing bench makes the outcome NON-CONFORMANT**. A bench that is not there is not a bench that passed.

---

3. Composition of the battery

90 benches, grouped by what they demonstrate.

3.1 Scientific data

Bench	Verifies	Checks
`audit_farmaci`	Structure ⟷ weight, duplicates, and the **formula rebuilt from the graph** against ChEMBL's for every entry declaring a provenance	263 entries, 36 with a source
`test_spettri`	Functional-group recognition on reference molecules	36
`test_spettri_ui`	Grafting of the engine into the app's spectrum-drawing points	16
`test_assi`	Axis conventions in the SVG spectra	13
`test_assi_canvas`	Axis conventions in the canvas spectra	6
`test_costanti`	Physical-constant lookup and refusal of ambiguities	45
`audit_dati`	Consistency of the tabulated data	29
`test_farm_ui`	That the Drug Atlas **draws** every drug it holds in memory, category by category, and that a **never-labelled** category still appears	11
`test_astro`	The astrochemistry data: molecular weight ⟷ formula for the 30 interstellar molecules against IUPAC atomic weights, no repeated celestial body, every quiz question with a valid answer among its options	13
`test_datasci`	The maths the Data Science section **displays**: R^2 on an exact relation, standardised weights in the true ratio, ROC on hand-computable values, behaviour on pure noise, and that **no section is left unreachable**	23
`test_simmetria`	Point groups, normal modes, IR/Raman selection rules	26
`test_cheminfo`	The cheminformatics engine: Tanimoto and Dice against hand-computable values, Morgan and MACCS fingerprints, Butina clustering, Bemis–Murcko scaffolds, PCA, kernel ridge, scaffold split, **null model by label scrambling**, grouped cross-validation, CSV and SDF exports, **contrast of the nine panels**, agreement between lab and engine, fragmentation, **matched pairs against known effects**, substructure search, SAR table, model comparison in both directions, enrichment (EF and BEDROC) on hand-computable cases, conformal intervals and their **refusal on classification**	189
`test_spettrolettore`	The spectrum reader and the models that follow from it: ASDF symbol by symbol, explicit and DIF/DUP-compressed JCAMP, prominence peaks on **built** spectra **and on pure noise**, curve versus peak list, neutral losses on toluene, ESOL against four experimental values, four drug-likeness filters reported separately, random forest both ways, **a single entry point** (one button, empty `accept`, routing checked on PDF/ `.docx` / `.txt` / `.jdx` / `.csv` /a disguised binary/an image) and the version badge compared with `BSI_APP_VERSION`. **And the relationship between the two IR tables**: `SPEC_DATA.ir` is for studying, `BANDE_IR` for assigning, they are not a duplicate — but the assignment umbrella must **cover** the reference for each of the 23 groups present in both, and not be arbitrarily wider (margin derived from the data, not invented)	66

Bench	Verifies	Checks
`test_nmr`	Per-atom NMR prediction: two separate molecule sets (tuning and **validation**, the latter never used to choose parameters), the signal count on molecules whose count is known, refusals on an unreadable SMILES and on an empty string, the **history** with duplicate detection by InChI key, and the **peak ↔ structure ↔ 3D** sync verified on the indices, not on how many atoms light up	64
`test_nmr2d`	Correlation maps counted on the graph: HSQC with inverted sign on CH_2, COSY between non-equivalent protons, HMBC two and three bonds away. **In both directions**: hexafluorobenzene gives no HSQC spots, benzene **no** off-diagonal COSY spots, methane **no** HMBC spots despite having four protons	40
`test_digitalizza`	A spectrum extracted from a **figure**, over seven constructed images whose Gaussian centres are known to the cm^{-1}: bare figure, with frame and labels, with a grey grid, with a black grid, with frame **and** grid, dashed, in %T. **Mean 2.2 cm⁻¹** deviation, **worst 7.3** — one column is worth ~4 cm^{-1}, which is the limit of the figure and not of the method. **In both directions**: from a blank sheet and from the app icon **no** curve comes out; the crude fallback on the same framed figure finds **one** peak instead of three; on a logarithmic axis the midpoint is the geometric mean and on a linear axis the arithmetic one. %T→A conversion on exact values (50 %T → A = 0.301) and a **full round trip** of the exported JCAMP, read back by the reader itself	45
`test_fisica`	The eight **physics** sections against literature numbers: Schrödinger (harmonic oscillator, infinite well, exact Morse, Crank-Nicolson norm), **STO-3G Hartree-Fock** (Szabo-Ostlund H_2 and HeH^+, Crawford H_2O to 10^{-6} Eh), rovibrational (HCl r_e, B from combination differences, odd J of O_2 Raman), **NIST-JANAF** standard entropies of eight gases, molecular dynamics (energy conserved, g(r) peak), Hückel, FFT, Runge-Kutta, Arrhenius, the 2θ angles of NaCl/Si/Cu/Fe with absences, Bateman (^{99m}Tc), ^{14}C, Weizsäcker; then in the browser every section and tab on two viewports, nothing overflowing at 390 px, the simulation stopped when the section is closed, the buttons translated. **92 checks**, declared minimum 90	92
`test_matematica`	The eight **mathematics** sections and the **symbolic engine** against independent values: simplifications, table derivatives and a complicated derivative checked by finite differences, verified antiderivatives (by parts, by substitution) and the Gaussian that must **not** have one, ∫e^(−x²) = √π, Taylor series (1/120, 17/315), standard limits and left/right ones; significant figures, pH, acetic acid from the quadratic formula, $e^x = 3 - x$ against Lambert's W; function analysis; **the integer inverse of the Hilbert matrix** and det 1/2160, a 4×4 det recomputed separately, eigenvalues 2 ± √2, **balanced reactions** (permanganate, redox with charges, Wöhler); the order of Euler, Heun and RK4 measured; Lotka-Volterra and Brusselator equilibria; critical points and the Maxwell relation; e^(iπ) = −1, Gibbs, Parseval; t, F and Grubbs from the tables, **the Miller & Miller calibration line**, propagation against Monte Carlo. Then every section and tab in the browser on two viewports, a real interaction and the translations	

Bench	Verifies	Checks
`test_esami`	The **exam solvers** for Chemistry (40 models on four levels), Mathematics (Istituzioni I and II plus the advanced level) and Physics (General I and II plus modern physics) and the new physical-chemistry sections: domains, limits, partial fractions, improper integrals, series, constrained extrema, double integrals, differential forms, ODEs with Cauchy checked against SymPy or hand calculation; the 48 physics models against independent formulas; Tafel, Eyring, BET, Langmuir, Mn/Mw, quantum yields; every exercise of the solved papers must produce a working. In the browser the four sections, an opened paper, the solver recomputing, translations	
`test_assistente`	The **assistant of the three macro-areas**: over 80 natural-language questions, the kind of exercise or model understood, the data read with units (M vs m, µC, km/h, solar masses, percentages), the result compared with SymPy or an independent formula; the inverse problem; the symbolic engine additions verified by differentiation; in the browser the three areas, a question typed into the box, catalogue, search, levels, Spectra's tools, 390 px and English	
`test_grafica3d`	**What is shown is right**: orbitals (nodes, exact ⟨r⟩ on the sampled cloud, lobe fraction), Kepler and Lorenz, crystals (content, packing, Miller planes, NaCl peak), fields (Gauss, loop); in the browser the 3D Laboratory on two screens, the gestures, chart animations ending identical to the direct drawing, the crosshair, «reduce motion», 3Dmol viewers with bar and quality	
`test_laboratori`	**Simulations recover the exact answer**: projectile in vacuum, collisions, pendulum with the elliptic integral, normal modes, lenses, grating, RLC transients (Kirchhoff), filters, cycles ($\Delta U = \Delta S = 0$, efficiencies), disc gas (Maxwell-Boltzmann, virial), numerical method orders, Runge, conics, skew lines, kinetic equilibrium and Le Chatelier both ways; in the browser the 22 tabs on two screens and a perturbation in the page	
`test_scienze`	**Astrophysics, relativity, fluids, vector calculus, complex kinetics** against known values (age of the universe, Planck = σT^4, Hohmann, transits, twins, Poiseuille, Green, vortex, Brusselator) and the **animation engine**: pause, speed, off-screen stop, «reduce motion», on three browser configurations	
`test_esplorazioni`	**Probability, visual mathematics, electromagnetism, sound, semiconductors, distillation, voltammetry** against known results (Bayes, central limit, Fourier, logistic map, dipole, n_i, V_bi, Antoine, azeotrope, Fenske, Randles-Ševčík, Nicholson); in the browser the 24 tabs on two screens, a hand drawing rebuilt by epicycles, dragged charges	
`test_approfondimenti`	**Numbers, graphs, gravitation, heat, Ising, phase diagrams, LEPS surface** against known results or a second method (π(x), RSA, Li(x), Floyd–Warshall, Prim, figure-eight, Lagrange, Onsager, IAPWS, Span–Wagner, measured eutectic, H_3 saddle); in the browser the 25 tabs on two screens; the **subject buttons** (visible groups, icons, remembered subject, no section unreachable)	

Bench	Verifies	Checks
`test_pareggio`	The **twelve sections** that bring mathematics and physics to the level of chemistry: runs the twelve scripts in `verifiche/` (462 checks against known results or a second method), opens the 47 tabs on two screens, checks every tab opens in under 100 ms and that mathematics and physics have the declared number of sections	
`test_v237`	The **twenty new sections** (ten astrophysics, five physics laboratories, five mathematics laboratories): runs their twenty scripts in `verifiche/` (705 checks against published values or a second method, §3.1-bis) and those of the study kits and resources; in the browser opens every tab at **1280 and 390 px**; tests Spectra's face, emotions and mind and the section names in English. Requires at least 900 checks	1061
`test_3d238`	The **new 3D tabs of 29 sections** of chemistry, physics and mathematics: runs their 29 scripts in `verifiche/` (311 checks against known values, a closed form or a second method, §3.1-ter); in the browser opens every tab of the 29 sections at **1280 and 390 px** and requires full tabs, no ⚠, canvases drawn and named, the last tab a 3D scene that draws, synchronous work on click under **250 ms**; nothing overflows at 390 px; no JavaScript error	488
`test_potenzia250`	The **four v250 sections**: Saha against the textbooks (half ionisation, Balmer maximum, Ca II in the Sun); the interstellar cloud against the exact solutions and the layered structure; deconvolution against spectra whose truth is known; t-SNE against PCA; and the pages at 412 and 1280 px	65
`test_sellingua`	The round language switcher, on **two viewports** (1280 and 390 px): the button is visible, round, top right, **at the measured title height** as the overlap of the two boxes, with an accessible name that says what it does; the disc with the fourteen flags on two shells **inside the screen**; and **both ways** the language actually changing, the flag following, the disc closing and, going back to Italian, the text identical to before. **30 checks**, declared minimum 30	30
`audit_statico`	The classes of bug this project has already had, searched across **all** files instead of where they showed up: `calc()` without spaces (the browser drops the rule), secrets in logs, API keys in the code, plaintext passwords, `target="_blank"` without `noopener`, duplicate ids in the markup, unbalanced scripts. The report **flags and does not block**, because those classes have false positives. Three measures **do** block, and have a written baseline: verbatim duplicated code blocks (**0**), redefined global functions (**11**, the file's declared "additive patch" style), of which with identical bodies (**0**). If they grow, the battery fails	3 gates
`test_articoli`	That no function has two entry points, and that the articles are not hand-written. Ten tabs inside a single Spectra centre, with each identifier present in the document **exactly once**; no repeated id, no duplicate button, no section reachable by two roads; Spectra out of the menu and with its dedicated button. Then the articles: the links tested as **pure functions** of the identifier, the Nobel explanations matched to the topic of the motivation and **none** invented when it is not recognised,	58

Bench	Verifies	Checks
	and the decisive proof — an **injected** response with a name that exists nowhere, which must appear on screen. **Three distinct failures**: failed request, empty answer, unrecognised shape	
`test_geom3d`	The 3D geometry built from the graph, measured against literature values: bonds, valence angles, alkyne linearity. **In both directions**: benzene comes out planar and cyclohexane does **not**. And the proof that makes the peak↔atom link legitimate: every heavy atom has **the same index and element** as the predictor's	29
`test_documento`	Opening any document and working through the problem inside: PDF, Word and text genuinely opened and the content verified; the data recognised **with their provenance**. **In both directions**: an image declares there is no optical character recognition and invents no working, an unknown format is refused rather than guessed, $CDCl_3$ does not become the compound's formula, without an MS label there are no masses	47

A bench that verifies a model must also verify it when the model is supposed to fail. `test_cheminfo` exercises the null model in both directions: on a learnable signal the true R^2 (0.394) beats all eight label-scrambled twins, with a margin of 0.287; on the **same molecules with random labels** the R^2 (-0.199) does **not** beat them. A check that verifies only the first case would pass even if the guard were wired to "true".

A "contrast 0" that did not cover the new surface. `verifica-accessibilita` measures what is visible, and the six panels of the Cheminformatics section stay hidden until the analysis has been run: it had never seen them. `test_cheminfo` now runs the analysis, opens the panels in turn and measures contrast over **460 text elements**, with the WCAG formulas rewritten inside the bench — asking the code under test how good its own contrast is, is not measuring. The bench also demands that more than 250 were measured: if the panels failed to open, the zero would be empty and the check would fail.

Three peaks found prove nothing on their own. A detector that is too sensitive finds thirty where there are three; one that is too strict finds none. Neither case raises an error. `test_spettrolettore` starts from three Gaussians **built** at 1715, 2950 and 3400 cm^{-1} under noise and demands that the reader find three — 1716, 2952, 3402 — and then, on the **same noise without the Gaussians**, that it find **zero**. Without the second check the first would pass even with a detector that marked every wiggle. The same holds for the random forest: out-of-bag R^2 0.90 on the real relationship and **−0.16 on pure noise**.

> **The defect was in my counter, not in the data.** The first run of `verifica_farmaci_v187` returned **0 formulas out of 36**: every molecule came out as `C<n>`, all carbons and no hydrogens. RDKit's `get_json` format is **commonchem**, where an atom carries only the fields that **differ** from the defaults — `z` is 6 and `impHs` is 0 — so that an atom written `{}` is a carbon. I was reading `a.element`, which does not exist in that format. Thirty-six data points that look wrong all at once are almost always a wrong counter.

3.1-bis The twenty v237 sections: what is verified

Each section has a script `tools/banchi/verifiche/verifica-<id>.js` that can be run on its own with `node` and that sets the computations against published values or an independent second method. Two or three examples per section, with the value the script prints:

Section	Script	Checks	Examples
🪐 Solar system 3D	`verifica-sastrosole`	36	Kepler solved with residual < 10^{-12} up to e = 0.97 and confirmed by bisection; Julian date of 1957 Oct 4.81 = 2436116.31 (Meeus, ex. 7.a); Mars 2003 at 0.3730 AU on 27 August (true: 0.3727)
🛰️ Orbital mechanics 3D	`verifica-sastroorbite`	26	elements → state → elements over 2000 random orbits (400 hyperbolic), error $2.8 \cdot 10^{-14}$; Curtis ex. 4.3 (h = 58,312 km²/s, i = 153.25°, e = 0.1712); Hohmann LEO → GEO 3.893 km/s
📐 Celestial mathematics	`verifica-sastromat`	34	Newton, fixed point and Bessel series agree within 10^{-12}; Laplace limit 0.6627434; q v q* = R(q) v over 500 random rotations
⭐ Life of stars 3D	`verifica-sastrostelle`	32	first Lane–Emden zeros: √6, π, 3.65375, 6.89685; numerical solutions equal to the exact ones within 10^{-13}; Chandrasekhar mass from the n = 3 polytrope, 1.4563 M☉
🕳️ Black holes	`verifica-sastrobh`	30	ISCO at 6M and photon sphere at 3M from the effective potential; Mercury's perihelion 42.982″ per century; GPS +45.72 and −7.21 μs/day
〰️ Gravitational waves	`verifica-sastrogw`	29	GW150914 chirp mass 28.096 M☉; f ∝ (t_c − t)^(−3/8), slope −0.375; Peters coalescence time equal to integrating da/dt
🌌 Galaxies 3D	`verifica-sastrogal`	32	modified Bessel functions against the tables ($I_0(1)$ = 1.2660659); Freeman disc peak at 2.150 R_d; NFW M(r) closed form against the integral, $7.6 \cdot 10^{-12}$
🌠 Cosmology 3D	`verifica-sastrocosmo`	38	age with Planck parameters 13.791 Gyr; Einstein–de Sitter $2/(3H_0)$; maximum D_A at z = 5/4 in Einstein–de Sitter
🔭 Planetarium 3D	`verifica-sastrocielo`	30	GMST of Meeus ex. 12.a and 12.b; sidereal day 86,164.091 s; equation of time: +16.45 min around 3 November, −14.20 min around 11 February

Section	Script	Checks	Examples
🌍 Exoplanets 3D	`verifica-sastroeso`	27	the Sun's habitable zone 0.95–1.67 AU; Earth's T_eq 254.8 K; transit depths of Earth and Jupiter 84.1 ppm and 1.056 %
🌀 Spin, qubits and entanglement	`verifica-sfisspin`	30	HZH = X, HXH = Z, $S^2 = Z$, $T^2 = S$; Bloch vector norm 1 after 2000 random gates; measured frequency against $\cos^2(\theta/2)$ within 3σ
🎲 Statistical mechanics	`verifica-sfisstat`	30	Schottky peak at kT = 0.41678 Δ, C = 0.43923 k; N_2 v_rms at 300 K = 516.84 m/s; energy conserved over 19,527 elastic collisions
☢️ Nuclei and reactions	`verifica-sfisnuclei`	28	B/A of ^{56}Fe 8.7903 MeV from the masses; ^{62}Ni with the highest B/A; Q of D + T 17.589 MeV, 14.049 of it to the neutron
🧱 Elasticity and materials	`verifica-sfismateriali`	27	elastic slope of steel = E = 207 GPa; closed-form moments of inertia against the strip integral; numerical cantilever deflections equal to the closed forms
🔌 Electronics	`verifica-sfiselettronica`	27	simulated ripple against the exact computation in the rectifiers; diode with Newton = closed form with Lambert's W (0.650463089 V); Zener with Newton = bisection
🌊 Partial differential equations	`verifica-smatedp`	46	$J_0(1)$ and $J_1(1)$ from the tables within 10^{-14}; twelve Bessel zeros; square membrane with $\omega_{12} = \omega_{21}$ and $\omega_{12}/\omega_{11} = \sqrt{(5/2)}$
🌀 Complex analysis	`verifica-smatanalcomp`	49	e^(iπ) + 1 = $1.2 \cdot 10^{-16}$; 2π jump of the logarithm across the cut; Cauchy–Riemann on nine functions
🔗 Markov chains	`verifica-smatmarkov`	51	stationary distribution by powers = linear system = Cramer's rule; Ehrenfest with the binomial and detailed balance; periods 2 and 3 recognised
🧩 SVD	`verifica-smatsvd`	41	$\|A - U\Sigma V^T\|/\|A\| \approx 1.6 \cdot 10^{-15}$ on matrices up to 12×12; σ of [[3,0],[4,5]] = 3√5 and √5; σ of the shear = φ and 1/φ
📡 Codes and elliptic curves	`verifica-smatcodici`	62	Hamming(7,4): distance 3, 112 single errors corrected, 336 double ones decoded wrongly; CRC-32("123456789") = 0xCBF43926

On top of these come `verifica-studio` (167 checks on the kit engine), `verifica-studio-fisica` and `verifica-studio-matematica` (schema and minimum quantities; for mathematics **264 answers recomputed** out of 346 questions) and `verifica-risorse` (87 checks: citation formats and table proofs).

> **Two time thresholds, and why.** In the browser `test_v237` measures how long the page stays blocked on the click of each tab. **2D** tabs must open within **250 ms**; those that create a **3D** scene within **6 s**. The difference is not leniency: in the battery the browser has no graphics card, and WebGL runs in the **SwiftShader** software rasteriser, where creating a context and compiling the shaders costs 0.2 to 4 seconds. On a device with a GPU it is a matter of milliseconds. A tight threshold on the 3D tabs would measure SwiftShader, not the app; a loose one on the 2D tabs would let a real stall through.

3.1-ter The v238 3D tabs: what is verified

Twenty-nine sections receive, at the end, a 3D tab built on the section's model. Each tab has a script `tools/banchi/verifiche/verifica-<id>-3d.js` (for Topology and Non-Euclidean geometries the checks were added to the section's own script) that runs on its own with `node`. Two examples per tab, with the value the script prints:

Section · 3D tab	Script	Checks	Examples
🏔️ Potential energy surface · 🧊 3D surface	`verifica-schipes-3d`	9	at the saddle a single negative curvature: 2116 and −2022 cm^{-1} (imaginary frequency); the model's ω(H_2) 4396.6 cm^{-1} against 4401 measured
🧊 Phase diagrams · 🧊 P–V–T surface	`verifica-schifasi-3d`	8	acentric factor from the vapour curve: water 0.3443, CO_2 0.2249; equal Maxwell areas on the dome within $5.5 \cdot 10^{-12}$
🌀 Complex kinetics · 🧊 Molecular collisions	`verifica-schicinet-3d`	7	kinetic energy conserved after 6000 steps and 2993 collisions ($1.8 \cdot 10^{-15}$); fraction of collisions above $E_0/kT = 2.5$ equal to e^(−2.5) within 3σ (0.0812 against 0.0821)
⚖️ Equilibrium and Le Chatelier · 🧊 3D box	`verifica-schiequil-3d`	7	A ⇌ B: measured K 2.9976 against 2.9951 theoretical (within 1.5 %); halving the volume lowers the dissociation degree of N_2O_4 from 0.177 to 0.129
💎 Crystallography and X-rays · 🧊 Cells in 3D	`verifica-scristalli-3d`	9	coordination numbers 6, 8, 12, 12, 6, 8, 4 for SC, BCC, FCC, HCP, NaCl, CsCl, diamond; d(hkl) = 1/|G| equal to `dSpaziatura` within $4.4 \cdot 10^{-16}$
🌊 Schrödinger laboratory · 🧊 3D cloud $\|\psi\|^2$	`verifica-sschrod-3d`	10	degeneracies of the 3D oscillator 1, 3, 6, 10; fraction of the cloud outside the classical region 0.1107 against $P(\chi^2_3 > 6) = 0.1116$
🧮 Live Hartree-Fock · 🧊 Orbitals in 3D	`verifica-shf-3d`	9	H_2 at 1.4 bohr: E = −1.116714 Eh (Szabo-Ostlund); $\int\|\psi\|^2$ over the grid = 1.0000 for σg and σu
🎻 Rovibrational spectroscopy · 🧊 3D rotation and vibration	`verifica-srotovib-3d`	10	T + V = G(v) along the classical motion within $9.5 \cdot 10^{-9}$; CO_2: ν_1 IR-inactive, ν_2 and ν_3 active

Section · 3D tab	Script	Checks	Examples
🎱 Molecular dynamics · 🧊 3D box	`verifica-sdinmol-3d`	7	NVE: $\|\Delta E\|$ per particle $9.6\cdot10^{-4}$ over 2000 steps; thermostat: $\langle T\rangle = 1.400$ like the bath
🎯 Mechanics in motion · 🌬️ 3D launch with wind	`verifica-sfismecc-3d`	7	in vacuum $R = v_0^2 \sin 2\theta/g = 91.774$ m even with 15 m/s of wind; terminal speed $\sqrt{(g/k)} = 22.145$ m/s (measured 22.1435)
⚡ Electromagnetism in motion · 🌀 3D fields and Lorentz	`verifica-sfisem-3d`	8	Larmor radius $mv\perp/qB = 0.61538$ (measured 0.61539); $E\times B$ drift $= (0, -0.5)$ the same for $q = +1$ and $q = -1$
〰️ Waves and optics · 🌊 3D ripple tank	`verifica-sfisonde-3d`	6	$d = 3\lambda$ gives 6 nodal lines, $d = 0.4\lambda$ none; far minima at $\arcsin((m + ½)\lambda/d)$ within $2.1\cdot10^{-5}$ rad
🪐 Gravitation · 🕳️ 3D potential well	`verifica-sfisgrav-3d`	8	$\|\nabla U\|$ at the five Lagrange points $1.1\cdot10^{-15}$; measured period 7.2779 = 2π a^(3/2)
⏱️ Special relativity · 🔦 3D light cone	`verifica-sfisrel-3d`	6	invariant interval in (x, y, ct) within $2.7\cdot10^{-13}$; boost composition $\Lambda(\beta_2)\Lambda(\beta_1) = \Lambda((\beta_1 + \beta_2)/(1 + \beta_1\beta_2))$ within $5.3\cdot10^{-15}$
🌊 Fluids · 🌀 3D flow past an obstacle	`verifica-sfisfluidi-3d`	7	sphere: $v = 1.5U$ and $C_p = -1.25$; cylinder: $v = 2U$ and $C_p = -3$; lift $-\rho U\Gamma = -5988$ N/m
🔥 Thermodynamics in motion · 🗻 3D p-V-T surface	`verifica-sfistermo-3d`	8	van der Waals critical point: $p = 1$, $\partial p/\partial V$ and $\partial^2 p/\partial V^2$ zero; at $T = 0.9$ p_sat 0.6470, V_l 0.6034, V_g 2.3488
💡 Semiconductors · 🔷 3D Brillouin zone	`verifica-sfissolido-3d`	7	fcc zone: 24 vertices, 36 edges, 14 faces ($V - E + F = 2$); m*_dos(Si) = 1.084 m_0
🧲 Ising model · 🧊 3D Ising	`verifica-sfisising-3d`	6	ground state $e = -3$, $m = 1$ (with $h = 0.4$: $e = -3.4$); $\|m\|$ 0.946 at $T = 3$, 0.052 at $T = 5.5$ ($T_c \approx 4.5115$)
⚛️ Particle physics · 🔭 3D detector	`verifica-sfispart-3d`	6	helix in 3.8 T: $R = p_T/(0.3\ B) = 0.8772$ m; invariant mass reconstructed from 200 $Z \to \mu\mu$ pairs equal to the generated one within $1.6\cdot10^{-5}$
🗻 Functions of several variables · 🧊 Tangent plane in 3D	`verifica-smatmulti-3d`	9	$x^2 + y^2$ at (1, 2): $f = 5$, $\nabla f = (2, 4)$; second-order deviation from the tangent plane (ratio 4.0445)
∇ Vector calculus · 🧊 3D field and divergence	`verifica-smatvett-3d`	8	$F = (x, y, z)$, $r = 1.2$: flux $= \iiint 3\ dV = 4\pi r^3 =$ 21.7146884216; point charge: flux 4π with the origin inside, 0 outside
∫ Derivatives, integrals and series · 🧊 Solids of revolution	`verifica-smatcalc-3d`	8	unit sphere $V = 4\pi/3 = 4.188790204786$; spindle of $\sin x$ $V = \pi^2/2$
🔲 Linear algebra · 🧊 Space transformations	`verifica-smatlin-3d`	9	det = triple product $= \Pi\lambda = 6$; reflection: det -1 and det $M(1/2) = 0$
🌀 Differential equations · 🧊 Chaos in 3D phase space	`verifica-smatode-3d`	9	Lorenz Lyapunov exponent 0.9036 against 0.9056; RK4: error ratio 16.04 when halving the step

Section · 3D tab	Script	Checks	Examples
🎲 Probability in motion · 🧊 Bivariate normal 3D	`verifica-smatprob-3d`	7	∬ xy f = ρσxσy = 0.546; samples inside Q ≤ 1: 39.37 % against 39.35 %
🍩 Topology and surfaces · 🧊 Surfaces in 3D: χ and orientability	`verifica-smattopo`	45 (3 from v238)	triangulations of five surfaces at four refinements with χ equal to the mesh's; Möbius: n(2π)·n₀ = −1 and n(4π)·n₀ = +1
🌐 Non-Euclidean geometries · 🧊 Sphere and hyperboloid in 3D	`verifica-smatnoneucl`	56 (4 from v238)	hyperboloid and Poincaré disc with the same angles and sides within $1.2\cdot10^{-15}$; Gauss–Bonnet on the hyperboloid: π − Σ = area (1.0700757)
ℂ Complex numbers and Fourier · 🧊 Riemann sphere	`verifica-smatcompl-3d`	7	sphere rotations = SU(2) Möbius maps on 4 axes × 3 angles × 30 points within $5.5\cdot10^{-16}$; rotation by π about x: f(z) = 1/z and f(0) = ∞
🎢 Calculus of variations · 🧊 Soap film in 3D	`verifica-smatvariaz-3d`	8	h*/R = 1.32549 (u* tanh u* = 1); h/R = 1.36 tears, h/R = 1.30 stays a catenoid

All the scripts, run while writing this document: **311 checks passed, 0 failed**. The counts for `smattopo` and `smatnoneucl` include the checks that existed before v238.

> **One time threshold for the new 3D tabs.** `test_v237` allowed the 3D tabs 6 s, because the scene was built inside the click handler and in the battery WebGL runs in SwiftShader (§3.1-bis). The v238 tabs build text and controls at once and the scene **after** the handler returns: the synchronous work on click must stay under **250 ms**, as for 2D tabs. Measured in the trial runs: a few tens of milliseconds at most.

3.2 AI agent

Bench	Condition simulated
`test_404`	The provider has retired the model
`test_503`	Temporary overload
`test_ko` · `browser_ko`	Provider unreachable from the browser
`test_firma`	Opaque reasoning signatures to be returned
`test_tetto`	Token ceiling learned from the error
`test_attesa` · `test_attesalunga`	Timeouts and imposed waits
`test_doppioinvio`	Two overlapping submissions
`browser_prova`	Reachability measurement without a key
`test_nucleo` · `browser_prov`	Agentic loop and provider selection
`caccia_ai`	Malformed answers, truncated stream, 502 with HTML, Stop during the answer, panel closed halfway, hostile text

3.3 Stability

Bench	What it puts to the test
`audit_stabilita`	194 sections opened 5 times, full storage, corrupted data, bursts of clicks
`audit_promesse`	Rejected and unhandled promises, with a healthy network and a dead one
`audit_quota`	`localStorage.setItem` forced to fail on 10 pages
`test_sw`	Offline, degraded network, update while working
`test_filemanager`	Store unavailable, space exhausted, failed writes
`test_visore3d`	WebGL context consumption while browsing molecules

3.4 Interface

`browser_reset` · `browser_proxy` · `browser_proxyui` · `browser_rdkit` · `browser_lab` · `browser_frontiera` · `test_aggiorna` · `test_controllaagg` · `test_spectra238` · `test_lezione` · `test_spectra_studio` · `test_spectra_rig` · `test_spectra_figura` · `test_spectra_esplora` · `test_spectra_olo` · `test_impronta` · `test_voce` · `test_guidaproxy` · `test_fluidita` · `test_avvio` · `test_giro` · `test_lingue` · `test_mol3d` · `test_spettrolettore`

`test_spectra238` — 66 checks. It comes from reports made with a phone screenshot: with the app installed, Android's status bar and navigation keys covered the controls of the Spectra Live console, and the model attribution overlapped the dialogue. Since v239 it also checks, first of all, that Spectra has **a single visible button**: the circle with the face (`#bsiJV-fab`) is visible, the «Spectra» pill (`#bsi-spectra-fab`) is not, and the circle is round (`border-radius: 50%`). The rest of the bench works in four parts:

Part	What it requires
Layout	On a simulated phone (412×915, touch, full-screen window) the safe-margin class is active, the first button at the top is at least **34 px** from the edge and the controls at the bottom at least **46 px**; the telemetry is in two columns and does not cover the face. On a large screen no forced margins. On both: header above the face, face above the state, state above the dialogue, dialogue above shortcuts and controls, attribution inside the control bar; at least four shortcuts with a section open
Navigator	For chemistry, physics and mathematics the 🧭 navigator lists **exactly** the sections of the bar (more than 20 per subject), at least four quick tools and ten phrases; opening a section of another subject switches subject, closes the console and leaves the Spectra Live button
Level and concentration	An advanced-course question weighs more than a sum (gap > 0.2); «in simple words» lowers it; the same question weighs more (gap > 0.3) in a PhD-level section than in a high-school one, and the declared level is the path's
Creator, eyes, sleep	«who created you?» names Samuele Pio Provenzano; «I am your creator» asks for proof and the field becomes a password field; a wrong phrase is rejected, the field goes back to normal and the phrase remains neither in the dialogue, nor in the page, nor in memory; after three attempts the wait; recognised (the bench writes the `bsi-spectra-creatore-v1` mark in `localStorage`: the phrase does not appear in the bench): greeting by name, telemetry «SAMUELE · CREATOR», context for the model and the mind; «forget that I am your creator» removes it. Eyes: joy narrows them (< 0.8), surprise widens them (>

Part	What it requires
	1.25), doubt makes them asymmetric, `#1a3050` at rest, four emotions give four tints, concentration narrows and brightens them, the blink closes the eyelids (< 0.2); after two minutes without contact it sleeps, at the first touch it wakes with surprise

Still outside, as `docs/08` §6 declares (AI-13): voice, continuous listening and the other local commands, which cannot be tested in the battery's headless Chromium.

`test_lezione` — 30 checks. It comes from the request for an interactive lesson «for every section» (`bsi-lezione.js`). Since the lesson is not written by hand but built from the section, the bench does not test one lesson: it tests them all.

Part	What it requires
All sections	It opens the sections of the bar one by one in the three subjects (switching subject when needed) and requires at least 190 of them to be visited (**194** at the time of writing). For each: the lesson exists and runs from the opening to the summary; the «🎓 Guided lesson» button is there; if the section has tabs (**99**), the chapters are one per tab plus opening, summary and the check if any. At least 60 lessons must have a check question (**79** at the time of writing). No JavaScript errors
The player, at 390 and 1280 px	On 〰️ Waves and optics: the button opens the lesson from the opening; floating buttons go from visible to hidden; at the first slider the value really changes (at least four different values in 4.5 s) and the spotlight is on; skipping ahead half-way the slider returns to its starting value; a select goes through at least two options and returns to the starting one; the check shows the right answer and an explanation longer than 20 characters; the summary lists the chapters; nothing overflows; on closing the panel disappears, the `bsiLZ-aperto` class is removed and the floating buttons are visible again; `BSISpectraLive.comando('fammi lezione')` answers and starts the lesson; no JavaScript errors

Still outside: the lesson's voice (there is no speech synthesis in headless Chromium) and the teaching quality of the texts, which depends on the sections' explanations and has to be judged by reading.

`test_spectra_studio` — 87 checks, on the real embedded body (`models/spectra-corpo.bsi`, since v241). It tests embodied Spectra (`spectra-studio.html`, `bsi-spectra-figura.js`, `bsi-spectra-corpo.js`, `bsi-spectra-voce.js`, `bsi-spectra-studio-lezioni.js`; design in `docs/spectra/02`).

Part	What it requires
Computations (23)	Runs `verifiche/verifica-studio-spectra.js`, also standalone with `node`: fall from 20 m in √(2h/g) = 2.0193 s for both spheres, range 12.116 m, with air (k = 0.02 m^{-1}) the thrown one lands later (2.1789 vs 2.1563 s), halved step changing the time by less than 10^{-4}; VSEPR: tetrahedron 109.471°, trigonal 120°, linear 180°, water 104.40° with w = 1.25 (calibrated), ammonia 106.69° with the same w (prediction), angle decreasing as w grows; Basel: S(100 000) within the tail of $\pi^2/6$, tail between 1/(N+1) and 1/N for N = 10, 100, 1000, Fourier series of x^2 at π and at 1, harmonic series as ln N + γ, coprimes → $6/\pi^2$ within 3σ; for each lesson exactly one right answer, every option explained, at least three proof steps, exercises on four levels
Body (11)	On the mannequin: bones 22/23 (no jaw), blendshapes 14/14, hair found and turned black (RGB 0.039 0.039 0.047); arms down from the T-pose; pointing (forearm on the target, 1.7° off); thinking (hand 20.8 cm from the head); gaze to the right; walking

Part	What it requires
	(0.96 m, legs at 24°); sitting (hips down 42 cm); joy → smile, worry → mouth corners down and smile at zero; lip sync from audio (jaw 0.75); blink (eyelids closed)
Voice (3)	In an offline audio context: at 80 Hz the bass comes out boosted (6.9 dB), the reverb leaves a tail after the sound ends; without a key `qualita()` is `'sistema'`
Lessons, at 1280 and 390 px	The «Noir» face resting on the neck (the reference is the neck bone plus 4.5 cm, where the chin rests on the collar); for fall, water and Basel the measurements in the scene (2.019 s, 109.47° and 104.4°, S_{1000} = 1.64393 and coprimes 0.6079), four predictions, formulas in the proof, the reveal with the right answer and four exercises of which one recommended, the memory remembering the prediction and the exercise opened; at 390 px nothing overflows; no JavaScript errors
Rigged body, at 1280 and 390 px (3 + 3)	On the **real embedded body**. **Face–hood sync**: while Spectra walks, makes the «spiega» gesture and looks at 24 different points, the face position measured in the head bone's frame moves by no more than 1 mm (measured 0.0 mm); **of the «Noir» model only the face is visible** (only `Object_2`, `Object_3`, `Object_4` are visible: mask, eyes, details); the face is **«solidale»** (head movement is done by the body, and the face does not do its own breathing inside the hood)
Embedded body, at 1280 and 390 px (4 + 4)	Spectra's body is always there, without loading it (embedded), and the «Spectra sta arrivando…» notice is gone; no «🧍 Corpo» button and no file choice; the published file is a packed block («BSIC» signature), not a reusable `.glb` model; a block that is not a model is rejected, without JavaScript errors
Voice and data (2)	The muted voice stays muted after a reload; «Clear data → Persistent memory» removes `bsi-spectra-studio-v1` and `bsi-spectra-studio-voce`
Without WebGL	No scene, but the lesson gives the measurements (2.019 s); no JavaScript errors

Since v241 the rigged-body checks (chin on the collar, face–hood sync, face only, «solidale» face) run on the real «Bounty Hunter» body, embedded in the app; the synthetic humanoid injected by the bench and the checks on a body loaded from the device are gone. Still outside: the original `.glb` (`models/spectra-corpo.glb`, out of git), which the studio tries only in development; the call to OpenAI and listening to the voice (the bench does not reach OpenAI and does not listen); the visual quality of the body and gestures, which has to be judged by looking.

`test_spectra_figura` — 111 checks (55 at 412 px, 54 at 1280 px). It comes from the author's request: Spectra's icon with the face only and, in every AI section, Spectra with the body, «and nothing else». It tests `bsi-spectra-figura.js` in the app, on a phone (412 px) and on a large screen (1280 px). Since v241 the bench uses the **real body**, embedded in the app (`models/spectra-corpo.bsi`).

Part	What it requires
Icon	Spectra's button is the face only; the «Spectra» pill is hidden
The four sections	In Spectra AI (`sai`) and in the areas with the assistant (`sareachim`, `sareafis`, `sareamat`) the figure is **inside the section, under the title**, at least 280 px tall; in the whole app there is **one figure only** (one WebGL context); for the sections outside `#main` no empty space above (`#main` at most 12 px tall; measured 0)

Part	What it requires
The embedded body	Spectra's body is always there, without loading it: embedded in the app (bones 20/23); no «🧍 Corpo» button and no file choice (the buttons are «🎙️ Parla con me», «🧠 Spiega», «🎓 Studio»)
With the body	The «Noir» face rests on the collar (0.0 cm, measured at steady state: the bench waits for the alignment of the first frames to finish); of the face model only `Object_2`, `Object_3`, `Object_4` are visible
Spectra Live console (3 per width, v242)	With the console open, **Spectra with the body (and the hologram) takes the place of the bust**: the figure is inside `#bsiJV-figura`, more than 150 px tall, and the bust `#bsiJV-volto` is hidden; the figure follows the console's state (working → «pensa» gesture, hand on chin); with the console closed, the figure returns to the section
Reactions	`bsi-spectra-stato` «pensa» → «pensa» gesture and concentrated emotion; `bsi-spectra-risposta` → Spectra first points at the hologram («indica» gesture), then accompanies the explanation with the hands («spiega»), joy, lip sync on body and face; a failed answer → doubt; Spectra Live speaking → the figure speaks with it; the local assistant solving «deriva x^2» → «Ecco lo svolgimento…» and a warm smile
«🧠 Spiega» (14 per width)	«🧠 Spiega» is the second tab of the Spectra window and «Lezione» is gone; the figure moves into the tab; with a spectrum open («Spettri Ref.» in the Spectra centre) it reads the screen: at least one graph, the control values, the image; five levels; the model receives the screen text, the question and **the image**; the request asks for an exhaustive, non-simplified explanation, with the quantitative reading of the spectrum (δ, multiplicity, J, integral) and of 3D scenes; the mandatory structure; the answer arrives formatted (titles and table) and Spectra thinks while it is written; the follow-up question continues the conversation without re-sending the image; at advanced PhD frontier, open problems and primary literature; without a model it does not pretend; reads SVG graphs as an image (since v242 the bench inserts a known SVG spectrum into the open section, deterministically, instead of relying on the spectra quiz, which is random); reads a 3D scene («Meccanica orbitale 3D»: 2 canvases, image); with that scene open behind «🧠 Spiega» the hologram shows it live; privacy: with «Includi lo schermo» unchecked, the named section opens but the model receives only the question, with no screen text or image
Hologram (7 per width)	It is **to Spectra's right** (screen x); there are **six live scenes**; Kepler: the ratio of perihelion to aphelion speed equals (1+e)/(1−e) = 1.94 within 8 %; **the subject follows the conversation** (an IR answer → spectrum, an orbit question → planets); the label leads to the section; a tap on the label opens «Sistema solare 3D» **behind** the «🧠 Spiega» window: Spectra stays visible in front and the hologram shows the section live; once the window is closed, the app returns to Spectra's section and the figure is there
Catalogue (5 per width)	The catalogue has **every** section of the bar (201/201 since v250: chemistry 78, physics 63, mathematics 31, shared 29); a question that names a topic finds its section (titolazioni → `stitr`, Schrödinger → `sschrod`, integrali per parti → `smatcalc`, spettro di massa → `sms`, spettro infrarosso → `sirvis`, meccanica orbitale → `sastroorbite`, «ciao come stai» → none); asking about titrations, Spectra opens the section behind the window, shows it in the hologram and explains it with its data; a section without graphs (Periodic Table) is projected as a card on the hologram; «🎞️ Mostrami tutto» runs through the subject's sections
Phone	The figure's buttons do not end up under the floating buttons
Errors	No JavaScript errors, at each width

The bench scrolls the figure into view before measuring: the figure renders, and aligns the face, only when it is visible.

Still outside: the visual quality of the figure and gestures, which has to be judged by looking; the voice; the quality of a real model's explanation: the model's answers are simulated, and the bench checks what is sent (screen text, image, level, requested structure) and how it is shown, not what a real model would explain.

`test_spectra_esplora` **— 75 checks (v242).** It comes from the author's request: «there is no section for each subject like every-joule». It tests `bsi-spectra-esplora.js`: in node the chapters and the slider numbers, in the browser (412 and 1280 px) the three «⚡ Explore with Spectra» sections.

Part	What it requires
Chapters (12, in node)	For chemistry, physics and mathematics: **ten chapters**; levels from 0 (high school) to 4 (advanced PhD · frontier), in order; every prediction has **four answers**, the index of the right one and **a reason for each**; the ids are unique
Slider numbers (22, in node)	Against the formulas: pH of HCl 10^{-8} M = **6.978** with water autoprotolysis, 10^{-2} M = 2; Henderson–Hasselbalch pH = pKa at ratio 1; $t_{½}$ = ln 2/k = **6.931 s** at k = 0.1 s^{-1}; Marcus, zero barrier at $\Delta G° = -\lambda$ and **λ/4 at −2λ** (inverted region); fall from 20 m in **2.019 s**; range at 45° with v_0 = 20 m/s **40.775 m**; 1 m pendulum **2.006 s**; Carnot between 600 and 300 K = 0.5; particle in a 1 nm box **0.376 eV**, ×4 when halved; tunnelling, double thickness → transmission squared; muon γ at 0.9995c = **31.63**; derivative of x^3 at 2 = 12; $(1+1/n)^n$ with $n = 10^6$ within $2\cdot10^{-6}$ of e; $\int_1^e dx/x = 1$; Basel with 10^5 terms = $\pi^2/6 - 10^{-5}$; harmonic with 1000 terms **7.4855**; eigenvalues of the 90° rotation ±i; Gibbs maximum with 201 terms ≈ **1.179**; heat mode k = 2 at t = 0.1 → e^(−0.4)
The bar (1 per width)	The three sections (`sesplchim`, `sesplfis`, `sesplmat`) are in their subject's bar, and every chapter links to a section that exists
Per subject (5 per subject and width)	Ten chapters and **Spectra with the body in the section,** `sticky`**, still in view after scrolling** (under the app header); the chapter in front (the fifth, brought to the middle) is **active** and the hologram takes its scene; predicting with a wrong answer marks the right and the wrong one, with the reason for each answer, and **Spectra reacts** (surprise); the memory (`bsi-esplora-v1`) remembers the prediction and the bar advances (1/10); the slider recomputes
Phone (3)	At 412 px nothing overflows, for each subject
Watch it live (2 per width)	«📺 Guardalo dal vivo» opens the linked section **behind** the Spectra window («🧠 Spiega» tab) and the hologram shows it live («schermo» scene); with the window closed, the app returns to the Explore section (`sesplmat`)
Errors (1 per width)	No JavaScript error

Left out: the voice of «🔊 Ascolta Spectra» (headless Chromium has no speech synthesis); the explanation of «🧠 Spiegamelo tutto», which depends on a model (the path is the one of «🧠 Spiega», tested by `test_spectra_figura`); the didactic quality of the texts, which has to be judged by reading.

`test_spectra_olo` — 69 checks (15 in node, 28 at 412 px, 26 at 1280 px; v242, v248). It comes from the author's request to select the hologram and interact with it, with every step explained. It tests `bsi-spectra-olointerattivo.js`.

Part	What it requires
Spectra (8, in node)	Six reference molecules, each with the five spectra (IR, ^{1}H, ^{13}C, mass, UV); molar mass of acetone **58.08 g/mol**; degree of unsaturation benzene 4, acetone 1, ethanol 0; the CH_3 triplet at 400 MHz **2J/ν₀** wide; at 60 MHz the multiplet is **10 times** wider than at 600 MHz; IR trace: 1715 → C=O, narrow 2950 → sp^3 C–H and **not** acid O–H, broad 3300 → O–H; MS trace: 58 → molecular ion, 43 → M − 15 (CH_3·) and the CH_3CO^+ acylium
Universe (7, in node)	Earth: Kepler period 1 year, vis-viva **30.29 km/s** at perihelion and **29.29 km/s** at aphelion; Mars: speed ratio = (1+e)/(1−e), surface gravity **3.71 m/s²**; Earth escape velocity **11.19 km/s**; a written orbit with a = 4 AU → T = 8 years
The window (3 per width)	The «⛶ Interagisci» button is on the hologram and opens the window in «spettri» mode (because the hologram was showing a spectrum); Spectra with the body moves into it; the app's floating buttons disappear
Live spectra (3 per width)	Changing type (IR) and writing a trace redoes the spectrum and the steps (1715 and 2950 cm^{-1} among the steps); a mass trace with two peaks gives two steps; tapping the peak at m/z 43 selects its step and Spectra says it
The ✨ menu in the hologram (v248: 3 per width)	The list also has the **15 ✨ menu entries** (8 pages and 7 windows: Astrochimica 3D, Lab Quantistico, Accademia, Glossario…) and the Centro spettri; «Astrochimica 3D PRO» opens **inside the frame** (its real page) and is removed on changing mode; for a window (Lab Quantistico) «↗ Apri la sezione» closes the hologram and opens it
«🗂️ Tutta l'app» (v248: 4 per width)	The list has **all** the sections of the bar (201/201, read from the catalogue, not hand-written), filterable by subject and by name; the Periodic Table, with no graphs, is **really hosted** in the frame (the real section, moved and alive); choosing another section puts it back and «Sistema solare 3D» is shown **live** with its lesson; once the window is closed the app is back in the starting section
Reading on the phone and audio test (v248: 2 at 412 px, 1 at 1280 px)	On the phone **the whole window scrolls**: the last step reaches at least 70 px from the edge (above the Android keys), the header stays on top, the steps are at 15.5 px; «🔔 Prova audio» plays a beep, says a test sentence and explains what to do if nothing is heard (mute, «Media» volume, phone voice)
Voice, tap and Spiega without a key (3 per width)	Once a trace is written, Spectra explains it **aloud on her own** after 1.4 s, without touching «🔊»; a tap **on the hologram itself** (the panel to Spectra's right, hit with a ray from the camera) opens it in interactive mode; «🧠 Spiega» **without a key** still answers the question (pH of 0.01 M HCl: the local assistant's full working) and says it aloud
Universe, mathematics, physics (5 per width)	Mars with its six steps, in the 3D scene; «studia f(x)=x^3-3x» → zeros −√3, 0, √3 and stationary points ±1, with at least eight assistant steps; the projectile with range **35.31 m**; the 2 m pendulum with period **2.837 s**
Voice, Spiega, return (3 per width)	«🔊 Spectra spiega i passaggi» says them all; «🧠 Approfondisci» opens «🧠 Spiega»; with the windows closed, Spectra returns to the section
Phone (1)	At 412 px nothing overflows
Errors (1 per width)	No JavaScript error

Left out: the real voice (headless Chromium has no speech synthesis: the bench checks that every step is spoken, not how it sounds); a real model's explanation after « 🧠 Approfondisci»; the chemical correctness of every value of the six molecules, which are typical literature values for teaching, not measured spectra; the trace interpreter proposes candidate assignments from tables, it does not determine a structure.

`test_impronta` — 25 checks (v243). It comes from two reports by the author on v242: «I can't hear the voice» and «I'd like it to be me, with a fingerprint recognition, but only with me». It tests `bsi-spectra-creatore.js` (WebAuthn) and the voice in `bsi-spectra-live.js`.

Part	What it requires
The fingerprint (12)	With Chromium's **virtual authenticator** (DevTools protocol, WebAuthn domain) standing in for the sensor: without the secret phrase the fingerprint is not bound; a wrong phrase does not recognise; bound after the phrase, only the credential `id` and the date are stored; in a new session the browser's memory is not enough and Spectra waits for the fingerprint; the right fingerprint with a verified user recognises the creator and the model knows; with an **unverified** user, or **another** authenticator, no recognition
The console (6)	«attiva l'impronta» binds it and Spectra says so; opened in a new session the console asks for the fingerprint with «👆 Sono io»; tapped, recognised and greeted by name; «prova la voce» answers; `statoVoce()` readable; «dimentica che sono il tuo creatore» also removes the fingerprint
The voice (3)	A fake synthesis that starts after 2.5 s **without «onstart»** (like some Android engines): **no sentence cancelled** and 2/2 reach the end (before v243 it was cancelled after 1.2 s: the cause of the silence); a synthesis that never starts does not block the speech: lip-sync carries it forward
Chat and overload (3)	«sono il tuo creatore» typed in the 💬 **Chat** tab does not go to the model: it closes the window and opens the console with the hidden field (0 calls to the model); Gemini answering **503** on the first model: it moves on its own to another model with the same key and the answer arrives, in under 5 s
Errors (1)	No JavaScript errors

WebAuthn does not accept an IP address as a domain: the bench opens the app from `localhost`. **Not tested**: a real phone sensor and audible sound.

`test_voce` — 21 checks (v246, v248). It comes from the author's request for a male robotic voice, «Ultron style», for every Spectra answer in the Chat, using only the Web Speech API. It tests `bsi-ai-hub.js` (`speakRobotic`, `stopSpeaking`, `toggleAutoSpeak`, `createSpeakButton`) with a **fake** speech synthesis that records every sentence and offers Italian and English, male and female voices.

Part	What it requires
Voice and parameters (5)	The Italian male «Luca» is chosen; every sentence at pitch 0.58, rate 0.88, volume 1; a 1228-character answer read **in full**, sentence by sentence (before: 600 characters); no markdown read aloud; no cancel when nothing is in progress
In the Chat (9)	The «🔊 Leggi automaticamente» switch is there and is remembered on and off (`spectraAutoSpeak`); when on, Gemini's answer (fake network) is read by itself; a 🔊 button on every answer, which becomes «■ Ferma» and stops; the voice stops on closing the window and on changing section

Part	What it requires
Console (1)	Spectra Live uses the same parameters
Android (4, v248)	With an Android phone's identity: no empty unlock utterance on first tap; no voice chosen by name, only the `it-IT` language; console and Chat speak; after a cancel the new sentence starts ≥ 120 ms later
Without synthesis (1)	No button, no error
Errors (1)	No JavaScript errors

Not tested: the real sound of a phone voice.

`test_spectra_rig` **— 30 checks.** It tests the automatic rig (`bsi-spectra-rig.js`), which gives a skeleton, skin weights and the app's colours to a humanoid figure without bones. The real model is not in the repository, so the bench builds in code a **synthetic humanoid in A-pose**, a single mesh: torso, arms and legs as continuous tubes, a head with a hood. The hard case is deliberate: the hanging hand is a few centimetres from the flank, and as the crow flies the flank is closer to the arm bones than to the torso's.

Part	What it requires
Skeleton (5)	Rig built in under 3 s; **27 bones** (20 weighted and 7 tips); Mixamo names (`Hips`, `Spine2`, `Neck`, `Head`, `LeftArm`, `RightForeArm`…); the body driver recognises them (at least 20 of 23) and can make it walk
Weights (5)	Weights sum to 1 on every vertex; the hand's vertices go to the hand and the forearm; **the torso flank next to the hanging hand gets no arm weight at all** (the defect that tore the torso when an arm was raised); a leg never takes weight from the other leg's bones; the left leg goes to its own bones (over 99 %)
Movement (3)	In the «spiega» gesture the hand moves (half a metre) and the flank stays still (1 mm); walking moves the body
Head and hood (8)	Without a head the triangles above the neck are removed; with the hood fewer are removed, because the back of the head stays; with the head none; **with the face only, no triangle is removed**: the opening is an ellipse computed per pixel in the shader, on the rest pose, so it follows the head; with the `visiera: false` option (available, not used by the studio, which keeps the glossy black visor) the helmet under the hood becomes matte shadow without removing vertices; through the opening the inside of the helmet is visible (double side), not empty space; the hood is rigid with the head bone (all its vertices)
Colours (5)	Mid grey becomes night blue; shadows stay almost black; highlights stay light, slightly blue; turquoise becomes teal emissive light `#00c9b7` (0, 201, 183); grey emits no light
Definition (3)	With Phong tessellation (`suddividi: 1`) every triangle becomes four; the weights stay normalised on the new vertices too; the skin stays closed: open border edges only double (each border edge becomes two), no seam opens
Errors (1)	No JavaScript errors

The real body, embedded since v241, is tested by `test_spectra_studio` and `test_spectra_figura`. Still outside: the visual quality of the folds (elbows, knees, armpits) and of the hood, which has to be judged by looking.

`test_controllaagg` **— 12 checks.** It comes from a report: «trying to update, nothing comes up». It was true. The 🔄 Updates window looked in `sw.js` for a `CACHE_VERSION` line that no longer exists (`sw.js` declares `var CACHE = 'bsi-vNNN'`) and looked for it on `raw.githubusercontent.com`: the published version was never read. `test_aggiorna` opened the window but did not press «Check», which is why the defect survived. The new bench presses the button in three cases, obtained by intercepting only the check request (`sw.js?controllo=`):

Case	How it is produced	Expected answer
Same version	`sw.js` served as is	«You already have the latest published version», with the number; no «Update now»
New version	`sw.js` served with a higher cache name (v999)	«New version available» with both versions, and «Update now»
Site unreachable	request aborted, and the GitHub fallback too	«Cannot check now»

In the same three cases it tests `BSIAggiorna.controlla()` (`'aggiornata'`, `'nuova'`, `'off-line'`), its pop-up notice and, in the second, that it asks the service worker to update.

`test_fluidita` **— 15 checks.** It measures how long the page stays *blocked* at every section switch, across all 91. It found three defects no other bench could see: the "3D PRO Viewer" froze the page for **797 ms** on the first click — the profiler attributes them to 3Dmol's first `render()`, which compiles the WebGL context's shaders; the **296 figures** in "Synthesis" (5,351 SVG nodes) were all drawn before anything was shown, while three are visible on opening; and **296 timers 40 ms apart** were scheduled when that section opened — eleven and a half seconds of wake-ups, each calling `getTotalLength()` on every stroke of every figure, including those off screen.

After the fixes: median switch **16 ms**, worst **87 ms**, **no section above 100 ms** (there were two, the worst at 1,166 ms).

At version `bsi-v188` the same bench caught two things. First: with 263 drugs instead of 233, «Pharmacology» blocked the page for **159 ms** and the count of sections over 100 ms went back to **three** — the battery had passed by a hair the time before, and a limit exceeded intermittently protects nothing. The cards are now drawn in 8 ms slices: **55 ms**, zero sections over 100.

Second concerns the bench itself. The viewer threshold was **80 ms absolute**; with the same published code, on different containers, the same click measured 14, 22, 98, 105 and 192 ms, because WebGL here is **SwiftShader** and the cost is shader compilation. Such a threshold measures the machine: it fails on identical code and would pass on worsened code. Even a **ratio** against a calibration taken in the same run does not hold — by the time the bench reaches the viewer WebGL is already warm (2.9 ms) while the viewer pays the cold cost: the ratio swung between 14× and 36×. The claim became **structural**, which is also the original defect: immediately after the click handler returns the WebGL context **must not exist**, shortly after it **must**. Measured 0 and 1; and the measurement is not vacuous, because on a second click — when the canvas is already there — it returns 1.

`test_avvio` **— 9 checks (v247).** It comes from the author's request to optimise the app. The start-up profile on a simulated slow phone (CPU ×4) showed a single ~2 s block: two preloads that started on their own (the name archive of «Linguaggi delle molecole» and the RDKit warm-up). The bench requires the **behaviours**, not the milliseconds, which are not repeatable on a test machine: with the app in use RDKit does not start by itself and the archive is not built; in Physics and with «Data saver» it does not start even when still; in Chemistry when still it warms up; the archive is built on opening «Linguaggi delle molecole» (283 molecules); whoever asks for it gets it at once; no JavaScript errors.

`test_giro` **— 24 checks (v249).** The inspection round of the whole app, born from the author's request to check everything to the utmost. It opens **every catalogue section one by one** (201) at 412 and 1280 px and for each requires: that the section asked for by identifier is the one that opens, that it is visible and not empty, that the page does not overflow horizontally, that the text shows no «undefined», «NaN» or «[object Object]», and no JavaScript error. Then it opens the **14 separate pages** (Astrochimica, Accademia, Chimica Organica, RDKit Lab, PRO Hub, Simulazioni, the guides, the Studio…) with the same checks. The external network is closed. The first time it found six sections that opened a similarly named one, 36 reactions printed as «undefined» and four pages overflowing on the phone.

`test_lingue` **— 64 checks.** It watches two new features and two different ways of lying without noticing.

The language. A switch that "translates the application" is easy to write and hard to keep honest: someone changes an Italian label and its translation is no longer found. The defect is invisible — the label stays Italian in the middle of English — unless someone counts. The bench counts: **160 skeleton elements, 160 translated, none left**. It also demands the two opposite proofs: that switching to English **actually changes** the text (an empty dictionary would pass "everything translated" without moving a letter) and that switching back to Italian restores it **character for character**.

The molecular languages. Conversions are checked against external facts — seven literature InChI keys **written inside the bench** — and the full SMILES → molfile → SMILES round-trip is required. Formulas are counted by hand.

The IUPAC namer. 26 names written by hand in two languages, and seven molecules that **must be refused**: a namer that never refuses is a namer that invents.

Tested in both directions: restoring the `trim()` that breaks molfiles and removing one translation makes the bench fail three checks out of twenty.

The doors. Since v184 the bench also checks that the two features are **reachable**: a navigation button, a section that really draws, and — a check born from a real defect — that neither shows the "🚧 Section under construction" placeholder that the empty-section fallback stamps 800 ms after a click into any section left empty. It also checks that the ✨ menu entry **navigates** to the section rather than opening an overlay window, and that no duplicate id exists with both sections drawn.

The six languages. For each one two things are demanded together: full skeleton coverage (**162 of 162**) and that the text **really changes** — an empty dictionary would pass "everything translated" without moving a letter.

`test_mol3d` — 46 checks. It watches the molecule-formation animation and the bond-angle measurement, two additions with two different ways of looking right without being right.

The animation. "It is there" or "it is not" cannot be seen by inspecting the code: a loop that spins and redraws the same thing passes any check on the function's existence. The bench looks at the **pixels**: at 130 ms the canvas must be nearly empty — dust — and at 2.3 s there must be a molecule. Measured: **532 lit pixels → 2,334**. If the two numbers were equal there would be no animation, however the program is written. It also checks that the loop **stops** when stopped: a removed canvas with the loop still running is a drain paid in battery.

The angles. A number with a "°" next to it looks like a measurement. They are checked on hand-built geometries whose value is known **by construction**, not taken from a table: regular tetrahedron **109.4712°**, water 104.47°, CO_2 180°, BF_3 120°, ammonia 106.13°. Tolerance 0.05°.

The refusal. On a **flat** structure (z = 0) the angles button must be disabled and the reason written: on 2D coordinates a bond angle is not a bond angle, and showing it would be the worse of the two errors because it looks like data. Tested both ways: a structure with $z \neq 0$ must not be mistaken for flat, or no angle would ever be shown.

The first draft of this bench read the control bar **after** stopping the loop, which removes it: it found zero buttons and then declared "each has an accessible name" over an empty set. A check that passes because it has nothing to look at is exactly the defect this project chases; the check now demands two buttons **and** two names.

The selection. The viewer chooses the angle by clicking atoms: one shows all the angles at that atom, two the bond length, three the angle with the vertex at the second click, four the **dihedral**. The bench checks all four cases on ethanol — C–C 1.509 Å, C–C–O 111.9°, H–C–C–O 60.7° — and demands that two **unbonded** atoms be declared as such rather than passed off as a bond: a distance is not a bond.

The dihedrals. Checked on four hand-built conformations: eclipsed 0°, gauche 60°, orthogonal 90°, anti 180°. Here too the hand-written table corrected an error of mine, not of the program: with the coordinates I had chosen the dihedral is 120°, not 60°, because looking along the central bond the projected vector sits at (−0.5, 0.866).

The arrival order. "Skeleton first, hydrogens after" is a **measurable** property, not an intention: the bench reads each atom's progress halfway through the formation and requires every heavy atom to be ahead of every hydrogen. That check found a real defect: the elastic function that produces the bounce is not monotonic, and using it as the progress too meant an atom that started earlier could look "less arrived" than one that started later. The two quantities are now separate — the progress is monotonic, the elastic factor only does the bounce — and the bounce, previously clipped away, is visible.

Half the checks are the *opposite proofs*, because a page that builds nothing is instantaneous: the WebGL canvas must appear anyway, all 296 figures must exist shortly after, no frame may stay empty, printing must not come out mute, and the search must filter even cards whose figure is not yet born. And the bench counts the sections it traversed: on ten instead of 89 the timings would be excellent and the measurement worthless.

3.5 Security and accessibility

Bench	Verifies
`verifica-sicurezza`	API keys in tracked files, clear-text passwords, secrets in `wrangler.toml`, telemetry, scripts from external domains
`audit_storia`	That no credential has EVER entered the repository: **1,551 distinct versions of text files** across **390 commits**, with 9 patterns. It refuses to pass on a shallow clone
`verifica-accessibilita`	WCAG AA contrast, accessible names, field labels, alternative text, heading hierarchy, `lang` attribute — across 13 pages
`audit_mobile`	That at **390 px** the page does not scroll horizontally, across all 194 sections — **and** that no section clips content into unreachability: an element wider than its section, with no scrolling ancestor, is text that is present and cannot be read. A wide table inside a scrolling container is not reported
`audit_rete`	That **no user data leaves the device**: a canary value seeded into 71 stores, the app used across 6 pages and 194 sections, the URL, headers and body of every request inspected
`audit_copertura`	How many bytes of JavaScript are **actually executed** while walking the application: 49.89 %, recorded and defended

`audit_storia` — 6 checks. `verifica-sicurezza` examines *tracked* files, i.e. the present state: it answers "there is no key in the repository today", not the question that matters for a public repository. A key added in one commit and removed in the next passes `verifica-sicurezza` forever and stays readable to anyone who clones, because git does not forget. If that happens, the remedy is not a fix-up commit: it is revoking the key.

This bench examines every *version* of every text file ever committed. Result: **zero credentials across 1,551 versions and 390 commits**.

Two defences against itself, because here a perfect number can come from less surface or from a blind pattern:

- **The clone must not be shallow.** A `--depth 1` holds a fraction of the

history: the bench would run, find nothing, and claim "no credential anywhere in history" having seen a tenth of it. If it finds `.git/shallow` it fails and names the command to run.

- **The patterns are tested before being trusted.** A wrong pattern finds

nothing and is indistinguishable from a clean repository: each of the 9 must match a crafted example and must *reject* a near-miss, and both are tested inside a tail of noise containing digits, as a real file would have.

The second defence was born from a defect in this very bench. The AWS pattern required a digit via `(?=.*\d)` — always satisfied in a large file, because the digit lies further ahead: it found 53 matches, all the same string `AKIAAAAAAAAAAAAAAAAA`, sixteen "A"s, the zero region of a base64 image. The near-miss did not catch it because it was too short to contain digits. An AWS identifier is now judged by the *variety* of its sixteen characters, and the near-misses carry their tail.

Tested in the other direction too: with a local commit containing a well-formed AWS key the bench fails naming the blob; once the commit is removed, it passes again.

> **The obstacle belonged to the tool, not to the problem.** For three versions code coverage was declared unmeasurable, with this reason: "instrumenting would require introducing a build, which the application does not have". That was true of `c8` and `istanbul`, which rewrite the source before running it. But **Chromium collects it by itself**, inside the engine, without touching a byte of the file served: `Profiler.startPreciseCoverage`, which Playwright exposes as `page.coverage.startJSCoverage()`. No build was needed: what was needed was noticing that the constraint belonged to the tools chosen.
>
> **The first result was 100 % on every file**, including 3.6 MB of `index.html`. That was the bench measuring nothing, in its most insidious form: a perfect number. V8 emits, for each function, an outer range with the number of entries, and inside it ranges with `count: 0` for what was not executed. Summing the positive ranges sums the whole file. One does the opposite: start from the total and subtract the union of the empty ranges.
>
> The real number is **49.89 %**, and it holds the same pact as the accessibility debt: recorded, and the bench fails if it falls. An absolute threshold ("80 %") would be an invented figure; "it must not get worse" is a promise that can be kept.

> **Why a canary and not a list of requests.** SEC-02 — "no personal data leaves the device" — was verified by checking the two *mechanisms* of exit: empty telemetry variable, no script from an external domain. That is real evidence but indirect: it says "I cannot see how it would leave", which is not "it did not leave". Looking instead at the list of hosts contacted would be just as weak, because it concludes from absence of evidence.
>
> The canary reverses the burden. An unrepeatable string is written into the user's own data — notes, chats, API keys, settings, progress — then the application is **used**, and every outgoing request is opened and read. If the string appears nowhere, it is because it really did not leave.
>
> ```
> BSI_PROVA_FUGA=1 node tools/banchi/audit_rete.js   # must FAIL
> ```
>
> The bench carries its own proof of working: with that variable it causes a leak on purpose, towards a **declared** host — contacting it is legitimate, sending it the user's data is not. If it does not fail then, the instrument is the thing that is broken.

> **Why a bench on horizontal overflow.** UI-03 says "the app must work at 390 px", and the matrix gave it as verified by `audit_stabilita`, which in a phone viewport looks at **JavaScript errors**: a page can have not a single one and still run a hundred and sixty-three pixels off the screen. It was a requirement declared covered and not measured. The bench measures the document's `scrollWidth` — exactly what makes the scrollbar appear — and not any element wider than the screen, which inside a container built to scroll is correct and would fill the output with noise until nobody reads it any more.

> **What the accessibility check does not cover**, and it must be said: text inside SVGs (the colour comes from `fill`, the background is a drawn shape) and text on gradient backgrounds (they do not have *a* colour). The skipped elements are counted and reported, not hidden.

> **One section at a time.** The inspection skips elements that are not visible — and rightly so: a hidden element has no contrast to measure. But `index.html` alternates 194 sections and shows only one: out of **19,751** text elements the bench was looking at **41**, and printing "0 defects". It was not a false result, it was a result on a sample nobody had declared. The sections are now opened one by one, and **the number of sections traversed is printed**.
>
> That count earned its keep on the very first run: the first attempt traversed **zero** — Playwright's click waits for the element to be visible, and the buttons sit inside collapsed navigation groups — and the bench said so ("87 sections present, none traversed") instead of printing another reassuring zero.

3.5-bis The accessibility debt, and the pact that it must not grow

Walking every section, **1,069** contrast defects surfaced. Today they are **zero** — the list of causes and remedies is in `docs/09` §4 — and unlabelled fields have fallen from 40 to **zero**.

The mechanism described below remains, and matters more than the number it guards. Demanding zero *from the outset* would have left only two roads, both bad: a verification red for ever, or one loosened until it turned green again. The third road is to **declare the number** and forbid it from growing, whatever it is. That is how 1,069 became zero: every step recorded, none of them reversible in silence.

File	Role
`docs/evidence/accessibilita-riferimento.json`	The recorded count

Situation	Bench outcome
The count **rises**	× FAILED — it is a regression
The count **falls**	✓ passes, and asks for the baseline to be updated
The count is **unchanged**	✓ passes silently: debt declared, not grown — today that value is **0**, and the bench defends it

```
node tools/verifica-accessibilita.js --aggiorna-riferimento
```

To be used **only after reducing** the defects, never to turn a regression green. Even at zero this is not full WCAG conformance: text inside SVGs and text over gradients stay outside, which automation cannot judge and which the bench **counts and reports** on every run (`docs/09` D-09). It is the honest measure of what can be measured, with the guarantee that it will not silently get worse.

3.6 Consistency

Bench	Verifies
`verifica_guida`	That the documentation's promises exist in the code; absence of conflict markers across 58 files
`verifica-documenti`	Internal links, version alignment, index consistency, justification of the deviations, **internal consistency of the traceability matrix** (no duplicated identifiers, coverage totals equal to the identifiers actually defined)
`genera-pacchetti`	Rewrites the references of the extracted documents and **verifies that no link is left broken** in the three delivery packages. It also verifies that **no PDF is older than the document it comes from** and that the bundled verification report declares **this** version: a typeset document or an evidence report from the previous version, inside a current package, is a silent lie
`genera-pdf`	Converts the documents to PDF and fails if a file falls below the plausibility threshold (empty conversion)
`verifica-affermazioni`	Compares every number declared in the documents — sections, drugs, diseases, tumours, strategies, modules — with the one **measured in the running application**, in Italian *and* in English
`test_stack`	That the React/FastAPI stack runs **the same files** as the application: it compares the SHA-256 fingerprints `GET /salute` declares with those of the files on disk, checks the numbers match the browser's, and that **no `.ts` / `.tsx` file contains a shift table** — because that is where the second engine would be born

> **Why numbers must be measured, not remembered.** The technical dossier opens with figures: "63 diseases (23 tumours)", "46 strategies", "25 modules". They are the first thing an assessor reads, and they are enough for everything else to be believed or doubted. Nobody was checking them: written once, they stayed put while the application changed. And indeed `chimorga.html` declared **"25 modules"** at the top and **"17 modules"** further down — two statements about the same object, in the same file, one of them false.
>
> **A lesson paid for while writing the bench.** I had already concluded that "63 diseases" was an exaggeration: in the source I had found a `DISEASES` array with 12 entries, and I was about to "correct" the document. Measuring in the DOM I saw that the menu really does list 63 — the array I had found was a different, smaller one. **A count on the source is not a measurement: it is a clue.** That is why the bench opens the page, opens the section, and counts.

> It also checks that the documents agree **with one another**: two documents saying different things about the same object are a defect even when one of them is right. The English translation counts as one of those documents — `docs/en/05` had been left at "84 sections" while the app had 87, and had been stuck at version `bsi-v146` for three releases — and no bench saw it, because none looked at the English set.

> **Why a check on the matrix.** The matrix declared `SCI-10` twice — once as a requirement verified at 100 %, once as an explicitly **unverified** one — and the summary table added up to 36 requirements while 37 were defined in the document. Both errors sat in the number an assessor reads first. The bench now counts the identifiers actually present and fails if the totals written do not match: the proof that the check really measures something is that, with the two errors reintroduced one at a time, it reports both.

4. Notable test cases

Some benches deserve a mention because they verify properties a functional test does not observe.

Case	Method	Measured result
Memory leaks	194 sections × 5 rounds, DOM nodes counted each round	+26,876 on the first round (construction), **+0** on the four that follow
Exhausted storage	`setItem` replaced with a function that always throws	10 pages out of 10 stay operational
Degraded ≠ absent network	Requests held for 20 s, interception at context level	Answer from cache in **3,507 ms** (threshold 3,500)
Spectrum reproducibility	Drawn twice, compared byte for byte	Identical
WebGL contexts	Viewer constructions over 10 consecutive molecules	From **7 to 1**
Corrupted history	Two `Enter` presses 500 ms apart	`user,assistant` instead of `user,user,assistant,assistant`
Text contrast	WCAG formula on every element with text of its own, 13 pages **and 194 sections**	19,751 elements examined (it was 41): **1,069** defects surfaced, **1,069 corrected**, **0 recorded** as a value that must not grow
Immediate cancellation	Stop pressed at 1.5 s, state sampled every second	Send button available from the **1st** second (it was the 10th)

5. How to add a bench

Minimal structure

```
/* A comment explaining WHAT this bench demonstrates and why it is needed. */
const { chromium } = require('playwright-core');
const CHROME = '/opt/pw-browsers/chromium-1194/chrome-linux/chrome';

(async () => {
  const b = await chromium.launch({ executablePath: CHROME, args: ['--no-sandbox'] });
  // serviceWorkers:'block' if the bench modifies files: without it the SW
  // might serve a cached copy and the old file would be measured
  const ctx = await b.newContext({ serviceWorkers: 'block' });
  const pg = await ctx.newPage();
  const err = [];
  pg.on('pageerror', e => err.push(e.message));

  let ok = 0, ko = 0;
  const att = (d, atteso, avuto) => {
    if (String(atteso) === String(avuto)) { ok++; console.log('  ✓ ' + d + '  → ' + avuto);
}
    else { ko++; console.log('  ✗ ' + d + '\n      expected: ' + atteso + '\n      got:
' + avuto); }
  };

  await pg.goto('http://127.0.0.1:8899/index.html', { waitUntil: 'load', timeout: 60000 });
  await pg.waitForTimeout(2500);

  // … measurements …

  att('no JS error', 0, err.length);
  await b.close();
  console.log('\n' + (ko ? '✗ ' + ko + ' FAILED, ' : '') + ok + ' passed');
  process.exit(ko ? 1 : 0);   // the exit code is what counts
})();
```

The name is then added to the appropriate family in `tools/genera-evidenza.js`.

The rule that matters

> **A bench that measures nothing passes.** It is the most insidious way to obtain a false guarantee: no alarm, no failure, and no verification.

Therefore **every bench that simulates a condition must count how many times the simulation actually fired, and fail if that count is zero.**

Traps actually met while building the battery:

Trap	How it showed itself
`page.route` does not intercept Service Worker requests (`context.route` is needed)	The "degraded network" test measured a healthy network and concluded in 16 ms instead of 3,500

Trap	How it showed itself
Probing exhausted `localStorage` with a one-byte write	After saturation one byte always finds room: the bench "passed"
Looking for `[data-tab]` where navigation uses `[data-s]`	Zero tabs found, therefore zero measurements — and zero failures
The ticks of an axis on `<canvas>` are not text	They must be collected by intercepting `fillText`; and the bottom-most line is the axis *title*, not a tick
A probe installed on a library loaded **afterwards**	It is replaced along with the library: zero calls counted while the code was working
A `var` inside an IIFE is not `window.something`	Assigning it from outside creates a different variable

6. Coverage gaps

Declared. The coverage reported in `08-Traceability-Matrix.md` measures **how many requirements have a bench**, not how many lines of code are executed: they are two different things and must not be confused.

Gap	Current situation	Recommendation
Code coverage	Measured: **49.89 %** of statements over the widest path. What stays outside is whole-battery coverage and **branch** coverage: an `if` entered from one side only counts as covered	Extend collection to every bench, and move from statement coverage to branch coverage
Accessibility	Automated over 13 pages and all 194 sections: WCAG contrast, accessible names, labels, alternative text, heading hierarchy. Text inside SVGs and over gradients stay outside, and are **counted** on every run	Add `axe-core` alongside, for the rules this bench does not implement (ARIA roles, tab order, focus management)
Security	`verifica-sicurezza` runs 10 checks over 616 tracked files and covers SEC-01, SEC-03, SEC-05, SEC-06; SEC-04 is covered by `verifica_guida`. **SEC-02 remains indirect**: see `docs/09` D-03	Observe the network traffic during real use, the only direct verification of SEC-02
Browsers other than Chromium	No automatic test on Firefox or WebKit	Extend the main benches to `webkit`, where the differences on IndexedDB and Service Worker are greatest
Performance	Manual cross-device testing	Automatic measurement of first-paint time
Visual regression	Absent	Screenshot comparison for the charts, which are the heart of the product
Spectra Live and quick tools	Since v238 `test_spectra238` covers the console layout, the navigator, the level, the face and the creator commands; since v239 the single button, and `test_lezione` the guided lesson and «fammi lezione»;	Test the local commands as pure functions (`BSISpectraLive.comando`) and

Gap	Current situation	Recommendation
	`test_spectra_studio` the body, the voice (without OpenAI) and Spectra Studio, `test_spectra_rig` the automatic rig on a synthetic humanoid, `test_spectra_figura` Spectra with the body in the four AI sections, in the Spectra Live console (since v242) and its reactions; `test_spectra_esplora` (v242) the three «⚡ Explore with Spectra» sections, their chapters and the slider numbers; `test_spectra_olo` (v242) the interactive hologram (spectra and written traces, universe, mathematics and physics from the assistant, the steps read by Spectra). Still without a bench: the rest of the local commands, voice, the tools that act on the page and the per-subject ✨ menu (UI-21, AI-13). Speech recognition and synthesis do not exist in headless Chromium	the quick tools in node, as is already done with the sections' computations

7. Acceptance criteria

A version is not published if even one of these is unsatisfied.

- ☐ `node tools/genera-evidenza.js` closes with **0 failures**
- ☐ `tools/verifica-farmaci.js`: no defect, every deviation recorded with its reason
- ☐ `verifica_guida`: no promise in the documentation without a counterpart
- ☐ No non-network JavaScript error on any page
- ☐ Evidence report regenerated against the commit being published

Document updated to version `bsi-v250`.